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Canada

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OECD ECONOMIC SURVEYS

2002-2003

Canada

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BASIC STATISTICS OF CANADA

THE LAND

Area (thousand sq. km)	9 976
Agricultural area (1991, per cent of total area)	6.8
Population of major cities (thousands, 2002)	
Montreal	3 549
Toronto	5 030

THE PEOPLE

Population (2002)	31 363 847
Number of inhabitants per sq. km	3.1
Population, annual net natural increase (average 1998-2002)	108 850
Natural increase rate per 1000 inhabitants (average 1998-2002)	3.6
Labour force (2002)	16 441 301
Employment in agriculture	329 608
Immigration (annual average 1998-2002)	214 149
Average annual increase in labour force (1998-2002, per cent)	1.9

THE PRODUCTION

GDP (million of Canadian dollars, 2002)	1 154 949
GDP per capita (Canadian dollars)	36 824
Gross fixed investment per capita (Canadian dollars)	7 243
Gross fixed investment (per cent of GDP)	19.7
Origin of gross domestic product (2002, per cent of total)	
Agriculture, forestry and fishing	2.1
Mining and quarrying	3.8
Manufacturing	16.8
Construction	5.2
Public administration	5.6
Other	68.5

THE GOVERNMENT

Government current expenditure on goods and services (2002, per cent of GDP)	19.0
Government gross fixed capital formation (2002, per cent of GDP)	2.6
Federal government current revenue (2002, per cent of GDP)	16.9
Federal direct and guaranteed debt (2002, per cent of current expenditure)	242.5

Composition of Parliament (June 2003)	Number of seats	
	House of Commons	Senate
Progressive conservative	15	31
Liberal	169	63
New Democratic	14	..
Bloc Québécois	34	..
Independent	4	4
Canadian Alliance	63	1

THE FOREIGN TRADE

Exports (2002)	
Exports of goods and services (per cent of GDP)	41.1
Main goods exports (per cent of total)	
Agricultural and fish products	7.5
Energy products	12.0
Forestry products	9.0
Industrial goods and material	17.0
Machinery and equipment	23.5
Automotive products	23.4
Other goods	7.7
Main customers (per cent of commodity exports)	
United States	87.2
EU	4.4
Japan	2.1
Imports (2002)	
Imports of goods and services (per cent of GDP)	36.7
Main goods imports (per cent of total)	
Agricultural and fish products	6.1
Energy products	4.6
Forestry products	0.9
Industrial goods and material	19.3
Machinery and equipment	29.7
Automotive products	22.8
Other goods	16.5
Main suppliers (percent of commodity imports)	
United States	62.6
EU	11.2
Japan	4.4

THE CURRENCY

Monetary unit: Canadian dollar	
Currency units per US$	
Year 2002	1.570

Note: An international comparison of certain basic statistics is given in an annex table.

This Survey is published on the responsibility of the Economic and Development Review Committee of the OECD, which is charged with the examination of the economic situation of member countries.

•

The economic situation and policies of Canada were reviewed by the Committee on 12 June 2003. The draft report was then revised in the light of the discussions and given final approval as the agreed report of the whole Committee on 30 June 2003.

•

The Secretariat's draft report was prepared for the Committee by Pietro Catte, David Rae and Richard Herd under the supervision of Peter Jarrett.

•

The previous Survey of Canada was issued in September 2001.

Assessment and recommendations

Recent economic performance has been robust…

Canada's recent economic performance has been stronger than in most other OECD countries. In particular, and despite close trade integration, the 2001 slowdown was shorter and the subsequent recovery stronger than in the United States. Canada's job creation performance, exceptionally robust in 2002 after only a pause the previous year, was also significantly more favourable. It has benefited from a less pronounced ICT investment cycle and a smaller impact of stock price declines, as well as from having a proportionately larger automotive sector, which enjoyed booming demand in 2002 thanks to the earlier monetary policy easings in both countries. But recent performance also indicates an increased resilience to external shocks. This is partly the result of past structural reforms that increased flexibility, but it is also due to fiscal consolidation and the establishment of a credible monetary framework in the 1990s, which together lowered sustainable real interest rates and increased the scope for, and the effectiveness of, counter-cyclical monetary policy.

… and the current slowdown is expected to be short-lived

Economic activity has gradually slowed since mid-2002, with the faltering US recovery curbing export growth and international uncertainties holding back business investment. Nevertheless, household expenditure is still strong, supported by earlier employment gains, high confidence and low interest rates. So far the expansion has been relatively well balanced: there is a comfortable current account surplus, while households and most firms are in healthy financial situations. Although the current slowdown could last a little longer in case of a delayed or sluggish US recovery or because of the impacts of SARS and BSE, economic activity is expected to gradually pick up again

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toward the end of the year and through 2004 as the global economy gathers strength. With some slack in labour and product markets, there is downward pressure on underlying inflation, and headline rates should soon converge on the 2 per cent midpoint of the target range.

Monetary policy has responded to changing economic conditions

The Bank of Canada, which had started to withdraw monetary stimulus in the spring of last year, recently resumed the process, partly in response to disappointing inflation outcomes. Last winter's temporary spike in consumer price inflation was due in part to one-off effects and signs of underlying demand pressure. A sharp fall in actual and expected inflation in the spring, the appreciation of the Canadian dollar, the fall in energy prices and temporary negative demand shocks relieved the need to continue to raise official interest rates in the short term. The Bank will need to closely monitor inflation developments, the impacts of SARS and BSE, signs of recovery in the United States and elsewhere, and the effect of the appreciation of the Canadian dollar on aggregate demand and price levels in Canada. If the negative shocks buffeting the Canadian economy prove larger or more persistent than expected, then a temporary monetary response may be called for. Looking forward, the Bank will need to resume the process of removing the remaining monetary stimulus once the effects of the current negative shocks dissipate.

To reduce the gap in living standards with the United States, the key challenges are to raise productivity growth and employment rates

Thanks largely to the long series of structural reforms and the improved frameworks for macroeconomic policy formation, medium-term potential growth has picked up to over 3 per cent per year, and the gap in real income per capita with the United States has narrowed from its 1997 peak. However, the gap remains around 15 per cent. Since it is mainly due to the difference in labour productivity, raising productivity growth is the main challenge facing Canada. To do so, it must focus on boosting innovation, competition and skills. But there is also scope to make better use of labour resources. Structural unemployment remains persistently high, partly as a result of various features of the social assistance and unemployment insurance systems. Improving work incentives could also lift participation rates further; these have recovered to beyond US levels but are still well

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below the very high rates achieved in, for example, the Nordic countries.

Employment rates can be increased through further welfare reforms

Major reforms since the mid-1990s have made work more attractive relative to social assistance, but "welfare traps" still exist. The introduction of the child tax benefit, which does not depend on work status, has significantly lowered disincentives to move off welfare. But other hurdles remain, including the loss of in-kind transfers such as subsidised health care and housing. More use of in-work or back-to-work benefits could reduce this problem, and provinces could learn from the innovative federal experiment with such a programme for sole parents. However, the phasing out of in-work payments raises marginal effective tax rates, so any reforms should be part of a package that addresses the uncoordinated stacking of abatement rates more generally. In this regard, the government has recognised the need to work with the provinces and territories to examine claw-back rates and other elements of the tax and benefit systems. The package should also include stronger training and job-search requirements and greater use of initial case management and diversion programmes, as the countries that have been most successful in cutting unemployment are those that have improved both incentives and enforcement.

Work incentives in the employment insurance system should also be improved

Features of the employment insurance (EI) programme also contribute to the high unemployment rate. EI has moved well beyond providing income support during unexpected spells of unemployment and has become a major vehicle for delivering family, social and regional assistance. Frequent use of the system in some, mainly seasonal, industries creates a large cross-subsidy between sectors and regions. Workers were experience rated between 1996 and 2000 in order to discourage recurrent use. The criticism of the abolition of worker experience rating contained in the previous *Survey* remains valid. Other aspects of the system also need to be improved. The qualification period is short by international standards, while variations in eligibility rules between high- and low-unemployment regions discourage internal labour mobility, leading to persistent differences in joblessness across the country and thus higher structural unemployment.

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Past structural reforms are paying off

Achieving faster productivity growth is nevertheless the main way to raise the standard of living. The government has recognised this in its *Innovation Strategy*, which focuses on developing human capital and fostering a strong research environment. Some encouraging signs of a pickup in productivity growth have begun to appear as the pay-off from the wide-ranging structural reforms of the past 15 years. However, trend productivity growth has risen only far enough to match the US post-1995 acceleration. It has not yet begun to close the gap. There is unfinished business in structural reform before that is likely to happen.

Barriers to competition in some sectors may be holding back growth and innovation

International experience shows that competition is an important driver of innovation and productivity growth. In general, Canada has one of the OECD's most business-friendly regulatory environments and has been a pioneer in ensuring its regulations are clear, cost-effective and directly linked to its policy goals. There are several glaring exceptions, however. Canada is one of the few OECD countries to have retained widespread foreign ownership restrictions in the *telecoms* sector, a strategy that is holding back investment and competition. This policy is currently under review, along with similar restrictions in the *cable* and *broadcasting* industries that are intended to safeguard Canadian culture. In the *airline* industry, the Competition Commissioner has recommended raising the foreign ownership limit from 25 to 49 per cent and allowing foreign-owned but purely domestic ("Canada only") carriers. In *banking*, the limit on a single shareholder's ownership of a bank has recently been raised from 10 to 20 per cent and now applies only to the largest banks. Additional moves in this direction would further promote potential competition. Restructuring of the *electricity* industry is also lagging behind many other countries. Ontario abandoned its retail deregulation when prices increased sharply, with the taxpayer now picking up the difference between the low fixed retail price and what generators receive in the spot market. Despite attractive wholesale prices, private investors are holding back from building new capacity due to uncertainty about the regime they will face in the future and the possibility of substantial nuclear power capacity being refurbished.

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The freeing up of internal trade needs fresh momentum

One obstacle to competition has traditionally been the array of differences in provincial regulations. Important progress in dismantling barriers to internal trade was made in 2001 with the elimination of most remaining restrictions on labour mobility. But other long-standing issues appear intractable within the current framework, including dealing with the energy sector, agriculture and investment incentives and extending the government procurement provisions. Indeed, the dismantling process now seems to be grinding to a halt, preventing firms from fully exploiting economies of scale. The *Agreement on Internal Trade* therefore needs to be broadened and strengthened. For example, it should be extended to cover the balkanised securities sector (two committees will report on securities regulation this year). More generally, new procedures for reaching agreement among the provinces may also be needed, such as majority rather than consensus decision-making and automatic application of mutual recognition principles in cases where meaningful agreement cannot be reached. In addition, the dispute resolution procedure also needs to be improved: it is slow, costly and lacks teeth because rulings are not binding.

Improving human capital should be a priority, especially through adult education

Human capital is a notable strength, with a well educated adult population and excellent achievement by students of school-leaving age. However, this good performance is concentrated in the middle of the distribution. Canada does less well at the two extremes. At the top end, too few school leavers go on to degree programmes at universities. Access to university education could be improved by extending eligibility for the tightly-targeted student loan scheme to more middle-income families. At the bottom end of the distribution, adult literacy is a problem. Although average literacy levels are good, a small number of people seriously under-perform by international standards. Provincial governments have taken many steps to improve the adult education system, but it remains underfunded, with long waiting lists in some areas and does not seem to be adequately catching those who need it most.

Immigrants are seen as bringing both economic and other benefits…

Immigration policy is seen as part of Canada's strategy to boost its human capital. Receiving substantial numbers of immigrants is in line with Canada's historical tradition, though the focus of selection policy has shifted over time: from settling land to recruiting industrial workers, then to

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filling specific occupational shortages and, finally, to attracting highly skilled workers who can integrate effectively in a knowledge-based economy. At present, most immigrants are admitted for their skills or are accompanying family members, although substantial numbers are also admitted on family reunification and humanitarian grounds. Immigration is widely seen in a favourable light: its potential economic benefits include positive spillovers from a larger and more culturally diverse pool of human capital, filling skill shortages and smoothing the fiscal burden of ageing, albeit modestly.

... but the economic performance of recent immigrants has deteriorated...

An essential condition for such benefits to be realised is that immigrants can integrate effectively into the labour force. Understandably, the deteriorating relative economic performance of the immigrants who arrived over the 1980s and the 1990s has been a cause for concern. The difficult labour-market conditions that prevailed during much of this period account for part of this deterioration, and some signs of an improvement have appeared in the last few years. However, the earlier negative trend also had structural causes. Although their educational levels have risen in step with those of their Canadian-born counterparts, recent immigrants seem to obtain lower returns than their predecessors for their foreign-acquired education and work experience. This is in part because their credentials are not fully recognised, but also in part because their proficiency in Canada's official languages is lower than that of earlier immigrants, a handicap whose importance may have increased in a knowledge-based economy.

... leading the government to step up efforts to help immigrants to integrate...

Both federal and provincial governments are attempting to address these problems. Programmes to help immigrants to integrate into the host society and into the labour market, mostly run in collaboration with local communities, are reasonably effective. Entitlements to government benefits and services provide effective social protection without creating a culture of dependency. Moreover, the school system does a good job in integrating immigrants' children, so that the second generation seems to experience no disadvantage. But improving the process by which immigrants get their credentials recognised has proven difficult. The steps that have been taken

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so far – setting up specialised agencies for the assessment of academic qualifications, putting pressure on professional bodies to make their standards and procedures more transparent, disseminating information about the latter among applicants and encouraging them to start the recognition process early on – all go in the right direction. These efforts should continue to be intensified, since the payoff is potentially large.

... and to better focus selection policies on education, language skills and adaptability

Immigrant selection policies now emphasise adaptability and general skills, including knowledge of Canada's official languages. This is sensible, as previous attempts at skill-based micro-management proved too difficult in a rapidly changing economy. The latest changes were introduced only last year and it will take some time to see their effects, but both the analysis of past immigrant outcomes and the experience of other countries suggest that Canada could go further in the same direction. For example, it would be consistent with the new emphasis on adaptability to target a younger age range and to make a minimum level of language skills a requirement for economic immigrants (rather than just one element in the points tally), while reducing the weight assigned to past work experience. The recent decision to make it easier for temporary foreign workers who have been in Canada for some time and already have a job to apply as permanent immigrants filled an obvious gap in the system; introducing a similar possibility for foreign students with a job offer could be explored.

Emigration flows are part of a global trend to greater mobility of the highly skilled

Canada is part of an increasingly integrated North American labour market, where the higher returns to skills offered by the US economy continue to exercise a strong enticement. Indeed, emigration increased in the mid-1990s, partly as a result of diverging labour-market conditions in the two countries, as well as specific push or pull factors in certain professions (*e.g.* health-care workers, computer specialists). But with the net flow of university graduates still clearly positive, concern about a "brain drain" seems to be overstated. In the context of the global trend to greater international mobility of the highly skilled, attracting such workers from abroad and retaining its own will be facilitated if Canada is successful in its efforts to raise living standards and remain an attractive place to live and work. This includes finding the right balance between lowering taxes and achieving better value for money in public spending.

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Fiscal policy should have a more medium-term focus

Managing longer-term fiscal pressures is a further challenge. Federal policy for the past few years has been to set a balanced budget, after putting aside contingency and prudence margins of around 0.3 per cent of GDP in each of the two following years, increasing spending or cutting taxes to aim to absorb whatever is left. In recent years, however, stronger-than-expected output growth has allowed debt reduction averaging nearly 1 per cent of GDP per annum. Strictly maintaining a policy of balancing the budget or better in the event of a sufficiently significant downturn would force the government to take pro-cyclical fiscal measures. Moreover, a future upturn could lead to unsustainable increases in baseline fiscal commitments because increasing spending or cutting taxes in an upswing is easier than reversing those actions in a downturn. Adding more medium-term elements to the framework by adopting a broader set of financial objectives over a longer planning horizon, including, for example, a medium-term debt target, would avoid this problem and better ensure that the stance of fiscal policy is appropriate to the business cycle. To the government's credit, however, it should be noted that it has taken several efficiency-enhancing steps to improve public expenditure management. These include re-introducing a programme review process, moving to accrual accounting and fixing the longstanding problem of an excessive premium rate on the EI scheme (partly by cutting the rate, partly by enriching benefits). It also intends to improve the governance of any future arms-length foundations, which are increasingly being used to implement public policy, and will attempt to upgrade arrangements for existing ones.

There is almost no fiscal room left over the next few years

This year's budget left virtually no fiscal room over the next few years for further spending increases, tax cuts or debt reduction beyond what is already planned. Increased spending – especially on health care – and further small personal and corporate tax cuts leave the projected federal balance around zero (after setting aside the safety margin) until at least 2008. With provinces also near balance, the healthy-looking general government position is largely due to the public pension surplus.

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Canada is better prepared than many to cope with an ageing population, but challenges remain

From a longer-term perspective, it is not clear whether a balanced budget is good enough. The fiscal position will come under heavy stress when the baby boomers begin to retire. The government has taken several steps to put its finances on a sound footing, including pre-funding the earnings-based component of the public pension. However, the pension burden is projected to rise a little over the next three decades, and several other fiscal risks remain. The pension system could be much more expensive than expected if the real rate of return on investments is less than the assumed 4.5 to 4.7 per cent per annum or if price indexation of the first-pillar pension cannot be sustained. These risks are dwarfed, however, by the risks to health spending. Long-term "baseline" fiscal projections are fairly favourable, with net debt being eliminated over the next 40 years, provided that spending on health care by each age group rises no faster than incomes (and also assuming that the government allows primary surpluses to grow over the next couple of decades). However, even slightly quicker growth in health-care costs could make the fiscal position unsustainable. Unless effective measures to control these costs are put in place, it would be prudent to save more now by targeting surpluses of perhaps ½ to 1 per cent of GDP.

Efficiency and expenditure control in the health sector should be improved

Canadians need to confront the longer-term issue of how much they are collectively willing to spend on health and through which mechanisms costs should be controlled and spending allocated. Core health services provided under the Canada Health Act (CHA) – roughly speaking, doctors' visits and most hospital-based care – are free of charge, so there is little apart from queuing that can currently be used to control spending. In response to public dissatisfaction with supply shortages, the federal government substantially lifted funding to provinces as part of this year's inter-governmental *Accord on Health Care Renewal*. However, the system is not demonstrably under-funded; in fact the expenditure share of GDP is among the highest in the OECD. Current supply problems are therefore more likely to reflect structural factors. Given these bottlenecks, the immediate challenge is to make sure the extra money buys an increase in output rather than simply pushing up wages and prices in the sector. The longer-term challenge, however, is to deal with the underlying structural problems. While there are no easy answers, there are several options for

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reform. *First*, considerable efficiency gains could be obtained by improving the institutional framework. Steps could include reform of primary care, using alternatives to fee-for-service payments for physicians once provider supply shortages have been resolved, adopting output-based funding mechanisms for hospitals, making more use of contracting out and enhancing the information base in order to improve management decisions. *Second*, demand could be controlled more effectively by making more use of cost sharing mechanisms such as user charges and co-payments, so long as this does not compromise equality of access nor health outcomes. This will become increasingly important if the government broadens the CHA to encompass home care and catastrophic drug coverage, as it plans to do.

Environmental problems are few but occasionally severe

Longer-term sustainability issues extend beyond public finances to encompass social and environmental developments. While large parts of Canada are free of many of the environmental difficulties faced by other OECD countries, there are nevertheless a number of sometimes severe localised problems, some of which call for careful co-ordination of policies across jurisdictional boundaries. In particular, the eastern provinces could usefully design a trading system for sulphur emissions in order to reach their targets, and this could be generalised to cover other local air pollutants. There are also localised water problems related to the use of manure as a fertiliser and more specific issues that have arisen with inadequately controlled water supply systems. It is difficult to determine whether relying on codes of good practice is sufficient to ensure high-quality drinking water in the absence of systematic monitoring; gathering such information should be a priority. Further investment is needed to reduce the chances of accidental pollution, and the cost of this would best be met by charging users on a full-cost basis.

More ambitious use of economic instruments would make reducing greenhouse gas emissions less costly

In ratifying the Kyoto agreement last year, Canada has signed up to an ambitious target for greenhouse gas emission reductions, especially as its major trading partner has set itself a much easier goal. Canadian emissions are currently high and by 2008-12 are projected to be 42 per cent above its Kyoto target. The challenge is to reduce them without imposing an unmanageable burden on the economy, or on particular industries or regions, as emissions are

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very unevenly distributed. The government's proposals include a permit trading scheme for large industrial emitters, but it covers just 40 per cent of projected emissions and will deliver only a small part of the required reduction. If this scheme is to be efficient, emitters will need to be able to buy permits abroad in order to back their emissions. Outside this sector, the government has ruled out the use of a uniform carbon tax linked to permit prices on world markets, which would have been an efficient way to guarantee the lowest-cost abatement. Instead, it has made a large number of additional proposals, including voluntary agreements, covenants with major industries, command-and-control measures and subsidies. Taken together, these will probably be unnecessarily costly as marginal abatement costs will differ widely and will, in some cases, far exceed the international permit price. These measures will almost certainly not ensure compliance with the target, so the government has indicated that it is developing additional measures and is willing to cover any excess emissions by purchasing permits on the world market, despite the fiscal risks involved.

The fundamental challenges

Canada's economy has done comparatively well over the past two years, demonstrating a noticeably improved resilience to unfavourable shocks. Over the medium term the economy is poised to achieve growth rates that compare favourably with past performance, but that will not be enough to eliminate the per capita income gap with the United States. The fundamental challenge is to make Canada an even more attractive place to live, work and invest. While past reforms have begun to pay off, there is still unfinished business. To boost the employment rate further the government should reduce disincentive effects arising from the tax and benefit systems by, for example, making greater use of in-work benefits and reintroducing experience rating of Employment Insurance users. Faster productivity gains are more likely to be achieved in an economic environment conducive to innovation. A key to greater dynamism is strengthening competition by eliminating remaining foreign ownership limits and barriers to internal and external trade and continuing electricity deregulation. Investing in skills should also remain a prior-

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ity, with a particular emphasis on adult education. The nation has benefited by importing human capital from abroad through immigration but needs to intensify its efforts to remove the obstacles that prevent immigrants from having their skills fully utilised in the labour market. Having introduced a sound fiscal framework and reformed its public pension system Canada is in a better position than most other countries in facing ageing-related fiscal challenges, but the trend rise in health-care costs remains a risk. With the recent budget the government chose to address the short-term needs of the existing system. But, ultimately, ways to control the rise in budgetary costs – whether through incentives to raise efficiency or through cost sharing – will need to be explored, otherwise the resources needed to finance the future burden should be set aside in advance, for example by setting a bolder debt reduction target. Finally, environmental sustainability will be enhanced with least cost to the economy if the ambitious greenhouse gas emissions reduction is achieved through market-oriented instruments rather than command-and-control measures or voluntary agreements.

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I. Recent developments and key challenges

In the two years since the publication of the last *Economic Survey*, the Canadian economy has recorded a remarkably strong performance, its output growth outpacing that of all other major developed economies, including the United States. The latest OECD projections (which were finalised at the beginning of April and released in *Economic Outlook* 73) indicate that Canada's relative performance should remain favourable this year, before its growth rate is overtaken by that of the United States in 2004. The key question, however, is whether the rate of growth can be sustained over the medium term. Canada still has a substantial gap in terms of income per capita relative to the United States, due mainly to lower productivity levels. The income gap had been widening during most of the past 20 years, and only in the last five has it started to narrow reflecting mainly more favourable developments in employment rates, in turn largely the result of past structural reforms and an improved macroeconomic policy framework. There is some scope for getting more people into work if the causes of Canada's higher structural unemployment are addressed. But over the longer term the income gap can be further narrowed only through stronger productivity growth.

Recent macroeconomic performance

Output growth has been comparatively strong

The 2000-01 global slowdown had only a relatively mild impact on Canada's economy. While output contracted for three consecutive quarters in the United States, it did so only in the third quarter of 2001 in Canada. Canada also had a more robust upswing when economic activity started to recover toward the end of 2001 and then gathered pace over the course of the following year (Figure 1, Panel A). The divergence is even more marked when labour-market developments are compared (Figure 1, Panel B). In Canada employment growth was virtually nil in 2001, but then resumed in 2002 at an exceptionally rapid pace (3.7 per cent during the year). Moreover, the expansion has been relatively broad-based across sectors, and the quality of the jobs created has been good, including substantial numbers of full-time, high-wage jobs. By contrast, in the United States

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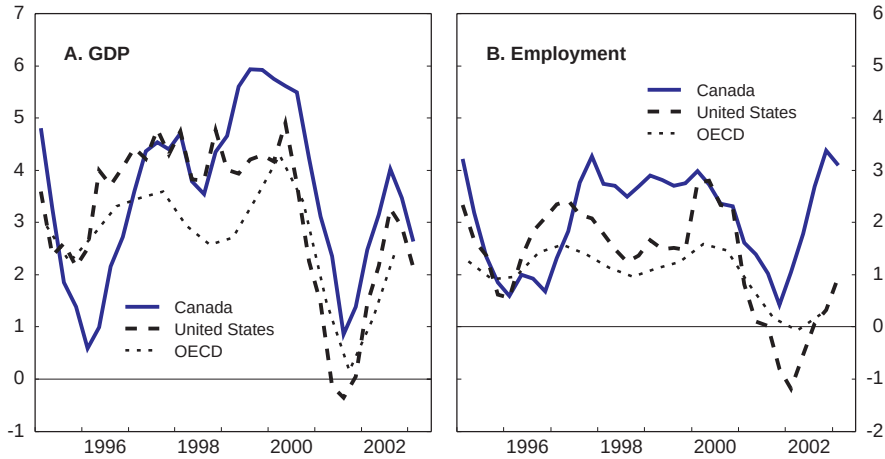
Figure 1. **GDP and employment growth**
Year-on-year percentage change



Source: OECD.

the number of jobs contracted sharply during the recession and then stagnated in 2002 due to the weakness of output growth.

The pace of expansion has slowed gradually since the spring of 2002, mainly as a result of the faltering global recovery and of an international climate dominated by weak equity prices and war-related uncertainties. While housing investment and consumer expenditure (particularly on durables) have continued to be buoyant, supported by strong consumer confidence and low interest rates, exports declined in the last quarter of last year and thus far in 2003, and the recovery of business investment has remained sluggish, reflecting the uncertain international climate (Table 1 and Figure 2).

Activity is believed to have remained relatively subdued in the first half of 2003, as the same factors nave remained at play. The recent appreciation of the Canadian dollar (by some 18 per cent *vis-à-vis* the US dollar since October 2002) has contributed to depress export demand. In the second quarter of this year, activity is also likely to have been negatively affected by the impact of the outbreak of Severe Acute Respiratory Syndrome (SARS). The output effect should be concentrated in the Greater Toronto Area which represents about one fifth of the Canadian economy. The impact will be felt mostly in the health care, travel and

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Table 1. **Demand and output**

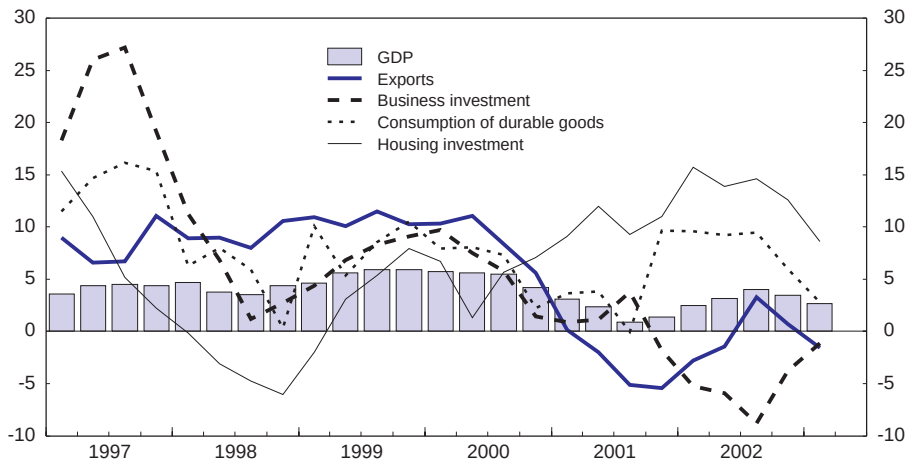
Percentage change, seasonally adjusted annual rates, volume (chained 1997)

	1995/1983	2002/1993	2000	2001	2002	Q4 2002	Q1 2003
Private consumption	2.8	3.2	4.0	2.6	3.4	4.4	3.5
Government consumption	2.3	1.2	2.6	3.7	3.0	2.4	4.0
Gross fixed investment	2.6	5.0	5.5	4.3	1.3	0.1	5.0
Public	4.3	3.9	3.4	10.3	11.8	–0.2	5.2
Private	2.3	5.2	5.8	3.5	–0.2	0.2	5.0
Residential	1.7	3.7	5.2	10.3	14.2	11.1	12.7
Non-residential	2.6	5.8	6.0	1.0	–6.0	–5.0	1.1
Final domestic demand	**2.6**	**3.1**	**4.0**	**3.2**	**2.9**	**3.1**	**3.9**
Stockbuilding[1]	0.1	–0.1	0.4	–1.7	0.6	1.2	2.1
Total domestic demand	**2.7**	**3.0**	**4.4**	**1.5**	**3.5**	**4.3**	**6.1**
Exports of goods and services	6.4	6.6	8.8	–3.1	–0.1	–8.7	–5.7
Imports of goods and services	7.3	5.4	8.0	–5.0	0.6	–1.4	2.6
Foreign balance[1]	**–0.1**	**0.7**	**0.7**	**0.6**	**–0.3**	**–3.2**	**–3.3**
Error of estimate[1]	–0.0	0.0	0.1	–0.2	0.1	0.6	–0.3
GDP at market price	**2.6**	**3.7**	**5.3**	**1.9**	**3.3**	**1.6**	**2.4**
Memorandum item:							
GDP per capita	1.3	2.7	4.3	0.9	2.3	. .	. .

1. Contribution to GDP volume growth.
Source: Statistics Canada.

Figure 2. **GDP and selected demand components**

Year-on-year percentage change



Source: OECD.

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tourism sectors, as well as on retail sales. It should be of a temporary nature, assuming that any effect on consumer confidence and on tourist preferences is short-lived. Another factor that could accentuate the current slowdown in Canada would be a prolonged ban on cattle exports to the United States following the discovery of one case of Bovine Spongiform Encephalopathy (BSE) in Alberta. Growth is projected to pick up towards the end of the year and then progressively strengthen in 2004, when the US upswing is expected to gather momentum (Table 2). Together with exports, business investment should contribute to drive the cyclical upswing, as capacity utilisation has increased, and profits and business confidence remain at relatively favourable levels. Consumer spending is

Table 2. **Short-term projections**[1]

Percentage change over previous period, seasonally adjusted annual rates, volume (chained 1997 C$)

	1997 C$ billion[2]	2003	2004	S1 2003	2S 2002	S1 2004	S2 2004
Private consumption	510.7	2.8	2.9	2.8	2.9	3.0	2.9
Government consumption	171.8	3.7	3.2	4.0	3.6	3.1	3.1
Gross fixed investment	174.8	3.4	5.2	2.6	3.8	5.3	6.2
Public[3]	20.1	5.4	5.2	6.4	5.9	5.0	4.9
Private residential	43.5	0.8	–1.7	–1.2	–3.8	–2.0	1.0
Non-residential	111.2	4.2	8.4	3.6	7.1	8.8	8.8
Final domestic demand	**857.3**	**3.1**	**3.5**	**3.0**	**3.2**	**3.5**	**3.6**
Stockbuilding[4]	8.1	0.2	0.1	–0.4	0.1	0.1	0.0
Total domestic demand	**865.4**	**3.3**	**3.5**	**2.5**	**3.2**	**3.5**	**3.6**
Exports of goods and services	348.6	4.4	7.3	4.1	6.2	7.9	7.3
Imports of goods and services	331.3	6.1	8.0	4.7	7.6	8.3	7.8
Foreign balance[4]	**17.3**	**–0.4**	**0.1**	**–0.0**	**–0.2**	**0.2**	**0.1**
GDP at market prices	**882.7**	**2.7**	**3.4**	**2.4**	**2.9**	**3.6**	**3.6**
Unemployment rate		**7.3**	**7.0**	**7.4**	**7.3**	**7.1**	**6.9**
Inflation and wages							
GDP implicit price deflator	. .	3.5	2.1	4.0	2.1	2.1	2.1
Private consumption deflator	. .	2.8	2.1	3.2	2.0	2.1	2.2
Private compensation per employee	. .	3.2	3.7	3.4	3.6	3.8	3.9
General government balance (per cent of GDP)	. .	1.1	1.0	1.2	1.0	1.0	1.0
Interest rates							
Short-term	. .	3.6	4.6	3.2	3.9	4.4	4.9
Long-term	. .	5.3	5.7	5.1	5.4	5.6	5.8
Current balance (per cent of GDP)	**. .**	**2.0**	**2.2**	**2.0**	**2.0**	**2.1**	**2.2**

1. The projections presented in this Table were finalised in April 2003 and published in the OECD *Economic Outlook* 73. Therefore, they do not reflect more recent information.
2. Current prices.
3. Excluding nationalised industries and public corporations.
4. Contribution to GDP volume growth.

Source: OECD.

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expected to continue supporting activity, as households' financial situation is healthy, confidence remains high, access to credit is easy and the debt service ratio remains below its historical average. Only housing investment, after growing at double-digit rates in 2002, should slow down as previous pent-up demand is exhausted. In summary, after slightly exceeding its estimated potential level in 2002, output is expected to increase slightly less than capacity during most of 2003, but should again move above its trend potential growth rate next year.

Inflationary pressures have proved to be temporary

Consumer price inflation picked up markedly in the second part of 2002 and remained high in early 2003, with the headline rate substantially above the Bank of Canada's 1-3 per cent target band and the core measure (which excludes the eight most volatile components) hovering around the upper limit (Figure 3, Panel A). Several special factors contributed to drive inflation higher at different times during this period. The most important were the rise in international energy prices and the sharp increase in home and auto insurance premiums, the latter due to a coincidence of rising claims and declining market value of insurance company assets. In the summer and fall of 2002 there was also a temporary sharp rise in electricity prices in Ontario during the brief price liberalisation experiment and, in the latter part of the year, a statistical base or "echo" effect connected to the heavy price discounting that had taken place in the fall of 2001. In addition, changes in indirect taxes contributed about ½ percentage point to headline inflation. Some of these factors have had a one-off effect on prices; others, like the oil and electricity price rises, were reversed in subsequent months. But even excluding these influences underlying inflation seemed to be rising over this period, as demand pressures started to run up against capacity constraints.

During the spring of 2003 inflationary pressures started to subside, thanks in part to the unwinding of the above-mentioned special effects. The recent appreciation of the exchange rate and the current weakness in demand have also helped and should continue to put downward pressure on prices, at least for a while. Headline inflation is thus expected to continue declining in coming months and should then remain near its 2 per cent target value through the remainder of 2003 and into 2004. The recent surge in inflation rates did not give rise to wage pressures (Figure 3, Panel B), and inflation expectations rose only marginally and temporarily (Figure 3, Panel D), a testimony to the overall credibility of the existing monetary policy framework.

Monetary policy has responded to changing economic conditions

The Bank of Canada, which had aggressively lowered interest rates in the second half of 2001, started withdrawing stimulus in the spring of last year and, after a pause, resumed doing so in the first half of this year (Figure 4). The policy

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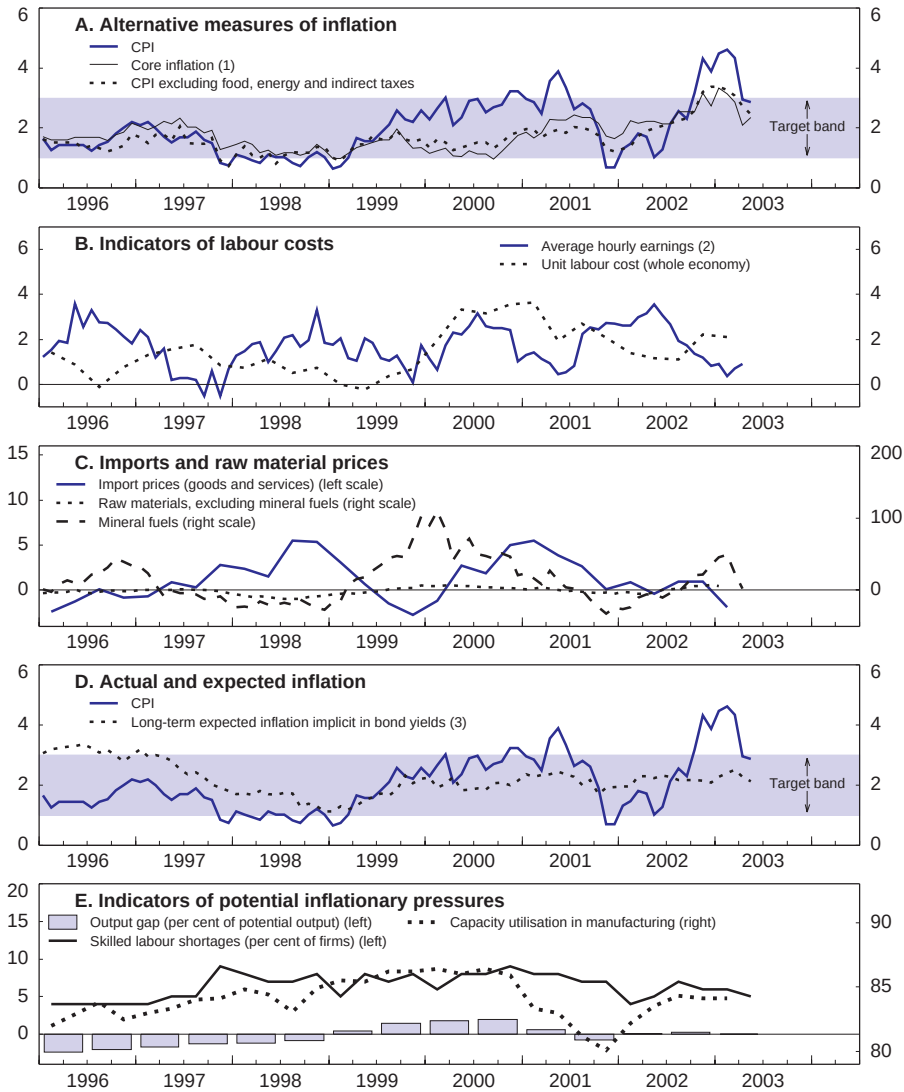
Figure 3. **Indicators of inflation**

Percentage change over 12 months



1. Excluding 8 most volatile components and indirect taxes.
2. Fixed-weighted.
3. Yield spread between long-term conventional and real return bonds; based on bonds maturing in December 2026.

Source: Statistics Canada; Bank of Canada.

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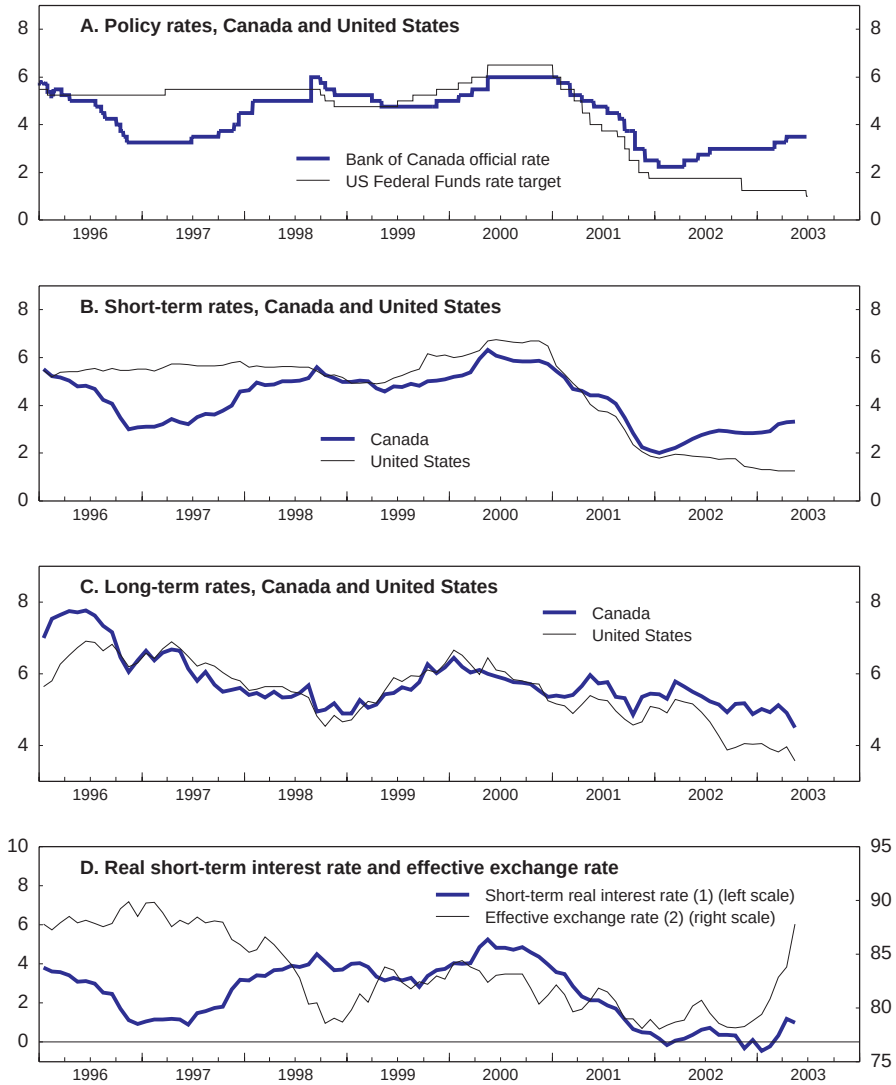
Figure 4. **Interest rates and other indicators of monetary stance**

Per cent



1. Deflated by the core inflation rate calculated over the previous 12 months.
2. Exchange rate against the 6 main trade partners; index 1992 = 100. A rise indicates an appreciation of the Canadian dollar.

Source: Statistics Canada; Board of Governors of the Federal Reserve System.

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easing in 2001 had begun already in the first part of the year in response to signs that the pace of economic activity in both the United States and Canada had started to slow down. But what had started as a gradual return of the monetary stance toward an approximately neutral position went further during the summer and became a much more aggressive policy easing after the terrorist attacks of September 11, as the Bank of Canada – like other central banks – was concerned that the confidence effect of the attacks could transform the US and global slowdown into a more protracted downturn, with potentially serious deflationary effects. These risks warranted decisive pre-emptive action. The Bank felt relatively comfortable with the unusual extent of the easing action because at the time it estimated that the economy had already been operating below potential in the first half of 2001 and that slack was continuing to build. For the same reason, when it started removing part of the stimulus in the spring of 2002, following signs that the most serious deflationary risks had dissipated and that the Canadian recovery was robust, it did so only to a limited extent, leaving short-term nominal interest rates only slightly above the current inflation rate.

However, by the time inflation started rising in the autumn of 2002, a number of downside risks to activity were starting to accumulate as well, with the US recovery losing steam and a rising probability of a conflict in the Middle East. Thus, the Bank of Canada chose to wait until early 2003 to resume tightening policy and, even then, preferred to proceed at gradual pace. The persistence of a considerable degree of uncertainty around economic projections throughout this period contributed to make the choice of the appropriate monetary stance particularly challenging. The Bank has had to balance the need to take insurance against the risks to the Canadian economy of a synchronised global downturn against its mandate to return inflation to its 2 per cent target.

Although real short-term interest rates are currently below historical averages, whether calculated on the basis of actual or expected inflation rates (Figure 4, Panel D), monetary conditions have actually tightened since late 2002 because of the rapid appreciation of the Canadian dollar. All other things equal, this has alleviated the need for higher interest rates. However, the relationship between interest- and exchange-rate changes in determining appropriate monetary conditions is not mechanical, depending on the factors that cause the exchange rate to move. In the case of the recent uptrend of the Canadian dollar, these include, *inter alia*, not only the strong cyclical performance of the Canadian economy *vis-à-vis* its US counterpart, but also an adjustment process that is related to the correction of global current account imbalances. Thus far in 2003 output growth has been running below its potential rate, and although the economy is expected to pick up later this year, no generalised capacity constraints are likely to emerge until well into 2004. Inflation is, therefore, expected to ease in the face of the factors mentioned earlier. The Bank will have to monitor inflation developments, the strength of domestic demand and signs of recovery in the US and

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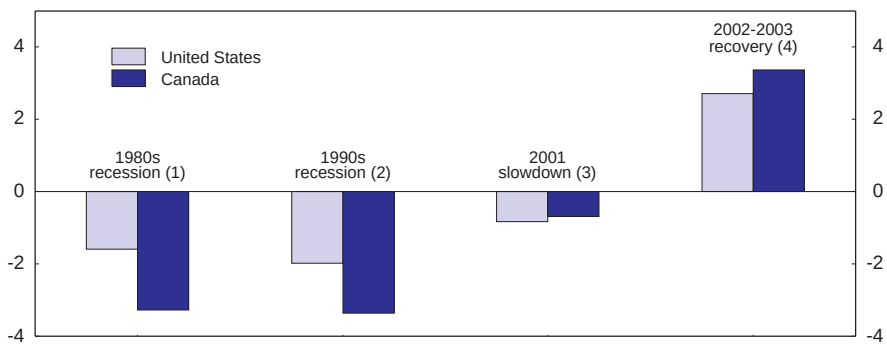
global economies, and to continue to assess the effects of the appreciation of the Canadian dollar on aggregate demand and price levels in Canada. In this regard, negative shocks buffeting the Canadian economy may require a monetary policy response, but any such response should be temporary as the effects of the shocks are expected to be short-lived. Although it is appropriate for monetary policy to remain easy in the short-run, the Bank of Canada will eventually need to remove remaining monetary stimulus. The appropriate timing of this move is impossible to define at this point, but the degree of pre-emptiveness required (*i.e.* how far in advance of the exhaustion of economic slack) depends in part on how well anchored are inflation expectations, itself a function of the credibility of the low-inflation targets. In this regard, good inflation performance in recent years should have served to increase the monetary authorities' margins of flexibility.

The improved performance in perspective: fortuitous or structural?

Canada's relative performance over the last few years contrasts sharply with previous business cycles. For example, both in the 1982-83 and 1990-91 recessions Canada had suffered substantially larger output and employment losses than the United States (Figure 5). This raises the question whether this difference can be attributed to a particularly fortunate set of circumstances specific

Figure 5. **GDP growth in the recent and previous cycles**

Percentage change, GDP in chained dollars, at annual rates



1. United States: Q1 1981 to Q3 1982; Canada: Q2 1981 to Q4 1982.
2. United States: Q2 1990 to Q1 1991; Canada: Q1 1990 to Q1 1991.
3. United States: Q4 2000 to Q3 2001; Canada: Q2 2001 to Q3 2001.
4. United States and Canada: Q3 2001 to Q1 2003.

Source: OECD.

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to the recent cycle or has more structural roots. The comparatively milder Canadian downturn in 2001 and stronger recovery in 2002 can be explained at least in part by the underlying drivers of the recent cycle. F*irst*, the contraction in business investment was both shallower and shorter than in the United States, mainly because of a less pronounced ICT investment cycle and a smaller weight in the economy. S*econd*, the protracted stock price decline had a smaller impact on Canadian households' financial situation, because equity holdings represent a smaller fraction of their total wealth. T*hird*, while in both countries the aggressive easing of monetary policy in the second half of 2001 boosted expenditure on consumer durables, Canada benefited also from US demand through stronger automotive exports. F*ourth*, the Canadian housing market was less buoyant in the late 1990s, and therefore low interest rates had a proportionately stronger effect on housing investment. F*inally*, the much more favourable employment situation also contributed to support personal income and expenditure.

In addition, there are reasons to believe that the Canadian economy has become more resilient to shocks than it was in the 1980s and early 1990s. This is not due to greater insulation from external shocks: indeed, between 1990 and 2002 the ratio of total exports to GDP increased from 26 to 41 per cent (the entire rise being accounted for by the rise in exports to the United States after the implementation of the FTA and then the NAFTA), although the rise is less – from 19 to 28 per cent – when the import content of exports is excluded. At the same time, however, structural reforms have helped make the economy more flexible, while the macroeconomic measures introduced during the 1990s – in particular, the considerable fiscal consolidation programme undertaken between 1994 and 1998 and the establishment of a credible inflation-targeting monetary policy – have helped create greater room for the counter-cyclical use of monetary and fiscal policy. The 2001 tax cut – although already scheduled and enacted for structural rather than for counter-cyclical reasons – luckily turned out to be well-timed in relation to cyclical developments. In the case of monetary policy, the greater room available was effectively used during the 2001 slowdown. The effectiveness of monetary policy has probably also increased, as a credible policy framework makes monetary policy signals easier to interpret. Moreover, with the sustainability of the fiscal situation restored and price expectations better anchored, long-term interest rate volatility has diminished and the differential *vis-à-vis* the United States has declined since the early 1990s, particularly in real terms.

The improved macroeconomic fundamentals have also been reflected in Canada's external balance and in the financial situation of the private sector. The current account of the balance of payments, which had been almost continuously in deficit over the previous four decades (on average, by about 2 per cent of GDP), has been in surplus since1999, so for the first time in a long while the country's external asset position is improving. The turnaround reflected in part the earlier gradual exchange rate depreciation, but also the structural adjustment of the gov-

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ernment's financial balance since the mid-1990s, offset only in part by the decline in household savings.[1] The current surplus declined by more than a percentage point of GDP in 2002, to 1.5 per cent, mainly reflecting Canada's cyclical position relative to the United States. Looking forward, it can be expected to be driven by two opposing factors: the favourable effect of the anticipated US economic recovery and the impact of the recent exchange rate appreciation, if sustained. At the same time, the financial situation of households and of non-financial corporations remains sound. While household debt levels have risen over the past decade, the ratio of interest costs to disposable income has not, thanks to lower interest rate levels. And given the smaller proportion of shares in Canadian households' wealth relative to their US counterparts, their net worth has not been hit as much over the past three years. The profitability of non-financial corporations has improved substantially since the mid-1990s owing to the combination of a favourable growth performance with relatively moderate wage growth, and is now at historically high levels. Over the last ten years the indebtedness of Canadian firms has risen much more moderately than that of their US counterparts, partly because they have not invested as much; and it has actually declined substantially as a ratio to equity, measured at book value.[2] Although the recent exchange rate appreciation can be expected to squeeze profit margins in the sectors most exposed to foreign markets, only the least profitable firms should experience difficulties. On the whole, structural and macro-economic reforms have helped to create a healthier environment for growth, with no significant imbalances building up. This has been reflected, among other things, in a much less volatile output growth.

Can higher growth be sustained in the medium term?

Canada's recent performance has been helped by the fact that the long period of sub-par growth in the first half of the 1990s had left substantial margins of unused resources, and part of the subsequent rapid growth represented a recovery toward trend. Not only did unemployment remain stubbornly above 10 per cent for several years after the recession of the early 1990s, but labour force participation declined as laid-off workers and new potential entrants alike were discouraged from seeking employment. Canadian participation rates, which were on a par with US levels at the beginning of the decade, diverged sharply. Workers in the younger and the older age groups suffered proportionally larger losses than those in the prime-age range. When structural reforms and macroeconomic stabilisation started to pay off in the second half of the decade, economic activity expanded at a sustained pace, and employment growth averaged 2.5 per cent in 1997-2002, twice as fast as the growth of the working-age population. Particularly among men, those groups whose employment rates had fallen most during the recession then achieved the largest gains. By the end of 2002 the employment rate had returned to its 1990 peak level, virtually eliminating the gap with its US counterpart (albeit with some help from the latter's cyclical decline).

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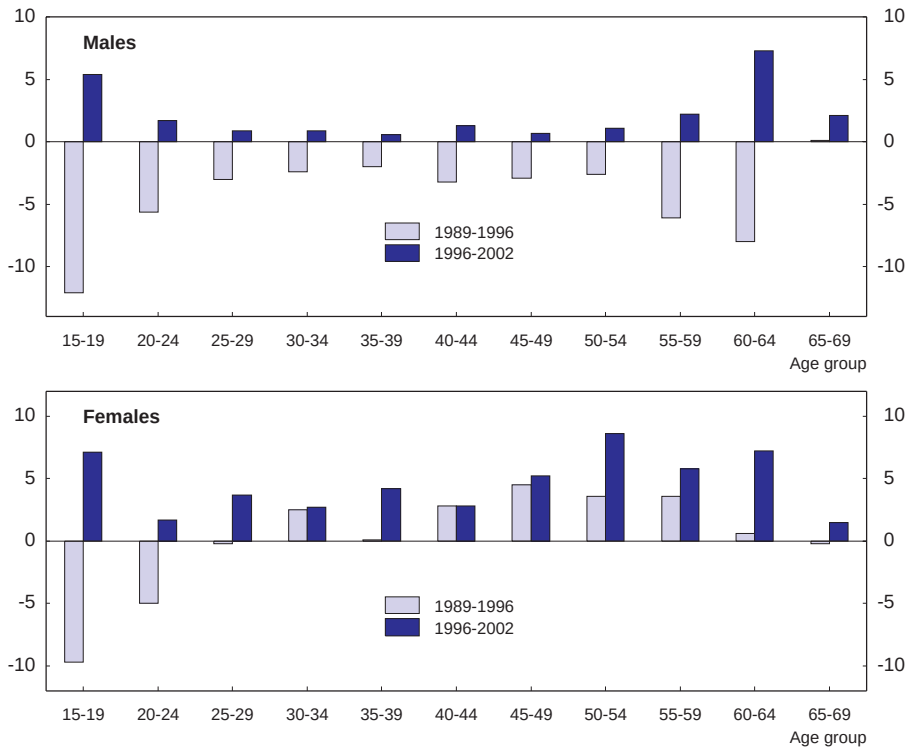
The scope for a further rise in participation rates is difficult to assess at present. The recent gains have been due in part to cyclical or other temporary factors and in part to longer-term trends. Improved macroeconomic conditions have helped to attract people back to the labour market. Indeed, the younger and the older age groups, who had been hit proportionately hardest by declining job opportunities in the early 1990s, have recorded the largest participation gains (Figure 6). In the last few years, the rise in older workers' participation may have also been supported by a desire to protect retirement income against the impact of falling share prices on wealth (during the 1990s, rising wealth had worked in the opposite direction). But longer-term factors, some of which should continue to

Figure 6. **Changes in participation rates by age group**

Percentage points



Source: Statistics Canada.

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operate in coming years, have also been influencing the degree of labour force attachment of the succeeding birth cohorts. For example, the large participation gains for women in all age groups have reflected both long-term changes in social attitudes and fertility and rising levels of educational achievement. Female participation should continue to rise, since the cohorts presently in the central age range (25 to 44), who are highly educated and have the narrowest participation gap with males, can be expected to carry forward their greater propensity to participate as they age. On the other hand, for males, the favourable influence of higher education levels has offset only in part the trend decline in participation observable for virtually all age/education groups. Looking forward, education-related cohort effects should contribute to raise overall participation rates by about one percentage point for women and ½ point for men over the next ten years. Another kind of cohort effect that has been at play in recent years is connected to the evolution of macroeconomic conditions. For example, some of the older male workers who had left the labour force after losing their jobs in the 1990s recession or through early retirement have been exiting the working-age range (15 to 64) and are being replaced by cohorts with tighter labour force attachment. Nevertheless, it is still unclear whether the long-term decline in participation for older male workers has been arrested; this will depend in part on the extent to which the earlier trend rise in wealth resumes after the stock market decline of the last few years. In coming years, the above-mentioned favourable cohort effects will need to offset the unfavourable composition effects resulting from the fact that ageing is shifting the composition of the working age population toward age groups with lower participation. Holding present age-specific participation rates constant, the aggregate participation rate would decline by 1 percentage point for males and 1½ points for females over the next 10 years. All in all, unless policies change there seems to be limited room for further gains in participation based on current trends.

As for unemployment, it is not far above its estimated structural level. Canada's unemployment rate has been nearly 3 percentage points higher than in the United States on average over the past five years.[3] Most of this is due to structural differences in the two labour markets. While it is difficult to estimate the structural rate of unemployment with much precision, the OECD's econometric estimates place Canada's rate nearly 2 percentage points above the United States, one point above the OECD average and higher than in any of the eight other OECD countries outside the euro area for which OECD estimates exist. It is possible that structural unemployment is still falling as the lagged effects of the cyclical bounce-back work their way through the system, but it is hard to argue that it could be much below 7 per cent at present. The lowest that actual unemployment has fallen over the past 25 years is 6.7 per cent (in 2000), and at that time most other indicators pointed to an economy that was over-stretched, with core inflation picking up a short time later.

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Table 3. **Medium-term potential output growth**[1]

Annual average, percentage points

	Potential GDP growth		=	Potential labour productivity growth (output per hour worked)		+	Potential labour input growth (hours worked)		Contributions to labour input growth: Working age population		Trend participation rate		Change in NAIRU		Hours worked per person	
	1996-2002	2003-08		1996-2002	2003-08		1996-2002	2003-08	1996-2002	2003-08	1996-2002	2003-08	1996-2002	2003-08	1996-2002	2003-08
Canada	**3.3**	**3.1**		**1.5**	**1.8**		**1.8**	**1.3**	**1.2**	**1.1**	**0.2**	**0.2**	**0.2**	**0.0**	**0.1**	**0.0**
Australia	3.7	3.7		2.0	2.4		1.6	1.3	1.4	1.2	0.1	0.0	0.2	0.1	–0.1	0.0
Austria[2]	2.3	2.0		1.9	1.9		0.4	0.1	0.3	0.0	0.0	0.1	0.0	0.0	. .	. .
Belgium	2.1	2.1		1.8	1.8		0.3	0.3	0.1	0.4	0.6	0.2	0.2	0.0	–0.6	–0.3
Denmark	2.4	2.1		2.0	2.0		0.4	0.1	0.2	0.1	–0.1	0.0	0.3	0.0	0.0	0.0
Finland	2.9	2.3		2.5	2.4		0.3	–0.1	0.3	0.2	0.1	0.1	0.3	0.1	–0.4	–0.4
France	2.1	2.0		2.0	1.7		0.2	0.3	0.3	0.4	0.4	0.1	0.2	0.1	–0.7	–0.2
Germany	1.5	1.5		1.8	1.6		–0.3	–0.1	0.0	–0.2	0.4	0.3	–0.1	0.1	–0.5	–0.3
Greece	3.0	3.5		2.3	3.0		0.7	0.4	0.2	0.1	0.5	0.2	–0.1	0.1	0.0	0.0
Iceland	2.9	2.9		1.7	1.8		1.2	1.1	1.3	1.0	0.0	0.0	–0.1	0.1	0.0	0.0
Ireland	7.5	4.8		4.7	3.5		2.6	1.3	1.8	1.1	0.8	0.3	1.0	0.1	–1.0	–0.3
Italy	1.8	1.8		1.6	1.5		0.2	0.3	–0.1	–0.1	0.3	0.4	0.2	0.1	–0.2	–0.1
Japan	1.2	1.2		2.0	1.5		–0.7	–0.3	–0.1	–0.4	0.4	0.3	–0.2	0.0	–0.8	–0.2
Netherlands	3.1	1.8		1.8	1.7		1.3	0.2	0.4	0.4	0.9	0.3	0.4	0.0	–0.4	–0.6
New Zealand	3.0	3.2		1.4	1.9		1.6	1.2	1.2	1.1	0.2	0.1	0.3	0.0	–0.1	0.0
Norway	2.8	2.0		2.3	1.6		0.5	0.4	0.7	0.5	0.1	0.0	0.2	0.0	–0.5	–0.1
Spain	2.9	2.8		1.1	1.0		1.8	1.8	0.6	0.3	1.0	1.1	0.2	0.4	0.0	0.0
Sweden	2.4	2.3		2.1	1.9		0.3	0.4	0.4	0.6	–0.3	–0.2	0.1	0.0	0.1	0.0
Switzerland	1.3	1.2		1.0	0.8		0.4	0.4	0.4	0.4	0.0	0.0	0.1	0.0	–0.2	0.0
United Kingdom	2.5	2.4		2.1	2.0		0.4	0.4	0.4	0.3	0.1	0.1	0.2	0.0	–0.2	–0.1
United States	3.4	3.1		1.9	2.1		1.4	1.0	1.3	1.1	0.1	–0.1	0.0	0.0	0.0	0.0
Euro area	2.1	2.0		1.8	1.6		0.3	0.3	0.2	0.1	0.5	0.3	0.1	0.1	–0.3	–0.1
Total OECD	2.6	2.4		1.9	1.8		0.7	0.6	0.7	0.5	0.2	0.1	0.1	0.0	–0.4	–0.2

1. The projections presented in this Table were finalised in April 2003 and published in the OECD *Economic Outlook* 73. Therefore, they do not reflect more recent information.
2. On a per-worker rather than per-hour basis.

Source: OECD.

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All this suggests that the process of reabsorbing the labour freed up by the long 1990s downturn has now been virtually exhausted. Any further significant gains in employment will need to come from structural policy measures aimed at reducing unemployment and raising participation rates, as discussed in Chapter II. Beyond that, output gains will be constrained by productivity growth. At present, output is estimated to be already close to its potential level. Thus, over the medium term it will be able to expand approximately in line with the growth of potential output According to the OECD's *Medium-Term Scenario* this should be about 3.1 per cent per year on average between 2003 and 2008, with employment gains of 1.2 per cent, only slightly in excess of the rise in the working-age population, and annual productivity increases of a little below 2 per cent (Table 3). This projected performance would not be sufficient to make much progress in narrowing the gap in GDP per capita *vis-à-vis* the United States, where potential output is expected to grow at rates very similar to Canada's and population growth to be only fractionally higher.

Closing the income gap with the United States

Despite being among the highest in the OECD, per capita incomes in Canada were around 16 per cent below those in the United States in 2002 if the comparison is made using the OECD multilateral purchasing power parities (15 per cent using Statistics Canada's bilateral PPPs) (Figure 7). However, comparisons based on any particular year reflect in part the relative cyclical positions of the two countries, and indeed Canada's was more favourable than the United States in 2002. The gap has narrowed by about five percentage points since 1997, but this was due in large part to the already mentioned recovery of Canadian employment rates from cyclically low levels, and the narrowing resulting from this source is unlikely to continue in the next few years. Arithmetically, the income per capita shortfall can be split into the difference in the number of hours worked and in the output produced each hour. Looking at underlying trend values, the average Canadian works slightly fewer hours each week[4] and fewer of them work (with the structural unemployment rate being higher than in the United States and the trend participation rate about the same). This gap in labour utilisation (in total, slightly more than 3 per cent) is largely offset by Canada's demographic advantage, as a larger proportion of its total population is of working age. Overall, therefore, the difference in living standards is almost entirely accounted for by the productivity gap (17 per cent). Nevertheless, raising the degree of labour utilisation to US levels would allow Canada to reduce the income gap by about one-fifth.

Raising productivity growth is therefore one of the central challenges facing the Canadian economy. It will be an especially daunting challenge given that the productivity shortfall has been both long-standing and widespread in the sense that it is not isolated in a handful of regions or industries. Comparing across industries,

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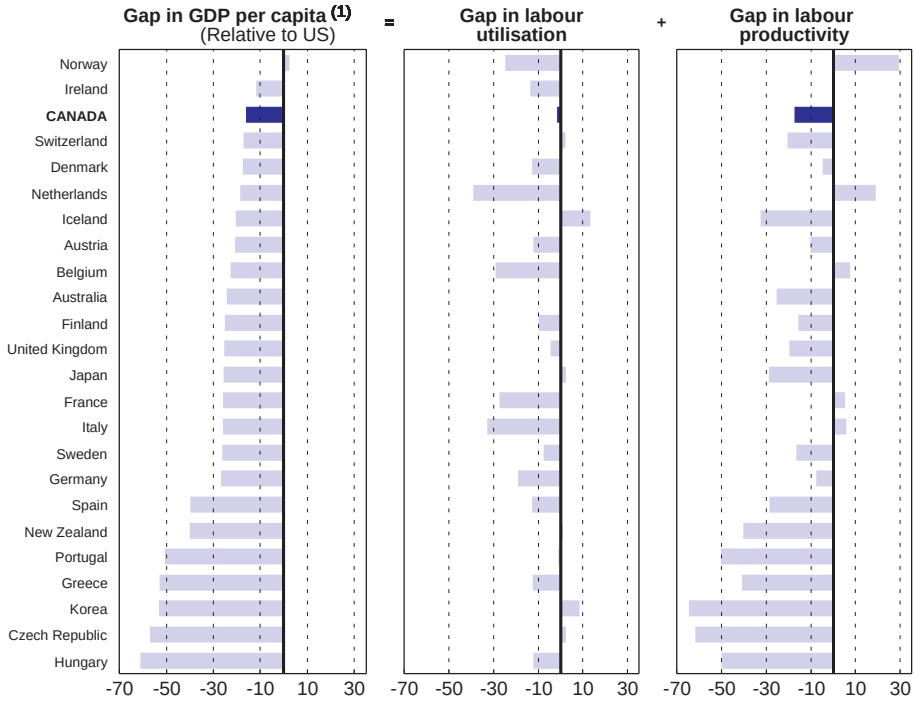
Figure 7. **Per capita income relative to the United States**

Percentage point differences in GDP per person relative to the United States, PPP-adjusted, 2002



1. The gap in GDP per capita is only approximately equal to the sum of the two components shown as there is a small additional demographic effect (differences in the share of population that is of working age).

Source: OECD.

Canadian productivity lags its southern neighbour's in most cases, and the gap is especially wide in manufacturing (Figure 8, Panel A). The disparity in labour productivity in the manufacturing sector can be partly attributed to the higher capital intensity of US industries, as the gap in multifactor productivity levels is smaller than the gap in labour productivity, but even here the difference remains significant. Comparing productivity levels in the service sector is more difficult, but productivity *growth* in services is estimated to have been much higher in the United States than in Canada over the past 10 to 15 years (see the next Chapter for more details).

There has been surprisingly little productivity catch-up over a span of several decades, especially considering the high degree of economic integration

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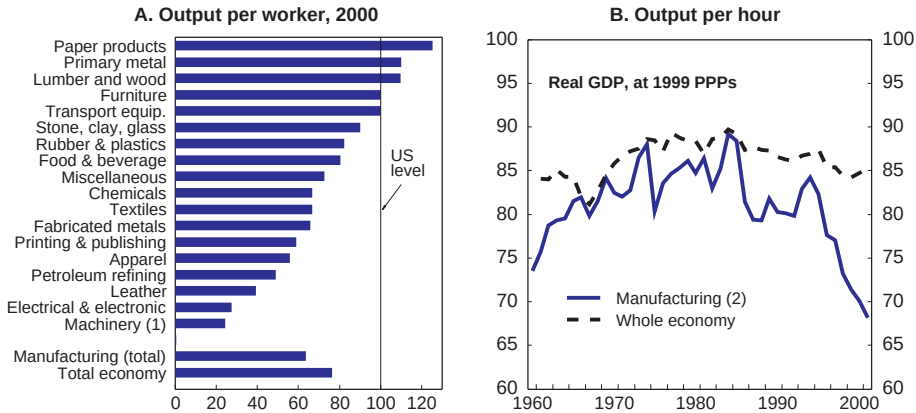
Figure 8. **Relative labour productivity of Canadian industries**
United States = 100



1. Includes computers and office equipment.
2. The level is benchmarked to a 1987 value estimated in de Jong (1996).

Source: Industry Canada computations based on data from Statistics Canada; US Bureau of Economic Analysis; US Bureau of Labour Statistics; OECD, STAN database.

in North America. Despite the fact that Canada started from a lower level 40 years ago, cumulative labour productivity growth since then has been almost the same in the two countries (Figure 8, Panel B), with the United States suffering a bigger productivity slowdown in the 1970s but a sharper pickup thereafter. This lack of convergence, however, is more than just a Canadian phenomenon. Throughout the OECD, labour productivity levels in manufacturing stopped converging in the 1980s, well before the "new economy" boom in the United States (Lee and Pilat, 2002). Since then, US productivity has accelerated away from both Canada and much of the rest of the OECD. There are increasingly clear signs that, looking through the cyclical effects, trend productivity growth in Canada picked up over the 1990s (see Annex I) but not by enough to start reducing the gap with US levels.

The productivity gap with the United States reflects genuine differences in efficiency within individual industries and firms, rather than being a statistical artefact caused by differences in industrial structure between the two countries. In fact, recent firm-level research, described in Chapter II, has located the shortfall more precisely. Comparisons across categories of firms within Canada tell us which firms' productivity performance has been superior: capital-intensive plants are doing better than others; small plants have been falling behind large ones;

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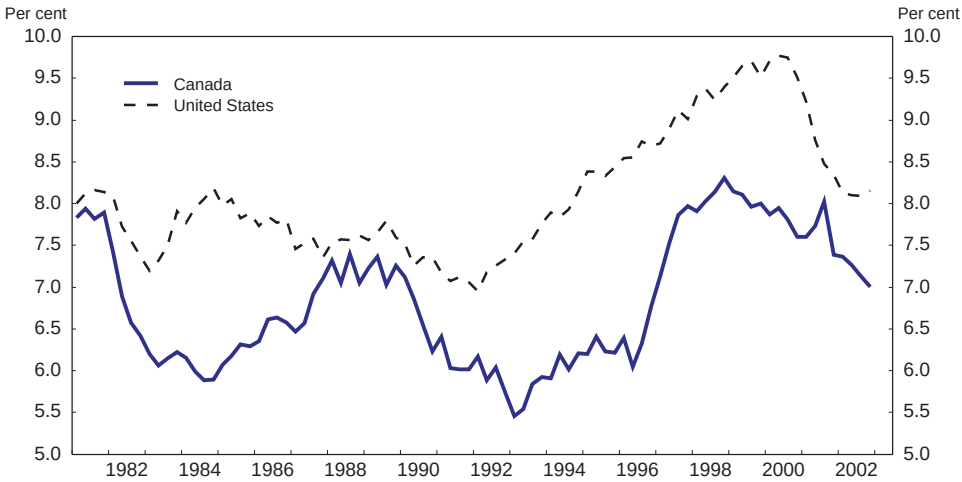
domestic owned firms have been falling behind those that are foreign-controlled; heavy users of ICT have been pulling away from less innovative firms; and exporters have been pulling away from non-exporters. While it is difficult to compare Canadian and US firms across the export-intensity and foreign-ownership dimensions, given the different size of the two countries, it would seem reasonable to conclude that the smaller size and lower capital intensity of Canadian firms relative to their US counterparts, as well as their more limited use of ICT, explain an important part of the productivity gap.

Capital accumulation, and especially ICT use, plays an important role. While both Canada and the United States devote about the same proportion of GDP to business investment, Canadian firms tend to invest less in machinery and equipment (M&E) and more in structures. The US M&E investment rate has been higher than Canada's every year since the 1980s, even during the recent ICT investment slump (Figure 9). The difference in M&E investment rates widened in the 1990s, mostly as a result of the US ICT boom in the second half of the decade. Indeed, the ICT component of capital deepening accounts for about ½ percentage point of the difference in labour productivity growth in 1995-2001 (Table 4). Apart from capital accumulation, the comparative analysis conducted in the context of the OECD Growth Project has identified a number of other key drivers of productivity

Figure 9. **Machinery and equipment investment**

Business sector, per cent of value added



Source: OECD.

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Table 4. **Sources of labour productivity growth**
Annual average percentage growth, business sector

	Canada			United States			United States minus Canada
	1981-88	1989-2001	1995-2001	1981-88	1989-2000	1995-2000	1995-2000
Labour productivity growth	1.3	1.5	1.7	1.9	2.0	2.7	0.9
Contributions from:							
Capital deepening	0.6	0.7	0.5	0.6	0.8	1.1	0.6
ICT	0.3	0.4	0.5	0.5	0.6	1.0	0.5
Non-ICT	0.3	0.2	0.1	0.2	0.2	0.1	0.1
Labour quality	0.5	0.5	0.3	0.3	0.3	0.3	0.0
MFP growth	0.3	0.3	0.9	0.9	0.9	1.4	0.3
From ICT producers (1995-2001)	. .	. .	(0.2)	. .	. .	(0.6)	(0.4)
Total contribution from ICT (ICT capital deepening + MFP of ICT producers)[1]	. .	. .	0.7	. .	. .	1.6	0.9

1. See Crawford (2003) for sources and precise time periods.
Source: Statistics Canada; US Bureau of Labor Statistics; and Crawford (2003).

growth, including investment in human capital and in R&D; regulatory policies that encourage competition; outward-oriented trade and foreign investment policies; and a strong science framework that supports the creation and diffusion of innovation. The productivity challenge is clearly a multi-faceted one, and there is no single, universally valid solution, but improvement will be required in a broad range of areas. One factor that makes the challenge more difficult is the need to ensure that the targets for the reduction in greenhouse gas emissions that Canada has committed itself to are achieved in a way that does not work against its agenda for higher productivity growth. An assessment of where Canada stands on that scorecard and a discussion of the policy implications are presented in Chapter II.

Longer-term challenges

As in other OECD countries, Canada faces the prospect of slowing population growth and of a population that will age rapidly over the coming decades. According to central official projections – based on unchanged fertility rates, migration flows at present levels and life expectancy continuing to rise gradually – population growth would slow to zero by 2041, from 0.8 per cent per year at present; the median age of the population would rise from 38 to 46; and the old-age dependency ratio would more than double. At the same time, the labour force

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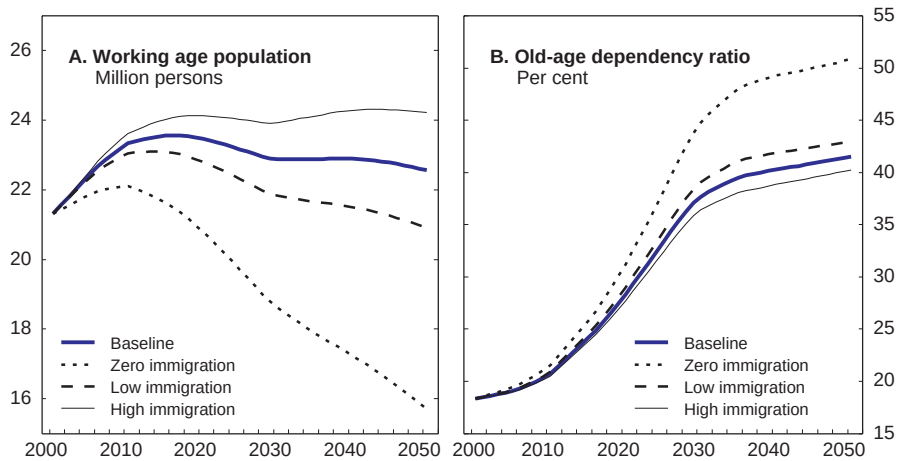
would peak in around 2016 and would also become older, with the share of people in the 45-64 age range rising from 35 to 42 per cent.

With regard to ageing trends Canada does not face a more difficult challenge than other OECD countries. The level and the projected pace of increase in old-age dependency are close to the average, and population growth will continue longer than in most other countries, despite the fact that Canada had a particularly pronounced "baby boom" and has one of the highest life expectancy levels, with a fertility rate that is about average. Part of the reason for this more favourable situation lies in the fact that Canada remains a high-immigration country. Annual inflows are equal to about ¾ of a per cent of population – a proportion that is three times as large as in the United States, but a little smaller that in Australia, New Zealand and some European countries – and net migration adds about 0.5-0.6 per cent to population every year. Without immigration, working-age population would decline considerably faster; the old-age dependency ratio would be about 5 percentage points higher by 2026 and 10 points higher by 2051 (Figure 10).

The contribution of immigrants to the economic welfare of Canadians cannot be assessed on the basis of sheer numbers. Canada selects 30 per cent of its immigrants on the basis of their skills (with their accompanying family members,

Figure 10. **Working-age population and old-age dependency ratios under alternative immigration assumptions**



Source: Statistics Canada.

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the economic categories make up about 60 per cent of the overall immigrant flow), on the assumption that those with higher educational achievements have a better chance of adapting to the host society and are more likely to make a positive economic contribution. But this contribution will materialise only if the newcomers are effectively absorbed into the labour market. The experience of the 1980s and 1990s, when new immigrants found increasing difficulty in obtaining jobs matching their skills, has shown the importance of having a dynamic, expanding economy to obtain the full benefits from immigration. Canada's current immigration policy and its recent experience are discussed in Chapter III.

The Canadian economy's ability to achieve higher productivity growth and to narrow the gap in living standards with the United States will also interact with long-term demographics by influencing the attractiveness of Canada for potential immigrants and its ability to retain its own highly skilled workers. In an increasingly integrated global labour market for the highly skilled – particularly in North America – Canada cannot take for granted its continuing ability to attract the most desirable immigrants and to retain its own "best and brightest" if the salaries it can offer continue to slip relative to those available in the United States. This is not only a function of average income per capita but also of income distribution, in particular the comparative return earned from higher skills in the respective labour markets. It is well known that this return is higher in the United States than in most other developed economies, including Canada. In addition, in Canada individuals with higher earnings normally pay higher personal income taxes, partly because of the need to finance higher public expenditure on social services, including health. Even though it is faced with competitive pressure from the increasing degree of integration with the United States, this social model involving a greater degree of income redistribution can be sustained provided that it supports a dynamic and innovative economy and that it continues to provide high-quality public services at a cost that does not imply an excessive tax burden.

Canada has made more progress than most other OECD countries in dealing with the fiscal challenges deriving from ageing trends, in particular by reforming pension systems and setting the ratio of public debt to GDP on a downward path. However, it still faces one other major consequence of those trends, namely the rise in health expenditures. Public spending on health is projected to rise by 4 percentage points of GDP by 2050 as a result of the changing age structure of the population. This projection does not take into account the trend rise in the relative price of health services and the increased volume of services people in each age group want, due both to the higher-than-unit income elasticity of demand for health and to scientific progress making available new and expensive treatments. So far, Canada's public health system has been able to deliver high-quality services, but cost containment was achieved during the 1990s in large part through funding cuts, which generated queuing and staff shortages. The government's recent decision to substantially increase public funding for health responded to

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widespread public pressure to fix those problems. At the same time it was decided, in line with a broad consensus in Canadian society, not to alter the main characteristics of the current system (free and universal coverage for core needs). Sooner or later, however, the federal and the provincial governments will need to face the longer-term issue of how much Canadian society is collectively willing to spend on health, and through which mechanisms costs should be controlled and spending allocated between alternatives. To the extent that these collective choices go in the direction of accommodating rising demand for health, long-term fiscal plans need to start creating in advance the financial room that will allow future governments to cover that cost without sharply rising taxes. In practice, this would mean envisaging more ambitious targets for the reduction of public debt over the next ten years, while the demographics are still favourable. These issues are discussed in Chapter IV.

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II. Raising living standards

Canada has undertaken wide-ranging and impressive economic reforms over the past 15 years. The regulatory environment and the incentives facing firms and individuals have been substantially improved through measures such as tax reform, the North American Free Trade Agreement, reducing and simplifying tariffs, clearing away barriers to internal trade, privatising public enterprises, shaking up the public sector, and revamping labour market programmes to promote job attachment. Macroeconomic policies are also now on a sound footing. The payoff is clear: the sustainable growth rate has risen, and unemployment has been reduced. When measured against the OECD as a whole, Canada is now performing well in most areas. However, it has the advantage of being closely integrated with the dynamic US economy and so tends to grade itself not against the OECD or even the G7 average but against its southern neighbour. By setting a higher benchmark, the comparisons are often less flattering. Most conspicuously, and as noted in Chapter I, per-capita incomes in Canada are around 15 per cent below those in the United States. This chapter looks at ways to close the gap in living standards. Over the long term, the most important requirement by far is to increase productivity. To do so, Canada must focus on boosting innovation, competition and skills. Income levels can also be raised by further boosting employment rates, and several policy options for improving the labour market are discussed. However, living standards ultimately depend on more than just per-capita incomes, and the chapter touches on several environmental aspects of sustainable development: water quality, air pollution, and climate change.

What explains the income gap with the United States?

Chapter I showed that most of the Canada-US income gap from 1997 to 2002 could be attributed to Canada's lower level of productivity. A shortfall exists to varying degrees in most industries and regions and thus cannot be attributed solely to a difference in industrial structure, such as the United States having a larger ICT-producing sector. In fact, industrial structure works in Canada's favour; if it adopted the US industry make-up but kept its own productivity levels within each industry, then GDP per person would be around 2 per cent lower than it currently is (Lee and Pilat, 2002). In terms of growth performance, the United States

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has pulled further away from Canada since 1995 in a majority of industries, most notably in the manufacturing industries that are not directly related to primary industries, but also in the service sector (Table 5). Productivity growth in the wholesale and retail trade industries has been much stronger in the United States.

Research undertaken as part of the OECD's *Growth Study*, and studies performed elsewhere, have shed light on some of the reasons why productivity levels differ across firms, industries and countries. One branch of research has looked at where, or at what level, most productivity growth actually takes place. Differences in aggregate productivity growth between countries can be split into three parts:

Table 5. **Labour productivity growth**

Real value added per worker, annual average per cent change

Industry	Canada			United States		
	1987-95	1995-2000[1]	1987-2000[2]	1987-95	1995-2000	1987-2000
Primary industries	3.2	4.5	3.7	1.1	0.4	0.8
Construction	–1.3	0.0	–0.8	0.1	–0.7	–0.2
Manufacturing	3.2	1.5	2.5	2.9	4.3	3.4
Food and beverage	2.0	–0.9	1.4	2.4	–6.4	0.8
Tobacco	2.8	2.4	2.7	–1.1	–4.6	–1.8
Rubber and plastics	2.0	1.8	2.0	4.2	5.2	4.4
Leather	1.0	0.4	0.9	4.3	–0.5	3.2
Textiles	2.5	0.9	2.2	2.9	2.9	2.8
Lumber and wood	–0.9	1.5	–0.5	–2.9	–2.6	–3.1
Furniture and fixtures	3.1	1.1	2.7	0.9	2.7	1.2
Paper products	3.3	2.8	3.2	–0.1	6.4	1.2
Printing and publishing	–2.0	–0.3	–1.7	–2.4	–1.6	–2.5
Primary metal	5.3	4.7	5.2	2.6	2.6	2.6
Fabricated metal	1.3	0.7	1.2	1.6	1.6	1.6
Machinery except electrical[3]	2.7	–0.3	2.0	6.6	10.5	7.4
Transportation equipment	5.8	2.3	5.1	0.8	–1.6	0.3
Electrical and electronic equipment[3]	7.7	3.8	6.9	12.7	16.5	13.5
Stone, clay and glass	–0.1	7.1	1.3	3.0	4.1	3.2
Petroleum and coal production	7.8	–1.8	5.8	0.4	0.1	0.4
Chemicals	4.0	1.6	3.5	2.3	5.7	2.9
Other manufacturing	2.0	1.8	1.9	0.4	–2.5	–0.3
Transportation and warehousing	1.9	1.7	1.8	1.7	1.6	1.7
Communications and utilities	1.0	3.7	2.0	4.0	1.7	3.1
Wholesale trade	1.8	1.9	1.9	2.9	7.2	4.5
Retail trade	0.1	3.8	1.5	0.9	4.8	2.4
Finance, insurance and real estate	1.4	1.0	1.2	1.7	3.4	2.4
Total economy	1.0	1.8	1.3	1.3	2.4	1.7

1. 1995-97 for sub-sectors of manufacturing in Canada.
2. 1987-97 for sub-sectors of manufacturing in Canada.
3. Computer and office equipment are included in machinery in the United States and in electrical and electronic equipment in Canada.

Source: Rao and Tang (2003).

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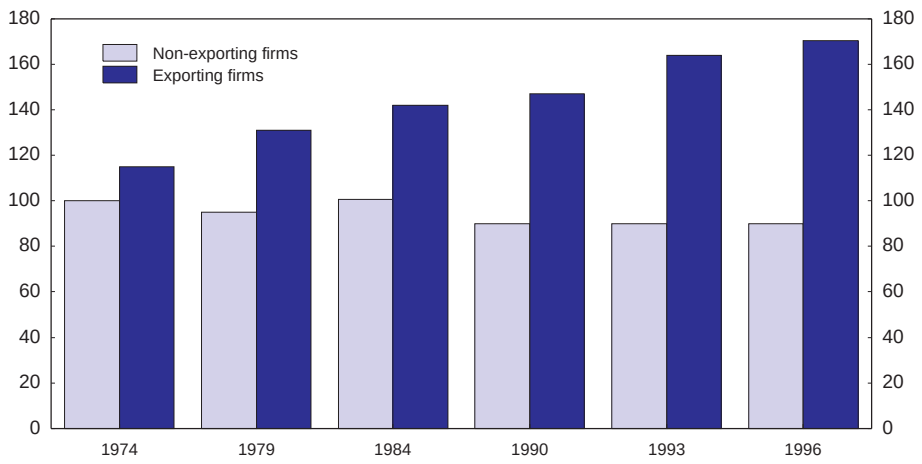
i) differences in the rate of productivity improvement within individual firms; *ii*) the pace with which resources shift into higher-productivity sectors with the birth and death of firms; and *iii*) differences in industrial structure, *i.e.* the concentration in high-growth industries. Firm-level evidence shows the first factor to be the most important: the majority of productivity growth occurs *within* firms,[5] with a smaller contribution from resource reallocation or differences in industrial structure.

A second conclusion from this research is that productivity performance is strongly affected by exporting (although the relationship obviously works in both directions). In manufacturing, virtually all Canada's productivity growth over the past three decades has been generated by exporters (Figure 11): average labour productivity levels of non-exporters have barely changed.[6] The reasons why exposure to trade can boost productivity include stronger competition, contact with foreign ideas and best practices,[7] greater scope to realise economies of scale and encouraging greater product-line specialisation and longer product runs.[8] Exporters also tend to be larger, especially those that are truly global exporters rather than just serving the North American market. They are also more likely to be foreign controlled: 60 per cent of Canadian manufactured exports come from foreign-controlled plants, a much higher proportion than for manufacturing sales overall. However, most firms are still heavily

Figure 11. **Productivity of exporters and non-exporters**
Labour productivity levels in manufacturing
Non-exporters level in 1974 = 100



Source: Baldwin and Gu (2002a).

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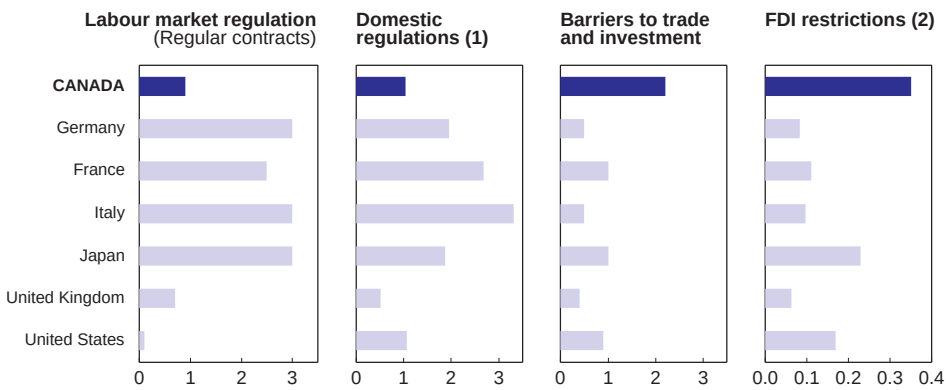
reliant on local sales. Amongst small and medium-sized manufacturers who exported in 1997, the local market generated three-quarters of revenues, with the United States accounting for 19 per cent and other countries just 5 per cent (Lefebvre and Lefebvre, 2000). This partly reflects the difficulties of breaking into and holding export markets. Of the manufacturing firms that were not involved in exporting in 1990, only 9 per cent had become exporters by 1996, despite the implementation of the NAFTA. Of those that were exporting in 1990, a quarter stayed in business but dropped out of the exporting business and another quarter either exited or went out of business by 1996 (Baldwin and Gu, 2002a). A sizeable proportion of firms who are recorded as exporters at a particular point in time are filling a one-off order. A continuous, sustained export focus is rare, especially to markets beyond the United States.

A third conclusion is that competition is also a key driver of productivity. There is increasing evidence that vigorous competition is a major spur to innovation and productivity. Restrictive product- and labour-market regulations lower productivity growth, with a stronger impact the further a country is from the technological frontier.[9] Bearing in mind the difficulties of comparing regulatory regimes, the OECD's synthetic indicators suggest that Canada has very low barriers to entrepreneurship[10] and a relatively liberal environment in which to do business, but it is among the most restrictive in the OECD when it comes to regulatory barriers to trade and investment (Figure 12). In particular, while discriminatory

Figure 12. **Product market regulation**

The scale of indicators is from least to most restrictive, late 1990s



1. Inward-oriented policies.
2. The scale of this indicator cannot be compared to the others.

Source: OECD Regulatory Database.

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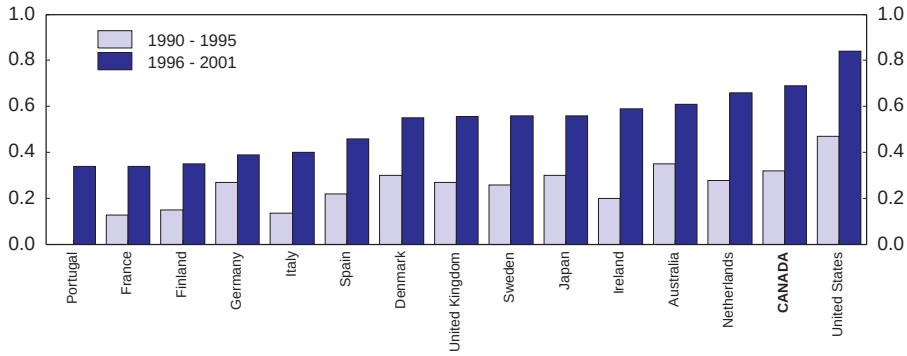
procedures are about average by OECD standards, foreign ownership barriers are high (Nicoletti, Scarpetta and Boylaud, 1999). The OECD's composite index of FDI restrictions shows Canada to be the second most restrictive in the OECD (OECD, 2003a).[11]

Fourth, productivity growth is strongly related to innovation and to ICT investment in particular. Canada appears well placed in this area. ICT production and use contributed 0.7 percentage points, or nearly half, of the labour productivity growth from 1995-2001 (Table 4). The use of ICT also contributed around 0.6 percentage point to annual GDP growth in the second half of the 1990s, up from 0.3 in the first half (Figure 13). This is high by OECD standards, though it still falls short of the US outcome, possibly because Canada's ICT investment pickup began a couple of years later. ICT take-up is higher in countries with more liberal product- and labour-market regulations, reinforcing the need to minimise barriers to competition and adjustment.

Other important factors that affect productivity include ownership and plant size. On average, foreign-controlled firms are more productive than those that are Canadian controlled, even after adjusting for differences in size, capital intensity and labour quality.[12] This points to the need to encourage foreign investment and to minimise entry and ownership barriers. Plant size also matters.

Figure 13. **The contribution of investment in ICT capital to GDP growth**

Percentage point contribution to annual average GDP growth, business sector



Note: The figures in this chart for Canada are similar to those of two recent Canadian studies. Khan and Santos (2000) estimated that ICT investment contributed 0.33 percentage point to business sector growth over 1991-1995, and 0.55 percentage point from 1996 to 2000. Armstrong *et al.* (2002) estimated 0.4 percentage point (1985-95) and 0.7 percentage point (1995-2000).

Source: OECD.

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Labour productivity of large manufacturing plants relative to small ones has risen continuously over the past thirty years. Interestingly, the relative productivity improvement of big plants over small ones has been almost the same in the United States, but Canada has fewer large plants, and this may explain part of the productivity gap.[13] If so, it points to the importance of removing unnecessary barriers to reaping economies of scale, such as barriers to inter-provincial trade.

To sum up, both labour and multifactor productivity are well below US levels – but high by world standards – with little catch-up over the past few decades. Within Canada, small plants have been falling behind large ones; domestically owned firms have been falling behind foreign-controlled ones; heavy ICT users have been pulling away from less innovative firms; exporters have been pulling away from non-exporters; and capital-intensive plants have been doing better than their low-investing counterparts. Ways to address some of these issues are taken up in later in the chapter. The first challenge, however, is to get more people into work, and this is discussed next.

Boosting employment

Persistently high structural unemployment is the comparative weak spot of Canada's labour market, which continues to under-perform relative to many other high-growth economies. In 2001, the last year in which the OECD-wide output gap was around zero, nearly half of OECD countries had unemployment below 5 per cent. Canada has not yet made it into this elite group. The gap in labour-market performance – employment rates in particular – has been largest for young men and for people close to retirement (Table 6). Compared with the OECD as a whole, employment rates for women are well above average, but the proportion of prime-age males in work is a couple of percentage points below the OECD norm. The worst-affected group has been the unskilled: less than two-thirds of adult men with less than upper secondary education were working in 2001, compared with 70 per cent in the United States and 76 per cent across the OECD.

On the positive side, the combination of high participation with unemployment spells of short average duration implies that the jobless remain fairly well attached to the labour market. The Employment Insurance (EI) scheme can take some of the credit for this, as the most generous employment, family and sickness benefits are available only to those who have recently worked. On the negative side, however, various features of the social assistance and EI programmes keep unemployment unnecessarily high because of their effect on work incentives. Canada still has plenty of headroom if it wants to be an above-average performer as, despite the undoubted progress of the past few years, employment rates in the OECD's top five labour markets are still more than 8 percentage points above Canada's.

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Table 6. **Employment rates**
Per cent of population, 2001

	Men				Women			
	Canada	United States	OECD	Top five[1]	**Canada**	United States	OECD	Top five[1]
	By age group							
15-24	**56.5**	59.4	48.8	64.2	**56.3**	56.2	40.2	63.1
25-54	**85.4**	87.9	87.5	94.2	**74.3**	73.5	63.9	81.6
55-64	**57.6**	65.8	59.8	80.3	**39.4**	51.6	37.8	62.7
Total (15-64)	**75.9**	79.3	75.6	84.4	**66.0**	67.1	55.2	74.0
	By skill level (of those aged 25-64)							
Less than upper secondary	**65.6**	69.8	75.8	88.9	**43.3**	47.1	40.4	64.6
Upper secondary	**82.2**	82.1	83.3	92.3	**68.3**	70.6	65.2	79.6
Tertiary	**85.9**	90.3	90.0	94.6	**79.5**	79.3	75.8	88.3
Unemployment rate (15-64 year-olds)	**7.5**	4.9	6.1	2.6	**6.6**	5.1	6.8	3.0

1. Simple average. The "top five" countries are different for each category, but Denmark, Norway, Sweden, Switzerland and Iceland are often in the top group.
Source: OECD, *Employment Outlook*, July 2002.

Minimising unemployment traps in social assistance

The share of the population on welfare more than doubled between 1980 and 1994, peaking at 11 per cent before declining to around 7 per cent in 2002. The rise reflected an increasingly generous social assistance system interacting with the recession of the early 1990s. Despite several years of recovery, long-term benefit dependency still seems to be a problem. Serious welfare reform began in 1996 when the federal government withdrew from its cost-sharing arrangement with provincial governments and moved to a block grant for health and social services (the Canada Health and Social Transfer, CHST). Benefit levels were cut almost everywhere, typically by between 10 and 35 per cent in real terms (see the table on page 199).

Benefit cuts were combined with reforms that made welfare harder to get, such as changes to eligibility rules and administration. Some provinces have been more pro-active than others. The most far-reaching changes have been in Alberta, which is the only jurisdiction that has aggressively moved to implement US-style welfare reforms. Its review of new and existing cases was stepped up, waiting periods were introduced, claimants had to attend information sessions before their applications were processed, and stricter work testing was put in place. These reforms have contributed to a halving in caseloads, with the result that Alberta is

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the only province with welfare rates comparable to US outcomes. Reforms in British Columbia centred around a greater emphasis on job search and ending welfare altogether for people under 19 years of age. More recently, as a result of drastic cuts to provincial government spending, British Columbia announced in 2002 that it will further reduce benefits and tighten eligibility. Ontario introduced mandatory "workfare" in 1998, where a beneficiary must enter one of three programmes: job search assistance; community work; or a wage subsidy programme. In contrast, Quebec eased eligibility requirements by raising the asset limit for recipients. Other provinces introduced changes of varying importance, but several have simply streamlined the bureaucracies that deliver benefits, rather than introducing deep structural reforms.

While significant progress has been made, there are several areas where further improvements to social assistance would be useful:

– *Improving work incentives by reducing high marginal effective tax rates* (METRs). Effective tax rates for someone moving from welfare to work can be high over some income ranges, including the amount that someone could expect to earn in a low-paying job. The main culprit is the combined claw-backs of social assistance, the Child Tax Benefit and the GST sales tax credit. For example, for a Canadian family with two children, the effective tax rate rises from around 80 per cent to a peak of 85 per cent as its income rises to about two-thirds of the average wage (Figure 14). However, Canada is not alone in this problem; many other countries have high peaks in effective tax rates over certain income ranges. Another major, but hard to quantify, contributor to the "welfare wall" is the loss of in-kind benefits such as subsidised health care and housing. Reducing METRs is not easy, since lowering abatement rates makes the programme more expensive, while lowering benefit levels can increase poverty. However, the National Child Benefit has made an important contribution to reducing the financial disincentives to leaving social assistance. Going forward, the federal government and the provinces will need to ensure that low- and modest-income families with children who increase their earnings – for example, by taking up better paying jobs – keep more of that extra money. This will include examining the claw-back rates from the Canada Child Tax Benefit as well as other elements of the tax and benefit structure. Improvements could be made at low cost through better co-ordination of abatement regimes (including non-cash benefits) across different federal and provincial programmes.

– *Make more use of in-work benefits*. If well designed, these can be an effective way of boosting employment without increasing poverty (Pearson and Scarpetta, 2000). Canada has few employment-conditional benefits, with most programmes consisting of provincial earnings supplement

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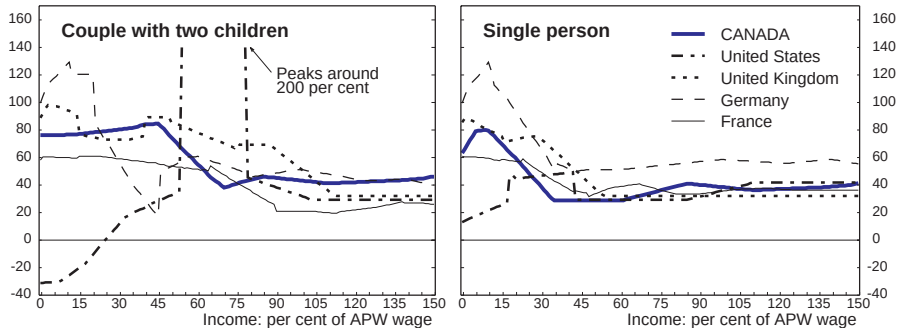
Figure 14. **Effective tax rates when moving from welfare to work**

Effective tax rate in per cent for a large increase in income[1]



1. The charts show the effective tax rate when moving from social assistance into work for an increase in income of one quarter of the average wage. For example, a two-child Canadian family faces an effective tax rate of 85 per cent when its income rises from 40 to 65 per cent of the average wage. The Canadian tax and benefit rates are for Ontario in 1999. Since then, statutory tax rates have been reduced while various benefits have been increased. Other countries are based on 2001 tax-benefit systems.

Source: OECD, Tax-benefit models.

schemes for families. In addition, all provinces pay small allowances for back-to-work expenses and, in some cases, day care, while British Columbia pays a bonus for participation in work activities. The federal government led an experiment in the 1990s with a back-to-work benefit for sole parents, paying a wage subsidy for up to three years provided they remained in work and off welfare. The subsidy was generous – doubling the income of someone in a minimum wage job – and had a significant impact on employment compared with a control group that did not receive the subsidy (Annex II). Provincial governments should look at implementing similar schemes, although they could probably be less generous than the federal pilot and still have positive effects. The major drawback of these policies is that the phasing out of in-work benefits increases METRs, so any reforms should be part of a package that addresses the uncoordinated stacking of abatement rates more generally.

– *Stronger training and job-search requirements.* While most provinces have toughened job-search requirements or beefed up what was generally lax enforcement, Canada has still not moved as far as many OECD countries in this area. Alberta is the only province where work requirements begin immediately; in Ontario, for example, mandatory enrolment in the

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Ontario Works programmes begins after a four-month waiting period, during which welfare operates much as it did in the past (Boessenkool, 1997). The evidence reviewed in Grubb (2000) showed that the European countries in which unemployment fell most sharply in the 1990s have considerably tightened eligibility criteria and their implementation. Also, the evidence on active labour-market policies suggests that training programmes – a key part of Canada's suite of labour-market measures – are effective only when combined with monitoring of job search and enforcement of the work test (Martin, 2000).

– *Introduce diversion programmes.* Alberta, like many US states, reduced welfare caseloads partly by reducing the number of people applying for welfare for the first time. For example, applicants in the US state of Wisconsin have an initial interview whose objective is to find a solution that does not involve going on to welfare. British Columbia experimented with a diversion programme in 1995 and 1996, in which 17 regional offices were given verification staff who double-checked both new and existing marginal cases. It found that initial benefit denial led to less benefit use in the future, with the effect lasting about 2½ years, although the programme had no impact on a "core" group of hard-to-treat beneficiaries (Green and Warburton, 2001). However, such measures need to do more than just discourage claimants; they must also encourage work if they are to be welfare enhancing.

Further reforms to the Employment Insurance *programme would lift labour supply*

The unemployment insurance system has undergone major improvements since 1990. The initial changes involved improving work incentives by tightening eligibility criteria and shortening the duration of benefits. Further reforms in 1996 were designed to reinforce the insurance principle: workers became experience-rated; the generosity of the system was reduced further; eligibility was tightened again, albeit by less than in the earlier reforms; and some income redistribution measures became better targeted. The system also changed from a weeks-based to an hours-based system, making it fairer for part-time workers and aligning incentives to ensure employees accept every hour of work on offer. The government should continue reforming the system in order to help the jobless and welfare recipients, thereby lifting labour supply. Various possibilities are reviewed below.

Re-instate experience rating for workers

A key part of the 1996 reforms was to experience-rate workers so that benefits were lowered and made harder to get for frequent users of the system. In 1996, frequent claimants – those who had drawn EI at least three times in the

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previous five years – accounted for around 40 per cent of all claims. The EI rules created an incentive for firms in some seasonal industries to temporarily lay off workers each year, turning the income-support principle into an implicit wage subsidy for frequently idle manpower.[14] The major flow of funds was from the service industries and towards the primary sector and construction, and from Ontario towards the provinces to its east (Corak and Chen, 2002). A significant amount of frequent claims was paid to workers with above-average weekly earnings. Transfers were also highly concentrated at the firm level. In the 10 years to 1996, only 6 per cent of firms consistently received a net positive transfer, but they were responsible for 28 per cent of all benefits paid and contributed only 4 per cent of levies. Almost three-quarters of claims in the firms that were subsidized every year were due to above-average rates of temporary layoffs.

The key change was the introduction of an "intensity rule" that slightly reduced benefit levels in line with the number of weeks of benefit collected over previous years.[15] The claw-back mechanism was also toughened. Recipients who go on to earn incomes above a certain threshold in the year following an EI claim are required to pay back up to 30 per cent of the benefits received. For frequent users, that income threshold was lowered and their repayment rate lifted to between 50 and 100 per cent.

Experience rating was cancelled in 2001. The government felt that the policy was having little impact on behaviour yet was excessively penalising recipients in regions where off-season job opportunities were few and far between. However, as discussed in the previous *Survey*, that conclusion seemed premature. It was based on an assessment covering only two years of a policy that was being slowly phased in. The intensity rule took effect in 1996-97, with the average reduction in weekly benefit rising from just C$5 in the first year to C$13 in 2000-01. In addition, everyone's EI history was reset to zero at the outset, so nobody started out as a frequent user.

Some of the evidence that has accumulated since the last *Survey* reinforces the assessment that experience rating should have been kept in place. While it had only a small impact in the first couple of years,[16] from 1997-98 to 2000-01 the number of claims by repeaters fell 18 per cent, compared with just 1.1 per cent for non-frequent claimants. In the last year of the intensity rule (2000-01) – when unemployment began to rise again – claims by infrequent recipients rose 4 per cent, yet those by frequent claimants fell by 4 per cent even though the intensity rule was in place for only half of that year. The impact has been even greater for non-seasonal repeat users: since 1997-98, their claims have fallen by 28 per cent. This suggests that people do respond to incentives provided alternative work opportunities exist. On the other hand, in 2001-02, the first full year in which the intensity rule was no longer in place, the number of claims from repeaters increased by just 0.5 per cent, while the number of claims from non-repeaters

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increased by a substantial 12.2 per cent. However, part of the reason for this may be that normal claims are much more sensitive to the economic cycle than are claims by repeaters (and 2001-02 was a period of rising unemployment).

If the intensity rule and experience-rated claw-back were judged to be too harsh for repeaters in some regions, they could have been modified rather than eliminated. For instance, the definition of a repeater could have been relaxed or made to depend on local unemployment rates, and the experience-rated claw-back threshold could have been raised rather than eliminated entirely. After all, the higher claw-back did not apply to special benefits or first-time claimants and was imposed only on those repeaters who went on to earn more than 110 per cent of the average wage. As well as undermining the insurance principle, the amendments have adverse implications on fairness and income distribution by allowing relatively well off users in specific sectors to regularly receive much more from the system than they contribute. In particular, it redistributes income from low- to high-unemployment regions and sectors, to the benefit of industries such as fishing, tourism and construction. The foregone cost savings translate into higher levies on other EI users. The deadweight loss from this cross-subsidy is large, with one estimate putting it at 16 per cent of EI benefits (Corak and Chen, 2002).

Two alternatives to reintroducing experience rating of workers would be to experience rate their employers instead or to significantly lengthen the stand-down period so that EI would not become payable for several weeks. The first approach could lead employers to place undue pressure on employees to not claim EI as it would affect the firm's rating; the second would deal with the problem in seasonal industries but at the cost of greater hardship on others who are genuinely and unexpectedly laid off. If it is felt that reinstating employee experience rating would have a disproportionate impact on certain regions, then this should be dealt with using regional policy and not by a major economy-wide labour market programme.

Continue to improve incentives to work part time

Incentives to work were improved by making "small weeks" a permanent and national feature of the EI programme in November 2001. The small weeks provision began in 1997 as a pilot project in high unemployment regions. It was designed to test ways of removing work disincentives within EI that had been identified by employers and employees. Results from evaluation studies indicate that the pilot was successful. Specifically, they estimate that participants worked two more weeks on average than they would have done otherwise. Prior to the small weeks feature, the benefit rate had depended on average weekly earnings in the weeks that were actually worked in the six months prior to making a claim. This meant that part-time work (a small earnings week) would lower the subsequent benefit, but a week without work (zero earnings) would not, because it

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would not count in the earnings calculation. Loosely speaking, the new rules ignore these weeks of low earnings ("small weeks").

However, a similar problem still exists for people who work while receiving a benefit. Regular EI claimants are allowed to earn up to C$50 or 25 per cent of their weekly benefit, whichever is greater, beyond which it is reduced dollar for dollar. This creates a disincentive to work part time because if the claimant instead worked a full week and earned enough to reduce the benefit to zero, it would not count as a week of EI entitlement used and would effectively extend the claim by a week. Hence, claimants prefer not to work while on EI unless they can reduce their benefit to zero and keep that week of entitlement for later use. Although working while claiming a benefit is common, repeat EI users have a much greater propensity to do so (Gray and de Raaf, 2002). The provision has therefore become one where repeaters can extend their claim and go back to their previous employer at the start of the next season, discouraging them from getting permanent, non-seasonal jobs. The government has signalled that it will continue to monitor this issue.

Lengthen the qualifying period to be closer to international norms

Despite significant tightening in the mid-1990s, access to EI remains relatively easy compared with many other countries. Entrance requirements vary depending on the local unemployment rate, ranging from 420 hours (12 weeks) of work over the previous year in areas of high unemployment, to 700 hours (20 weeks) in low-unemployment regions. Eligibility periods vary across countries but are typically longer than in Canada. The usual requirement is to have worked for at least half the time over the previous one to three years, and often significantly more than half.[17] However, in Canada entrance requirements are tougher for new entrants and re-entrants to the labour force than they are for workers with prior attachment. This policy is designed to encourage continued workforce attachment, especially for young people. For this group, the entrance requirement is 910 hours (26 weeks). In 2001, this higher threshold was eliminated for re-entrant parents, since it unfairly penalised those who had been on a prolonged parental or maternity absence. The entry requirement in all regions is 600 hours (17 weeks) for special benefits, such as maternity, parental and sickness benefits.

Continue to monitor sickness benefits

Abuse of sickness benefits is not a major problem, as the only available sickness benefit is under the EI scheme and therefore has a limited duration. The numbers involved are low. Even so, the number of beneficiaries rose significantly after EI eligibility rules were tightened in 1995: by 44 per cent between 1996 and 2002, despite a 22 per cent fall in regular claims. While an ageing population has probably contributed to this trend, it is likely that sickness benefits have

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increasingly been used to extend overall benefit duration since they were exempted from the intensity rule and can be drawn in combination with regular benefits. An official study (CEIC, 2001) attributed some of the rise in sickness claims to the economic recovery as stronger employment growth raises eligibility. However, it is possible to correct for this effect by looking at the ratio of sickness beneficiaries to the numbers employed. This ratio still shows a significant increase shortly after the intensity rule was imposed on regular benefits, and a decline after the rule was removed. This, combined with the fact that about 10 per cent of sickness claims follow hard on the heels of a maternity or parental claim, suggest that it may be easy to misuse the benefit, in which case better gate-keeping may be needed. The government has indicated that it will continue to monitor and assess this issue.

Boosting internal migration would lower structural unemployment

Structural unemployment could be lowered by encouraging greater labour mobility within the country. While the internal migration rate is fairly high (not as high as within the United States, but higher than between regions of individual European countries – see OECD, 1999 and 2000a), it has nearly halved since the early 1970s, despite the fact that regional differences in employment rates are as large as ever (Cousineau and Vaillancourt, 2000). As a result, differences in regional unemployment rates are larger and "stickier" in Canada than in many other countries.[18] The fact that Canadians are more willing to stay where they do not have a job probably reflects labour-market institutions rather than simply being due to Canada's stretched-out geography, as mobility is still higher among the 12 US states strung along the south side of the border (Coulombe and Day, 1999).

The mobility problem is concentrated among the less skilled. About 13 per cent of bachelors graduates migrate to another province, compared with only 2 per cent of youth overall (Finnie, 1998). Similar disparities are seen in most countries but the difference is especially large in Canada. Compared with most other OECD countries, skilled Canadians are much more evenly spread across the country, but the less educated are much more heavily concentrated in a few areas (Table 7). So while it is useful to remove barriers to the mobility of professional workers (see the discussion of internal trade later in this Chapter), it is even more important to encourage the less skilled to move. While there are few overt barriers – for example, almost all entitlements to government social assistance, health services, and pensions are fully portable across provinces – the level and eligibility requirements of EI reduce the incentive to move, since benefits are easier to get in regions with high unemployment. That does not necessarily act as a disincentive for occasional users of EI, because once a claim has been established in one province that person can then move to another province with no change to their

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Table 7. **Variation of skills across regions of a country**

	Number of regions	Coefficient of variation	
		Low educational attainment[1]	High educational attainment[1]
Canada	**66**	**19.7**	**9.6**
Belgium	11	6.8	20.0
Czech Republic	8	13.1	46.6
Finland	5	9.7	22.3
France	22	10.2	31.5
Greece	13	16.9	28.2
Hungary	7	13.9	29.4
Italy	20	5.1	18.4
Korea	15	23.5	25.2
Mexico	32	19.2	24.1
Netherlands	12	7.8	18.0
New Zealand	12	13.2	7.2
Norway	7	12.8	26.4
Portugal	7	7.1	26.3
Spain	17	7.6	24.6
United States	51	15.7	14.0

Note: The lower the coefficient of variation, the more similar are regions within a country.
1. Low and high education attainment correspond to less and more than upper secondary education, respectively.
Source: OECD (2000a).

benefit entitlement or level. However, it may act as a disincentive for frequent EI users, since they will be better off staying in a high-paying province if they expect to claim EI again in the future.

Minimise incentives to retire early while maintaining retirement income sustainability[19]

The Canadian retirement income system has made impressive progress in ensuring income adequacy for the retired population, reducing the poverty rate from one of the highest to one of the lowest in the OECD (Figure 15), partly due to the maturing of the public pension system (Myles, 2000). By 2000 the percentage of seniors living on less than Statistics Canada's after-tax low-income cut-off had dropped to 7 per cent, a third of the 1980 share. Increasing pension income for the needy has been achieved at a relatively small budgetary cost with total outlays held down by a ceiling on pensions for people with above-average earnings. As a result, higher income groups rely on private savings in old age, and their income sources are well diversified, although employment is a relatively unimportant source of income for the elderly (Table 8). Unlike many countries, adequate incomes for the elderly have been achieved without creating strong incentives to

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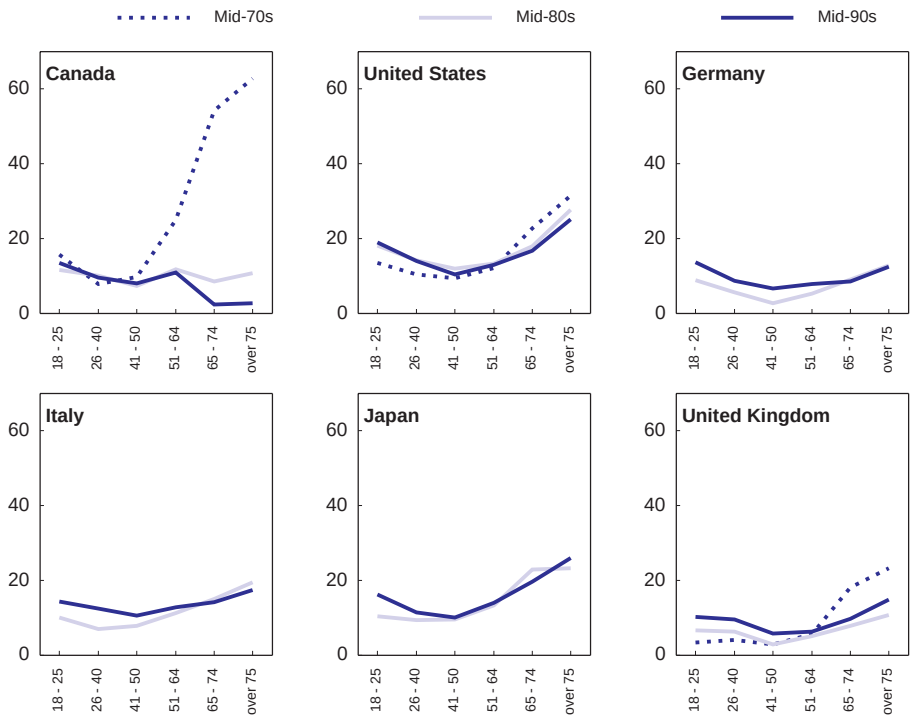
Figure 15. **Percentage of population that is below half median income**[1]
Mid-1970s, mid-1980s and mid-1990s, per cent



1. Half median disposable income of the entire population. So, the rate might be changed by population ageing, even if there were no change in income levels within sub-population groups.

Source: OECD questionnaire on distribution of household incomes (1999).

retire early. It is the case that, until recently, employment rates for older men were well below OECD norms, while for older women they were only average, and both rates remain below their counterparts in the United States. However, this has probably been due to factors other than the public pension (such as high unemployment among seniors in certain regions as a result of collapsing employment in primary industries, and perverse incentives created by some private pension plans). The main challenges facing the government are deciding for how long it can maintain price indexation of the safety net component of the public pension without jeopardising the income adequacy of lower income groups; whether private saving will be sufficient to offset future lower incomes from the state; and whether remaining early-retirement incentives can be reduced further.

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Table 8. **Sources of income for the elderly**

Pre-tax

	1980	1985	1990	1995	1999
Employment income	26.4	20.1	17.3	16.7	10.0
Investment income	23.4	22.6	21.2	14.9	13.1
Retirement income	11.6	13.0	16.2	21.0	28.5
Old-age security and GIS/SPA	25.9	28.0	25.1	23.6	24.0
CPP/QPP benefits	7.7	10.9	14.3	17.3	19.0
Other	4.9	5.5	6.0	6.4	5.3

Source: Statistics Canada (2000b).

Policies

The public pension system in Canada has accounted for a growing proportion of total income of the elderly, but its share is set to decline in the coming decades. The system is composed of several components. The first tier provides the safety net, and consists of a basic flat rate Old Age Security pension (OAS), worth around 14 per cent of the average wage which is clawed back gently from richer pensioners, combined with a Guaranteed Income Supplement (GIS), a means-tested benefit that is phased out quickly with income. The second tier, known as the Canada/Quebec Pension Plan (C/QPP), consists of a compulsory earnings-related benefit with a replacement rate of 25 per cent of average pensionable earnings. However, there is a relatively low ceiling on pensionable earnings (around 125 per cent of average earnings), and some pensioners do not qualify for full pensions, so the average new pension in this tier is equal to only 15 per cent of average earnings. This component of the system is being prefunded by a contribution rate that, at 9.9 per cent of pensionable earnings, is 1¾ percentage points above that which would be currently needed for a pay-as-you-go system, with the surplus placed in a variety of financial instruments. This component can be taken at any age between 60 and 70, with age 65 as the normal age for pension eligibility; the payment rate is reduced for early retirees and increased for later retirees in order to maintain parity between them. Overall, the sum of the components results in a roughly flat rate system paying about a third of average earnings. But at half average earnings, and if the person has no savings, the replacement rate is around 70 per cent (Figure 16).

A key element limiting the growth of expenditure on the safety net pension is the policy of indexing these benefits to prices: under price indexation, the increase is fairly sustainable from a fiscal point of view (Table 9).[20] The result is that the average payment would drop to 14 per cent of average earnings when expenditure peaks relative to GDP in 2030, down from 20 per cent in 2001. While this might be manageable for those at or above average earnings, the fall in the replacement rate will be greater for those on low earnings due to their greater

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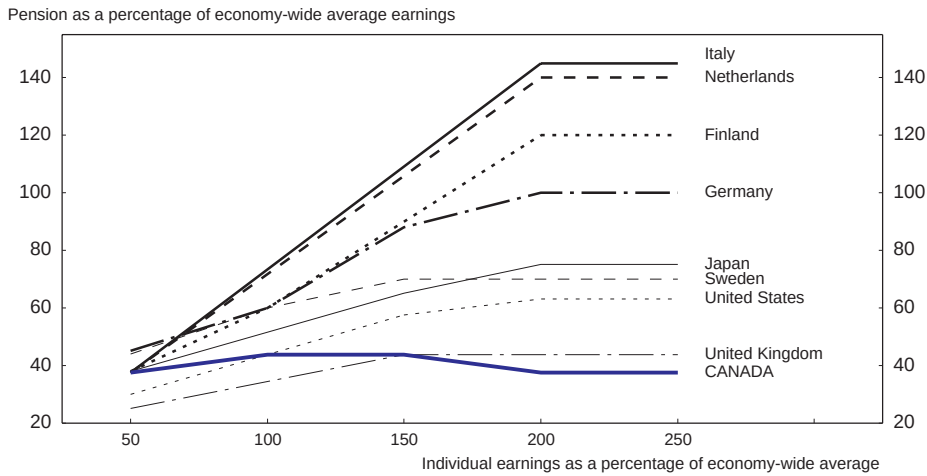
Figure 16. **Public pension profiles**

Pre-tax pension values for typical workers, late 1990s



Source: OECD, *Ageing and Income*, 2001.

reliance on this component of the pension scheme. Indeed, at half average earnings, the replacement rate might fall by 12 percentage points, to around 57 per cent, and even more over the longer term. For low-income groups, the income-tested supplement to the benefit affects saving incentives by lowering the net return to tax-deferred retirement saving, though this group may save little as, at present, public pensions replace a significant amount of their income, much of which is needed to meet living expenses.

Individuals can lessen the impact of the moderate level of public pensions by contributing to the third tier of the Canadian pension system. Nearly all public employees and about one-third of private employees are enrolled in employer-sponsored registered pension plans (RPPs), nearly exclusively of a defined benefit form (Table 10). In addition, individuals can contribute to individual registered retirement savings plans (RRSPs) up to a limit that takes into account the value of their, and their employer's, contributions to a RPP.[21] The dollar limits for RPPs and RRSPs are being progressively raised over the next three years and will then be indexed to average wage growth. The popularity of such plans for saving purposes arises from the deferral of tax contributions and investment earnings until benefits are drawn from the plan, which raises the net return to saving. However, since 1997 the proportion of people in employment covered

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Table 9. **Performance indicators: sustainable retirement income**

	Projected increases in old age pension spending[1]	Low income rate of the elderly[2]	Relative disposable income of the elderly[2]	Private pension funds 1999	Age of withdrawal from labour force, 1994-99[3]		Participation rate, 2001, per cent		
								Aged 55-64	
	Change in per cent of GDP 2000-50	Per cent of the elderly with income less than 50 per cent of median disposable income	Per cent of the disposable income of all individuals	Per cent of GDP	Male	Female	Aged over 65	Male	Female
Australia	1.6	16.1	67.6	63.8	59.7	61.3	6.0	60.0	36.9
Austria	2.2	14.9	86.6	3.6	. .	. .	2.8[2]	42.1[2]	17.5[2]
Belgium	3.3	13.8	77.9	6.1	. .	. .	1.3	36.6	15.7
Canada	**5.8**	**2.5**	**97.4**	**111.0**	**62.6**	**61.1**	**6.0**	**61.3**	**41.7**
Czech Republic	6.8	. .	. .	3.8	. .	. .	4.0	55.0	24.5
Denmark	2.7	9.2	73.0	24.4	62.4	61.5	4.6	65.5	51.8
Finland	4.8	7.5	79.0	10.7	59.8	60.0	3.7	51.2	49.5
France	3.9	10.7	89.7	6.3	59.3	59.8	1.2	43.8	34.1
Germany	5.0	10.4	85.6	3.2	60.5	60.8	3.0	50.6	32.4
Greece	. .	29.2	76.8	4.6	61.7	62.2	5.0	57.0	23.6
Hungary	1.2	6.0	85.2	2.2	. .	. .	3.1	36.3	15.4
Iceland	. .	. .	. .	86.0	. .	. .	19.9	92.8	81.7
Ireland	. .	16.7	74.6	57.8	. .	. .	7.9	66.1	29.5
Italy	–0.3	15.3	84.1	3.0	59.3	58.4	3.4	57.8	26.6
Japan	0.6	. .	. .	18.7	69.1	66.0	21.8	83.4	49.2
Korea	8.0	. .	. .	3.2	67.1	67.5	29.6	71.3	47.9
Luxembourg	. .	6.7[4]	. .	. .	. .	. .	0.0	38.1	14.3
Mexico	. .	32.9	85.3	2.4	. .	. .	30.5	80.5	27.6
Netherlands	4.8	1.9	86.3	119.3	61.6	60.1	3.1	52.0	26.9
New Zealand	5.7	. .	. .	. .	. .	. .	8.6	74.6	51.7

Table 9. **Performance indicators: sustainable retirement income** (*cont.*)

	Projected increases in old age pension spending[1]	Low income rate of the elderly[2]	Relative disposable income of the elderly[2]	Private pension funds 1999	Age of withdrawal from labour force, 1994-99[3]		Participation rate, 2001, per cent		
								Aged 55-64	
	Change in per cent of GDP 2000-50	Per cent of the elderly with income less than 50 per cent of median disposable income	Per cent of the disposable income of all individuals	Per cent of GDP	Male	Female	Aged over 65	Male	Female
Norway	8.0	19.1	74.1	7.4	64.2	64.7	13.2	73.6	63.2
Poland	–2.5	8.4[4]	. .	. .	. .	. .	7.5	41.5	24.1
Portugal	. .	. .	. .	11.4	65.3	66.5	19.0	63.7	41.9
Slovak Republic	. .	. .	. .	. .	. .	. .	1.1	43.0	11.2
Spain	8.0	11.3[4]	. .	2.3	61.1	61.1	1.6	61.4	23.6
Sweden	1.6	3.0	89.2	. .	63.3	61.8	9.4	73.5	67.4
Switzerland	. .	8.4[4]	. .	97.3[5]	. .	. .	11.4	82.4	56.1
Turkey	. .	23.1	92.7	. .	. .	. .	18.1	50.8	18.4
United Kingdom	–0.7	11.6	77.8	84.1	62.0	61.2	4.8	64.4	44.6
United States	1.8	20.3	91.7	74.4	65.1	64.2	13.1	68.1	53.0

1. Dang *et al.* (2001). French official reports suggest a 4.4 per cent increase on unchanged labour market policies for the period 2000-40 (Conseil d'orientation des retraites, 2001).
2. Förster and Pellizzari (2000).
3. Scherer (2001).
4. Jesuit and Smeeding (2002).
5. 1998.

Source: Förster and Pellizzari (2000); Jesuit and Smeeding (2002), *Luxembourg Income Study*; OECD *Labour Force Statistics*; Scherer (2001).

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Table 10. **Defined benefit and defined contribution pension plans**

	1989	1991	1993	1995	1997	1999	2001
	Per cent of employees in sector						
Number of adherents							
Employment based registered pension plans							
Total	45.7	48.5	48.1	45.7	44.6	43.6	43.5
Defined contributions	3.8	4.3	4.7	4.8	5.6	5.9	. .
Defined benefits	41.4	43.6	42.9	40.3	38.3	36.9	. .
Mixed	0.4	0.7	0.6	0.7	0.7	0.8	. .
Public	83.5	89.4	87.4	84.6	83.9	82.2	. .
Private	33.6	34.1	33.6	32.1	31.4	31.1	. .
Registered retirement saving plans							
Defined contributions	31.9	36.6	39.7	42.5	44.7	42.7	41.4
	Average contribution C$, 1997 prices						
Employment based registered pension plans							
Total	3 366	3 534	4 093	3 943	3 845	3 546	. .
Public	4 905	5 375	5 615	5 180	5 460	4 741	. .
Private	2 141	1 834	2 630	2 808	2 406	2 522	. .
Registered retirement saving plans							
Defined contributions	3 880	3 517	3 966	4 178	4 456	4 351	4 258

Source: Statistics Canada (2000a).

by such plans has eased back, and average contribution levels have fallen. Middle- and higher-income groups account for the bulk of private retirement income. By contrast, the high income replacement rates provided by public pensions and the income-testing of the low-income supplement component of OAS reduce savings incentives for low-income earners. However, for those who need to save privately to obtain sufficient income replacement rates, participation in RPPs and RRSPs is high. About 76 per cent of those earning over C$15 000, and 93 per cent of those earning over C$45 000, contribute to a RPP or a RRSP.

The pension system does not represent a source of major disincentives for continued work. Only the second-tier pension can be taken at age 60, and deferment leads to increases in the value of pensions for those continuing to work beyond the age of 62 (Figure 17). Nonetheless, between 62 and 65, the discouragement effect tilts in favour of retirement, though the effect is much smaller than in some other OECD countries. However, the C/QPP pension is low, so any income effect that would draw people into retirement is limited. Beyond age 65, when the means-tested supplement is paid, the disincentive increases substantially, as the benefit is reduced by 25 per cent of additional earnings for a married couple but by 50 per cent for a single person.

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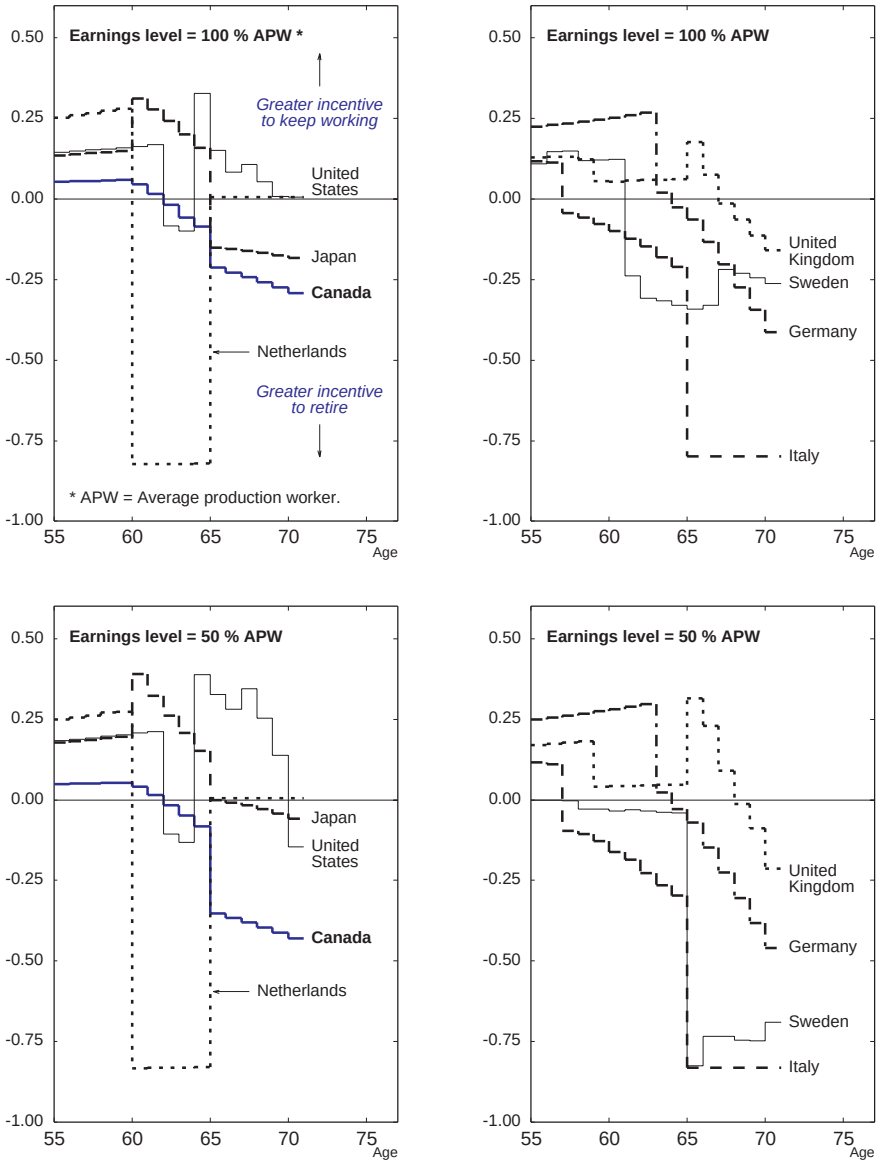
Figure 17. **Change in pension wealth from working an extra year**

Proportion of current annual salary



Source: OECD, *Ageing and Income*, 2001.

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Disability insurance, included in the C/QPP, represented a significant route by which individuals withdrew from the labour force, but it has become less important over time. The average benefit under this system is 75 per cent greater than the average C/QPP pension, since it includes a large flat-rate component. Between 1970 and 1994, the incidence of disability claims rose six fold for men and nine fold for women. At that point there was a major reform of the definition of disability, and much tighter and uniform administrative procedures were put in place. The role of socio-economic factors was de-emphasised, and medical factors were given primacy. Moreover, medical status is now regularly reviewed and vocational retraining offered. Consequently, the incidence rate for the payment of disability insurance payments for men fell back to its 1970 level by 2000. The change in the disability regime may have been one of the factors that led to a rise of 7 percentage points in the employment rate for men between 55 and 60 between 1995 and 2002. Nonetheless, the age-specific incidence rates suggest that disability benefit is still being used a route into retirement by half the male population.

Assessment

Canada has made considerable progress since the 1970s in ensuring that the retirement system reduces the threat of poverty in retirement while limiting the growth of pension expenditure. The system currently provides an effective backstop against low income in retirement while providing a framework for the better-off to accumulate their own savings. In the longer term, continued price indexation of the safety net component would threaten income adequacy at the lower end of the income distribution. Moreover, steep means-testing discourages lower-income individuals from saving for retirement. If the overall cost of indexing benefits to wages is considered too high, it could be moderated by increasing the age of retirement to 67 over the next few decades, in line with current policy in the United States. A further possibility would be to raise the minimum income guarantee in line with wages but index the basic pension to prices. While this would increase the extent of means-testing of benefits, it would help avoid placing a significant burden of adjustment on poorer groups. In any case, once the demographic bulge has passed, the official projections show that there would be little fiscal reason for continuing with price, rather than wage, indexation for the first-tier pension. Despite pro-work reform, the disability system still appears to serve as route to early retirement and needs some review, especially for those aged between 60 and 65.

Improving human capital

Raising skill levels of the workforce is a key part of Canada's *Innovation Strategy* and is important not only for increasing the long-term growth rate but also for getting the unemployed back to work. Human capital is a considerable

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strength for Canada, with high average performance of the adult population and excellent achievement by 15 year-old students who will soon enter the workforce. However, there are still relative weak spots. Canada's strong educational attainment is concentrated in the middle of the distribution. It does less well at the two extremes; at the top end, fewer go on to university, while at the bottom end adult literacy outcomes can be very poor. The most pressing challenges, therefore, are to improve the system of adult or "second chance" learning and to minimise barriers to university education, including enhancing the pathways from community colleges to universities.

The school system delivers good results at a reasonable cost

International comparisons frequently show that Canadian school students are among the best in the world. Most recently, the OECD's PISA study of 15 year-olds in 32 countries (OECD, 2001b) ranked Canada second in reading, fifth in science and sixth in mathematics (Table 11). Alberta and Quebec were especially strong performers, both in PISA and in other international studies,[22] but even the weakest provinces matched or beat the OECD average. Not only are the various provincial systems producing high-quality results, they are doing so at a reasonable cost, with spending per student close to the OECD average. They are also delivering greater *equality* than elsewhere: as in most countries, Canadians from low socio-economic backgrounds perform worse than their richer counterparts, but the gap in outcomes is much smaller in Canada than in other countries.

While this is good news for future productivity growth, the studies have revealed some areas that may need further attention. For example, it is not obvious why Alberta and Quebec consistently perform so well, especially as their school systems are quite different, but one reason may be that both provinces have relatively few students who speak neither English nor French at home. There may also be an issue with francophone minorities: comparing across provinces, students in francophone schools outside Quebec had significantly worse results than their anglophone counterparts, although outcomes in Quebec were fairly similar for the two linguistic groups (Bussière *et al.*, 2001). Second, both provinces have relatively tough teacher certification requirements[23] and have been considerably more active in province-wide testing of students. The possible importance of these factors should be researched further. Third, the impact of competition and school choice on education outcomes could also be investigated further. It is notable that private or charter schools are more prevalent in the three best-performing provinces (Quebec, Alberta and British Columbia). Within each province, outcomes for students in private schools are not significantly different from those in public schools after controlling for other factors such as socio-economic background and school resources (Bussière *et al.*, 2001). However, that may be because greater school choice lifts performance levels throughout a province. The federal government has done research only com-

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Table 11. **PISA scores**
Selected countries, OECD average = 0

	Reading	Mathematics	Science	Average
Alberta	**50**	**47**	**46**	**48**
Japan	22	57	50	43
Quebec	**36**	**50**	**41**	**42**
Korea	25	47	52	41
Finland	46	36	38	40
British Columbia	**38**	**34**	**33**	**35**
Canada	**34**	**33**	**29**	**32**
New Zealand	29	37	28	31
Manitoba	**29**	**33**	**27**	**30**
Australia	28	33	28	30
United Kingdom	23	29	32	28
Ontario	**33**	**24**	**22**	**26**
Saskatchewan	**29**	**25**	**22**	**25**
Nova Scotia	**21**	**13**	**16**	**17**
Ireland	27	3	13	14
Newfoundland	**17**	**9**	**16**	**14**
Sweden	16	10	12	13
Prince Edward Island	**17**	**12**	**8**	**12**
France	5	17	0	7
New Brunswick	**1**	**6**	**–3**	**1**
United States	4	–7	–1	–1
Denmark	–3	14	–19	–3
Germany	–16	–10	–13	–13
Spain	–7	–24	–9	–13
Italy	–13	–43	–22	–26
Greece	–26	–53	–39	–39

Note: Scores in blue are significantly above the OECD average.
Source: OECD (2001b).

paring outcomes within provinces; it has done no cross-provincial studies of the impact of school choice. It may be informative to compare institutional arrangements and outcomes across provinces in order to learn more about what factors influence education outcomes. Fourth, in seven out of ten provinces a large proportion of principals (compared with other countries) said that teacher shortages were a serious problem, especially in the Maritimes. Governments should consider options that increase pay flexibility, such as allowing higher pay for teachers in subjects or locations with shortages and allowing performance-based pay. Greater pay flexibility would be a better solution than across-the-board pay increases as Canadian teachers are already among the highest paid in the world.[24] Finally, the causes of the gap in performance between Aboriginals and other Canadians need to be uncovered: in 1996, nearly half of Aboriginals aged 20 to 29 did not go on to upper secondary education, and only 4 per cent had a university degree (Canadian Education Statistics Council, 2000).

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Supporting lifelong learning

Because youngsters entering the labour market have good skill levels, the policy focus is shifting to those who have already passed through the education system. The key problem here is adult literacy. Overall literacy levels are about average when compared with the richer OECD countries, and a comparatively large proportion of adults perform at the highest levels, but 40 per cent are below the level commonly regarded as adequate in a modern information-based economy.[25] Most countries face basic literacy problems among their working-age populations, and a proportion as high as 40 per cent is not unusual, but where Canada stands out is in the *depth* of its problem. Of Canadians with poor literacy, a comparatively large fraction has *very* low scores on international tests.[26] The causes in the past have included low levels of schooling in large swathes of the country, particularly in the regions dependent on primary industries where children went to work at a young age, and the practice of girls dropping out of school to care for younger siblings in some regions.

Improving access to and the quality of adult education is one of the government's key goals. At present, participation in continuing education and training is a little below the OECD average, especially in the private sector. The vast majority of courses are job-related, usually supported by an employer, and are comparatively intensive (as measured by the high number of hours per participant).[27] As in other countries, however, those who need education the most are the ones who receive it the least. While more than a quarter of all adults receive some formal continuing education, those at the highest literacy or educational attainment levels are nearly four times as likely to undertake adult training than those with the fewest skills (Table 12). This partly reflects the prevalence of employer-sponsored training, which is biased towards white-collar workers and is less common in the private sector than in the public sector, possibly reflecting language training. Participation in the Atlantic provinces was considerably below the national average, at least in the second half of the 1990s.

There is some evidence that the supply of adult education is falling well short of demand. In its recent *Thematic Review of Adult Learning*, the OECD found long waiting lists for most programmes, with the problem being especially serious for certain key groups: those lacking basic literacy, Aboriginals and the working poor (OECD, 2002b). For example, basic literacy programs in Saskatchewan have a six- to nine-month waiting list, and officials estimated they could easily double enrolments if more resources were available. The system seems to be under-funded, and the available resources need to be used more effectively. In addition, the courses that are available tend to be insufficiently intensive for those with serious literacy problems. Many have been described as "warm and supportive" but with little pressure to advance, so it is too easy for students to remain stuck at a particular level (OECD, 2002b).

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Table 12. **Adults in continuing education and training**
Per cent of population aged 25-64, mid-1990s

	By literacy level			
	Lowest level	Next highest	Next highest	Highest levels
Employer-sponsored	6.6	16.5	26.4	40.0
Not employer sponsored	14.3	19.9	20.3	27.1
Both types	16.5	29.4	39.6	60.4
	By highest qualification			
	Elementary	Secondary	College	University
Employer-sponsored	9.8	18.9	35.1	38.8
Not employer sponsored	12.1	19.7	24.3	26.5
Both types	16.4	31.5	51.9	59.3
	By employment type			
	Private employees	Public employees	Self employed	Unemployed
	(Per cent of those who are employed)			(Per cent of adult population)
Job-related	27.6	41.8	18.1	20.6
All types	34.4	49.9	24.3	27.0
	By province			
	Atlantic provinces[1]	Quebec	Ontario, Alberta, British Colombia	Canada
Job-related	18.1	14.6	24.0	21.1
All types	23.3	20.6	31.3	27.7

1. Simple average of Newfoundland, Prince Edward Island, Nova Scotia and New Brunswick.
Source: International Adult Literacy Survey (1996) and Adult Education and Training Survey (1998).

Provincial governments have diverse approaches to lifelong learning, including British Columbia's attempts to provide a "seamless" system and Quebec's policy of funding a smaller number of longer-duration courses combined with a "use it or lose it" payroll tax-break for employer-based training. This diversity should be encouraged, provided there is enough information on the effectiveness of the training programmes on offer; this is not the case at present, although formal assessments of active labour market programmes, as required under the federal-provincial Labour Market Development Agreements (LMDAs), may address that weakness. Recognition and portability of training can also become important issues in a diverse, non-integrated system (this applies equally to Canadians and new immigrants). The 13 provincial and territorial education minis-

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ters currently work together in a council (CMEC) to try to minimise any such problems, and they have developed the Prior Learning and Assessment Recognition (PLAR) protocol. The CMEC has also improved the ability to transfer credits for first- and second-year university courses between provinces, but some gaps remain, including a limited ability to move from community colleges to universities. Overall, however, more needs to be done to knit together the disparate parts of the learning system in order to make it easier to make lifelong learning a reality. Finally, the system may not be giving enough of a push to at-risk groups such as those with low literacy. Clients must select a course, find a training provider and negotiate a price; approval then depends on whether funds are available and whether the training matches the skill requirements of the local economy. It may be necessary for employment counsellors to push more actively some of the less skilled or motivated into training and to give them more assistance in finding suitable courses.

Post-secondary education

Although Canada is producing talented 15 year-olds, comparatively few go on to university – especially to advanced degrees.[28] While post-secondary enrolment rates are around average, an unusually large proportion goes to community colleges or trade schools that usually provide shorter duration courses. Of the university graduates, not enough go on to use what they have learned: the unemployment rate among graduates was still a relatively high 5 per cent as recently as March 2003, while less than four out of five working-age men with a university degree were in full-time employment. Despite this middling performance, Canada's tertiary education system is one of the most expensive in the world (Table 13), possibly because it has to pay high North American salaries to attract good staff.

Improving access through the student loan scheme

University tuition fees increased sharply in the 1990s, but the federal student loan scheme was made more generous around the same time. This funding helps provide significant support to students with limited resources and may have contributed to a modest increase in participation by students from low-income backgrounds.[29] Eligibility for federal loans is assessed by the provinces on behalf of the federal government and is based on need – a simple equation of costs minus resources – which in practice means they are targeted on students with low incomes (or from lower income families) and students with high costs. They are intended to cover the amount by which course costs and living expenses exceed a student's personal resources, which includes a presumed contribution from parents. The repayment rate is chosen by the student, over and above a minimum pace, but interest rate relief and partial debt write-offs

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Table 13. **Spending on post-secondary education**
Expenditure in 1999, adjusted using economy-wide PPPs

	Spending per student on:			Proportion publicly funded (Per cent)	Spending as per cent of GDP	Capital spending as per cent of GDP
	Teaching and related services	Research	Total			
	(C$ per student)					
United States	20 384	2 507	22 891	48	2.3	0.21
Canada	**14 993**	**3 123**	**18 116**	**64**	**2.5**	**0.12**
Germany	7 668	4 710	12 378	91	1.1	0.12
Japan	. .	. .	12 241	50	1.0	0.19
United Kingdom	7 289	4 090	11 379	73	1.1	0.03
France	7 905	1 466	9 371	91	1.1	0.12
Italy	8 994	. .	8 994	88	0.8	0.14
OECD	8 027	2 696	10 969	77	1.3	0.17

Source: OECD, *Education at a Glance*, 2002.

are available for people facing hardship. In that sense, the scheme has moved towards a targeted income-contingent repayment (ICR) approach (Finnie, 2001). It may be worth taking the extra step and making repayments fully income-contingent. Apart from being cheaper to administer, it may also boost participation by those students worried about taking on debt, since an ICR system makes it clearer that the loan does not have to be repaid unless they earn a high enough income after studying. The government should also consider raising the loan limit, which has not changed since 1994, and extending eligibility to more middle-income families: university participation by young people from middle-income families actually fell slightly over the 1990s (Corak *et al.*, 2003). By doing so, universities and colleges could raise tuition fees, thereby increasing resources in the sector. Finally, the role of grants relative to loans could be reassessed. Direct provincial grants were phased out over the 1990s but are being slowly reintroduced on a targeted basis. For example, the federal government introduced the Millennium Scholarships programme in 1998 to provide up to roughly C$3 000 per year for students beyond their first year of study. These scholarships are targeted using a combination of needs and merit-based criteria and are available to people of any age at virtually all types of post-secondary institution. While such grants have some advantages, income-contingent loans are, dollar-for-dollar, more effective at boosting participation. No matter how good the *ex ante* selection criteria, at least some of the grants will go to those who would participate anyway and would be able to pay them back after graduation had they instead been given in the form of a loan.

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Encouraging competition and improving regulations

As discussed in the introduction, a competitive business environment is an important – possibly the most important – driver of innovation and productivity growth. Canada scores well here, having one of the OECD's most business-friendly regulatory environments. However, there are some glaring exceptions, usually related to its past penchant for protecting "national champions". Several unnecessary barriers to competition need to be removed and the freeing up of internal trade given fresh momentum.

A *sound regulatory environment but with too many foreign investment barriers*

Canada has been a pioneer in ensuring that regulation is clear, low-cost, effective and, most importantly, directly linked to its public policy goals.[30] It was also one of the first to insist on regulatory impact assessments (RIAs) to encourage law-makers to consider whether the benefits of a proposed regulation were likely to outweigh costs, and whether other, less costly ways could be used to achieve the same goals. These RIAs are taken seriously: there is ample training in RIA techniques for regulators, and public consultation is extensive as RIAs are published at both the draft and the final stage. On the whole, regulatory reform has been thorough, wide-ranging and efficient.

Nevertheless, there is still room for improvement. Some parts of the economy remain shielded from competition. The issue is not just the restricted application of competition law. Some sectors are still effectively sheltered from competition, and in many cases this is the result of foreign ownership restrictions. Additionally, the regulated conduct doctrine may, in certain circumstances, protect anti-competitive business conduct from the application of the *Competition Act* if that conduct is required or authorised by statutes or regulations enacted by the federal or provincial governments. In addition, some sectors are exempted or enjoy special treatment under the Act, including airlines, fishing, professional services, broadcasting, book publishing, amateur sport, performing arts, wheat distribution and the export trade. Price controls remain in other areas, such as rail freight, telecommunications and electricity. Book publishing is a clear example of problems created by ownership restrictions, since the absence of foreign bidders at the time of a recent bookstore merger has resulted in a virtual monopoly. There are more effective ways than that to safeguard Canadian culture. A further notable weakness of competition policy is that it is difficult to attack cartels. While the law on collusion appears tough because it imposes criminal penalties, the burden of proof is difficult and complex. It would be better to move to a two-track approach with criminal penalties for hard core cartel conduct combined with an easier-to-enforce civil alternative for less extreme cases.

Reforms to *telecommunications* legislation in 1993 have been notably successful, leading to rapid deregulation in the sector. The weak spot, however, is that

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Canada is the only rich OECD country that has retained widespread foreign ownership restrictions in this sector.[31] The policy is intended to protect national sovereignty and security but risks reducing investment, competition and innovation in this dynamic industry. In April 2003, a House of Commons Committee recommended removing these restrictions and also recommended that any changes to the telecommunications sector should apply equally to the cable companies, as the two sectors have become so intertwined. However, the cable and broadcasting sector was the subject of a separate review, which in June recommended against the removal of the ownership restrictions. Those restrictions are intended to protect Canadian culture and heritage, but that goal could be achieved more efficiently by regulating content rather than the carriers, since there is no reason to think that local content regulations would be less effective if broadcasters were foreign-owned. In both cases, the government should show that the barriers are achieving their public-policy objectives cost-effectively and that benefits outweigh the costs. If not, they should be removed.

Competition and efficiency could be lifted in the domestic air travel industry

According to one measure, the regulatory and market environment for the air travel industry was the least competitive in the G7 in the mid-1990s.[32]. Despite recent changes in the industry, such as the fast growth of "no-frills" carriers, Air Canada retains its dominant market position despite recent financial troubles that led to its filing for protection from creditors in order to restructure in April 2003.[33] It continues to have very little competition for business travellers. Low-cost carriers now serve all the major cities in Canada but with limited frequency. Many routes involving smaller communities remain monopolies (Commissioner of Competition, 2001; OECD, 2002c). Two reasons for its dominance are the 25 per cent limit on foreign ownership and control of Canadian airlines and a ban on cabotage.[34] The ownership rules have had a significant impact on market concentration. Air Canada's main competitor, Canadian Airlines, collapsed in 1999, but, rather than relax restrictions and allow it to join with a US airline, the government suspended the Competition Act to facilitate the merger. Some regulatory changes were made at the same time to reduce the ability of Air Canada to abuse its dominant position but, in the view of the Competition Commissioner, they were not sufficient to create a competitive domestic market. To ensure a well-functioning market when the airline emerges from restructuring, the government should implement the Commissioner's three main recommendations. *First*, lift the foreign ownership limit to 49 per cent, as was also recommended by the Independent Transition Observer on Airline Restructuring (Ward, 2002). This may in fact be necessary in order to provide the capital needed for a financial rescue. *Second*, remove ownership restrictions to allow "Canada only" carriers, as Australia has done successfully in its own internal market. *Third*, allow "modified sixth freedoms" in which a foreign carrier could carry passengers from one Canadian city to another *via* a for-

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eign hub (*e.g.* Toronto-Chicago-Vancouver). The government's policy is to negotiate access on a bilateral basis rather than unilaterally opening up the market. However, there is strong evidence that a strict regulatory environment significantly reduces productivity in the industry (Gönenç and Nicoletti, 2001). Hence, the government should move unilaterally unless it can show that the current cost of this policy to Canadian travellers is outweighed by the hypothetical payoff from using the local market as a strategic trade lever.

The government electricity monopolies need to be reformed

Deregulating and promoting competition in the electricity industry is becoming increasingly common in the OECD. Once regarded as a natural monopoly, several countries are separating out the pieces that are genuine natural monopolies (such as transmission and low-voltage distribution) from those that are not (generation and retail supply). There are large differences between provinces in the extent of reform but, on the whole, Canada is lagging behind in this area (Steiner, 2001). The industry remains highly concentrated, with typically one or two provincially owned generators providing most of the power and having responsibility for transmission and distribution.

Deregulation has been most aggressive in Alberta. A wholesale spot market was created in 1996, competition in generation was increased in 2000 by auctioning operating leases on government-owned power plants, and full retail competition began in 2001. Higher prices now appear to be stimulating private-sector investment, and an emerging forward market is improving volatility management (NEB, 2001). No other province plans to create retail competition, although some allow access to the transmission network (with regulated retail prices). The British Columbia government intends to separate the transmission and generation arms of BC Hydro, but the generator will still control 90 per cent of the market and sell to households at a price that has been frozen by regulation since 1993.

Ontario began a significant reform programme in the late 1990s but made a major policy reversal in 2002. In April 1999, the provincial monopoly (Ontario Hydro) was split into a generator (OPG) and a transmission and distribution firm (Hydro One). Its C$38 billion debt was transferred to the province's books; customers face a debt-repayment charge that accounts for around 40 per cent of electricity bills. To increase competition in generation, OPG was required to immediately sell some of its price-setting (marginal) plants and to control no more than 35 per cent of all capacity within ten years. The distributor, Hydro One, was to be fully privatised, but the government now intends to retain total ownership.

Ontario's retail market was opened in May 2002. Prices fell from 4.3 Canadian cents per kWh to 2.9 cents in the first month of deregulation, but an increase in demand caused by an unusually hot summer, combined with supply

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shortages due to delays in restarting a nuclear plant, pushed up prices to a peak of 7.5 cents in September. Although that level was low by North American and international standards, the government re-regulated in the autumn to fix prices at the old level of 4.3 cents until 2006 for residential and small industrial customers. The fixed price was made retrospective to May 2002, with rebates being given to consumers for anything they had paid above that price. By doing so, the retail market was effectively shut down. Generators continue to sell into the still functioning wholesale market, with the provincial government paying them the difference between the market price (which has averaged around 6 cents since 1 May 2002) and the fixed price that consumers pay. This is expected to cost the taxpayer around C$300 million in each of the next two years. In principle, the low retail price need not deter private firms from building new capacity if they are convinced that the taxpayer subsidy is sustainable over the long term. In practice, the policy reversal has put several projects on hold despite an offer of subsidies and tax breaks. The main danger of this bungled deregulation and re-regulation is that future attempts to deregulate will be difficult, as investors will be wary of committing to an unstable regulatory environment.

With the evidence that, on balance, a deregulated electricity supply industry improves industry performance (Steiner, 2001; Productivity Commission, 2001), remaining provincial governments should begin the process but learn from Alberta's comparative success and Ontario's failure. Effective competition requires the generation monopoly to be broken up at an early stage if the private sector is to be enticed into the industry. Ontario also needs to urgently reconsider its low price cap.

Competition can also be boosted by further removing barriers to internal trade

There has been important progress since the last *Survey* in implementing the 1995 *Agreement on Internal Trade* (AIT). From 1 July 2001, restrictions on labour mobility were largely removed. All local residency requirements have been eliminated, and mutual recognition agreements, in which professional qualifications granted in one province are recognised nation-wide, have been signed for most regulated occupations. Around 97 per cent of regulated professional workers are now covered by these agreements.

Some trade barriers remain, but these are more of an irritant than a deep structural problem. The net cost of inter-provincial restrictions is probably very small since, for the most part, they are re-distributive, benefiting firms in one province at the expense of those in another. And judging by the amount of trade that takes place, inter-provincial impediments are much lower than those between Canada and the United States,[35] despite NAFTA making the border one of the world's freest. Even so, the remaining barriers add to the costs of doing business and make it more difficult for firms to grow.

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An ongoing issue is *government procurement*. The *Agreement* requires equal treatment of bidders from all provinces, but some large holes remain. Crown corporations are still exempt, years after the 1996 deadline for their inclusion; adding them would widen the scope of the procurement provisions by nearly half. Contracts with non-profit providers are also excluded. This exemption will become increasingly important if Canada follows the global trend towards contracting out the delivery of social services to third parties. A second major issue is the failure to remove barriers in the *energy sector*. This is nearly eight years behind schedule, with the most recent attempt – in July 2002 – failing again. There has also been little progress on reviewing and broadening the *agriculture* chapter, while the *code of conduct on investment incentives* (which tries to stop bidding wars between provinces) is unfinished and incomplete harmonisation of *consumer standards* and *trucking regulations* remains a nuisance, although good progress has been made in both areas.

The agreement appears to have gone about as far as it is likely to go in its current form. Consequently, the focus should now shift to including those areas that were initially excluded and to strengthening the framework itself. The main omitted sector is financial services. Securities regulation is a provincial matter, so any nation-wide issue of securities, including mutual funds, must be approved by each of the 13 regulators. A "one stop" financial company that takes deposits, offers securities and sells insurance throughout Canada may need to deal with 30 different agencies and their different reporting requirements. The costs of such a balkanised system are hard to estimate but include the legal and advisory costs for firms, the large number of people employed by regulators in Canada, and the fact that it makes prudential supervision more difficult. The situation is likely to get worse over time as economies of scale in the industry are becoming more important (Freedman and Goodlet, 2002).

However, some improvements have been made over the past few years. Quebec is amalgamating its eight financial agencies into a single non-independent body that reports to the Minister of Finance. The provincial regulators have also banded together to form the Canadian Securities Association (CSA); its two main achievements have been to implement an electronic filing system for regulatory documents, and a mutual reliance approach where a lead regulator makes an assessment that the remaining 12 can then rubber stamp. However, this is one step removed from full harmonisation of securities law, since local regulators can opt out of any decision, and securities must still be registered in all jurisdictions. The CSA is currently trying to harmonise securities regulation across Canada, but British Columbia has recently published its own draft legislation that diverges from the regulatory model in other provinces. One alternative worth considering is a single regulator. Australia, which faced similar regulatory problems, has successfully made this transition. An advantage of this approach, apart from the obvious economies of scale, is that the rules can be changed quickly by one agency rather than trying to get an agreement among all provinces. Alternatively, the mutual reli-

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ance approach could be taken a step further to become a passport system in which approval by the local regulator is automatically recognised in all other jurisdictions. Two committees – one federal, one provincial – are looking again at these issues and are due to report in the second half of 2003.

Getting the most from the *Agreement* may also require further institutional changes. *First*, there is wide frustration with the dispute resolution mechanism, which is slow, costly, hard to access and lacks teeth as rulings are not legally binding (ITS, 2001). So while in principle the labour market is now largely barrier-free, in practice anecdotes continue to crop up about trades people being unable to work in other provinces; in most cases, it is too difficult for an individual to fight these restrictions. *Second*, consensus decision-making has left some issues essentially intractable. The parties could consider some form of majority voting (as in Australia and the EU), or a default rule of mutual recognition of standards in the event that the parties cannot agree on how to harmonise (as in the EU). Pending agreement on a revised decision-making process, the focus of negotiations might constructively shift to moving ahead with those jurisdictions ready to do so, rather than waiting for agreement by all.

Encouraging innovation

Innovation is a key driver of productivity growth, as recognised by Canada's new *Innovation Strategy* (Annex III). Despite having a skilled workforce, Canada's innovation performance is mixed when compared with other G7 economies. Expenditure on R&D is low (but growing fast), the number of researchers per capita is a little below the G7 average and well below US levels, patenting rates are low, and so are overall machinery and equipment investment levels. However, Canada does have several strengths: its scientific publications are widely cited, it is a clear leader in the adoption of ICT including broadband (Figure 18), and it is well integrated with the United States, both in business terms and in the number of Canadians taking advanced degrees at US universities.

The science system could be more focussed and integrated

As in most countries, the Canadian government supports science in several ways, including tax incentives for industry, direct and indirect funding of university research and by maintaining its own scientific laboratories. In recent years, various changes have been made to the science system in order to increase the effectiveness of federal spending. The government's advisory bodies are now focussed on scientific fields rather than on specific industries; some steps have been taken to better align spending with policy priorities; and more of the research is being done by universities rather than government labs (Table 14). There have also been improvements in the way science is funded, with more use of contracts and grants. Most university research, for example, is paid for out of the

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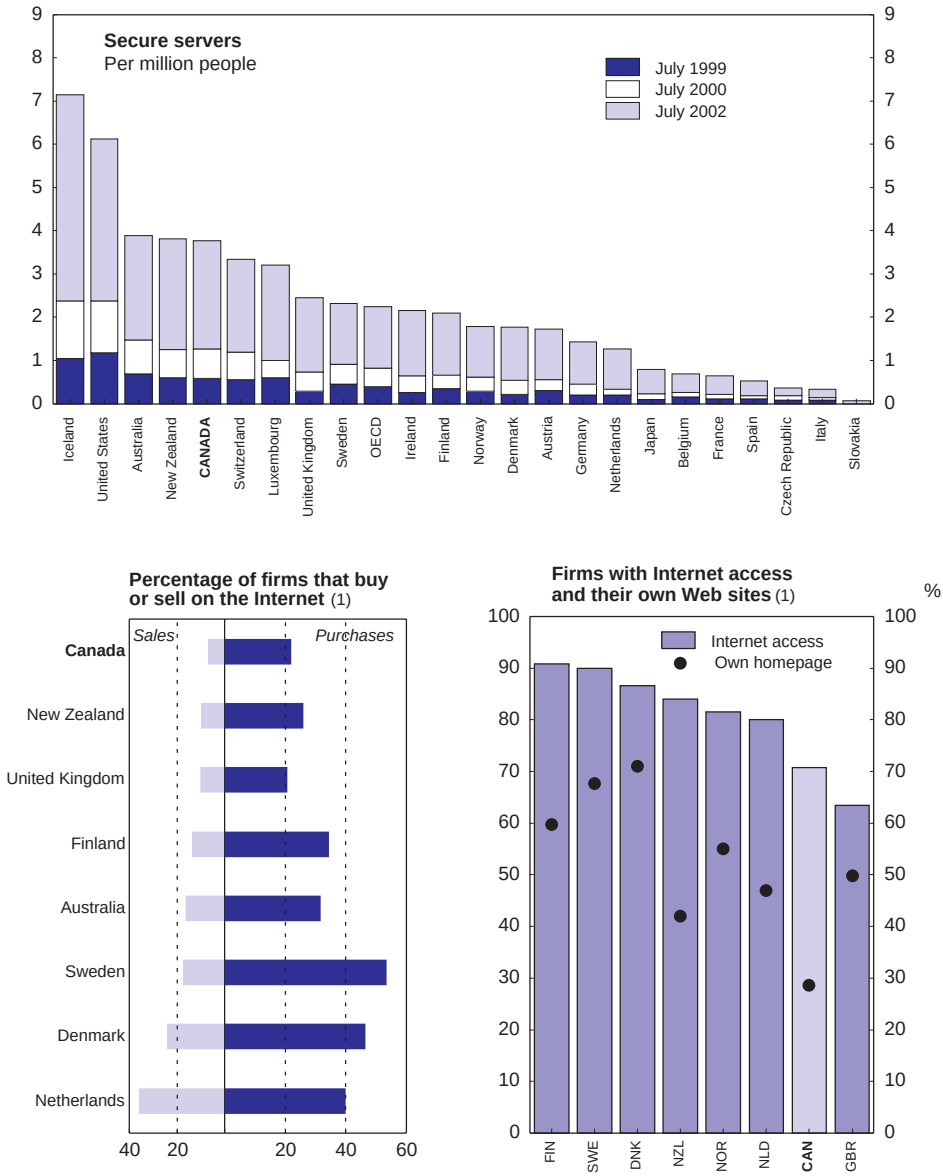
Figure 18. **Internet infrastructure and take-up**



1. 2001 or latest available year. Excludes smallest firms (typically fewer than ten employees).
Source: Statistics Canada; OECD, *Measuring the Information Economy, 2002*.

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Table 14. **Research and development spending by sector**
Millions of Canadian dollars, 2001

Funding sector	Performing sector					Per cent funded, 2001	Per cent funded, 1990-95
	Federal government	Business	Universities	Other	**Total**		
Federal government	1 907	361	1 431	33	**3 732**	17.9	25.6
Business	44	8 078	603	45	**8 770**	42.0	41.1
Foreign	. .	3 147	75	18	**3 240**	15.5	10.0
Universities	. .	. .	3 609	. .	**3 609**	17.3	15.3
Other[1]	2	70	1 097	351	**1 520**	7.3	8.0
Total	**1 953**	**11 656**	**6 815**	**447**	**20 871**	100.0	100.0
Per cent performed	9.4	55.8	32.7	2.1	100.0	. .	. .
Average share, 1990-95	14.5	52.9	29.1	3.5	100.0	. .	. .
G7 average share, 2001	12.8	64.5	21.1	1.7	100.0	. .	. .

1. Includes provincial governments, provincial research organisations, and private non-profit organisations.
Source: Statistics Canada; OECD.

institutions' block-grant budgets, or from one of three federally funded Research Councils (Natural Sciences and Engineering, or NSERC; Social Sciences and Humanities, SSHRC; and Health, CIHR). Additional funding is available from other bodies including the National Research Council (NRC), which is a federal agency and is not to be confused with the NSERC despite the similarity of their roles in some respects. Funding for research infrastructure is available from a different agency (the Canada Foundation for Innovation) from the one that funds the research itself.

Value for money would be improved with more use of competitively awarded funding, a further shift towards project-based rather than institution-based grants and better governance in the granting councils. F*irst*, contestable funding is common in the university sector, but federal labs are mainly institutionally funded and, along with industry, are shut out of the three main granting councils' resource pools. This lack of competitive pressure means that, despite changes to internal planning processes in some agencies, there are few signs of federal science institutions refocusing their activities to be more in line with government priorities (CSTA, 2000). A possible solution is to shift a portion of their institution-based funding into the competitive funding pools, but then allow government labs to bid for those funds, as is becoming increasingly common in other countries. Failing that, it should rethink the balance between universities and government labs. Some evidence suggests that publicly funded R&D has a bigger impact on productivity growth in countries that channel more of their budget to universities (Guellec and van Pottelsberghe de la Potterie, 2001). This may be

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because labs' specialised nature limits their ability to undertake interdisciplinary research, or because they typically do not teach students who can then transfer that knowledge to industry. *Second*, clarification is needed over how much attention granting councils should pay to the government's social and economic priorities. Currently, councils behave differently: the health council (CIHR) has separated its resources into 13 priority pools, while the NSERC's approach is to spread money as widely as possible, approving three-quarters of proposals. *Third*, more weight needs to be given to quality and performance when allocating money, along the lines of the research assessment exercises in the United Kingdom and the Netherlands. For instance, in the guidelines for choosing Networks of Centres of Excellence, which are virtual networks of public and private researchers, excellence itself is given a weight of only 20 per cent. Much more weight is given to "inter-disciplinarity, inter-institutional relations, management structure, relevance, training, and socio-economic impact" (Polanyi, 2000).

Some improvements have been made in these areas, but further progress would be useful. The government strengthened its approach to programme evaluation with guidelines aimed at improving accountability and data gathering. In the past the Auditor General has criticised the governance of many of these programmes, noting that they often lacked clearly defined objectives, their expected outcomes were not expressed in terms of innovation performance, and there were few strong links between spending and the overall objective of improving Canada's innovation outcomes. The government plans to have all of its grant and contribution programmes evaluated and new terms and conditions approved before March 2005.

R&D *spending by the business sector is low but has been growing fast*

R&D spending as a proportion of GDP is low, and so the government has set a target of lifting Canada's ranking from fifteenth to fifth place in the OECD by the end of the decade. To do so, it will need to focus on the business sector. Public research is already above the OECD average and not much short of the top few countries. Business R&D, on the other hand, is very low despite having the OECD's second most generous tax break (Figure 19). It has also been highly concentrated, with a single telecommunications firm accounting for around 40 per cent of all such R&D in 2001.

The low level of R&D may not be as disturbing as it looks, since it partly reflects the large number of US-owned firms operating in Canada (multinationals do most of their research at home, but share the results with their foreign affiliates; of course, this begs the question of whether they do this because Canada is not seen as an attractive place to do research).[36] But it may also reflect other factors, such as barriers to competition or a shortage of researchers. A competitive business environment – with low barriers to entry in particular – is a powerful spur

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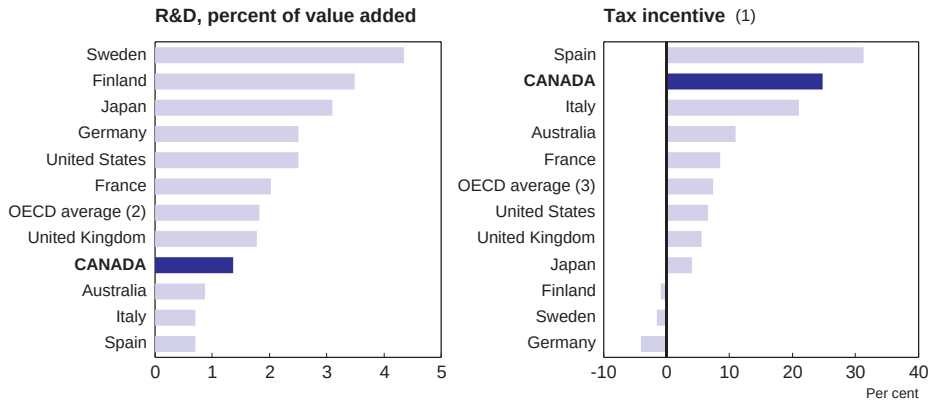
Figure 19. **Business sector research and development expenditure**
2000, or latest available year



1. For example, one dollar of R&D spending in Canada results in 25 cents of tax relief.
2. 20 countries.
3. 23 countries.

Source: OECD, MSTI Database, 2002-I.

for R&D (Bassanini and Ernst, 2002). In any case, it is not caused by a lack of financial incentives. A 20 per cent tax credit is available for eligible R&D spending by large firms and a refundable credit of 35 per cent for smaller companies. Provincial governments offer additional investment tax credits, typically worth 10-15 per cent of expenditures. It is not clear whether Canada is better off by having such a generous system.[37] International evidence suggests that tax incentives do not significantly alter corporate R&D strategies. They do not appear to encourage non-researching firms to begin investing, nor do they encourage firms to pursue projects with higher social returns or with more significant spillovers. Instead, they act more like a loosening of the budget constraint to enable existing projects to be expanded (OECD, 2002d). That conclusion is consistent with a survey of Canadian firms which found that internal cash flow was a key determinant of R&D decisions and that the tax break improved cash flow.

Some environmental aspects of sustainable development

In the long run, productivity is the most important driver of per capita incomes. However, living standards depend on more than GDP per person. Countries need to ensure that a balance is maintained between economic, environmental and social outcomes. The following sections look at three important environmental

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issues: air pollution, water quality and climate change, following consideration of the sustainability of retirement income earlier in this chapter. The section also considers whether institutional arrangements are in place to integrate policy-making across the different dimensions of sustainable development (Box 1).

Box 1. Policy integration[1]

The structure of government makes policy integration across different dimensions of sustainable development particularly challenging in Canada. The environment is an area of shared jurisdiction between federal and provincial governments, and the dividing lines between respective jurisdictions are not always clear. In order to improve policy integration within the existing system, a Harmonisation Accord was signed in 1998 by all jurisdictions. The Canadian Council of Ministers of the Environment also plays an important role in trying to coordinate actions between the federal and provincial governments. The federal government also relies on Administrative agreements with provinces and territories under the Canadian Environmental Protection Act (CEPA) in order to manage toxics. At the federal level, the Commissioner for the Environment and Sustainable Development, under the Auditor General, monitors the implementation of the sustainable development plans produced by each government department and must present annual reports to Parliament on key sustainable development issues. As well, the National Round Table on the Environment and the Economy (NRTEE), an independent organisation, also provides decision-makers, opinion leaders and the Canadian public with advice on sustainable development.

A key to integrated national sustainable development polices is developing an information base that can be used to assess past policies and highlight areas where change is needed. The NRTEE has reported that the availability of such information is sparse, even in areas of key environmental concern. It has recommended that additional support should be made available for developing a Canadian information system for the environment and to expand the System of National Accounts to include measures of natural, human and social capital.

One shortcoming in sustainable development policy formation is the limited use of cost-benefit analysis. At the federal level, new regulations have to be subject to such analysis and can be implemented only if benefits exceed costs. On the other hand, the environmental impact analyses that are required for projects and policies with important environmental effects do not have to include cost-benefit analysis. The use of such a framework in the area of transport policy, in particular, could help generate sustainable policies at lower costs.

1. The sections in this report dealing with air pollution, water pollution and sustainable retirement incomes are inputs into the Organisation's follow up on Sustainable Development as mandated by the Ministerial Council decision in May 2001. The issue of retirement incomes is dealt with earlier in this chapter in the section on labour markets, while a discussion of climate change has been added to this section.

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Air pollution

Main issues

The principal objectives of policies regarding air pollution have been to reduce the concentrations of pollutants in urban areas and to control acid rain. Though significant progress has been achieved over recent decades in reducing air pollution, public policy aims at further improving air quality in a number of areas. In view of the split of responsibilities between federal and provincial governments, the main issue is how to design economically efficient policies that balance the environmental benefits against their financial costs and that allocate abatement commitments in a cost-effective way.

Performance

As in most OECD countries, economy-wide emissions intensities have fallen for the main air pollutants in recent decades (Table 15). However, in the cases of sulphur dioxide and volatile organic compounds (VOCs), the decoupling of emissions trends from economic growth has been significantly weaker than in the OECD area on average, while the economy-wide nitrogen dioxide emissions intensity has declined in line with the OECD average. As a result of these trends, the levels of emissions per unit of GDP continue to be higher than in most other OECD countries.[38] Emissions trends did not permit a reduction in the concentrations of the main air pollutants in the 1990s. Moreover, the nationwide data mask a situation where peak concentrations of small particulates in many urban areas (notably the Windsor-Quebec corridor) are well above the limits set in air quality standards. Another pollutant of concern in urban areas is ozone. Average ambient concentrations have been rising and, once again, are highest in southern Ontario.

Policies

Canada is a signatory to a number of international agreements designed to limit emissions of sulphur and nitrogen dioxide. Following earlier agreements, Canada agreed to limit nationwide emissions of SO_2 and NO_x in the 1991 Canada-US Air Quality Agreement,[39] which also aimed at reducing the flow of these pollutants from the United States to Canada. Until the late 1990s the emissions caps in Canada were mainly limited to the eastern provinces. However, a new strategy for combating pollution in 1998 extended their coverage to most western provinces and tightened limits in the eastern provinces.[40] As a part of the strategy to reduce atmospheric concentrations of particulate matter and ozone in Canada, the 1991 agreement was supplemented by the December 2000 Ozone Annex.[41]

International agreements have provided a stimulus to improving urban air quality in the most affected provinces. In particular, Ontario has started to

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Table 15. **Performance indicators for air pollution**

	Change in emissions per unit of GDP, 1990-99[1]			Level of emissions, 1999[2]		
	Sulphur dioxide	Nitrogen dioxide	VOC	Sulphur dioxide	Nitrogen dioxide	VOC
	Per cent, per year			Grams per dollar of GDP (1995 US$ PPP)		
Australia	3.1	–1.9	–2.9	3.9	5.5	4.1
Austria	–10.2	–3.5	–6.5	0.2	0.9	1.2
Belgium	–9.2	–3.6	–4.8	0.8	1.2	1.1
Canada	**–5.3**	**–2.8**	**–2.9**	**3.2**	**2.6**	**3.5**
Czech Republic	–19.1	–6.6	–5.7	2.1	3.0	1.9
Denmark	–14.2	–4.9	–5.2	0.4	1.6	1.0
Finland	–13.0	–3.8	–4.1	0.7	2.1	1.4
France	–8.3	–3.8	–5.1	0.5	1.2	1.4
Germany	–20.1	–7.1	–8.8	0.4	0.9	0.9
Greece	–1.4	–0.3	–0.2	3.6	2.5	2.6
Hungary	–6.1	–1.1	–3.8	5.5	2.1	1.4
Iceland	–1.1	–1.7	–4.9	3.7	3.9	1.4
Ireland	–8.1	–6.3	–7.9	1.7	1.3	1.0
Italy	–7.6	–4.3	–4.4	0.8	1.2	1.4
Japan	–1.7	–1.3	–2.4	0.3	0.5	0.6
Korea	–8.9	–3.0	. .	2.1	2.0	. .
Luxembourg	–18.7	–9.1	–7.9	0.2	0.9	0.9
Mexico	–0.4	–0.9	–9.3	1.6	1.6	1.4
Netherlands	–10.1	–6.5	–8.8	0.3	1.1	0.7
New Zealand	–0.4	–0.5	–1.6	0.7	3.0	2.6
Norway	–9.7	–2.8	–1.8	0.2	2.0	3.0
Poland	–10.0	–6.6	–4.8	5.1	2.8	2.2
Portugal	–2.1	–0.9	0.1	2.4	2.3	3.1
Slovakia	–13.7	–7.4	–5.6	3.3	2.4	2.0
Spain	–5.8	–2.1	–3.5	2.2	1.7	3.6
Sweden	–8.3	–4.3	–3.9	0.3	1.3	2.1
Switzerland	–6.0	–5.4	–6.3	0.1	0.5	0.9
Turkey	3.2	1.0	0.0	3.4	2.3	1.6
United Kingdom	–13.9	–7.9	–6.6	1.0	1.3	1.4
United States	–5.5	–2.5	–4.7	2.0	2.7	1.9
European Union	–11.1	–5.0	–5.3	0.8	1.2	1.5
OECD Europe	–10.1	–4.8	–5.0	1.2	1.4	1.5
OECD	–6.7	–2.9	–4.4	1.5	1.9	1.6

Note: For the OECD average, missing data for either 1990 or 1999 has been replaced by nearby years, such estimates represent about 5 per cent of the area total for 1999.

1. Australia: 1995-99 for sulphur dioxide; New Zealand: 1990-98; Mexico: 1994-98 for sulphur dioxide and VOC and 1990-98 for nitrogen dioxide; Slovakia: 1990-98 for sulphur dioxide and nitrogen dioxide and 1990-97 for VOC.
2. 1998 for Mexico, New Zealand and sulphur dioxide and nitrogen dioxide in Slovakia; 1997 for VOC in Slovakia.

Source: Cooperative Programme for Monitoring and Evaluation of Long-Range Transmission of Air Pollutants in Europe (EMEP); World Health Organisation; OECD (2002a): OECD Environmental data 2002.

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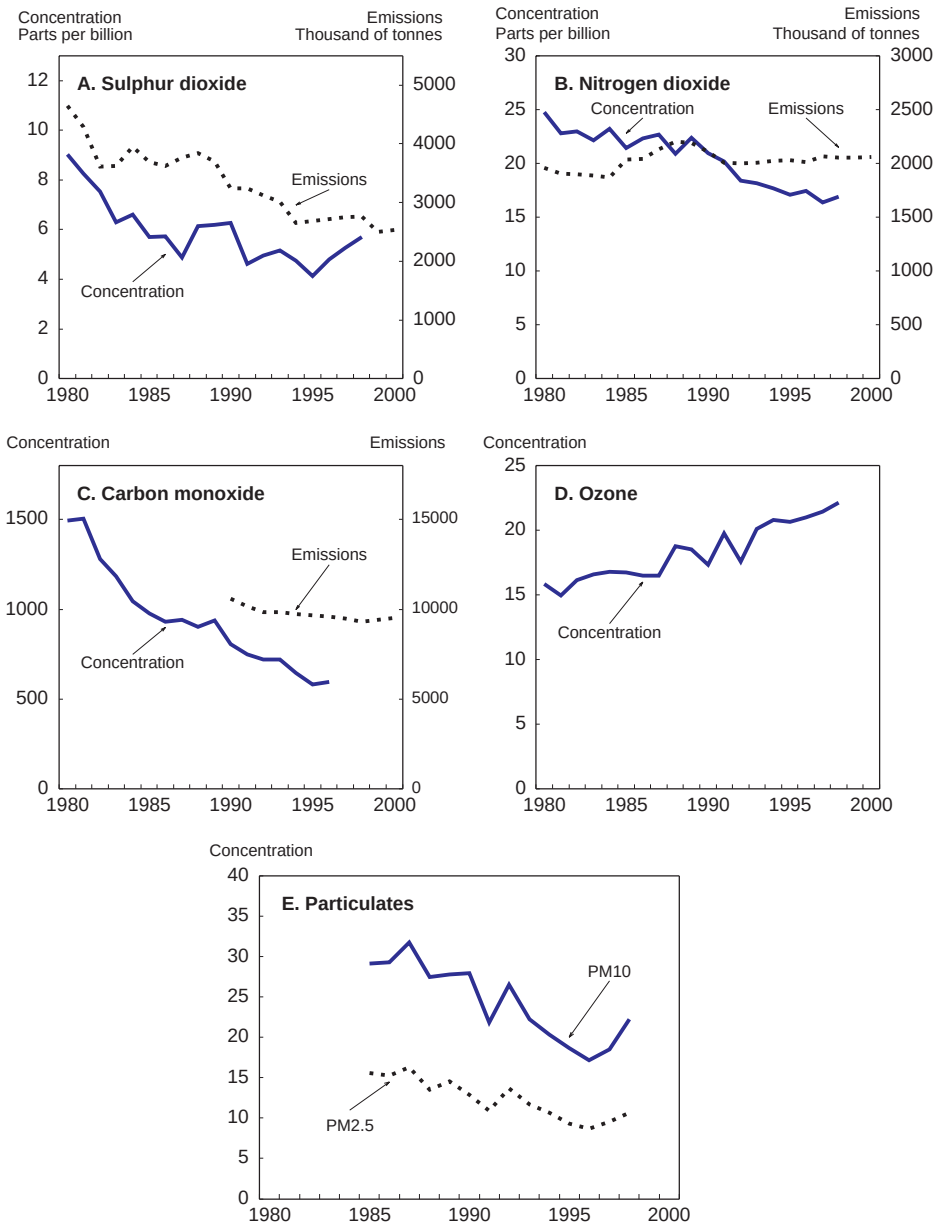
Figure 20. **Atmospheric concentrations of selected air pollutants**



Source: Environmental Technology Centre, Environmental Protection Service, Environment Canada.

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implement emissions trading for nitrogen dioxide – one of the precursors of ozone. While this cap-and-trade programme was limited to the electricity generation sector, it provided part of the means for reducing NO_x emissions to the internationally agreed limits. However, one problem with the scheme was that electricity generators were able to buy credits from companies that had reduced emissions in other sectors that were not subject to any cap. While the government has to certify that specified investments did reduce emissions, it is difficult to determine whether such reductions were offset by other increases, so limiting the programme's effectiveness. The provincial government is now considering a range of programmes to lower NO_x emissions outside the electricity sector, including widening the boundaries of the trading system. This option is likely to result in a more cost-effective reduction of NO_x emissions.

Progress under the 1998 agreement on a Canada-Wide Acid Rain Strategy appears to have stalled. Under this agreement, Ontario, Quebec, New Brunswick and Nova Scotia announced that their sulphur dioxide emissions would be lowered by 50 per cent by 2010 from their levels in 1994-95. However, marginal abatement costs vary greatly from one province to another (Table 16). The distribution of efforts among provinces should take these marked differences into account and suggests that an objective of imposing equal percentage cuts may be excessively costly. While about half the required reduction in emissions had occurred by 2000, little further progress is projected on the basis of current policies (Environment Canada, 2002).

Further progress in reducing sulphur dioxide emissions is required to meet the targets set in the Acid Rain Agreement. At the moment, only Ontario has introduced such a cap-and-trade system to limit emissions, but it is restricted to the electricity generation system. As this sector produced only 13 per cent of all sulphur dioxide in the province in 1995 (the last available data), it is unlikely that it alone could meet the reductions required to meet the targets. At the moment, a widening of controls is being considered by the provincial government. As with NO_x emissions, the most economic outcome is likely to be achieved by utilising a cap-and-trade policy. Widening the scope

Table 16. **Marginal abatement costs for a 50 per cent reduction in sulphur dioxide emissions**

1997 Canadian dollars per tonne of SO_2

Ontario	835-1 000
Quebec	312-512
New Brunswick	167-217
Nova Scotia	600-648

Source: AETG (1997).

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of the existing trading regime in Ontario to cover other provinces and parts of the United States would also have the advantage of introducing participants with a wider range of abatement costs, so increasing the likelihood that a successful market will develop and improving the cost effectiveness of the policy.

The ability of the federal government to act in the area of air pollution was increased by the classification of particulates as a toxic substance in May 2001. This classification gives more legislative power to the federal government under the Canadian Environmental Protection Act. Vehicle emissions and fuel standards are being aligned on those of the United States. The latter move, which involves reducing the sulphur content of fuel, will be particularly important for lowering particulate emissions from large trucks and off-road vehicles used in agriculture and construction, but, as yet, controls over pollution from railroad engines have not been introduced. While cost-benefit studies have shown that particulate reduction brings substantial benefits, it is not clear that focussing on reductions in the transport sector alone is necessarily the most cost-effective strategy. The use of wood, either as a fuel for domestic heating or as raw material in various industries, results in the emission of more than five times as many small particles as road transport. The government is investigating whether a link can be established between particles emitted from wood fires and stoves and health outcomes.

Conclusions

Substantial progress still needs to be made in reducing air pollution emissions. The co-ordination procedures among provincial governments appear to result too often in a sub-optimal allocation of abatement efforts. For problems like acid rain and smog precursors involving long-range transportation of pollutants, the use of cross-provincial economic instruments (tradeable emissions permits or emissions fees) would reduce the costs of improving the environment, when compared to the current approach that relies heavily on command-and-control regulation. While country-wide instruments would not be appropriate because the marginal costs of emissions reductions should be equalised only in geographic zones where pollutants have similar effects, the creation of an overall emissions target for the Sulphur Oxide Management Area of eastern Canada, which will be incorporated into international agreements, could provide the basis for a trading system that would be wider than the existing Ontario scheme. At the same time, the coverage of the Ontario cap-and-trade system for NO_x should be widened. Finally, for particulates, the government should not concentrate on reducing emissions just from diesel vehicles. Residential wood-burning and industrial emissions are far more important sources of pollution that need to be dealt with using economic instruments.

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Water quality

Main issues

Canada has extensive water resources, and a large part is concentrated in freshwater lakes close to the major areas of population settlement. These lakes have been subject to stress in the past, and so ensuring that their watersheds are kept in a sustainable state has been seen as a key issue in the past two decades. Industrial pollution, the extent of excess nutrients in surface water and the presence of pollutants that accumulate in the food chain being major areas of concern. In the past three years, there has also been concern about the ability to maintain the quality of drinking water at its usual high level, following the occurrence of fatalities caused by the water supply system (see below). In all of these areas, the challenge is to determine both the extent of the problem posed by these threats and to find policies that address the problems in a cost-efficient way.

Performance

It is difficult to assess the country-wide performance in the area of water pollution, given the lack of statistical information for the whole of Canada.[42] Some information is available for a number of the Great Lakes, for varying time periods, and it suggests that nitrate concentrations have stabilised but that phosphate concentrations have fallen. For four major rivers where data are available nitrate levels are amongst the lowest in the OECD area (Table 17). Nutrient pollution from agriculture is highly localised and comes mainly from intensive livestock farms.[43] Nationwide, the nitrogen balance has been rising on agricultural land. However, since this is the result of rapidly growing application to nutrient-deficient soils in western Canada, this has not posed a problem for water quality.

Policies

At the federal level, water pollution abatement efforts have been concentrated on reducing discharges to the Great Lakes, working in co-operation with the United States. Joint measures in the 1970s were successful in improving water quality in Lake Erie by reducing the use of phosphates, improving farm management practices and increasing the treatment of sewage waste. More recently, there has been less progress. For example, only one of the 17 Canadian sites of concern in the Great Lakes area identified in 1985 had improved sufficiently by 2002 to no longer be considered as a problem. This lack of progress appears to stem from weaknesses in the associated abatement plans agreed with the provincial authorities, few of which set priorities and none of which identified indicators that could be used to judge outcomes. Absence of information also bedevilled the zero discharge demonstration programme that aimed to eliminate pollutants in Lake Superior. In general, lack of basic information makes it difficult to assess the effectiveness, let alone the efficiency, of past policies in this area.

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Table 17. **Performance indicators: water pollution**
Selected rivers

	Biochemical oxygen demand		Nitrates		Total phosphates		Nitrate balance on agricultural land
	Mg O2/litre		mg N/litre		mg P/litre		Annual average change
	Average 1980-85	Average last 3 years	Average 1980-85	Average last 3 years	Average 1980-85	Average last 3 years	1985-87 to 1995-97
Australia	. .	. .	. .	. .	. .	. .	0.8
Austria	2.2	2.2	1.2	1.3	0.2	0.1	–2.4
Belgium	6.6	. .	3.8	. .	0.7	. .	–0.5
Canada	**. .**	**. .**	**0.1**	**0.1**	**0.1**	**0.0**	**8.4**
Czech Republic	8.5	4.6	5.1	3.5	. .	0.3	–5.8
Denmark	3.9	2.0	3.5	2.6	0.3	0.1	–2.7
Finland	. .	. .	0.2	0.3	0.0	0.0	–1.9
France	5.2	3.2	2.8	3.1	0.4	0.4	–0.9
Germany	3.6	2.4	3.5	3.3	0.5	0.2	–3.6
Greece	. .	. .	1.3	1.5	0.4	0.4	–5.3
Hungary	4.4	2.9	2.2	1.6	0.4	0.3	n.a.
Iceland	. .	. .	. .	. .	. .	. .	–0.2
Ireland	1.7	1.8	2.2	2.9	0.1	0.1	2.5
Italy	. .	. .	2.0	2.1	0.3	0.2	–3.7
Japan	2.3	1.4	. .	. .	. .	. .	–0.7
Korea	. .	2.8	. .	2.6	. .	0.1	3.9
Luxembourg	3.8	2.7	4.1	4.1	0.6	0.4	
Mexico	3.7	15.9	1.9	0.6	. .	0.1	–2.3
Netherlands	2.8	3.1	4.4	3.3	0.5	0.2	–1.8
New Zealand	. .	. .	. .	. .	. .	. .	3.3
Norway	. .	. .	0.3	0.3	0.0	0.0	0.1
Poland	5.0	4.3	1.8	1.9	0.3	0.3	–4.9
Portugal	. .	. .	. .	. .	. .	. .	3.8
Slovak Republic	5.1	3.3	2.0	2.2	0.1	0.2	1.0
Spain	4.7	3.6	1.7	3.1	0.6	0.2	–3.2
Sweden	. .	. .	0.4	0.5	0.0	0.1	–2.7
Switzerland	. .	. .	1.5	1.5	0.1	0.1	–3.4
Turkey	2.1	2.7	1.3	1.0	0.3	0.2	–2.1
United Kingdom	3.4	2.9	4.6	5.2	0.8	0.9	2.2
United States	1.8	1.6	1.1	. .	0.2	0.1	. .

Source: OECD.

At the provincial level, governments have generally adopted advisory or regulatory instruments to control pollution from agriculture. Policies have focussed on control of manure spreading, where there is concern about pathogens, phosphates, nitrates, acidification and heavy metals. Policies adopted to control pollution from manure production in intensive livestock farming vary con-

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siderably across the country. In some cases, control is delegated to the municipal level, which may be too small an area for effective policy implementation, given that the costs of pollution can spread well beyond a given community. The most common instrument used is the implementation of voluntary best practice plans for the use of manure. Such voluntary approaches are, in most cases, inferior to economic instruments, both from the view of environmental effectiveness and economic efficiency (OECD, 2003c) though recent research has shown that nutrient management plans have the potential to lead to sharp improvements in both surface and groundwater quality where hog production is prevalent, as in Quebec (Savard and Bohman, 2003). Indeed, a study of a small lagoon off Lake Ontario suggested that phosphorus trading between farmers and sewage plants might be a better route to pursue than further regulation of point sources (Conservation Ontario, 2001). The study suggested that trading of permits could lower abatement costs by 85 per cent relative to further regulation of point sources, supporting the evidence from other countries that economic instruments can reduce the cost of meeting discharge targets for agriculture and elsewhere.

A series of reforms have been introduced in Ontario to improve drinking water quality following an accident in which seven people died from drinking contaminated water. The 2002 Safe Drinking Water Act requires the certification of all drinking water system operators, including municipal operators. It holds local government responsible for ensuring that statutory standards are achieved, thus potentially creating a conflict of interest for local governments as owners and monitors of water companies. A fundamental problem with drinking water quality has been the low price paid by consumers for clean water. The policy of keeping the price of water below its long-run costs has resulted in municipal water companies not always having the funds necessary to upgrade their plants and so improve water quality. Indeed, it has been estimated that water prices would have to rise between 25 and 100 per cent to allow for full replacement of assets over 50 years. In order to bridge the funding gap, the provincial government passed a Sustainable Water and Sewage Systems Act in 2002. The first actions envisaged will be to quantify the full cost of water and sewage treatment. In a second stage, plans will be developed to move towards full cost recovery. In addition, the Ontario government is also addressing water quality concerns through the Manure Spreading Act, passed in 2000, that bans the spreading of untreated sewage on land and creates guidelines for the application of animal manure, in order to lessen the risk of spreading pathogens to the water supply.

Conclusions

The evaluation of policy effectiveness in reducing water pollution is hampered by an absence of national information on the quality of surface water. There has been no lack of plans aimed at reducing pollution, but without such informa-

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tion results cannot be assessed and judgements made on the best use of public funds or, indeed, the areas that merit priority attention. Creating a systematic information base should be a priority for the federal government. In addition, following the example of one government, other provincial governments should consider whether a more positive attitude to full-cost pricing of water might allow quicker renewal of existing assets and therefore more widespread use of best-practice technology.

Climate change

Main issues

As part of its commitment to the Kyoto agreement (ratified in late 2002), Canada has agreed that its average greenhouse gas (GHG) emissions in the period 2008 to 2012 will be held to 6 per cent below their 1990 level. However, the level of emissions has continued to rise quickly, such that business-as-usual (BAU) projections point to emissions being more than 45 per cent above target in the commitment period. Moreover, they are not distributed evenly across the country but are concentrated in a few provinces (Table 18). The challenge will be to cut emissions in the lowest-cost way possible, without imposing an excessive burden on a small number of industries or regions, especially as the United States, its major trading partner and competitor, has declined to ratify the Kyoto protocol and has a set voluntary target that is much less ambitious than Canada's.

Performance

In the decade to 2000, GHG emissions rose by a fifth, an increase exceeded by only four other OECD countries (Table 19). Consequently, in 2000 Canada had the OECD's sixth-highest rate of GHG emissions per unit of output (Table 20). Rising emissions intensities in the electricity generation and vehicle transport sectors have been partly responsible for this adverse trend. Another key factor has been the rapid growth of fossil fuel extraction and growing transformation in the emissions-intensive tar-sands sector, combined with more transportation of natural gas through pipelines from which leakages to the atmosphere occur. On the other hand, emissions intensities have fallen in the household and industrial sectors. Overall, since 1997, the growth of GHG emissions has been less than that of economic activity.

Policies

The government has already announced or implemented policies that it hopes will close about one-quarter of the gap between the target and the likely level of emissions. These measures include improvements volunteered by large industrial emitters, a negotiated agreement with car manufacturers to lift

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Table 18. **Emissions in 1990 and 2010 under business as usual**

Megatonnes of carbon dioxide equivalent

By sector	Level in 1990	Per cent of total	Increase, in level 1990-2010	Per cent of total	By province	Level in 1990	Per cent of total	Increase in level, 1990-2010	Per cent of total	Emissions intensity in 2010[1]
Transportation	147	24.5	58	27.9	Ontario	182	30.3	44	20.9	402
Mining and manufacturing	125	20.8	9	4.3	Alberta	166	27.6	97	46.4	1 479
Electricity generation	95	15.8	36	17.1	Quebec	86	14.3	10	4.7	347
Oil and gas	74	12.3	68	32.9	British Columbia	50	8.3	24	11.4	455
Buildings	75	12.5	10	4.6	Atlantic provinces	47	7.8	15	7.2	823
Agriculture	61	10.1	9	4.3	Saskatchewan	47	7.8	17	8.2	1 543
Landfill gas	21	3.5	3	1.4	Manitoba	21	3.5	2	0.7	538
Other	3	0.5	16	7.5	Territories	2	0.3	1	0.5	531
Total	**601**	**100.0**	**208**	**100.0**	**Canada**	**601**	**100.0**	**208**	**100.0**	**604**

1. Emissions in grams of GHG equivalent per Canadian dollar of GDP.

Source: National climate change secretariat and Statistics Canada.

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Table 19. **GHG emissions and sectoral indicators**

	Total GHG emissions		CO_2 emissions per kWh electricity	Manufacturing CO_2 emissions per unit of industrial production	Residential CO_2 emissions per unit of private consumption	Road transport CO_2 emissions per vehicle	Electricity use per unit of GDP	Industrial output per unit of GDP
	Level million tonnes CO_2 equivalent 2000	Annual average percentage change						
		1990-2000	1990-2000	1990-2000[1]	1990-2000[2]	1990-99	1990-2000	1990-2000[1]
Canada	**726**	**1.8**	**0.7**	**–2.4**	**–2.7**	**1.2**	**–0.5**	**0.9**
Australia	502	1.7	0.1	–0.8	–1.6	–0.6	–0.5	–1.6
Austria	80	0.3	–2.8	–2.8	–3.4	–1.0	–0.4	1.5
Belgium	152	0.7	–0.8	–0.7	–1.6	–0.1	–0.5	0.0
Czech Republic	147	–2.7	1.1	–10.1	–11.5	2.6	1.4	1.3
Denmark	69	–0.1	–3.7	–3.4	–5.3	0.3	1.0	1.1
Finland	74	–0.4	–0.5	–7.2	–7.8	–0.6	0.3	3.2
France	550	–0.2	–3.2	–1.8	–2.3	–0.1	0.7	–0.2
Germany	991	–2.1	–2.1	–2.6	–5.1	–0.4	–1.5	–0.4
Greece	130	2.2	–1.9	–0.9	8.2	–1.5	2.0	–1.2
Hungary	84	–1.8	–2.6	–13.4	8.8	0.0	1.3	7.3
Iceland	3	0.7	. .	. .	–7.5	–1.9	2.9	. .
Ireland	67	2.2	–1.1	–11.4	–5.4	2.5	–1.8	5.5
Italy	547	0.5	–1.2	–2.0	–0.7	0.3	0.8	–0.3
Japan	1 386	1.1	–0.9	0.5	–2.2	0.1	1.0	–1.6
Luxembourg	6	–7.8	–20.3	–12.5	–2.4	1.2	–8.5	–2.8
Netherlands	218	0.4	–0.3	–1.6	–3.9	0.2	–0.7	–0.8
New Zealand	77	0.5	3.7	1.8	–4.3	–1.2	–0.8	–0.8
Norway	55	0.6	0.4	0.2	–9.4	0.3	–2.0	–2.1
Poland	386	–3.7	–3.7	–6.4	11.7	–0.7	–2.9	3.8
Portugal	85	2.7	–0.7	1.3	1.1	–2.2	1.5	–1.4
Slovakia	49	–4.0	–0.3	–9.1	–2.1	0.3	1.3	–1.3
Spain	386	3.0	0.0	–0.2	1.8	0.2	1.2	–0.4

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Table 19. **GHG emissions and sectoral indicators** (*cont.*)

	Total GHG emissions		CO_2 emissions per kWh electricity	Manufacturing CO_2 emissions per unit of industrial production	Residential CO_2 emissions per unit of private consumption	Road transport CO_2 emissions per vehicle	Electricity use per unit of GDP	Industrial output per unit of GDP
	Level million tonnes CO_2 equivalent 2000	Annual average percentage change						
		1990-2000	1990-2000	1990-2000[1]	1990-2000[2]	1990-99	1990-2000	1990-2000[1]
Sweden	69	–0.2	0.2	–4.1	–3.7	0.2	–1.7	1.9
Switzerland	53	–0.1	–4.8	0.2	–3.1	–1.0	1.0	1.5
United Kingdom	649	–1.3	–3.6	–2.1	–1.2	–0.6	–0.7	–1.6
United States	7 001	1.3	0.2	–4.7	–1.9	0.9	–0.9	1.1
Total of above OECD countries	14 543	0.6	–0.7	–2.8	–1.8	0.2	–0.3	0.2
OECD excluding United States	**7 542**	**–0.1**	**–1.6**	**–1.9**	**–1.8**	**–0.3**	**0.3**	**–0.4**
EU countries	4 073	–0.3	–2.1	–2.0	–2.4	–0.2	–0.3	–0.3

1. 1995-2000 for Czech Republic; 1991-2000 for Germany; 1992-2000 for Hungary and Slovakia; no data for Iceland.
2. 1991-98 for Czech Republic; 1993-2000 for Slovakia.

Source: GHG National submissions to UNFCCC, national sources and UNFCCC; carbon dioxide data, IEA; industrial production, private consumption, OECD.

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Table 20. **Main indicators: climate change**

Indicators of greenhouse gas (GHG) emission intensity, grams of CO_2 equivalent per $PPP of GDP, in 1995 prices

	Total GHG emissions	CO_2 emissions, electricity	CO_2 emissions, transport	Other GHG emissions	Total GHG emissions	CO_2 emissions, electricity	CO_2 emissions, transport	Other GHG emissions
	Level, 2000				Average annual percentage change 1990-2000			
Canada	**888**	**156**	**183**	**549**	**–0.9**	**–0.2**	**–0.9**	**–1.3**
Australia	1 061	360	159	542	–1.8	–0.5	–1.5	–2.7
Austria	403	66	96	241	–2.1	–3.2	–0.0	–2.4
Belgium	600	105	97	398	–1.5	–1.3	–0.2	–1.8
Czech Republic	1 082	468	100	514	–2.8	2.5	6.3	–6.6
Denmark	501	171	88	242	–2.4	–2.7	–0.8	–2.6
Finland	597	178	99	321	–2.6	–0.2	–1.8	–3.8
France	402	30	102	271	–2.0	–2.6	–0.0	–2.6
Germany	519	168	91	260	–3.9	–3.6	–1.1	–5.0
Greece	819	275	122	422	–0.2	0.1	–0.0	–0.3
Hungary	747	192	79	476	–2.6	–1.3	–0.2	–3.4
Iceland	398	0	84	314	–1.8	. .	–2.5	. .
Ireland	643	152	98	392	–4.6	–3.0	0.2	–6.0
Italy	432	108	89	235	–1.1	–0.4	0.0	–1.7
Japan	441	132	81	229	–0.3	0.1	0.9	–1.0
Luxembourg	314	6	249	59	–12.5	–27.1	0.6	–23.1
Netherlands	553	138	80	335	–2.5	–1.0	–1.1	–3.3
New Zealand	1 078	82	179	817	–2.2	2.9	0.8	–3.1
Norway	454	3	97	354	–2.9	–1.6	–2.9	–2.9
Poland	1 109	458	74	576	–7.1	–6.5	–1.4	–8.1
Portugal	516	129	111	276	–0.1	0.9	3.5	–1.5
Slovakia	846	249	70	526	–5.2	1.0	2.0	–7.6
Spain	536	130	127	278	0.4	1.2	1.0	–0.3
Sweden	340	35	110	195	–1.9	–1.5	–0.8	–2.6
Switzerland	267	2	78	187	–0.9	–3.8	–0.4	–1.1
United Kingdom	512	137	106	268	–3.6	–4.3	–1.4	–3.9
United States	779	273	192	315	–1.9	–0.7	–1.3	–3.0
OECD total	639	201	137	307	–1.9	–0.8	–0.6	–2.9
EU	491	120	100	272	–2.4	–2.4	–0.4	–3.1

Source: Greenhouse gas emissions: national submissions to the UNFCCC and national publications. Carbon dioxide emissions for electricity and transport: IEA (2001). GDP: OECD, SNA database.

new-vehicle efficiency by a quarter by 2010, upgrading insulation on existing buildings and financial incentives for wind power research and production and a wide variety of other actions (see Action Plan, 2002). Many of these initiatives, which are being carried out in partnership with the provinces, territories and private sector, have been in operation for less than a year. The federal government will be carrying out an assessment, and adjustments will be made to ensure effectiveness in reducing GHG.

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In November 2002, the federal government released a plan with the goal of reducing Canada's greenhouse gas emissions in a manner that supports economic growth and avoids placing an unreasonable burden on any region. A key element of the plan is an initiative aimed at large industrial emitters (firms active in thermal electricity, oil and gas, or mining and manufacturing) that involves the setting of targets for emission reductions established either through covenants with a regulatory basis or through a financial backstop system. There are also a number of complimentary measures. Under both the backstop system and the covenants, the government will allocate emission permits to large emitters free of charge up to an amount equivalent to an average of 85 per cent of the product of each firm's business-as-usual emissions intensity and its expected output. Under the backstop system, permits will be allocated using an emission intensity approach. Under a covenant system, negotiations may lead to variations in the allocation system to reflect special circumstances the company may face. Under both systems, individual companies will be able to achieve their targets through in-house abatement measures or the purchase of domestic emission permits, domestic offsets or international permits.

The permit allocation rates will be set to achieve a total emissions reduction of 55 MT, assuming projected business-as-usual output levels, representing about one-quarter of the emissions needed to meet the target. In the event that the aggregate output of large industrial emitters exceeds projected business-as-usual levels, the government has stated that it is prepared to ensure that the necessary Kyoto compliance units will be made available to support the emissions due to the overrun of output. In addition, the government has also stated that companies will be able to meet their emission reduction responsibilities, during the first commitment period, at a price no greater than C$15 a tonne. The government is currently in the process of examining options for implementing this commitment.

In the energy-producing sector, which is projected to account for half of emissions growth until 2010, a key requirement is to reduce barriers to inter-provincial electricity trade. This would allow more trade of hydro energy from coastal provinces towards hinterlands that depend on fossil fuel for electricity generation. However, as noted in a previous section, negotiations on the Energy Chapter of the *Agreement on Internal Trade* have consistently failed to make progress, including the most recent attempt in 2002. In the longer term, a sustainable strategy is likely to require the development of new technologies that enable the production of electricity from fossil fuels with very low, or even zero, greenhouse gas emissions. To achieve this, the government is sponsoring a research and demonstration programme with the objective of lowering the costs of carbon dioxide capture and storage.

Additional reductions will be achieved through a variety of measures, including command-and-control instruments and subsidies. It is hoped that these

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steps will provide an extra 20 per cent emissions reduction. In the transport sector, measures may include incentives for the purchase of fuel-efficient vehicles and an initiative to expand and promote the uptake of bio-diesel. For households, steps include a "One Tonne Challenge" for reductions by consumers, and incentives for homeowners to improve energy efficiency in their homes. In addition, other measures include a Federal House in Order initiative, assistance for Aboriginal and Northern communities for energy efficiency, and a forestry initiative ("Forest 2020") to maximise reductions associated with sinks.

The government has ruled out a carbon tax, which raises the possibility that the measures undertaken may be relatively costly. In the transport sector such an approach could be justified as existing taxes on gasoline and diesel[44] are already well above the price cap on the large emitting sector if the purpose of those taxes is assumed to be solely for carbon abatement. However, such taxes also represent an internalisation of other costs, and so some increase could be justified if the component of this tax related to global warming was below the price of carbon permits. In other sectors, however, the absence of taxation leads to the adoption of a central-planning approach whose efficiency hinges on having accurate estimates of the marginal cost of emissions reduction for each option. Overseas estimates suggest that building public transport infrastructure is a very costly way to cut GHGs, while most other measures rely on the goodwill of manufacturers and users in the absence of financial or even regulatory incentives to change behaviour. In all, this part of the package risks being an expensive and possibly ineffective way to reduce emissions. However, the government has indicated that it will buy permits on the international market if target reductions are not met. It is considering a minimum purchase of 10 million tonnes of carbon dioxide permits (less than 2 per cent of expected emissions by 2010), favouring the acquisition of permits through projects to reduce emissions in developing countries.

Conclusions

The extent to which greenhouse gas emissions are likely to be above target makes it essential to use economic instruments to achieve the necessary reduction. The government's proposal for creating a trading system amongst large emitters moves in this direction and ensures efficiency by allowing large emitters to buy permits on a possible world market. The government has limited uncertainty for firms by setting a price cap on domestic permits, at the expense of introducing a budgetary risk for itself as it would have to purchase equivalent permits on the world market to stay within the agreed target. Outside this sector, the decision not to use a carbon tax but to rely on a series of regulatory command-and-control policies and voluntary initiatives means that, while some programmes may be successful, it is very unlikely that sufficient domestic reductions in emissions can be made at a reasonable price. The government would then need to buy per-

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mits on the international market to ensure compliance with the Kyoto agreement. With no mechanism to transmit world market prices to the bulk of the economy, the steps that are taken are likely to have varying abatement costs and, therefore, to be excessively costly.

Summary

Canada has several policy options for lifting living standards even further, and some specific recommendations are summarised in Table 21. Structural unemployment can be reduced, and participation rates lifted further, through reforms to the social assistance and EI programmes that aim to improve work incentives while not creating undue hardship on those in genuine need of support. In particular, the government needs to address high marginal effective tax rates wherever possible, make more use of in-work benefits, re-instate experience rating for employees and look at whether it is too easy to qualify for EI's increasingly-generous benefits. Encouraging greater internal labour mobility should also be a focus. The challenge of raising productivity growth is a tougher nut to crack. Priority should be given to reducing barriers to competition, improving the efficiency of the science sector and continuing to improve skills and human capital. The focus should be on the adult education system, since the compulsory school sector is already doing very well. Human capital can also be enhanced by a well-designed migration framework, which is the subject of the next chapter.

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Table 21. **Recommendations for structural reform**
Based on previous and current *Surveys* and actions taken

Previous and new recommendations	Action taken since the 2001 *Survey*
INCREASING EMPLOYMENT RATES THROUGH LABOUR MARKET REFORMS	
Reform unemployment benefits and social assistance systems	
Lengthen the qualifying period for Employment Insurance (EI) to be closer to international norms, and eliminate variations across regions.	New recommendation.
Re-introduce experience rating of workers.	None.
Continue to monitor the reasons behind the rise in the number of sickness and disability benefits to ensure that they are being used for their intended purpose and not to avoid tighter welfare provisions.	Ongoing.
Determine whether federal and provincial tax expenditure programmes offered to social assistance recipients could be simplified in order to improve work incentives.	None.
Make more use of in-work benefits.	New recommendation.
Strengthen training and job-search requirements, and consider more use of diversion programmes.	New recommendation.
Active labour market programmes (ALMPs)	
Adopt a more rigorous system of evaluation.	None.
Improve data comparability across provinces and expand their scope to improve evaluation of ALMPs.	None.
Shift the focus of ALMPs towards those of a shorter-term nature and away from costly long-duration training schemes.	None.
Labour mobility	
Reduce barriers to mobility by implementing mutual recognition agreements (MRAs) for all occupations.	MRAs were implemented in almost all professions by the deadline of July 2001. MRAs for the remaining few professions need to be completed and all agreements enforced effectively.
INCREASING HUMAN CAPITAL	
Assess how rising education expenses are affecting participation in tertiary education, and how well government financial assistance is meeting needs.	The Canada Graduate Scholarship Programme was introduced in 2003. It will provide 4 000 scholarships each year at the Masters and Doctoral level.
Continue research on what can be done about Canada's high drop-out rates, especially any link between low literacy and dropping out.	Ongoing.
Look at whether underfunding is leading to a shortage of adult training places for at-risk groups.	New recommendation.
Extend the student loan scheme to more middle-income families, make repayments income-contingent and favour student loans over scholarships.	New recommendation.

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Table 21. **Recommendations for structural reform** (*cont.*)
Based on previous and current *Surveys* and actions taken

Previous and new recommendations	Action taken since the 2001 *Survey*
INCREASING PRODUCTIVITY THROUGH INNOVATION	
Broaden the contestable funding pools to include all players, in particular government labs.	New recommendation.
Continue to simplify the application process for the R&D tax credit.	New recommendation.
Quickly implement the priority recommendations developed in the November 2002 National Summit on the *Innovation Strategy*.	New recommendation.
INCREASING PRODUCTIVITY BY IMPROVING MARKET PERFORMANCE	
International trade and investment	
Pursue efforts made for further opening of international trade.	Canada is supporting the Doha round of WTO negotiations. It concluded an FTA with Costa Rica in 2001 and is negotiating FTAs with four other Central American countries, Singapore and EFTA.
Reduce foreign ownership barriers wherever possible, including telecommunications, airlines and broadcasting.	New recommendation.
Internal market	
Complete the Agreement on Internal Trade (AIT) and set up new targets.	Progress has been slow.
Strengthen the AIT by improving the dispute resolution process, and consider majority voting and mutual recognition of standards as a default solution.	New recommendation.
Electricity	
Accelerate deregulation of the electricity industry, and break up the generation near-monopolies in each province.	New recommendation.
Agriculture	
Continue to improve market incentives, especially in the dairy industry where support remains high.	Over the last decade Canada has improved the market orientation of its policies, except in the dairy sector.
Establish a fair framework for income support that does not prevent economic adjustment, and stick to it.	An agricultural policy framework was announced in June 2002, aiming to improve farming practices and to renew the income stabilisation programmes.
Electronic transactions	
Reduce barriers to expansion of electronic commerce and tax issues by implementing the recommendations of the advisory groups on e-commerce.	Progress is being made on a considerable number of the recommendations.

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Table 21. **Recommendations for structural reform** (*cont.*)
Based on previous and current *Surveys* and actions taken

Previous and new recommendations	Action taken since the 2001 *Survey*
FINANCIAL SECTOR	
Set out the policy for how financial institutions would be wound up in the event of any failure.	OSFI has a guide to intervention on this issue.
Clarify the criteria that would determine whether a large bank merger would be approved.	Following a March 2003 House of Commons Committee report, the government said it would not consider any mergers before September 2004.
Continue harmonising regulatory standards, and reduce the number of regulators.	Two committees will make recommendations in autumn 2003.
TAXATION	
In future tax measures, give priority first to corporate tax relief and then to personal tax reductions.	The 2003 budget announced the gradual elimination of the corporate capital tax along with further corporate and personal tax reductions.
Consider shifting the tax base from payroll, personal and corporate income toward value added or consumption.	The share of consumption taxes in federal revenue should rise following the 2003 budget but will stay low by OECD standards.
Phase-out the preferential tax treatment of the resource sector.	Changes announced in budget 2003 will lead to a consistent tax treatment across resource projects and between the resource and other sectors by 2007.
PUBLIC EXPENDITURE	
Adopt a medium-term framework and present the full five-year impact of all decisions at budget time.	None.
Publish in advance a ranking of contingent year-end spending proposals.	None.
Re-instate key elements of the Programme Review process.	Re-instated in the 2003 budget.
Reduce federal/provincial disagreements over the alleged vertical fiscal (im)balance by achieving consensus on the facts.	The CHST will be split into separate health and social transfers. This may clarify the federal government's contribution to health care.
Review the rate-setting process for EI, and bring down the rate to the level required to meet programme objectives. Avoid moving the programme further from its insurance principles.	The 2003 budget re-affirmed the government's intention to start the review of the rate-setting process. EI premium rates have been reduced and are now close to break-even, achieved partly by reducing rates and partly by enriching benefits. Recent enrichments (notably the parental leave provisions) do move the scheme further from insurance principles.
Improve governance of arms-length foundations and clarify their objectives.	This process is ongoing.
Put more weight on rate of return when selecting infrastructure investment.	None.

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Table 21. **Recommendations for structural reform** (*cont.*)
Based on previous and current *Surveys* and actions taken

Previous and new recommendations	Action taken since the 2001 *Survey*
Modernise the human resource management process. Increase employee accountability and pay flexibility (including bonuses).	None. The federal government intends to eliminate geographical hiring restrictions but has no timetable for doing so.
HEALTH CARE	
Introduce a mixed fee-for-service/capitation payment system as a way of reducing costs once the physician shortage problem has been solved. Improve information on costs per procedure.	None.
Create a database to track key data, such as waiting lists, in order to target expenditure more effectively.	None.
Accelerate primary care reform based on what has been learned from pilot projects.	New recommendation.
Reform hospital funding mechanisms in order to increase incentives to raise efficiency. Consider output-based funding mechanisms.	New recommendation.
Increase the use of cost-sharing devices in areas where it will not have detrimental impacts on health outcomes.	New recommendation.
Clarify the reasons why supplementary private insurance is effectively ruled out by regulation, and look at whether those reasons are valid in all the cases where the ban is currently imposed.	New recommendation.
SUSTAINABLE DEVELOPMENT	
Make more use of cost-benefit analysis and economic instruments in order to make more progress on policy objectives.	None.
Make water rights transferable and price their use through metering with full cost recovery.	Some municipalities are applying a sewer surcharge to water bills, which also generates water savings.
Increase compensation for fishing license retirements. Encourage a reduction in the size of the fishing sector.	The east coast cod fishery was closed in May 2003 following a collapse in stocks.
Increase the share of fisheries managed by individual quotas. Adopt a rules-based approach to setting quotas and strictly enforce rules against overfishing.	The Newfoundland government plans to assist cod fishers mount a defence if they are prosecuted for overfishing.
Use economic instrument with the largest base possible in order to reduce the overall costs of GHG emissions reduction.	The climate change plan includes a limited domestic emissions trading scheme with access to international permits. It also includes a large number of potentially costly and inefficient reduction proposals. A broader carbon tax or trading scheme would have been preferable.
Apply the polluter pays principle more systematically.	None.

Source: OECD.

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III. International migration: economic context and implications

Introduction

During most of its history Canada has received large numbers of immigrants, and a policy of encouraging immigration has been an important element of the country's growth strategy. The foreign-born share of the population reached 18.4 per cent in 2001 (up from 14.7 per cent 50 years earlier) and is higher than in nearly all other OECD countries. Canada traditionally receives mainly permanent immigrants: on average in 2000-02, 235 000 of them were admitted each year, about ¾ per cent of the population. There is also some temporary immigration, but the net flow is relatively small. Emigration, directed mainly to the United States, has risen in the 1990s, but – at about 0.2 per cent of the population – remains much smaller than immigration (Figure 21). Thus, net migration is positive, as it has been almost continuously during the past century, and now represents about 70 per cent of annual population increase, a proportion that is set to rise further, assuming unchanged fertility rates.

Public opinion toward immigration tends to be more positive in Canada than in most other OECD countries (Bauer *et al.*, 2000). This favourable attitude is helped by the fact that most immigrants are skilled and are expected to integrate as their predecessors did in the past. Moreover, the level of illegal immigration remains modest and less of an issue than in other OECD countries thanks, in part, to Canada's geographical distance from source countries and sharing a border only with the United States. Thus, the authorities are generally perceived as being effectively in control of admissions. Public support for immigration policy tends to weaken during recessions, as was the case for example in the early 1990s, and has proved vulnerable to incidents that undermine public confidence in policy integrity. Similarly, it was negatively affected by security concerns in the months after 11 September 2001. But these effects tend to be temporary.

In Canada, immigration and the labour market have complex institutional environments. Under the Canadian Constitution, immigration is an area of shared jurisdiction between the federal and the provincial/territorial governments. Bilateral immigration and labour market agreements further define the respective roles and responsibilities. Therefore, close collaboration and partnerships with

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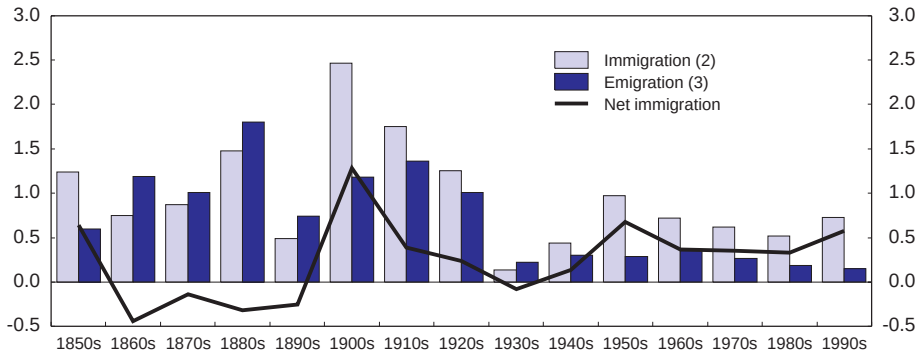
Figure 21. **Immigration, emigration and net migration**
As a percentage of population, average annual rate, by decade[1]



1. Estimates of inter-censual flows. Decades are defined as intervals between census dates (April 1991-April 2001, and so on).
2. Includes only permanent immigration.
3. Total emigration (of Canadian-born and foreign-born permanent residents) net of returning Canadians.

Source: Statistics Canada.

provinces and territories are essential to moving forward on immigration initiatives and related labour market issues.

This chapter examines the economic implications of international migration for Canada. It discusses immigration policy to the extent that its main elements – immigrant selection and the programmes supporting immigrant settlement and integration – influence the economic outcomes of immigrants in the Canadian labour market and their broader impact on the economy. While recognising that the ultimate objectives of immigration policy are not only economic, the chapter investigates the potential welfare effects, not only for the immigrants themselves – presumably positive – but also for the receiving population, which are more difficult to identify and to quantify. The economic implications of emigration from Canada are also discussed in the context of the increasing mobility of highly skilled workers, particularly within the North American labour market.

Immigrant selection policies

Present selection policies emphasise skills and adaptability

Canada's current immigration legislation envisages three main categories of permanent immigrants: economic immigrants, which include skilled workers,

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business immigrants and a few smaller categories; the family reunion class; and refugees (see Annex IV). In 2002 skilled workers and their families represented 54 per cent of all immigrants admitted, with business immigrants comprising another 5 per cent (both numbers include dependants as well as the principal applicants). The share of the family class was 28.5 per cent, and refugees made up most of the remainder (some 11 per cent) (Table 22). Skilled workers are selected on the basis of a "points system", where points are assigned on the basis of immigrant characteristics, and applicants are admitted if total points reach a minimum threshold. Thus, obtaining zero points for one characteristic does not disqualify an applicant if enough points are obtained for the others. However, to meet the definition of skilled worker, the applicant must have one year of experience in a skilled occupation in the last ten, making zero points for experience an effective bar. The new Immigration and Refugee Protection Act and the related regulations that came into effect in June 2002 have revised the points system, increasing the weight of education and language skills, eliminating occupation-specific criteria and stressing immigrant adaptability instead (see Annex IV).

There are three specific programmes for business immigrants aimed, respectively, at investors, entrepreneurs and self-employed persons. They are intended to attract immigrants who, by using their entrepreneurial skills and/or investing capital, will make a positive economic contribution. Hence the requirements in terms of capital invested and number of jobs created. In terms of meeting formal requirements the programmes seem to have been relatively successful.[45] However, it is difficult to assess whether the jobs created have been truly additional and, more generally, whether the programmes have in fact helped foster immigrant entrepreneurship on a sustained basis. Part of the problem may be inherent to the policy's rationale: immigrant entrepreneurs are supposed to set up a business upon arrival in Canada, although initially they lack much of the local knowledge required and more than half of them speak neither English nor French. Naturally, most will tend to choose a business activity in the ethnic enclave of a large city and probably take over an existing small business rather than start a new one. Between the mid-1980s and the late 1990s the programmes seem to have provided an avenue for relatively well-to-do immigrants from Hong Kong (China) who would not have qualified as skilled workers (half of business immigrants came from there). The overall number of business immigrants has declined since 1997.

In line with its long humanitarian tradition Canada receives significant numbers of refugees. Since 1978 it follows a policy of admitting a steady flow every year, as well as responding to emergency situations. In the last few years between 24 000 and 30 000 refugees have been admitted annually (between 11 and 13 per cent of all permanent immigrants). In proportion to the population the refugee influx is in line with the average of other OECD countries (Figure 22). Close to half of them were selected from abroad, sponsored by the government or by private organisations, while the rest are people who requested protection after

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Table 22. **Actual and planned immigration by entry category**

Immigrant class	Average 1996-2000		2001		2002		2003 plan
	Level	Per cent	Level	Per cent	Level	Per cent	
Family class	**58 993**	**28.5**	**66 646**	**26.6**	**65 277**	**28.5**	**59 000-64 500**
Immediate family	40 740	19.7	45 385	18.1	42 775	18.7	. .
Parents and grandparents	18 253	8.8	21 261	8.5	22 502	9.8	. .
Economic immigrants	**119 375**	**57.8**	**155 596**	**62.2**	**138 528**	**60.5**	**132 000-147 000**
Skilled workers	99 127	48.0	137 119	54.8	123 379	53.9	121 000-131 000
Principal applicants	43 317	21.0	58 860	23.5	53 448	23.3	. .
Spouses and dependants	55 810	27.0	78 259	31.3	69 931	30.5	. .
Business immigrants	16 567	8.0	14 580	5.8	11 041	4.8	7 000-9 000
Principal applicants	4 618	2.2	4 082	1.6	3 047	1.3	. .
Entrepreneurs	2 222	1.1	1 612	0.6	1 177	0.5	. .
Investors	1 376	0.7	1 766	0.7	1 235	0.5	. .
Self-employed	1 020	0.5	704	0.3	635	0.3	. .
Spouses and dependants	11 950	5.8	10 498	4.2	7 994	3.5	. .
Provincial/territorial nominees	402	0.2	1 274	0.5	2 127	0.9	2 500-4 000
Live-in caregivers	3 279	1.6	2 623	1.0	1 981	0.9	1 500-3 000
Other immigrants[1]	**2 384**	**1.2**	**210**	**0.1**	**164**	**0.1**	**900-1 000**
Refugees	**25 940**	**12.5**	**27 894**	**11.1**	**25 122**	**11.0**	**28 100-32 500**
Government sponsored	8 215	4.0	8 693	3.5	7 504	3.3	7 700
Privately sponsored	2 628	1.3	3 570	1.4	3 055	1.3	4 200
Landed in Canada	11 808	5.7	11 891	4.7	10 544	4.6	13 000-15 600
Dependants abroad[2]	3 288	1.6	3 740	1.5	4 019	1.8	4 500-5 000
Total	**206 692**	**100.0**	**250 346**	**100.00**	**229 091**	**100.0**	**220 000-245 000**

1. Includes Post-determination refugee claimants, Deferred removal orders and Retirees.
2. Dependants of a refugee landed in Canada who live abroad.

Source: Citizenship and Immigration Canada.

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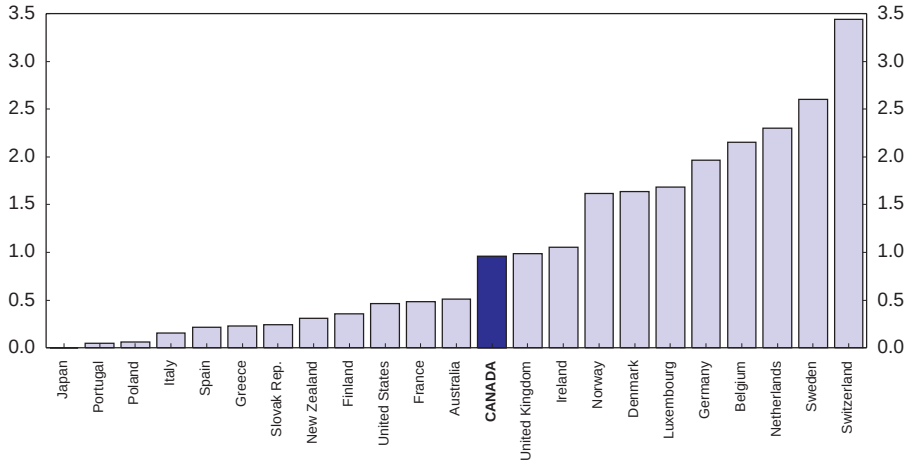
Figure 22. **Inflows of asylum seekers into selected OECD countries**
Per thousand of population, average 1992-2001



Source: OECD, *Trends in International Migration*, SOPEMI 2002.

landing in Canada and whose refugee status has been recognised. The number of claimants has increased substantially over the last few years. The recognition rate, about 49 per cent in 2002, is one of the highest among OECD countries. After a preliminary determination that their claims have a credible basis, refugee claimants are considered temporary residents while their claim is under consideration. They are allowed to work and have access to basic health coverage and to some public assistance. Once their status has been recognised, they have 180 days to apply for permanent residence. The new legislation has streamlined the procedure for assessing asylum claims, prescribing shorter processing times, even for applicants who arrive without papers, and rationalising the appeals process.

Net inflows of temporary foreign workers are relatively modest compared to those of permanent immigrants, and the number present in Canada in 2001 was not much greater than it was in the previous (1990) cyclical peak. By contrast, in Australia and New Zealand – like Canada, traditional countries of permanent immigration – the number of immigrants admitted on a temporary basis has recently been increased considerably, and the latter now represent about half of the overall immigrant inflow. Normally a Canadian employer wanting to hire a foreign worker needs to make a job offer and to obtain from the labour department (Human Resources Development Canada) an opinion that the job cannot be filled

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by a permanent resident or citizen. Certain categories of workers – including entrepreneurs and intra-company transferees – are exempted from this requirement. In addition, information technology workers and a few other occupations benefit from facilitated procedures. This is also the case for some broader categories (professionals, traders and investors) under the North American and the Canada-Chile free trade agreements. Temporary workers from the United States (mostly professionals) and from Mexico (most of them lower skilled) represent about one-quarter and one-tenth of the total, respectively. The 2002 legislation has introduced measures to speed up the authorisation process. Moreover, workers initially admitted with a temporary permit, who can obtain extra points for their Canadian experience if they subsequently apply for permanent resident status as skilled workers, can now receive landed status while in Canada.

Canada also receives substantial numbers of foreign students. Their number has risen substantially in recent years, more than doubling between 1995 and 2001, and they now outnumber temporary workers. Foreign students can be allowed to work under certain conditions, and their spouses can obtain temporary work permits. After graduation, they can obtain temporary authorisation to work in their field of study for one year (two in New Brunswick under a new pilot programme). Under the present points system it can be relatively difficult for them to qualify for permanent immigration as skilled workers if they do not have work experience in their field, even though they may have high academic qualifications and a job offer. They can, however, meet the requirement to have a year of work experience by staying in Canada as temporary workers.

Every year the government sets a target or "planned" range for total admissions of permanent immigrants, broken down into planned ranges for the individual entry classes. The immigration plan for 2003 envisages between 220 000 and 245 000 new arrivals (see Table 22), only slightly above the plan for 2002, with 60 per cent of the total in the economic class. In the past, the annual plans have usually been met with a reasonable degree of accuracy.[46] The government has indicated a medium-term objective of raising immigration to one per cent of the population, which would mean admitting over 300 000 new immigrants a year. However, annual planned levels are being set below that objective because they also take into account the absorption capacity of the Canadian economy and the resources available, not only for processing applications but also for supporting immigrants' integration. In particular, it is felt that to absorb larger numbers of immigrants it would be necessary to achieve greater progress in this latter area, for example addressing the issue of credentials recognition and moving toward a more balanced geographical distribution of immigrants. These issues are discussed further below.

The immigrant selection process and the refugee scheme use considerable public resources, but a significant share of the cost is covered by processing

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fees.[47] Most applications are examined through the missions Citizenship and Immigration Canada maintains in source countries. Processing times can differ significantly across immigrant categories (and also across country missions). For example, until recently family class applications were usually processed in less than a year, while for skilled-worker applications the average time was over two years and reached four years in some countries. In 2002 the total backlog of applications represented 500 000 individuals, 60 per cent of which were skilled workers principal applicants and their dependants. Recognising that long processing lags risk discouraging some of the more qualified applicants, the government has recently assigned a higher priority to skilled worker applications and will fast track those with job offers. The creation of an easier entry route for temporary workers seems to go in the same direction. In addition, the more selective admission standards introduced in 2002 are already contributing to reducing the backlogs.

Learning from experience: past immigration policies and their effects

The composition of immigrant flows has changed dramatically, partly as a result of selection policies

The size and geographical composition of migration flows has undergone significant changes over the course of the country's history, reflecting both global migration trends and the evolution of immigration policy. The most profound change occurred in the mid-1960s, when an admissions policy that limited access to immigrants of European stock was replaced with one selecting them according to their skills (see Box 2). Australia and the United States, two other traditional settlement countries, implemented a similar shift around the same period, although in the latter family reunification, rather than skill-based selection, has been the main immigration channel.

The composition of immigrant flows has changed considerably as a result. A most obvious result of the removal of race-based selection rules in the mid-1960s was the dramatic change in *source-country composition*, with the progressive reduction in the share of European immigrants and a rise in those of developing countries, especially Asian countries (Figure 24). The skill level of immigrants has also changed significantly. The new cohorts of immigrants have higher levels of *education* than earlier ones, but on average have not gained ground relative to the Canadian-born, remaining slightly ahead of them according to some measures (Figure 25). But there are important differences across entry categories. Over 80 per cent of skilled worker principal applicants admitted in 2001 had a university degree, against an average of 25 per cent in the 1980s. For family-class immigrants the percentage was much lower (20 per cent), but still above the Canadian average, which is 15.4 per cent. It was lower (9 per cent) for refugees.[48] The *occupations* declared by immigrants at entry –

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Box 2. The evolution of Canadian immigrant selection policies

In the late nineteenth century the central goal of immigration policies was to contribute to population growth and to the settlement of the western provinces. Recruitment agents initially sought mainly farmers, but over time immigrants were increasingly steered also toward railway construction, manufacturing and mining. From the start recruitment was selective in terms of race and countries of origin, with a preference for immigrants from Britain, the United States and North-western Europe. After World War I the government started to take into account current labour market conditions in setting immigration targets. Thus, the inflow was virtually brought to a halt during the 1930s depression. When it resumed in the post-war period, the supply of immigrants from the traditional ("preferred") countries had declined, while inflows from Eastern and Southern European countries had become predominant. Although most of these immigrants were unskilled, they were readily absorbed by the fast-growing post-war Canadian economy. By the late 1950s, however, it was felt that the insufficient supply of skilled labour was increasingly constraining growth, and a conflict developed between the desire to use immigration to increase the skill level of the labour force and the hitherto prevailing race-based selection criteria.

In 1962 Canada eventually abandoned country of origin and race as selection criteria and moved to skills as the key trait for economic immigrants. A points system was introduced in 1967 for that purpose, while immigrants in the family class and refugees were not points-tested. In addition to general characteristics (age, education, language skills), it assigned points according to the immigrant's specific occupation (see Table A2 in Annex IV). In certain periods additional requirements were introduced.[1] The system was designed to allow the government to fine-tune both the total number of immigrants (by setting annual targets) and their skill composition (through the attribution of occupation-specific points) to prevailing labour market conditions.[2] However, since the applications of family-class immigrants received priority in processing, the skill-assessed component was determined as a residual and was squeezed whenever the annual targets were reduced (or the eligibility criteria for the family and refugee classes were relaxed). This is what happened, for example, after the recession of the early 1980s (Figure 23). While the fundamental features of this system (no discrimination on the basis of country of origin, selection of certain categories based on a points system, overall annual targets) are still in force, some important elements have evolved, particularly since the mid-1980s. *First*, after 1986 the government stopped adjusting the annual immigration targets in light of current labour market conditions and started to set them according to longer-term demographic or economic considerations. Targets were progressively increased, in spite of the early-1990s recession. *Second*, in the early 1990s eligibility criteria for the family class were tightened, with the result that the share of skill-assessed immigrants started to increase. *Third*, in the parameters for the attribution of points there was a shift in emphasis from sector-specific to general skills. This change, which started in the mid-1990s and was completed with the new legislation enacted in 2002, reflected the growing evidence that in a fast-changing economy immigrants' adaptability,

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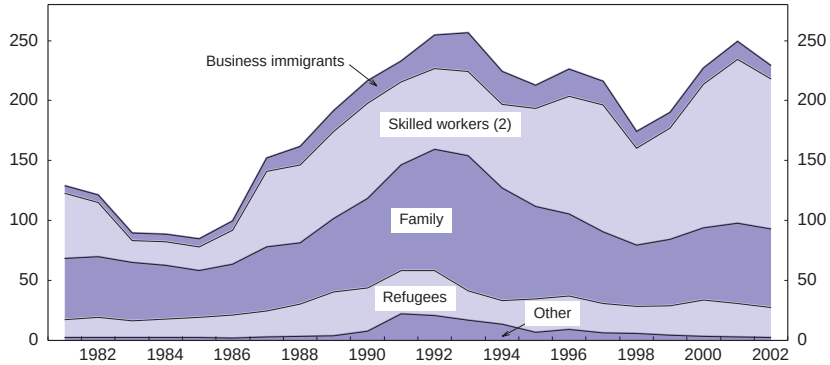
Box 2. **The evolution of Canadian immigrant selection policies** (*cont.*)

rather than possession of skills in immediate short supply, is the key to their successful integration. Taken together, these changes indicated a move from short-term or sector-specific labour market considerations toward longer-term growth and demographic objectives.

Figure 23. **Permanent immigrants to Canada by entry class**[1]

Annual flows, thousand persons



1. Data for all categories include both principal applicants and dependants.
2. Including provincial nominees.

Source: Citizenship and Immigration Canada.

1. For example, in 1982-86 the government limited the admission of economic immigrants to those with pre-arranged employment.
2. Occupation-specific points were assigned according to a list of occupations prepared by the labour department based on the Canadian Classification and Dictionary of Occupations (CCDO) at the seven-digit level of detail.

which need not correspond to those actually taken up in Canada – have also shifted toward those requiring greater skills (Figure 26), an evolution that has broadly matched the shifts in the composition of jobs created by the Canadian economy. On the other hand, their average *language skills* have not been improving (Figure 27), a fact connected in large part to changes in their source-country composition. As expected, the proportion of immigrants who can speak neither

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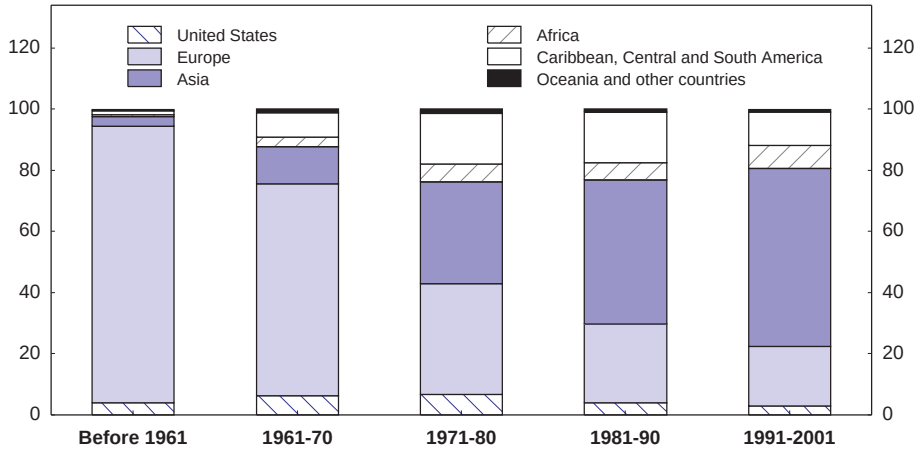
Figure 24. **Source countries of the foreign-born population in 2001 by period of immigration**

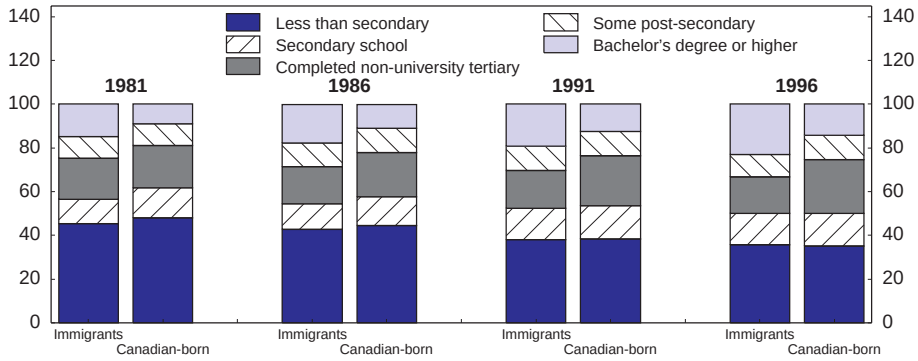
Per cent of total



Source: Statistics Canada.

Figure 25. **Educational attainment of recent immigrants and of the Canadian-born**

Per cent composition[1]



1. Recent immigrants are defined as those who arrived in Canada in the five-year period prior to each census. The comparison is age-adjusted: the shares for recent immigrant population are recalculated using the same age structure as the Canadian population.

Source: Zhao *et al.* (2000).

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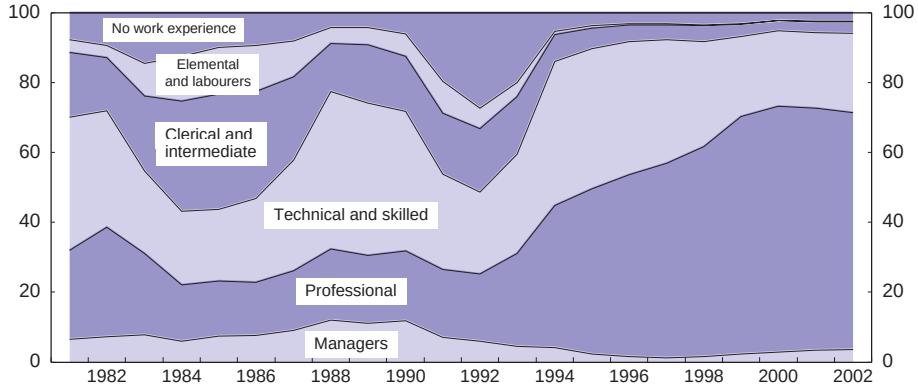
Figure 26. **Immigrants' occupations declared at arrival, by skill level**

Skilled workers, principal applicants, percentage of total[1]

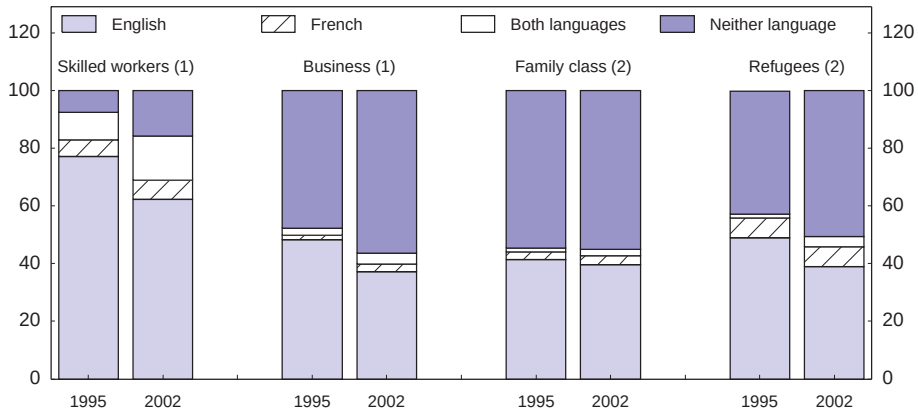


1. Percentage shares are calculated out of the total of adult immigrants (over 15 years old) who declared an intention to work. Occupations are classified by skill level, based on the National Occupation Classification (NOC).

Source: Citizenship and Immigration Canada.

Figure 27. **Language skills of immigrants upon arrival by entry class**

Per cent composition



1. Principal applicants.
2. Principal applicants and dependants.

Source: Citizenship and Immigration Canada.

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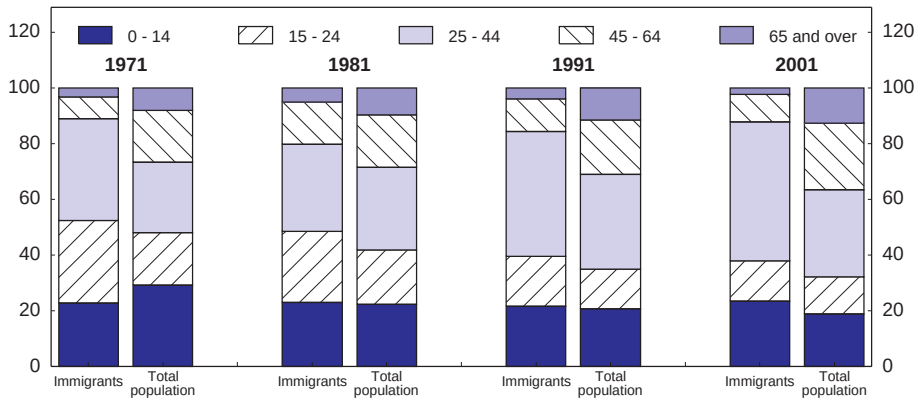
English nor French is higher among refugees and family-class immigrants, but it is also surprisingly high among business immigrants. Language skills presumably improve once the immigrants have settled in Canada, but there is little information on how quickly fluency is achieved, since available statistics measure only self-reported language abilities. Finally, immigrants tend to be younger, on average, than the Canadian population (Figure 28), and although for both groups the average age has been rising over time, the difference has increased.

Figure 28. **New immigrants and Canadian population: composition by age group**

Per cent[1]



1. The immigrants considered are those who immigrated in the year indicated. Data on the total population are based on the Census held the same year.

Source: Statistics Canada; Citizenship and Immigration Canada.

Changes in immigration policy were not alone in influencing the composition of immigrant flows over time, as they interacted with a host of other factors, both on the demand side (*e.g.* rewards for skills in the Canadian labour market) and on the supply side (*e.g.* demographic, political and economic changes in source countries). These factors also need to be taken into account, particularly when comparing the effects of immigration policies in different countries (see Box 3).

The main determinants of immigrants' economic outcomes

The skills immigrants possess when they arrive interact with conditions in the local labour market to determine their economic outcomes. For immigrants, as

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Box 3. Immigrant self-selection and cross-country comparisons

To the extent that immigration responds to economic incentives, wage differentials – corrected for travel and adjustment costs – can be expected to be the main determinant of the direction and size of immigration flows between two countries. However, economic theory has also tried to analyse the determinants of the composition of immigration flows. As Borjas (1987 and 1994) has shown, immigrants self-select, and if the destination country has higher returns to skills than the source country – that is, for example, if more educated workers receive a comparatively larger wage premium – the more skilled will have a stronger incentive to move, and this will be reflected in a greater share of them in the overall flow. The opposite will happen if the destination country has comparatively lower returns to skills. This theoretical framework has also been used to analyse return migration (former immigrants who go back to their country of origin) and on-migration (those who move on to a third country after accumulating human capital and/or wealth), as well as the determinants of the composition of immigrant flows to alternative destination countries. In the latter case, however, self-selection mechanisms interact with the immigrant selection policies in place in each country. The latter, if based on skills, can be thought of as "truncating" the skill distribution out of which immigrants are allowed to self-select. Other factors (geographical proximity, historical links) can also have an effect.

For example, immigrants to Canada are, on average, more highly educated than those entering the United States and have better language skills. However, these differences are entirely accounted for by the much larger proportion of Latin American immigrants that the United States receives. If they are excluded, US immigrants have higher education (see Figure 29, Panel B). Based on these findings it may appear that the Canadian points system has raised the skill level of immigrants relative to that of their US counterparts mainly by affecting their source-country mix, rather than by selecting more skilled individuals from each source country. However, this interpretation, suggested by Borjas (1993), is incomplete, for several reasons. *First*, historical and geographical factors, more than selection policies, explain why the United States attracts large numbers of immigrants from Latin America, many of which, in any case, enter illegally. Thus, it cannot be assumed that in the absence of the points system similar flows would have reached Canada. *Second*, as stated above, immigrants (or applicants) self-select according to the rewards given to skills in the potential destination countries as compared with their source country. Canada has a lower return to skills than the United States, reflecting both smaller wage differentials in favour of workers with higher education and greater income redistribution by the state (Figure 29, Panel A). Therefore, it can be expected, *ceteris paribus*, to attract less skilled immigrants than the United States. Thus, it is not surprising to find that, from each source region, the average education level of immigrants to the United States was equal or higher than that of immigrants to Canada (Borjas, 1993). This does not necessarily mean that the Canadian points system was ineffective. For the categories to which it was applied, it presumably filtered out the lower part of the skill distribution, but probably did so out of a distribution of self-selected applicants from each source country that already had lower average skills than that of would-be immigrants to the United States.

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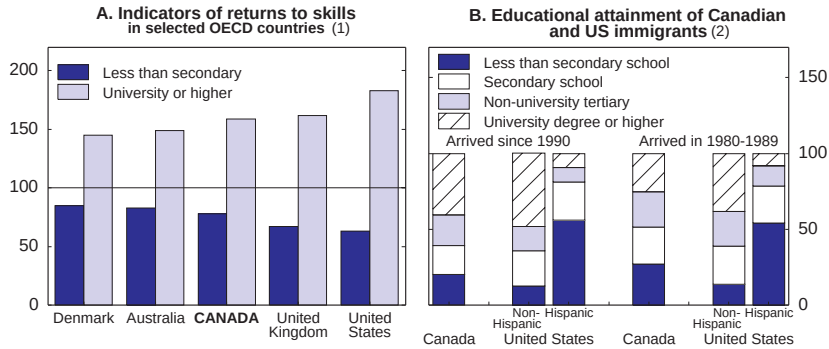
Box 3. **Immigrant self-selection and cross-country comparisons** (*cont.*)

Figure 29. **Returns to skills and skill level of immigrants**



1. Relative earnings of males aged 30-40 with income from employment, by level of educational attainment (upper secondary and post-secondary non-tertiary education = 100). Data refer to 1999 for Australia, Canada and Denmark; to 2001 for the United Kingdom and the United States.
2. Foreign-born persons of both sexes aged 25-64. Data refer to 2001 for Canada, to March 2002 for the United States.

Source: Statistics Canada; US Census Bureau; OECD (2002e).

for the Canadian-born, levels of education are positively correlated with labour market outcomes, but the returns to education – that is, the earnings advantage enjoyed by workers with more years of schooling or an academic degree – tend to be lower for foreign-educated immigrants than for the Canadian-born. This is in part because Canadian employers do not value foreign educational credentials, particularly from countries they know little about, as much as they do their domestic counterparts. For similar reasons, according to most empirical studies, foreign work experience appears to receive little or no reward in the Canadian labour market. Moreover, because migration represents a fracture in an individual's work career, recent immigrants tend to have considerably lower job tenure than Canadian-born workers, which also helps explain their initial earnings handicap (Zhang, 2002).[49] On the other hand, the return to Canadian work experience is higher for immigrants than for the Canadian-born, although it is difficult to separate this effect from the more general process of assimilation after arrival, which also involves other forms of human capital investment.

An important factor behind the lower valuation received by foreign qualifications is connected with inadequate language skills. Empirical studies indicate

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that, for equivalent schooling and experience, immigrants who are not fluent in either English or French have an earnings gap of around 10-12 per cent relative to those who are; the gap is higher for those who completed their education prior to immigration (Chiswick and Miller, 2002).[50] Moreover, language skills are complementary with other skills: immigrants who speak the local language are able to extract a much higher return from their years of education and from work experience in the country of origin. These gains are even larger for immigrants whose mother tongue is English or French or who normally speak it at home.[51] This represents an important qualification to the previously mentioned findings on the value of foreign education and work experience. An immigrant's age at the time of arrival in Canada also matters, because, like language, it interacts with most of the other factors mentioned above. Immigrants who arrive at a younger age accumulate most of their work experience in Canada and in some cases receive part of their education there, which facilitates their assimilation. They also face fewer difficulties in learning the local language and adopting the local culture. Indeed, it is found that the returns to education and experience for people who immigrated as children are equal to or even higher than those earned by the Canadian-born. Moreover, these immigrants seem to face no disadvantage for not having English or French as their mother tongue or for belonging to a visible minority (Schaafsma and Sweetman, 2001).

To the extent that it is not connected to the immigrant's language skills, the lower return to education and work experience acquired abroad could be the result of genuine imperfect transferability of these credentials to the Canadian labour market, of risk-averse behaviour on the part of employers given uncertainty as to their actual worth, or of outright discrimination. It is usually difficult to distinguish between these alternative explanations. Some empirical studies using direct measures of literacy and numeracy skills have found that the levels of such skills are lower for immigrants from some source regions than for Canadian-born workers with nominally equivalent educational qualifications, even after controlling for mother tongue, and that these differences account for at least part of the earnings gaps not explained by standard measures of education (Finnie and Meng, 2003). It is also reasonable to assume that certain professional skills are less transferable than others. In addition, immigrant characteristics connected to ethnicity, such as language skills and certain cultural differences, may condition the transferability of qualifications in ways that are often difficult to distinguish from discrimination. An interesting issue in this regard is whether, after controlling for education and other characteristics, Canadian-born visible minorities (the term used in Canada to denote non-white ethnic groups) have lower earnings than white Canadians. The evidence on this is mixed, but even studies that find an earnings gap indicate that it is not very large (except for Aboriginals, who are not immigrants) and that differences across the various visible minorities are very relevant.[52] Conversely, immigrants from Western Europe or other English-speaking countries perform as well as the Canadian-born or better.

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Thus, if there is a handicap due either to discrimination or to cultural obstacles to integration, it seems to affect mainly immigrants who are also of visible minorities, rather than immigrants or visible minorities *per se*.

Immigrants' economic performance deteriorated until recently

Immigrants' integration in the host-country labour market, both in Canada and in other immigrant-receiving countries, has traditionally followed a well-known pattern. Initially the newly arrived immigrants face a disadvantage *vis-à-vis* Canadian-born workers of comparable age and education, due to their lack of certain country-specific skills (in particular, language skills), local work experience and labour market knowledge. This disadvantage typically shows up in higher rates of unemployment, lower earnings and greater receipt of public transfers. Over time, however, as they learn the local language, acquire local work experience and upgrade their skills, immigrants gradually catch up to the natives' employment rates and earnings levels and even surpass them if they possess superior skills and/or greater motivation to succeed. Thus, the negative "entry effect" is gradually offset by a positive "assimilation effect". Indicators of economic performance for immigrants to Canada have broadly conformed to this pattern.

In the 1980s and at least until the mid-1990s, however, the performance of immigrants in the Canadian labour market showed a progressive deterioration. From the point of view of immigration policy, the fact that this negative trend appeared to parallel what was observed in the United States was seen as a source of concern, since it seemed to put in question the effectiveness of Canada's skill-based selection policy. The deterioration was reflected in lower participation rates and higher unemployment rates (Figure 30), as well as lower earnings profiles; and it was observed for all entry categories (Figure 31, Panel A).[53] The groups that in the past had performed better were those who lost more ground in relative terms: male workers, principal applicants in the economic immigrant categories (who are individually selected according to the points system), workers with a university degree and those with better language skills at arrival. For example, the employment earnings of economic principal applicants entering in 1980 had been about 25 per cent above the Canadian average (not adjusted for age and education) after one year in Canada and 35 per cent above after five years.[54] Those of immigrants entering in 1990 were almost 20 per cent lower than the Canadian average after one year and only slightly above them after five. Immigrants arriving in the early 1990s not only had lower starting earnings and employment rates, but also appeared to catch up more slowly than the previous cohorts. Also immigrants who had arrived earlier experienced a slower earnings catch-up during that period. The patterns shown in the charts are not adjusted for age structure and education levels, which differ not only between immigrants and the average Canadian population, but also across immigrant cohorts. However, econometric studies controlling

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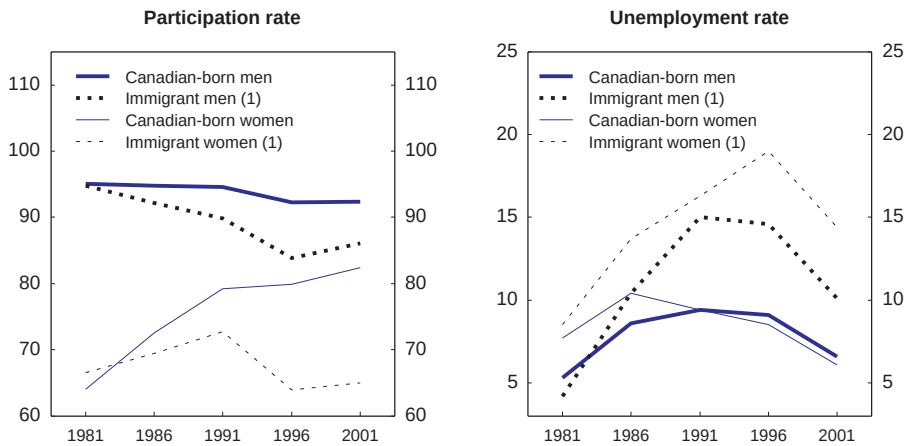
Figure 30. **Participation and unemployment rates of recent immigrants and of the Canadian-born**

Per cent; age group 25-44



1. Immigrants landed within the previous five years.
Source: Statistics Canada, various Censuses.

for age and education (Baker and Benjamin, 1994; Bloom *et al.*, 1995) also found clear evidence of a deterioration over time across immigrant cohorts with respect to both earnings and unemployment probabilities. The relative decline seems to have been particularly pronounced during recessions. Moreover, the pace of earnings assimilation relative to Canadian-born workers of comparable age and education was found to have become considerably slower over the 1980s and early 1990s, not only for the most recent arrivals but also for earlier cohorts.

Since the mid-1990s, however, some signs have emerged that the negative trend has stopped and may have started to reverse, although evidence of this is still tentative. The latest immigrant cohorts have higher participation and lower unemployment rates and, although their starting earnings remain significantly below the Canadian average, they seem to be catching up more rapidly. Recent econometric studies controlling for immigrant characteristics (Grant, 1999; Hum and Simpson, 2002) seem to confirm this. However, the evidence as to the extent of the recovery is mixed. The data shown in Figure 31 probably overestimate it, since the comparison between skilled workers and the Canadian average does not control for the level of educational achievement, which has risen dramatically for that immigrant category over the last few years. Census data

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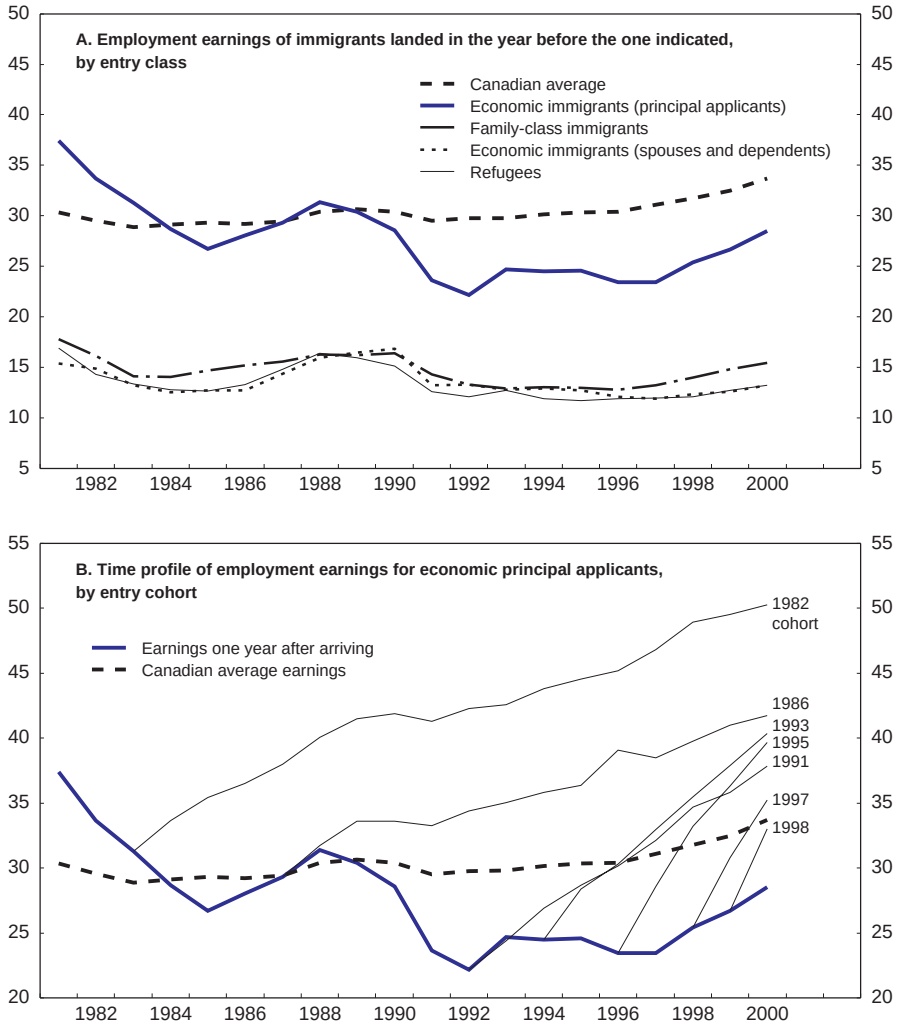
Figure 31. **Initial earnings and catch-up profiles of immigrants relative to the Canadian average**

Constant dollars, 2000 prices, thousands



Note: The data shown are elaborations based on the IMDB longitudinal database, which links immigrant entry records with tax records filed by immigrants in subsequent years. Average earnings for the Canadian population are calculated from the Canada Customs and Revenue Agency Tax Files. Earnings comparisons between immigrants and the population average, as well as between different immigrant cohorts, are not adjusted for education levels, age and other demographic characteristics.

Source: IMDB (Citizenship and Immigration Canada and Statistics Canada); Canada Customs and Revenue Agency.

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indicate that in 2000 the earnings gap between immigrant university graduates and their Canadian-born counterparts (over 50 per cent for men aged 25-54 arrived one year earlier, almost 30 per cent for those who had been in Canada for 10 years) was still larger than in 1990 (Statistics Canada, 2003a). And low-income rates for immigrant households continued to increase during the 1990s, while they were falling for non-immigrants (Picot and Hou, 2003).

From a policy standpoint it remains very important to correctly assess the evolution of immigrant economic success and to understand to what extent its underlying determinants are connected to immigrant characteristics, to the functioning of the Canadian labour market, or to both. A substantial body of economic literature, in Canada and elsewhere, has been devoted to these issues. The stylised facts described above have led researchers to test two main explanations, not necessarily mutually exclusive. The first one is that there has been a deterioration in the "quality" of immigrants, whether reflected in their observable or unobservable characteristics. The second is that changes in the Canadian labour market – as a result of either macroeconomic conditions or structural factors such as technological progress – have had a particularly adverse effect on immigrants, possibly in connection with some characteristics that made them more vulnerable than Canadian-born workers.

The fact that recent immigrants get a lower return from their education than the Canadian-born in terms of earnings, and that the difference increased between 1981 and 1996, especially for more recent immigrants (Reitz, 2001), could be consistent with the first explanation. There is evidence that immigrants from regions other than Northern Europe and the United States get less skilled jobs than Canadian-born workers with similar educational levels, and that this under-utilisation of their measured skills became more severe for the cohorts arriving in the 1980s and early 1990s (Thompson, 2000). Those immigrants also experienced lower participation rates during that period, a phenomenon not observed among male immigrants from the more traditional source countries. Of course, what appears as skill under-utilisation could be due to a decline in the average relative "quality" of immigrant education, not adequately measured by years of schooling or academic qualifications, or to changes in some other immigrant characteristics complementary to education, such as language skills. Indeed, immigrant language skills have declined over the past two decades, mainly as a result of the shift in the source-country mix, but also, in the 1980s, due to the expansion of the family class at the expense of the skill-assessed economic immigrant classes. At the same time, structural changes in the composition of labour demand (for example, a greater emphasis on communication skills in a knowledge-based economy) may have reduced the value of the immigrants' typical mix of skills in the eyes of employers. The fact that the return to work experience acquired before immigration has also declined (Green and Worswick, 2002) lends some indirect support to this hypothesis.

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However, there is also substantial evidence that the deterioration of immigrants' relative position is connected to the weak conditions that prevailed in the Canadian labour market during the 1980s and the first half of the 1990s. In this period Canada experienced two protracted recessions, a low average rate of job creation and rising unemployment. The position of the more vulnerable categories in the labour market, among which recent immigrants, weakened. This was the case also for highly skilled immigrants, partly as a result of the fact that during the 1980s and 1990s growing numbers of Canadian-born university graduates were entering labour market reflecting rising educational levels and female participation rates. Some econometric studies show that, when the aggregate unemployment rate is higher, the probability of unemployment rises more for immigrants than for natives, and their relative earnings position also deteriorates (McDonald and Worswick, 1997 and 2000; Aydemir, 2002). This differential effect is probably due to employers' reluctance to lay off more senior workers, in whom they have invested greater firm-specific human capital, and possibly to the fact that discriminatory or risk-averse behaviour in hiring is less costly in a recession than in a tight labour market. Indeed, similar patterns have been found for young workers and for Canadian-born visible minorities. The above mentioned studies also show that the rate of immigrants' earnings assimilation – that is, the speed at which they catch up – is negatively correlated to macroeconomic conditions.[55] In the early 1990s, two additional factors probably helped to weaken the labour market position of recent immigrants: *first*, the number admitted continued to be rise substantially throughout the recession, in contrast with previous practice; *second*, a large proportion of them continued to settle in Toronto, where the downturn was especially pronounced.

In sum, both structural changes and economic conditions seem to be part of the story. The finding that immigrant performance started to turn around in the second half of the 1990s, when macroeconomic conditions improved would seem to be consistent with the latter explanation. However, the recovery in relative earnings has been at best partial, which suggests that structural changes, probably interacting with the evolving mix of immigrant characteristics, have been responsible for an important part of the observed deterioration.

Implications for selection policy

The analysis of immigrant outcomes over the past 20 years suggests a number of conclusions for immigration policies. The evidence on the main factors that influence immigrant assimilation supports the direction of the recent changes in selection policies, which emphasised general skills and adaptability and raised the standards for the assessment of language skills. Some issues, however, deserve further consideration. *First*, the selection policy could do more to target younger immigrants, who tend to be the most adaptable. At present, applicants in

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the 21-49 age range get maximum points. Moreover, as an indirect effect of the points assigned for higher education and work experience, applicants between 30 and 49 probably have an advantage within that range. By comparison, Australia takes a much more aggressive approach, assigning top points to the 18-29 age range and setting an age ceiling at 45. *Second*, although language assessment has been tightened, as long as language remains just one factor in the point system it is possible to offset a low language score with high points for other factors. Other countries (*e.g.* New Zealand) make a minimum standard of language knowledge a condition for admission. Given the growing importance of language skills, both directly and through their effect on the transferability of other skills, such an alternative approach could be considered. *Third*, research seems to show that foreign work experience has limited value. Although these results should be taken with caution – especially given the qualification that, when combined with good language skills, foreign experience does matter – assigning the same number of maximum points for work experience as for education and making one year of work experience a necessary condition for admission as skilled worker may be excessive. And, *fourth*, Canadian immigration policy could make greater use of temporary work permits, not only to address skills shortages in specific professions, but also as a way of "feeding" the permanent immigration flow with workers who have already gained Canadian work experience and adjusted to the local way of life. The same argument applies, to some extent, to foreign students.

Immigrant integration policies

The evidence of the difficulties encountered by recent immigrants in the labour market have also led the government to focus greater attention on its policies and programmes designed to support their integration. Foremost among the issues that have attracted attention is that of immigrant credentials recognition. But the range of relevant policies is much wider, including not only integration programmes and entitlement to public services – which are discussed in this section – but also the question of the geographical distribution of immigrants within Canada, which is addressed in the next. More generally, it is increasingly recognised that the host country's institutional context – in particular, its labour market institutions and educational system – is at least as important as immigrant selection policies in determining immigrants' economic outcomes (Reitz, 1998).

Immigrants have full access to health care and social benefits

Immigrants with the status of permanent resident have a right to equal access to employment, except in the public sector. They also have the same access to public education, health care and social benefits (including employment insurance) as Canadian citizens, as well as to all other public services. They pay taxes like other Canadians and are eligible for the same tax benefits. Eligibility for

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most services and benefits applies from when the permanent immigrant arrives. However, in the case of health care four provinces (including British Columbia, Ontario and Quebec, which receive the largest numbers of immigrants) impose a three-month waiting period. To cover this period, resettled refugees receive essential and emergency health services through the Interim Federal Health Program, while other immigrants are asked to pay for private insurance. Refugee claimants receive coverage through the Interim Federal Health Program, but once they are recognised as Convention refugees or persons in need of protection, they have access to provincial health systems.

To be eligible for an old-age pension after age 65 an immigrant needs to have resided in Canada for at least ten years. For those who qualify, the Old Age Security (OAS) component of the pension is proportional to the number of years of residence in Canada, but the Guaranteed Income Supplement (GIS) is not, and tends to offset the shortfall for those who have been in Canada for less than 40 years. Immigrant workers – like all Canadians – also contribute to the Canada or the Quebec Pension Plan, whose benefits are based on years of contribution, with no minimum number of years. Canada has agreements with more than 30 countries to allow residents who worked and paid contributions abroad to claim pension benefits.

As with other countries that encourage permanent immigration, Canada offers relatively easy access to citizenship. Permanent residents can become citizens after they have been in Canada for three years. A large majority of them do so: at the time of the 2001 census, 84 per cent of all the immigrants who had arrived before 1998 and were still in the country had obtained citizenship. And most of them tend to make use of this possibility quite soon after they become eligible: 79 per cent of those who arrived in 1991-95 and 57 per cent of those who arrived in 1997-98 were already Canadian citizens by 2001.

The federal and provincial governments run several immigrant settlement programmes

A number of government programmes are in place to facilitate the integration of immigrants. The Immigrant Settlement and Adaptation Program (ISAP) aims to provide direct, essential services (such as reception, orientation, translation and interpretation, counselling and employment-related services) for newly arrived immigrants. The Host Program helps immigrants make contact with their local communities. And the Language Instruction for Newcomers to Canada (LINC) provides basic language training. These programmes are managed at the federal level by Citizenship and Immigration Canada (CIC), except in those provinces whose agreements on migration with the federal government also cover immigrant settlement. But the actual delivery of services is, in most cases, carried out through provincial and local administrations, private businesses, non-profit organisations, educational institutions and community groups. In addition to funding,

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CIC provides the entities involved with orientation, benchmarks (for example, for language teaching) and support facilities.

Specific support programmes exist for refugees whose status has already been recognised. Government-assisted refugees receive income support from the federal government for up to one year, and are eligible for essential initial services such as temporary accommodation, orientation and information. In the case of privately sponsored refugees, the sponsors undertake to provide them with financial assistance and guidance for the first 12 months. Both categories are eligible for emergency medical assistance and basic orientation, including referrals to settlement and other general services. On the other hand, refugee claimants whose status has not been finally determined are not eligible for federal integration and settlement services, but have access to medical coverage (which is federal) and to provincial welfare assistance, and can obtain a temporary work permit to support themselves.

The recognition of foreign credentials remains a problem

A key issue emerging from the analysis of immigrants' performance in the labour market is the difficulty they find in obtaining recognition of their foreign educational credentials and work experience. For example, an immigrant wanting to be admitted to a profession would have to obtain recognition of his/her professional education, of any applied training or work experience and of any certification exam taken. If any of these qualifications is not recognised by the competent licensing body, it will need to be repeated in Canada. Since the cost can be prohibitive, many immigrants end up accepting low-skill jobs and never using the specific human capital obtained in their home countries. This seems to be confirmed by Census data on the occupations held by highly-skilled immigrants, especially women (Figure 32). There is anecdotal evidence that these barriers are greater in Canada than in the United States, partly because more professions and trades are regulated, but also as a result of a more conservative attitude of Canadian employers *vis-à-vis* foreign work experience. To the extent that this problem reflects restrictive practices by professional organisations, discrimination or lack of information on the part of employers – rather than genuine differences in the quality or the relevance of foreign qualifications – it involves welfare losses both for the immigrants themselves and for the Canadian economy.

This issue is particularly complex because in Canada professions and trades are regulated at the provincial level (through legislation in the case of trades, by delegation of authority to self-regulating bodies in the case of professions). The obstacles to labour mobility deriving from the multiplicity of jurisdictions affect the Canadian-born as well as immigrants but can be more obstructive for the latter due to their lack of familiarity with the system. Some progress in

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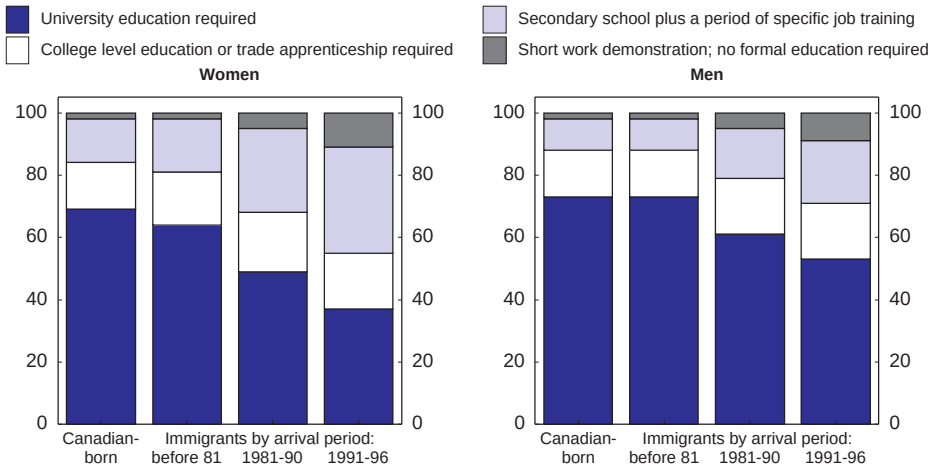
Figure 32. **Skill level of jobs held by immigrant and Canadian-born university graduates aged 25 to 64**

1996, percentage distribution



Source: Statistics Canada, 1996 Census.

eliminating them has been made through the implementation off the Labour Mobility Chapter of the 1994 Agreement on Internal Trade (see Chapter II).[56]

Both the federal and several provincial governments seem to have recognised the importance of addressing this issue, particularly for a country targeting highly skilled immigrants. The federal government has created the Canadian Information Centre for International Credentials (CICIC), which functions as a sort of information clearing house and referral service to support the recognition and portability of Canadian and international educational and occupational qualifications,[57] and Quebec, British Columbia, Alberta and Ontario have set up or supported the establishment of specific credential evaluation services. Visa officers now encourage immigrants to contact such services before they arrive in Canada, and federal and provincial agencies provide accessible information on qualification standards in the various professions. Moreover, a number of professional organisations have taken steps to make the credentials recognition process more transparent.[58] Recognition of academic and other "paper" credentials may not be sufficient, however. Employers stress the importance of on-the-job experience, as confirmed by a recent survey (Sangster, 2001) and tend to discount foreign experience, particularly when obtained in countries very different from Canada.

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Efforts are also being made to facilitate the re-training of immigrants whose academic qualifications are inadequate for access to professions and trades in Canada.[59] A specific aspect of re-training concerns specialised language training, a need not met by most language courses on offer (Brower, 1999). When they arrive, many skilled immigrants need not only to upgrade their general language proficiency but also to acquire profession-specific language skills for complete skills recognition. A promising initiative to address this issue has recently been undertaken in Ontario.[60]

The policies being followed seem to be going in the right direction. A recent survey (Ontario Ministry of Training, Colleges and Universities, 2002) showed that, although highly skilled immigrants in general continue to encounter significant difficulties in entering professions in Canada, those who received advance information on licensing requirements, have better profession-specific language skills and have used credentials assessment services are more likely to be working in their chosen professions. Specialised agencies are probably the most efficient way to handle credentials evaluation, but the validity of their assessments cannot be imposed on individual educational institutions, professional bodies and employers, and will ultimately depend on their reputations. Nevertheless, public authorities should continue to urge the self-regulating bodies to make their licensing practices more transparent and to make sure they are not used in a discriminatory way.

The educational system helps immigrants integrate

The educational system seems to do a relatively good job of fostering the integration of immigrants, both with regard to those who arrive as children and to Canadian-born children of immigrant parents. Moreover, several studies have found that second-generation immigrants of all ethnic groups tend to exceed the levels of educational achievement of other Canadians, and do not display the persistent segmentation along ethnic lines found in the United States (Boyd, 2002; Guppy and Davies, 1998). Children of immigrant parents have an initial disadvantage in terms of reading, writing and mathematical skills relative to those of Canadian-born parents but fully overcome this gap by the end of elementary school (Worswick, 2001). This favourable picture seems to be broadly confirmed by the results of the OECD PISA comparative study on the performance of 15 year-olds (OECD, 2001b) (Figure 33). That study indicated that foreign-born students do experience a disadvantage relative to their Canadian-born counterparts, but the gap is much smaller in Canada than in most other OECD countries. Indeed, foreign-born students in Canada outperformed the average student with non-immigrant parents in the OECD in both reading and mathematical skills. Immigrants also seem to make ample use of the opportunities offered by the educational system for upgrading their human capital.

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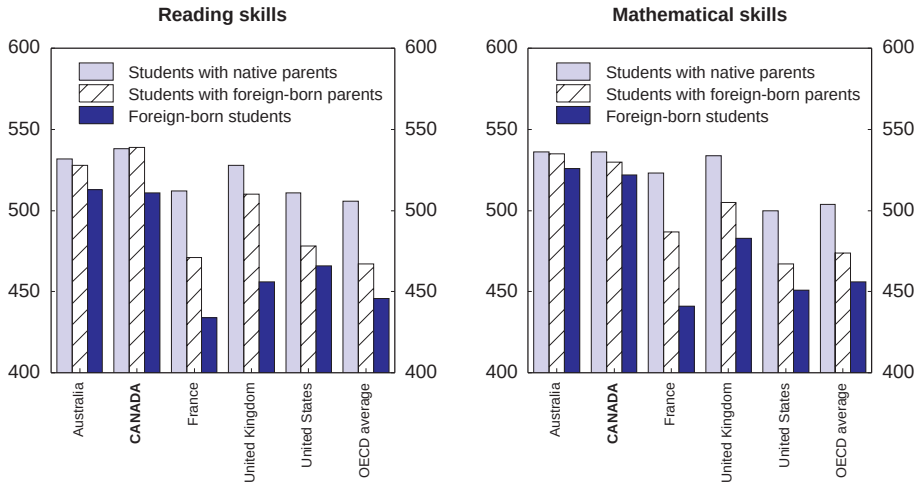
Figure 33. **Relative performance of 15 year-old immigrant students**

Mean scores



Source: OECD, *Knowledge and Skills for Life: First Results*, from PISA (2001).

According to 2001 Census data, a higher proportion of immigrants aged 20 to 34 who arrived in the last 10 years attend school, as compared to other residents of the same age, both among men and women.

The geographical distribution of immigrants

Immigrants have become increasingly concentrated in large cities

Immigrants are highly concentrated in cities and have become more so over the past 20 years (Table 23). Nearly three-quarters of new immigrants are now settling in Toronto, Montreal and Vancouver, which together have about one-third of Canada's population. A predominantly urban and selective settlement pattern is not unique to Canada; it is found to a similar degree in the United States. Geographic concentration reflects not only initial settlement decisions, but also subsequent migration patterns: immigrants who move between provinces tend to go predominantly to the two that already have the highest immigrant concentrations (Ontario and British Columbia). Some of the other provinces "lose" a high proportion of the immigrants who initially settled there. For example, it is estimated that by 1995 the Atlantic provinces had seen a net outflow of 35 per cent of immigrants

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Table 23. **Geographic distribution of immigrants and of visible minorities**

	Immigrants arrived since 1991	Population	Foreign-born			Visible minorities (Foreign-born and native)		
	Share of total		As a proportion of population					
	2001	2001	1981	1991	2001	1981	1991	2001
Canada	100.0	100.0	16.0	16.1	18.4	4.7	9.4	13.4
Newfoundland and Labrador	0.1	1.7	1.9	1.5	1.6	0.5	0.8	0.8
Prince Edward Island	0.0	0.5	3.7	3.2	3.1	0.6	1.0	0.9
Nova Scotia	0.6	3.0	5.0	4.4	4.6	1.7	3.4	3.8
New Brunswick	0.2	2.4	3.9	3.3	3.1	0.6	1.2	1.3
Quebec	13.4	24.0	8.2	8.7	9.9	2.6	5.6	7.0
Ontario	55.8	38.1	23.6	23.7	26.8	6.4	13.0	19.1
Manitoba	1.8	3.7	14.2	12.8	12.1	3.9	6.9	7.9
Saskatchewan	0.6	3.2	8.7	5.9	5.0	1.8	2.6	2.9
Alberta	7.1	9.9	16.3	15.1	14.9	5.3	9.4	11.2
British Columbia	20.2	13.1	23.1	22.3	26.1	8.4	14.2	21.6
Yukon Territory	0.0	0.1	12.4	10.7	10.6	2.1	2.7	3.6
Northwest Territories	0.0	0.1	6.0	6.6	6.4	1.5	3.5	4.2
Nunavut	0.0	0.1	. .	1.9	1.7	. .	0.9	0.8
Ten largest CMAs[1]	89.0	50.4	. .	25.3	29.2	. .	. .	23.9
Montreal	11.8	11.4	. .	16.4	18.4	5.2	11.0	13.6
Toronto	43.3	15.7	. .	38.0	43.7	13.6	25.8	36.8
Vancouver	17.7	6.5	. .	30.1	37.5	13.9	24.0	36.9

1. Census metropolitan areas. The ten largest are: Toronto, Montreal, Vancouver, Ottawa-Hull, Calgary, Edmonton, Windsor, Winnipeg, Hamilton and Kitchener.

Source: Statistics Canada.

who had arrived since 1980 and Saskatchewan of 45 per cent, with much higher proportions (50 to 65 per cent) for business immigrants and refugees (Citizenship and Immigration Canada, 2000). Concentration in certain cities reflects in part job-creation trends but is also driven by the tendency of newcomers to settle where immigrant communities from the same source countries already exist. There is evidence that in the case of Toronto and Vancouver, which receive the largest inflows of immigrants relative to their population, the size of these inflows is larger than what could be expected given local labour market conditions, and is also out of line with internal migration flows (Figure 34). Their magnitude seems to be better explained by the fact that they already have large concentrations of immigrants. This behaviour reflects the economic support ethnic communities can provide (such as information about jobs and business opportunities), as well as a desire to preserve ethnic ties and culture. In the case of family class immigrants, this inertia is virtually built into the system. While the support provided by ethnic networks probably facilitates new immigrants' initial integration, it is an open question whether in the long run the very "sheltering" they offer may slow down the acquisition of crucial country-specific human capital, in particular language skills.

A high degree of geographic concentration of immigrants can have other undesirable effects. If it is a result of inertia, it will negatively affect the employment prospects of both immigrant and Canadian-born workers in some regions while skill shortages persist in others.[61] It can also add to congestion in cities and put under strain the capacity of local administrations to provide immigrant integration, health and education services. The Toronto, Montreal and Vancouver metropolitan areas all had net outflows of Canadian-born inhabitants between 1996 and 2001.[62] This is not sufficient evidence of a displacement effect, but, together with the declining public acceptance of immigration in those cities, it suggests that some of these negative effects may already be present.

Provinces are making considerable efforts to achieve a more balanced distribution

However, the policy tools available for steering immigrants toward a more balanced settlement pattern are quite limited. Under the Canadian Charter of Rights and Freedoms permanent residents have full mobility rights. Even though the government can influence the initial assignment of some immigrant categories (such as government-sponsored refugees), a policy of greater dispersion will have little effect unless provinces and local communities increase their ability to retain them. This involves both immigrant selection and integration policies. In Canada, immigration policy is a shared responsibility between the federal government and the provinces. The federal government sets the general legislative framework, defines the immigrant categories and is responsible for health, safety and security matters. Within this framework the provinces have some degree of autonomy as regards the selection of certain categories of immigrants and in

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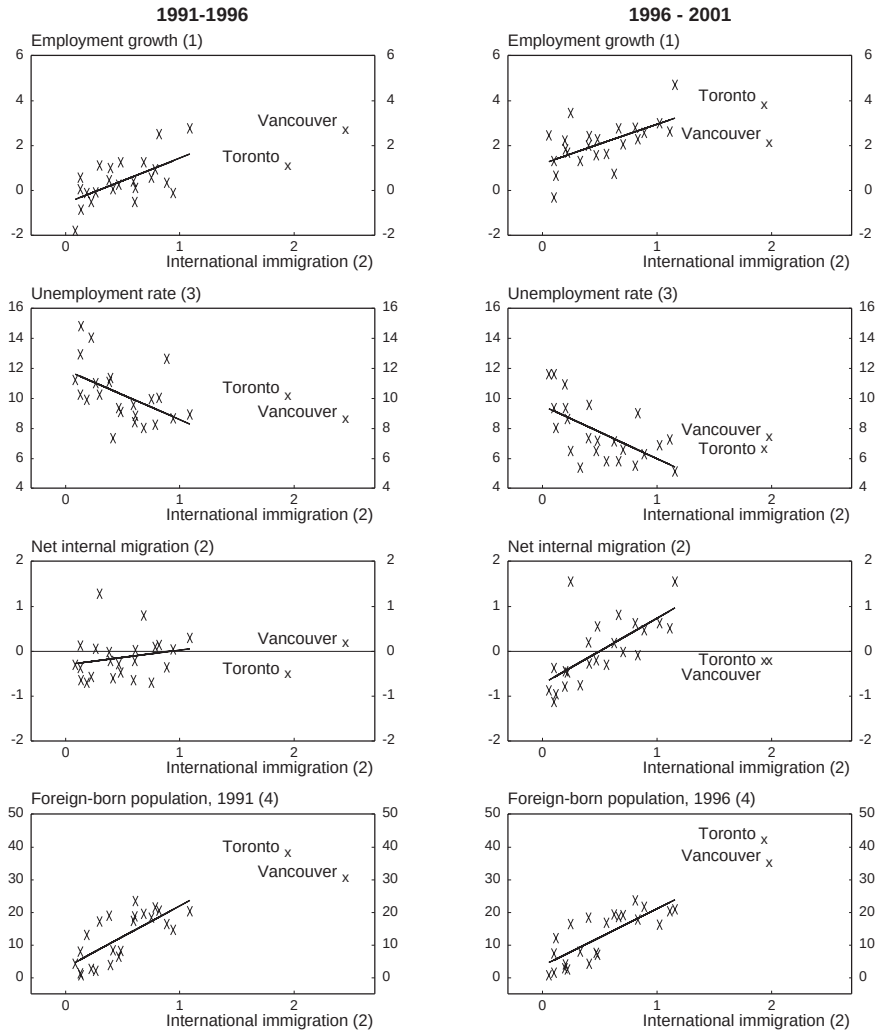
Figure 34. **Immigration rates for Toronto, Vancouver and other CMAs**

Correlation with employment growth, unemployment rate, internal migration and share of foreign-born population



Note: Data refer to 25 Census metropolitan areas. The lines indicating the correlation of the two variables shown in each chart are drawn based on 23 CMAs, excluding Toronto and Vancouver.

1. Annual percentage growth.
2. Average annual inflow as a percentage of population.
3. Annual average.
4. Stock in the indicated year.

Source: Statistics Canada.

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Box 4. Decentralisation of responsibilities for immigrant selection and integration: the case of Quebec

Quebec was the first among the Canadian provinces to negotiate an agreement on immigration with the federal government (in 1978). It is also the province with the broadest responsibilities in this domain. The present agreement, signed in 1991, gives the provincial government extensive powers in the selection of immigrants as well as responsibility for running immigrant integration programmes.

The Quebec government uses its own points system to select skilled workers and also has its own investor programme (with more attractive conditions than its federal counterpart). It is allowed to select the refugees to be settled in Quebec among those who apply from abroad, once the federal government has determined their refugee status. And it has its own immigration offices in source countries to process applications. The federal government sets the admission rules for family class immigrants and has a joint decision on temporary immigrants (workers and students). Overall, the provincial government is able to select over 60 per cent of the immigrants who settle in the province. It uses its selection powers to choose the immigrants considered most suitable for settlement in the province, emphasising in particular French language skills. This has influenced the source-country composition of immigrants, with higher shares from Africa, the Middle East and Latin America. The Quebec government has indicated the objective of raising the share of immigrants who speak French at arrival to 50 per cent in 2003, from about 45 per cent in 2000-01. Partly as a result of this focus on French-speaking immigrants, but also as a result of a reduction in Quebec's target level of immigration, its share of total Canadian immigrants has declined to 15 per cent, from 20 per cent in the 1980s. The retention rate (the proportion of immigrants who stay in the province after initial settlement) has continued to decline, but less than in other provinces except Ontario and British Columbia. Retaining immigrants has traditionally been a problem for Quebec, which also faces negative interprovincial migration of native-born Canadians. And retention rates are generally lower for refugees, who in Quebec represent a larger proportion of immigrants (about 20 per cent, against a national share of 11 per cent). Specific integration programmes are aimed at them, including an effort to encourage their settlement in communities outside the Montreal metropolitan area. Nevertheless, 86 per cent of all immigrants still go to Montreal. For the selection of skilled workers the Quebec points system, like the present federal version, emphasises education and adaptability, having already shifted from occupation-specific to general skills in 1996. While there are encouraging signs on immigrants' integration (Renaud *et al.*, 2001), an improvement in their economic performance is not yet visible: although the unemployment rate of immigrants to Montreal has declined, in 2001 both its level and the difference relative to that of the Canadian-born remained much higher than in other cities (Statistics Canada, 2003b).

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managing programmes to support their integration. The first province to take an active role in this area and the one that has developed it furthest is Quebec (see Box 4). Over the last ten years, most other provinces and territories have also signed agreements on migration with the federal government. Most of them have established "nominee programmes", which allow them to nominate a certain number of skilled immigrants. They are using these programmes to increase the number of immigrants they receive and in some cases to cover specific skill shortages.

Most provincial nominee programmes are of limited size so far. The largest is Manitoba's, which has been successfully used since 1997 to increase the number of skilled immigrants coming to the province and to settle some of them in rural areas. The selection of candidates based on the specific skills needed in the province is combined with assistance in finding employment, and emphasis is being put on supporting immigrant integration, in co-operation with local communities. The federal Minister of Citizenship and Immigration has recently suggested tying work permits for skilled temporary immigrants to a commitment to work for three to five years outside the main cities, after which they would be eligible for permanent status. The success of these efforts is likely to be greater with skilled immigrants, for whom it is easier to line up jobs, than with refugees. Retaining immigrants will remain a greater challenge in smaller communities as compared to medium-sized cities, which offer a greater variety of opportunities and where ethnic clusters can more easily reach a critical size.

Emigration from Canada

Canada is also a source country of migration. Both emigration and immigration flows were extremely high as a percentage of Canadian population in the second half of the nineteenth century, at a time when Canada and large sections of the United States were still being settled and labour mobility was therefore extremely high (see Figure 21 above). Because of the stronger attraction of the United States, net flows were negative for Canada for most of that period. Since the beginning of the twentieth century, with only the exception of the 1930s, emigration has been significantly smaller than total immigration, on average by a ratio of one to six in the post-war period. Moreover, emigration rates showed until recently an almost continuous downward trend (Figure 35). But the outflow started to pick up again in the early 1990s.

Historically, a large fraction of Canadian emigration has been directed toward the United States: on average, about one-third for temporary emigrants and one-half for permanent ones since 1980. Emigration to the United States has often received particular attention from Canadian policymakers, in part because of a periodic concern about a "brain drain". This issue was widely debated in the 1960s, when large outflows coincided with a sizeable lag in the development

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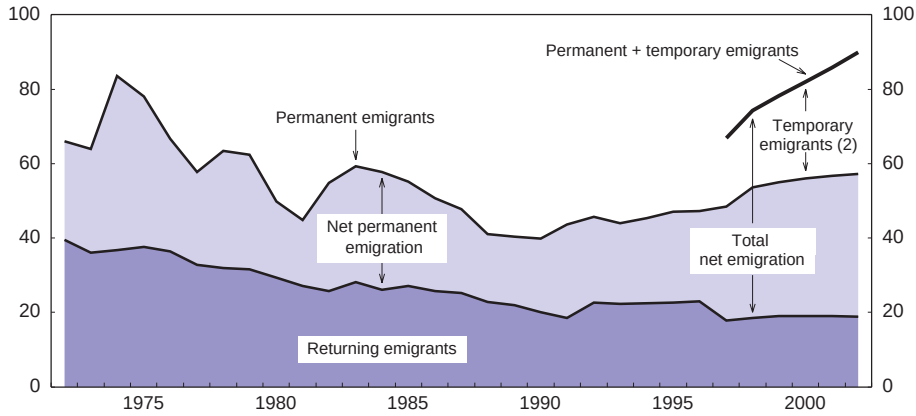
Figure 35. **Emigration from Canada, 1972-2002**
Annual data, thousand persons[1]



1. Periods from 1 July (of previous year) to 30 June.
2. Estimates of temporary emigrants are available only from 1997.
Source: Statistics Canada.

of Canadian academic and research institutions relative to their US counterparts. It has emerged again in recent years, since the US leadership in the ICT revolution has been accompanied by a rise in emigration of highly skilled workers to that country. Academic studies have also focused on emigration to the United States, partly because of data availability. Indeed, data quality is one of the main problems faced in studying Canadian emigration: as no direct record of emigrants is taken, emigration statistics are obtained combining multiple indirect sources and some unverifiable assumptions.

The composition of emigration to the United States by age and educational achievement has often been cited as an indication that Canada is suffering from a brain drain. Canadian tax filers who moved to the United States are more likely to come from the higher income brackets (Figure 36). Moreover, emigrants tend to be better educated than the average population: about half of those over the age of 16 who emigrated to the United States in the 1990s had a university degree, against only 18 per cent for the corresponding age group in the Canadian population. A Survey of 1995 Canadian university graduates (Frank and Belair, 1999) found that 1.5 per cent of respondents were living in the United States in 1997, a figure not out of line with the overall proportion of Canadians living there,[63] but the percentage was considerably higher (12 per cent) for those with a PhD degree. And these figures do not include Canadians who complete their tertiary studies in the

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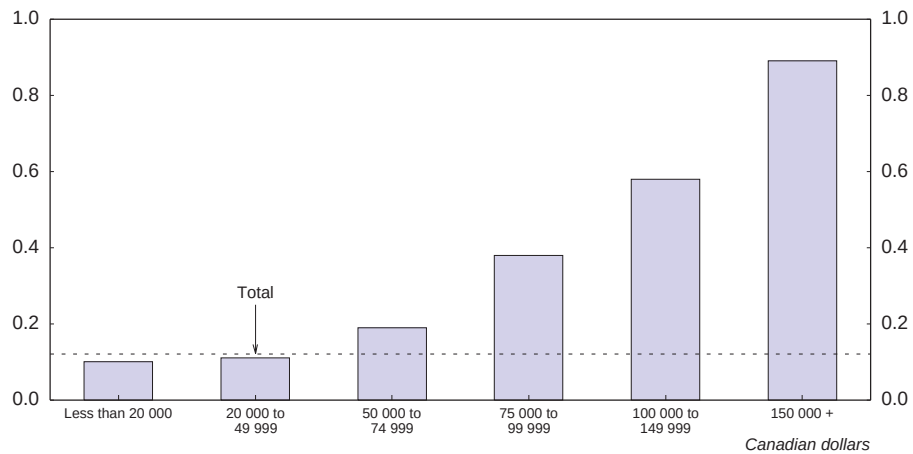
Figure 36. **Emigrants from Canada by income level**

Tax filers who ceased to reside in Canada in 1996, as a percentage of all 1995 tax filers by income group



Source: Zhao *et al.* (2000).

United States. The above-average education of emigrants is partly connected to the fact that they tend to be young (over half of them fall in the 25-44 age group, against 30 per cent for all Canadians). But it must be recalled that the same features are common to all migrants, including Canadian immigrants and inter-provincial migrants, since young, well-educated and higher-income individuals in general tend to be more mobile. The main reason why Canadians who migrate to the United States are mostly highly skilled, however, is probably self-selection, driven by the higher wage premium for more highly educated workers (as discussed in Box 3 above) and a less progressive tax system. Moreover, in the United States this wage premium increased substantially in the 1980s and the 1990s, while it remained virtually unchanged in Canada, where the increased demand for university-educated workers was offset by an increase in supply (Murphy *et al.*, 1998; Heisz *et al.*, 2002). Some sector-specific factors have also influenced the propensity to emigrate. In the 1990s workers moving to the United States tended to be concentrated in certain knowledge-intensive professions. For example, in 1996-97 physicians, nurses, natural scientists and engineers had higher-than-average emigration rates (Table 24). Indeed, for physicians and nurses, the number of permanent emigrants to the United States exceeded not only that of

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Table 24. **Emigration to the United States and total immigration, for selected professions**

Annual averages, in per cent of 1996 labour force by occupation

	Emigration to the United States[1]			Immigration (total)[1]		
	1986-89	1990-95	1996-97	1986-89	1990-95	1996-97
Physicians	0.25	0.45	0.78	0.69	0.70	0.51
Nurses	0.13	0.31	0.33	0.34	0.39	0.16
Teachers – post-secondary	0.13	0.17	0.13	0.38	0.49	0.35
Teachers – except post-secondary	0.06	0.07	0.06	0.25	0.32	0.23
Computer scientists	0.06	0.08	0.07	0.55	1.53	4.11
Engineers	0.27	0.31	0.27	1.01	2.11	5.21
Natural scientists	0.22	0.30	0.39	1.28	2.32	6.49
Managers	0.09	0.13	0.12	0.44	0.56	0.70
All other occupations[2]	0.06	0.05	0.03	1.23	0.92	0.72
All occupations	0.07	0.08	0.06	1.06	0.86	0.80

1. Permanent migration.
2. Includes cases where occupation was not identified.

Source: Zhao *et al.* (2000) and Citizenship and Immigration Canada.

immigrants from that country but also briefly the number of such immigrants to Canada from all countries. The exodus of these medical professionals during the mid-1990s was probably related to the health spending cutbacks enacted by most provinces in those years. At least for physicians, the outflow seems to have abated toward the end of the decade (Barrett, 2001).

A significant part of the recent increase in emigration is accounted for by temporary migrants. For those going to the United States, one explanation of this trend is the introduction of simplified procedures for obtaining temporary work permits under the FTA and later the NAFTA.[64] But this is not the whole story: temporary emigration to other countries has risen just as fast as that to the United States in the 1990s. Interestingly enough, while the share of migration to the United States has remained approximately constant, the geographical composition of the flows to the remaining countries (both of permanent and temporary migrants) shifted from Europe toward Asia in the past decade, mirroring the earlier changes in immigrant sources. While this undoubtedly reflects the rising importance of trade links with Asian economies, it probably also signals substantial return migration.

Although there are no reliable comprehensive statistics on the composition of emigrants, there is indirect evidence that former immigrants represent an important component. For example, DeVoretz *et al.* (2002), using data from

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the 2001 Hong Kong census, show that about 34 000 people born in Hong Kong (China), Macao or mainland China who had been in Canada at the time of the 1996 census (14 per cent of all pre-1996 immigrants from Hong Kong, half that percentage considering all those from China as well) had returned to Hong Kong by 2001.[65] What is particularly interesting is how those who returned self-selected: half of them had post-secondary degrees, against 26 per cent of those who stayed in Canada; 59 per cent of them were in the 20-39 age bracket, against 43 per cent for those who stayed; and they earned 2.3 times as much, on average, as those who stayed.[66] There are also former immigrants who choose to move on to a third country. According to US immigration data, in the 1990s the non-Canadian-born accounted for about 30 per cent of all immigrants to the United States whose last country of permanent residence was Canada, a much larger share than that of the foreign-born in the overall Canadian population (18.4 per cent in 2001).

The increase in emigration toward the United States reflects to a large extent the growing economic integration within NAFTA and rising labour mobility of the highly skilled. But the fact that the increase in labour flows has been asymmetrical points to a number of underlying factors, some of which may have been temporary, while others are structural. Among the temporary factors was the unfavourable relative performance of the Canadian labour market during most of the 1990s, which however has been reversed in the last few years. In addition, as mentioned above, the outflow of some categories of professional workers was influenced by occupation-specific factors (*e.g.* in the health sector). But there are also structural reasons why the net flow of highly skilled workers between Canada and the United States has always been negative, which also explain why the imbalance widens when – for other reasons – the overall mobility of the highly-skilled increases. F*irst*, having higher levels of labour productivity the United States can offer higher wage and salary levels. S*econd*, returns to skills are higher in the United States than in Canada. This tends to bias the composition of emigrants toward the highly skilled (as mentioned in Box 3 above). T*hird*, given the US labour market's larger relative size, it offers a greater variety of outlets for specialised skills, and much more information about job opportunities travels northward than southward across the border. And, *fourth*, personal tax rates are lower in the United States than in Canada, particularly for higher income brackets. Their counterpart is a higher level of public spending in Canada, and greater availability of publicly provided health care, tertiary education and other social services, but it is doubtful that this fully offsets the higher tax rates from the point of view of those Canadian emigrants who are in the higher income brackets. In other words, because Canadian taxation and public spending involve a greater degree of income redistribution, they tend to reinforce the effect of the lower pre-tax reward to skills that characterises Canada's labour market.

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The economic impact of migration

There is a wide consensus among economists that the main economic gains from international migration accrue to the immigrants themselves. But public discussions usually focus on the question whether, in addition to these, there are also gains to the destination country.[67] For such a discussion to be meaningful, in this context "the country" should be defined as the pre-existing residents, including earlier immigrants but excluding the new arrivals whose effect is being assessed.[68] There are three main channels through which immigration can affect the economic welfare of existing residents: *via* the supply of factors of production and its impact on the return to existing factors; *via* effects on economic efficiency (*e.g.* through economies of scale or other types of spillover effects); and *via* immigrants' net fiscal impact. Since most immigration is permanent, the focus of analysis will be on the long-run effects, although in some cases short-term effects are relevant. Since the demographic effects of immigration are important for each of the channels mentioned above, they are discussed first.

Immigration contributes to population growth, but has a limited impact on ageing

The contribution of net immigration to Canadian population growth has risen steadily since the 1950s, mainly as a result of the slowing rate of natural increase. While in the early post-war period it accounted for about a quarter of total population growth, that share has risen to over two-thirds (Figure 37, Panel A).[69] According to baseline official projections, which assume fertility rates unchanged at present levels and life expectancy continuing to rise according to recent trends, deaths would start to exceed births around 2025, after which the population would continue to grow only if there is a net inflow from abroad (Statistics Canada, 2001). Assuming constant immigration around current levels and emigration moving in proportion with the population, the Canadian population would peak at slightly over 37 million around 2040 and then start to decline slowly. In the absence of migration the population would grow very little until 2020, and by 2050 would be 4 million smaller than present levels.

Because most immigrants are young, immigration can also help to offset the impact of population ageing. Of all new immigrants entering in 2002, 64 per cent were in the 15-44 age range, against 44 per cent for the Canadian population. But the impact on overall ageing trends is relatively small if changes in the annual flow of immigrants within a "reasonable" range are considered. Nevertheless, cumulative effects over long periods of time are not negligible: the proportion of people 65 and over in the population, which increased from 7.6 per cent in 1961 to 12.7 per cent in 2002, would double by 2051 assuming current net migration levels; it would rise by 4 percentage points more under zero migration. Immigrants' younger age also means that they have a stronger effect on the working-age population than on total population (Figure 37, Panel B). In coming decades, as the

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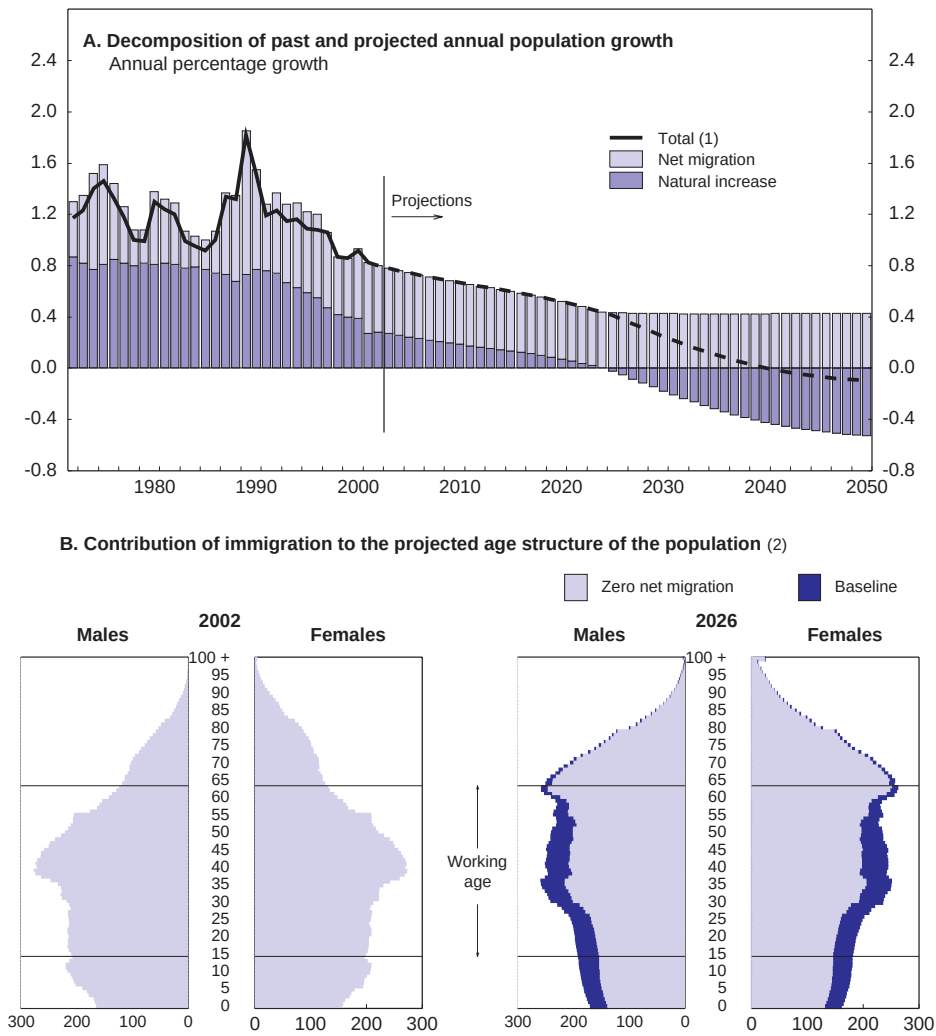
Figure 37. **Contribution of immigrants to population growth**



1. Total population increase may differ from the sum of components due to a statistical residual.
2. The 2026 chart shows the population age structure under the baseline projection and under an alternative scenario with zero net migration. The baseline assumptions are: *a)* fertility rates constant at 1.48, slightly below the level observed in 1997; *b)* life expectancy rising by 4.5 years for males and by 2.8 years for females between 1996 and 2026; *c)* permanent immigration constant in absolute terms at the target level for 2000 (225 000); *d)* no net temporary immigration (stock of temporary residents constant at the 2001 level); *e)* emigration growing in proportion to the population. The age distribution of immigrants and emigrants is assumed to be in line with historical averages. In the zero net migration scenario net overall migration is set equal to zero. The other assumptions are unchanged.

Source: Statistics Canada.

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baby boom generation retires, Canada's working-age population is expected to grow more slowly than the overall population. At present levels of net migration, it is expected to peak at 23.5 million around 2016 (9 per cent larger than in 2002) and then gradually decline (by about 3 per cent over the next 30 years). However, it would start shrinking already around 2010 if net immigration were to stop entirely and would then decline by almost 30 per cent over the following four decades (see also Figure 10 in Chapter I).

The effects of immigration on income distribution are probably negligible

According to standard economic theory, an inflow of labour not accompanied by a parallel inflow of capital will tend to lower the local wage rate if immigrant labour is a relatively close substitute for the native factor of production; if wages are not flexible downward, unemployment will result. However, empirical studies, both in Canada and elsewhere, have usually found it difficult to confirm this prediction by comparing cities or regions with different immigration rates, in part because immigrants tend to be attracted precisely to the areas where most jobs are being created. Moreover, in a world with mobile capital, any effects of immigration on local wages would normally be short-lived.[70] The available studies on Canada find relatively small effects of immigration on wages and unemployment levels, consistent with the results in the US literature.[71] DeVoretz and Laryea (1998) find no displacement or wage effects of immigration on Canadian-born workers at the aggregate level but some negative effects in industries that use a higher-than-average proportion of foreign-born workers. Potentially more relevant are the effects of immigration on the relative wage levels of different types of labour. For the United States, Borjas *et al.* (1992) estimated that the large 1980s immigration flows (whose skill content was lower than that of the US-born population) could explain about a quarter of the increase in the wage gap between skilled and unskilled workers in that period. A similar analysis has not been conducted for Canada.[72] It would probably not find equally large effects, since in Canada immigrants have higher skill endowments relative to natives than in the United States.

Immigrants are especially able to bring entrepreneurial skills. Canada has specific programmes to attract immigrant investors and entrepreneurs, as discussed above. In addition, many immigrants who originally arrived under other entry classes eventually start a business. Unfortunately, statistics are available only on immigrants who choose self-employment (often alongside employed work), but not on entrepreneurs. Using longitudinal tax return data Li (2001) finds that the proportion of immigrants declaring self-employment income is relatively low in the first year after migration but doubles after four years and triples after ten. Immigrants as a whole are 1½ times as likely as the Canadian-born to be self-employed. In many cases immigrant entrepreneurial activity is facilitated by the existence of ethnic enclaves, which can help offset the language handicap but can also provide a network of business connections, a protected market and a source of cheap labour.

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About 40 per cent of all self-employed immigrants from visible minorities work in retail trade or in the accommodation, food and beverage services sector, the typical businesses found in ethnic communities.[73]

Many immigrants also bring capital with them, either for investment or to bridge the initial settlement phase. The capital inflows attributable to immigrants are not negligible (on average, C$30 000 per immigrant since 1980, for a total annual inflow of 0.7 per cent of GDP). The size of the inflow was particularly high in the 1985-97 period, in coincidence with the arrival of large numbers of immigrants from Hong Kong. Ley (2000) reports evidence that a substantial fraction of their wealth was invested in real estate. From the point of view of welfare effects on the existing population, these inflows are largely irrelevant: the return to the immigrants' capital will accrue to them, and in a situation of very high capital mobility the return on other capital should be unaffected.[74] But in the short term capital inflows can have effects on asset prices: for example, in Toronto and Vancouver foreign investment in real estate contributed to raise property prices, resulting in capital gains for owners.[75]

Possible gains in economic efficiency from scale economies and spillover effects are hard to quantify

In Canada it is a widely held view that by increasing the size of the population, immigration allows the country to achieve economies of scale. This idea is rooted first of all in Canadian history. For a long time, population increase was seen as necessary not only to tap the country's large natural resources, but also to support the build-up of an adequate transport and communications infrastructure. There is some econometric evidence that the strong, immigration-induced increase in population between 1896 and 1913 had a substantial effect on per-capita income.[76] At that time, the country was still very sparsely populated and mostly rural, and there was presumably ample scope for achieving economies of scale connected to infrastructure and agglomeration. If such a scope still exists now, it is probably more limited. A study conducted more than ten years ago (Economic Council of Canada, 1991) found that potential economies of scale were still significant in the transport, communications and public utilities sectors and to a lesser extent in manufacturing. But the resulting overall gains from immigration were nonetheless small: it was estimated that an increase in population by one million (about 3 per cent) would raise income per capita by 0.1 per cent.[77] For a country facing the prospect of demographic decline at present fertility rates, an alternative way of looking at this issue is to ask how much immigrants would contribute by preventing the need to raise the per-capita cost of investment required to maintain the existing infrastructure. This type of calculation yields rather small estimates of the potential gains.[78]

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A specific type of efficiency gain could be connected to immigration's effects on the functioning of the labour market. As mentioned above, Canada faces the prospect of the size of its labour force peaking around 2016. As discussed in Chapter I, there may be scope for faster labour force growth through participation gains, but it is limited. The labour force will also become older, and its rate of turnover will decline, as the new Canadian-born generations reaching working age are smaller. An older labour force and a lower turnover could imply an overall decline in geographical and occupational labour mobility, given that older workers are in general less mobile and more costly to retrain than new entrants. This could constrain productivity growth in a phase of rapid technological change requiring occupational reallocation. Immigration can alleviate this constraint by providing a steady flow of new labour market entrants.[79] At present, immigrants represent about 30 per cent of all new labour-market entrants (Beaujot, 2002). If the geographical pattern of immigrant settlement is consistent with that of employment growth – which, to a large extent, seems to be the case (see Figure 34 above) – immigrants can contribute to "greasing the wheels" of the labour market, offsetting natives' inadequate geographical mobility (Borjas, 2001). Once they are in Canada, immigrants are found to be as mobile geographically as the Canadian-born, but they are more mobile across occupations. In part, this reflects the fact that in their assimilation process they move up from the lower-skilled jobs taken initially, but immigrants' occupational mobility seems to remain higher even many years after migration and is higher than for Canadian-born labour market entrants (Green, 1999). Not surprisingly, the most skilled immigrants and those with better knowledge of Canadian languages are the most mobile.

Other sources of efficiency gains could be any favourable spillover effects connected to immigration. It is often argued that highly skilled immigrants have beneficial spillover effects because of the network externalities connected to human capital (having a larger pool of highly skilled individuals multiplies the opportunities for exchanges of knowledge and increases the scope for specialisation). Moreover, the greater cultural diversity and wider variety of consumption goods and services that usually accompany immigration can be seen as welfare improving. Canada's multiculturalism policy, adopted in 1971, is predicated on the idea that multiculturalism, if properly managed, is an important social and economic resource (Department of Canadian Heritage, 2002). However, needless to say, all these potential spillover effects are extremely difficult to quantify.

A possible spillover effect of immigration is on trade. The economics literature has found that, in a number of countries, immigrants contributed to developing trade links with their home countries.[80] The explanation is that, through their knowledge of their home country, immigrants can reduce the transactions costs that stand in the way of an expansion of trade.[81] This is likely to be important for establishing trade relationships with non-traditional trading partners. Existing studies for Canada confirm the existence of this type of effect. Looking at trade

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and migration flows with the different trading partners, Head and Ries (1998) find that a 10 per cent increase in immigration raises exports by 1 per cent and imports by 3 per cent; the larger effect on imports could be attributed to a preference for home country goods on the part of immigrant consumers. According to these estimates, the effect is particularly large for immigrants from East Asia. These estimated effects appear rather large but could be biased upward by the fact that, during certain stages of development (*e.g.* the shift from a rural to an industrial economy), rising income levels in immigrant source countries can be associated both with expanding trade and with emigration.

Immigrants' fiscal impact is probably positive but depends on their economic outcomes

There are different ways of estimating the effects of immigrants on public finances. A *first* one is through a static calculation of the taxes paid and public goods, services and transfers received by foreign-born residents at a point in time. A study of this type was conducted by Akbari (1995), who made estimates for 1990. He concluded that immigrants' net contribution was positive, largely as a reflection of the relatively high income levels of earlier cohorts. However, the results of this type of study provide only limited indications as to the impact of present and future immigrants over their lifetimes.[82] A *second* methodology is to consider the present value of an average immigrant's net fiscal contributions over his/her lifetime and compare it to that of an average non-immigrant. For Canada, no studies of this type are available. This kind of calculation would in general give a positive estimate of the relative fiscal impact for those immigrants (a majority) who have completed their education before immigrating. Such an effect would be large in Canada, where education is heavily state-subsidised. A *third* approach is to compare the long-term effects of alternative demographic scenarios constructed on the basis of different immigration assumptions. Such a study was conducted in 1991 (Economic Council of Canada, 1991) but unfortunately has not been updated. It used demographic projections based on two alternative assumptions on the net immigration rate (0.4 and 0.8 per cent), combined with information on the age distribution of taxes and of public expenditure on health, education, pensions and social welfare. According to the results of the simulation, the projected increase in per-capita public expenditure connected to the rise in dependency ratios was 20 per cent smaller in the high-immigration scenario than in the low-immigration one after 25 years; about one-third of this gain was offset by lower per-capita income tax revenue and higher expenditure on immigrant-specific programmes. The net gains were found to be even larger after 50 years (30 per cent reduction of a much larger dependency-related expenditure increase, only minimally offset by taxes and other expenditures).[83] However, the simulation assumed that immigrants would differ from the existing population only with respect to their age structure, not for economic characteristics or their propensity to use individual public programmes at any given age.

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Table 25. **Government transfers to immigrant and non-immigrant households**[1]
Average amount of transfer per receiving household in 1995 (Canadian dollars)

Category of household[2]	Age of household[3]				All households
	15-24	25-44	45-64	65 and over	
Canadian born	3 900	4 700	5 400	15 300	7 300
Earlier immigrant	3 700	4 800	5 400	16 100	9 100
Recent immigrant	4 100	5 400	5 700	11 600	6 100
1980s immigrants	4 100	5 300	5 600	12 700	6 200
1991-94 immigrants with others	4 400	5 600	6 800	12 700	6 900
1991-94 immigrants only	3 900	5 500	5 200	6 300	5 400

1. Data refer to all kinds of government transfers to households, including old age security, Canada and Quebec Pension Plan benefits and the Guaranteed Income Supplement, as well as Employment Insurance, student assistance and other programmes. Data are based on tax returns connected to the IMDB database.
2. A household is defined as Canadian-born if all its adult members (*i.e.* those aged 15 and over) were born in Canada, as immigrant otherwise. Immigrant households are classified according to the period of immigration of the most recently arrived adult member. The category "1991-94 immigrants with others" refers to households whose adult members include both immigrants who arrived in 1991-94 and others (either earlier immigrants or Canadian-born), while "1991-94 immigrants only" refers to households where all adult members are 1991-94 immigrants.
3. The age of the household is defined as the age of the oldest parent in the case of a family, as the age of the oldest member in the case of a non-family household.

Source: Citizenship and Immigration Canada (2001).

Looking at participation in specific public programmes, Baker and Benjamin (1995) find that immigrants are less likely than non-immigrants to receive Employment Insurance (EI) and Social Assistance (SA) benefits, although they are more likely to receive rent subsidies. This, however, does not apply to elderly immigrants, who are much more likely than their Canadian-born counterparts to have low income and to receive EI and/or SA. Differences with natives are largest in the early years after the immigrant arrives and tend to narrow with assimilation, probably reflecting a combination of learning, eligibility and evolving socio-economic conditions. However, the more recent immigrant cohorts show a generally higher propensity to participate in these programmes, in line with the more general deterioration in their economic performance.[84] Data on the total transfer payments received by recent immigrants as compared with their predecessors are consistent with this, although they do not allow the separation of such cohort effects from the entry effect (the fact that immigrants are more likely to need assistance in the early years after arrival) (Table 25). The proportion of SA recipients among recently arrived elderly immigrants is 20 per cent, twice as large as among the Canadian-born (Baker, 2002). Since most of them are likely to have been admitted under the family class, this seems to indicate problems with the enforcement of sponsorship agreements (under which the sponsor undertakes to support the applicant for 10 years, see Annex IV).

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Conclusions

Canada has traditionally sought, selected and successfully integrated large numbers of immigrants. The goals of immigration policy have changed during the country's history. At present, together with long-term demographic objectives, they emphasise attracting immigrants with the types of fungible skills and adaptability that will facilitate their integration into a rapidly changing knowledge-based economy. Both the increase in the proportion of skill-assessed immigrants in the 1990s and the more recent shift toward general rather than occupation-specific skills are consistent with the finding that the younger, more adaptable and better educated immigrants are those who fit most easily into the Canadian labour market.

However, the economic gains associated with demographic objectives are not self-evident, since immigration flows of realistic proportions can have only a limited impact on population ageing trends. And while there may well be efficiency gains associated with a larger population – such as those connected to economies of scale or to spillover effects – it is hard to prove their existence or to measure their size. In any case, the goals of Canada's immigration policy are not only economic. In the choice to admit refugees, humanitarian motivations are of course prominent. In the case of family-class immigrants, an important goal is to support an effective and balanced long-term integration of immigrant families. But also for economic immigrants, their successful participation in the labour market is part of a broader objective to continue developing Canada as a successful, dynamic, multicultural society. Hence the considerable emphasis placed on immigrant integration programmes and the relatively generous eligibility conditions for social benefits and citizenship. In spite of this, there is evidence that, on the whole, immigrants do not represent a burden on public coffers.

That could change if the decline in the economic performance of immigrants observed in the 1980-95 period were to continue. There is some evidence that, as economic conditions turned around in the mid-1990s, immigrant performance has also started to improve, but it is still tentative and the recovery has been partial at best. Some concerns remain with respect to recent immigrants' lack of language skills, as these are becoming increasingly important in the Canadian labour market, and the difficulties they encounter in having their educational and work credentials recognised. These are areas that will require increased attention of policymakers.

The emphasis on the long-term sustainability of immigration policy seems to be paying off. It is significant that the policy enjoys a high degree of public consensus. Even factors that in other receiving countries have made immigration less easily accepted by the local population – such as the rising proportion of immigrants belonging to visible minorities and the heavy concentration in a few metropolitan areas – have had only a muted impact in Canada. This is probably attributable in part to the perception that the government is reasonably in control

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of the number of arrivals – with the aid of Canada's distance from the main source regions, which helps contain illegal immigration – but also to the success of immigrant integration programmes, usually managed with the extensive participation of local communities.

As part of an increasingly integrated North American labour market, Canada is in direct competition with other destination countries in its attempt to attract highly skilled workers. For much the same reason, it has also traditionally suffered an outflow of its own university graduates to the United States. But overall emigration is not large, and a significant part of it seems to be of a temporary nature; typically, returning emigrants bring with them enhanced human capital and valuable personal connections. Overall, given that the order of magnitude of outflows remains smaller than that of inflows, Canada remains a net importer of highly skilled workers. But the greater return to skills in the United States will leave Canada vulnerable to emigration, particularly for those near the top of the skill distribution. In part, this is a consequence of differences between the Canadian and the US social model, which reflect long-standing social preferences. The outflow of a relatively small number of highly-skilled workers can be seen as a small price to pay for a widely shared set of goals regarding the distinguishing features of Canadian society. But emigration flows will need to continue to be monitored as a signal of relative economic success and of Canadian society's comparative attractiveness. Thus, global competitive pressures will continue to require a constant emphasis on cost-effectiveness in the public sector, so that tax rates do not need to be raised to levels that would negatively affect economic performance. These fiscal challenges are discussed in the next chapter.

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IV. Managing fiscal pressures in the medium and long term

The 1990s was a decade of enormous fiscal consolidation, with the general government balance swinging from a deficit of 9 per cent of GDP in 1992-93 to a surplus of 3 per cent in 2000-01. The federal government achieved a balanced budget in 1997-98, with the provinces in aggregate following two years later (Table 26). As a consequence, general government net debt has fallen from a peak of 68 per cent of GDP in 1995 to around 40 per cent in 2002, compared with an (unweighted) OECD average of around 30 per cent.[85] This retrenchment was achieved through a combination of expenditure and revenue measures, but aggressive cutting of programme spending was a key factor (Figure 38). Significant reforms to federal and provincial income support programmes in the mid-1990s also played a major role. While expenditure cuts occurred in almost all areas, they were not indiscriminate: a Programme Review process ensured that the lowest-priority areas were hit hardest.

The federal government's current approach to fiscal planning is now one of the most prudent in the OECD. The budget's macro-economic assumptions are based on an average of private-sector forecasts. Each autumn, several private-sector forecasters produce detailed five-year fiscal projections based on *status quo* tax and spending policies. These are used as the basis for the *Economic and Fiscal Update*, and show the amount of fiscal room available over the medium term. An annual contingency reserve of C$3 billion (around ¼ per cent of GDP) is set aside and used to reduce debt if it is not needed. Normally, an extra prudence margin that rises by around C$1 billion a year is also set aside. If not required, this can go towards debt reduction or for spending or revenue measures. Budget decisions are then made over a two-year planning horizon. In other words, only the initial fiscal impact of each decision is shown at budget time; the medium-term consequences are sometimes not shown until the following *Update*, half way through the year.[86]

By design, the budget surplus usually comes in higher than "planned". In recent times, some of the fiscal room that has emerged during the fiscal year has been spent towards year-end. This has frequently been done *via* an arms-length

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Table 26. **Federal and provincial/territorial budget balance**

Per cent of GDP, national accounts basis, year ended 31 March

	1990	1991	1992	1993	1994	1995	1996	1997	1998	1999	2000	2001	2002	2003	Net debt 2002, per cent of GDP[1]
Federal government	–4.2	–4.8	–5.6	–5.8	–5.5	–4.7	–4.1	–1.6	0.5	0.3	0.7	0.8	0.7	0.5	49.1
Provinces, territories and local	–0.5	–1.3	–3.5	–3.6	–3.2	–2.0	–1.2	–0.8	–0.4	–0.5	0.6	1.2	–0.7	–0.5	21.8
Ontario	0.5	–0.8	–3.8	–4.2	–3.8	–3.0	–2.1	–1.5	–1.0	–1.2	0.3	0.5	–0.2	–0.3	23.5
Quebec	–1.2	–2.2	–3.1	–3.7	–3.6	–3.9	–2.5	–2.3	–1.6	0.1	–0.1	0.5	–1.2	–0.9	33.7
Alberta	–2.9	–1.5	–2.9	–4.2	–3.0	1.1	1.4	2.7	2.7	1.1	2.3	5.3	–0.5	1.0	–3.4
British Columbia	1.0	–0.2	–2.4	–1.8	–0.5	–0.2	–0.1	–0.4	–0.2	–6.8	–0.8	0.5	–2.1	–3.0	10.0
Manitoba	–1.0	–1.2	–1.5	–2.1	–1.9	–0.5	0.9	0.4	0.1	0.4	–0.3	0.6	–0.6	0.0	19.3
Saskatchewan	–2.5	–0.7	–5.7	–0.6	–3.9	0.3	0.0	0.7	0.9	0.7	0.0	3.3	–1.4	0.2	20.3
Nova Scotia	–4.0	–1.8	–2.3	–4.2	–3.1	–0.8	–0.9	0.2	0.5	–0.2	0.4	1.1	–0.4	0.1	44.0
New Brunswick	–0.2	–1.5	–3.2	–1.9	–1.8	–1.3	–0.8	0.2	0.2	0.6	–1.7	0.0	–0.1	–0.7	31.2
Newfoundland and Labrador	0.0	–1.8	–2.6	–3.4	–1.5	–0.2	–0.1	–0.2	3.5	0.7	–2.1	–2.0	–3.4	–1.7	35.5
P.E.I.	–1.2	–1.1	–2.4	–3.5	–7.4	0.9	1.1	–0.5	–1.2	–0.2	0.3	–1.4	–1.2	–1.3	27.9
Yukon	1.6	1.1	–1.9	–1.9	3.3	3.1	2.8	0.1	0.3	5.2	0.5	4.4	4.0	1.0	–5.3

1. Public accounts definition.

Source: Statistics Canada.

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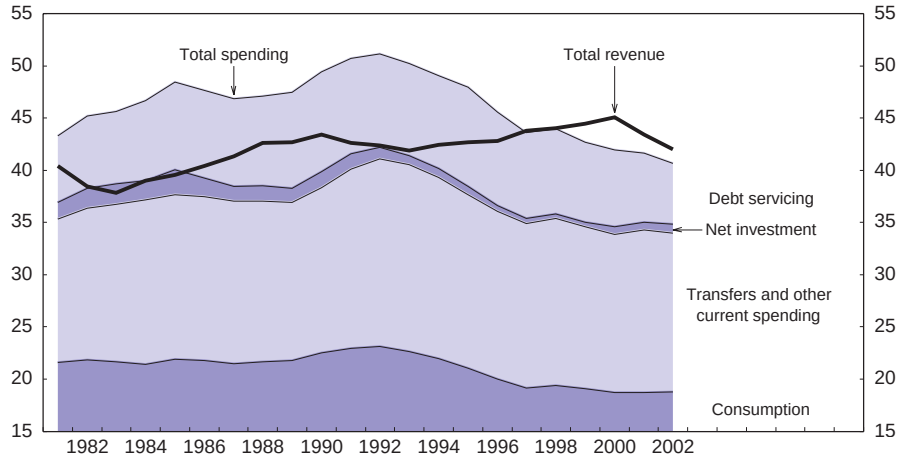
Figure 38. **General government spending**
Per cent of potential GDP



Source: Statistics Canada; OECD.

trust or foundation (see below). On the positive side, this means that lower-priority budget proposals are funded only if the government has enough money at the end of the year. In principle, this is a fairly cautious approach. In practice, it runs the risk that end-of-year spending decisions could be unfocused and *ad hoc*, rather than carefully traded off with other options (including debt reduction) as part of the budget process.

In most cases, provincial and territorial finances are also in good shape. Deficits in most regions were eliminated by the end of the 1990s, although only Alberta has recorded a sizeable surplus. Since then, weak commodity prices, tax cuts and the aftermath of the Asian crisis have pushed British Columbia back into deficit, while Ontario would have recorded a deficit in 2002-03 had it not been for C$1 billion of federal health transfers that came through very late in the year. The provinces and territories also have reasonably prudent fiscal frameworks. Most have some form of balanced budget requirement (Ontario and Quebec can run deficits so long as they are offset in the next fiscal year; New Brunswick and Saskatchewan must balance their budgets over a four-year period); some have legislated debt-reduction schedules (*e.g.* Manitoba and Alberta, although Alberta is considering dropping its rule); and several require a public referendum to approve major tax increases (Ontario, Alberta, Manitoba and the Yukon).

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A short-term balanced-budget target

The federal government's fiscal policy since 1998 has been to plan for a balanced budget after setting aside the contingency and prudence reserves. In other words, it estimates how much fiscal room is available over the medium term and increases spending or cuts taxes to ensure that the expected or planned budget balance excluding the reserves is zero in the first two years of the forecasting horizon. The contingency reserve is normally C$3 billion and is earmarked for debt reduction if it is not required. This implies the federal government has a debt-reduction target of C$3 billion each year from its stock of around C$500 billion of net liabilities. However, better-than-expected fiscal performance can lead to a faster pace of debt pay-down; from 1997-98 to 2001-02, debt reduction has averaged just under C$10 billion (0.9 per cent of GDP) per year.

This fiscal planning approach was followed in the budget of February 2003, which made spending and taxation commitments worth around ½ per cent of GDP over the period 2002-03 to 2004-05 (Table 27). Approximately half of this amount was related to health care, while the rest was made up of generally small allocations to a wide range of areas. It also included C$1 billion of permanent but unspecified savings that were to be found by May 2003. The bulk of the increased health spending is in the form of higher transfers to the provinces and territories: by 2008, this transfer will have increased by C$5.5 billion per annum, which is only slightly less than the amount recommended in the November 2002 Romanow report on health reform (see Annex V). Funding is also being provided for the purchase of diagnostic medical equipment, creating nation-wide electronic health records, encouraging reforms to the way primary care is organised, and improving information gathering to try to enhance reporting by governments to citizens on how health care dollars are being spent.

The system of inter-governmental transfers was also changed in the latest budget. Since 1996, the main health and social transfers from the federal to provincial governments have been amalgamated into a single block grant to each province, the Canada Health and Social Transfer (CHST). The CHST consists of cash payments worth around 1.7 per cent of GDP in 2002, plus "tax points" worth another 1.5 percentage points.[87] It can be used to fund any provincial social programmes, including health, welfare, post-secondary education and early childhood programmes. In recent years, this system has generated a vigorous yet pointless debate about how much the federal government is contributing to health care. Because the CHST is an untied block grant, no particular share of federal transfers is earmarked for health care, giving provinces the flexibility to allocate funding according to their own priorities among health care, social services and post-secondary education. From 1 April 2004 the CHST will be split into separate health (CHT) and social (CST) transfers, with a roughly 60/40 split in favour of health. However, this is simply an accounting reform as all provinces currently

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Table 27. **Highlights of the 2003 budget**
C$ billions

	2002-03	2003-04	2004-05
Projected surplus before budget	**9.4**	**8.8**	**11.5**
Minus contingency reserve	3.0	3.0	3.0
Minus prudence	. .	1.0	2.0
Surplus for planning purposes	**6.4**	**4.8**	**6.5**
Spending decisions in 2003 budget	6.4	4.1	5.0
Health care	4.7	1.4	2.1
of which:			
CHST supplement[1]	2.5	. .	. .
Health reform fund	. .	1.0	1.5
Diagnostic/medical equipment fund[1]	1.5	. .	. .
Other healthcare spending	0.7	0.4	0.6
Science, education and innovation	0.6	0.6	0.6
Defence and international assistance	0.7	1.3	1.7
Unspecified savings	. .	−1.0	−1.0
Other	0.4	1.8	1.6
Revenue decisions in 2003 budget	0.0	0.7	1.5
Child benefit supplement	. .	0.2	0.3
Increase RRSP limits	. .	0.1	0.2
Eliminate federal capital tax	. .	0.1	0.4
Other	. .	0.3	0.6
Total spending and revenue initiatives	6.4	4.8	6.5
Surplus after budget	**0.0**	**0.0**	**0.0**

1. These transfers were paid to a third-party trust but were accounted for in 2002-03.
Source: Finance Canada.

spend well above their allocated health and social transfers. This restructuring of the CHST may increase political pressures on the federal government in the long term by putting a floor under the federal share of total health spending. Provinces may therefore be laying off some of the risk of higher health spending onto the federal government and thereby be more able to behave as some of them are at present: cutting taxes, yet demanding more federal funding for their health systems.

The elimination of the federal deficit has enabled the federal government to switch its focus away from fiscal consolidation and towards funding its economic and social priorities. Looking back over the whole period since 1998, tax and spending measures costing an average 3.3 per cent of GDP have been put in place (Table 28; Box 5). How these have been split between spending increases and tax cuts depends on whether tax expenditures are counted on the revenue or expenditure side; if treated as tax reductions, revenue measures account for around 1.8 per cent of GDP on average, while spending increases account for the other

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Table 28. **Expected cumulative spending and tax cuts, 1997-98 to 2004-05**
Per cent of cumulative GDP over the period

	Tax expenditures counted as:	
	Tax reduction	Spending
Spending initiatives	**1.5**	**1.8**
Health and social transfer (CHST)	0.6	0.6
Other health-care	0.1	0.1
Human capital: science and education	0.3	0.3
Other spending	0.5	0.8
Tax changes	**1.8**	**1.5**
General tax cuts	1.2	1.2
EI premium reductions	0.3	0.3
Child tax benefit	0.2	. .
Other tax expenditures	0.1	. .
Total	**3.3**	**3.3**
Debt reduction		
C$ billion per year	6.0	6.0
Per cent of average GDP	0.6	0.6

Source: Finance Canada.

1.5 per cent. Net debt has been reduced by an average C$6 billion (0.6 per cent of GDP) each year. Increasing transfers to provinces has clearly been a priority, and a large fraction of this money has gone to the health-care system. There has also been a focus on human capital development, with increased funding for the science sector, university research and scholarships for advanced-level tertiary students. On the revenue side, the five-year tax reduction plan has been particularly important, as has the reform of the child support system.

Virtually no fiscal room is left over the medium term

The October 2002 medium-term fiscal update showed the federal surplus rising to 1½ per cent of GDP by 2007-08, or around 1 per cent of GDP after subtracting off the contingency reserves and prudence margins. Almost all of that was eliminated in the 2003 budget. Although the five-year impact of its budget decisions will not be published until October 2003, an estimate can be constructed from information in budget documents and other sources. This points to a federal surplus of around ½ per cent of GDP in 2007-08 when measured on a public accounts basis (*i.e.* virtually zero after subtracting the prudence and contingency margins), compared to the originally projected 1½ per cent.

This forecast does not include upcoming spending by arms-length foundations and trusts. For example, the 2003 budget allocated C$1.5 billion for prov-

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Box 5. Recent tax reduction initiatives

The federal government announced in 2000 a five-year tax reduction programme, and these reforms have been enriched in each budget since then. The impact of the plan as it now stands is that personal and corporate taxes will be around a fifth lower than they would otherwise have been by 2004-05, which is equivalent to a reduction in the size of the federal government of 2½ per cent of GDP (Table 29). The key measures have included:

- Restoring full price indexation to the personal income tax system.
- Cutting all personal tax rates as from 1 January 2001.
- Cutting the general corporate income tax rate from 28 per cent to 21 per cent by 2004, combined with various reductions in capital gains taxes.
- Increasing the child tax benefit. This has been the most significant tax expenditure initiative over recent years. By 2007, spending under this programme will have increased by 0.1 per cent of GDP, to stand at 0.7 per cent, but will also have become much more tightly targeted, partly through parametric reforms and partly because the various income thresholds are indexed to prices rather than wages (Figure 39). Without those changes, spending on this programme would have dropped to 0.4 per cent of GDP.
- Introducing a benefit for children with disabilities, tied to the national child benefit scheme.
- Eliminating the federal capital tax effective in 2004 for medium-sized companies and by 2008 for all companies. This tax was levied on firms with more than C$10 million of capital and was deeply unpopular as it had to be paid regardless of profitability. It may also have discouraged investment, since it increased up-front investment costs (because it had to be paid in the early years of an investment, before a project began to generate profits). Its elimination will cost around C$1.7 billion per annum by 2008, when the policy is fully phased in. However, the similar capital tax levied on large financial institutions has not been eliminated.

Table 29. **Five-year federal tax reduction plan**

Calendar years

	2001	2002	2003	2004	2001	2002	2003	2004
	Per cent of GDP				Per cent of tax base			
Personal income tax	1.3	1.4	1.5	1.8	15.2	15.7	17.3	19.8
Corporate tax	0.0	0.1	0.2	0.3	2.0	6.0	9.3	12.7
Employment insurance	0.2	0.2	0.3	0.3	11.9	13.3	17.0	20.0
Total	**1.6**	**1.7**	**2.0**	**2.4**	**12.3**	**13.8**	**15.8**	**18.6**

Source: Finance Canada and OECD.

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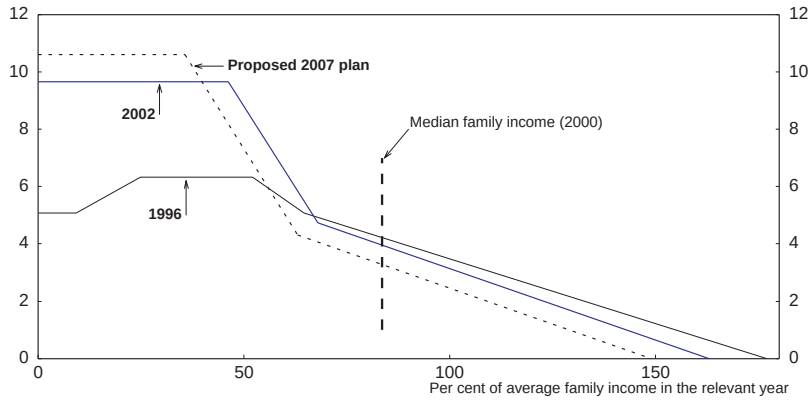
Box 5. **Recent tax reduction initiatives** (*cont.*)

- Reducing the corporate tax rate in the resource sector from 28 to 21 per cent from 2003 to 2007. This will bring that industry's tax rate into line with other sectors.
- Increasing the annual dollar limits for contributions to RPPs and RRSPs from their pre-2003 budget levels of C$14 500 and C$13 500 to C$18 000 by 2005 and 2006 respectively, and indexing those limits to average wage growth in 2006 and 2007.

Figure 39. **Child tax benefit**

For a two-child family, per cent of average after-tax family income[1]



1. Average income of all family types, not just two-child families.
Source: Finance Canada; OECD.

inces to spend on medical diagnostic technology, with the money being set aside in a trust fund to be drawn down over the next three years. The trust was set up near the end of 2002-03 and, consistent with government accounting policies, the federal government "booked" the expenditure in that fiscal year, although the provinces have until 2006 to draw down the money and buy new machinery. Similarly, the 2003 budget allocated C$2.5 billion for provinces to spend on health, placing that amount in a trust and "booking" it as an expenditure in the 2002-03 fiscal year with the funds to be drawn down over the next 3 years.[88] As to the foundations, at the end of 2002 approximately C$7 billion that has already passed

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Table 30. **Medium-term fiscal outlook**
Budget balance, per cent of GDP, national accounts basis, fiscal years ending 31 March

	2001	2002	2003	2004	2005	2006	2007	2008
Federal government								
October 2002 update	1.6	0.8	0.6	0.6	0.7	1.0	1.3	1.5
After Budget 2003	1.6	0.8	1.0	0.0	0.1	0.2	0.3	0.5
Provincial/territorial and local	1.1	–0.2	–0.4	0.2	0.0	–0.3	–0.3	–0.3
Canada/Quebec pension plans	0.5	0.6	0.7	0.9	0.9	0.9	0.9	0.9
General government	3.2	1.3	1.3	1.1	1.0	0.8	0.9	1.0
General government excluding pension plans	2.7	0.6	0.6	0.2	0.1	–0.1	0.0	0.1
Structural balance, general government	2.4	1.8	1.2	1.2	0.9	0.8	0.9	1.0

Source: Finance Canada; Conference Board of Canada; OECD.

through as spending in the federal accounts will appear as expenditure in future national accounts measures of the general government. An estimate on this basis shows essentially a balanced budget over the medium term, *before* the contingency reserve is set aside (Table 30). With provincial and local government sectors in balance or small deficit, the healthy-looking medium-term *general* government surplus is almost entirely due to the Canada/Quebec pension plans.

The policy implications are clear. There is no fiscal room over the next few years for further spending increases, tax cuts, or debt reduction beyond the economic prudence and the C$3 billion a year contingency reserve unless macroeconomic outcomes are better than predicted. Nor is there a great deal of room for uncertainty. A significantly weaker-than-expected economy may push the federal government back into deficit (certainly on a national accounts basis, and possibly using the public accounts measure, depending on the extent of the slowdown).[89] In any case, it may record a national accounts deficit this year even without an economic slowdown if the arms-length foundations spend their funds relatively quickly.

Is a balanced budget good enough? Long-term fiscal pressures

From a fiscal point of view, Canada's demographic profile is currently as favourable as it has been for a generation and more favourable than it will be for at least another fifty years. The number of people paying taxes relative to the number drawing pensions or drawing heavily on the health-care system is close to its peak. At this point in the demographic cycle net debt levels remain higher than the OECD average, and policy is aiming for something close to a balanced budget on a year-by-year basis. Whether the government should be content with a balanced budget, or should instead aim for more aggressive debt reduction while it

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has a fiscal window of opportunity, depends on whether current fiscal policy is sustainable over time.

Federal and provincial governments have already taken significant steps to put their finances on a sound long-term footing. The most important has been to ensure that the main pillar of its pension scheme is fully funded. As described in detail in Chapter II, the Canada/Quebec pension plan (C/QPP) pays a retirement allowance based on lifetime contributions. The compulsory levy, split equally between workers and employers, reached 9.9 per cent of earnings in 2003, up from 6 per cent in 1998. The Chief Actuary estimates that this contribution rate is sustainable. There is an additional safety net for people who have not accumulated enough earnings over their lifetimes. This federal Old Age Security and Guaranteed Income Supplement (OAS/GIS) scheme is unfunded (*i.e.* pay-as-you-go), with benefits indexed to prices.

With the bulk of the pension system already sustainable, the long-term fiscal position of the general government depends largely on the outlook for health-care spending and the unfunded OAS/GIS safety net. Assessing whether finances are sustainable is difficult, partly because there is more than one way to think about sustainability, and partly because results can be extremely sensitive to the underlying assumptions. The discussion that follows is based on an analysis of debt projections out to 2040.[90] The starting point is the long-term projections of federal and provincial budgets carried out by Jackson and Matier (2003). Their analysis is based on the 2001 federal budget, while this chapter uses an updated starting point that is consistent with the 2003 federal budget. The key assumptions in the baseline scenario are:

- Within each age group, all categories of expenditure grow in line with labour productivity in real per-person terms. This is equivalent to assuming wage-indexation for all spending, including the OAS/GIS (Jackson and Matier argue that it is unreasonable to assume that price indexation would be sustainable indefinitely, as it would imply a significant fall in pension incomes relative to those in the rest of the economy). This assumption also implies that the share of income devoted to health care remains constant within each age group. Even this is probably over-optimistic, so an alternative scenario is described below.
- Revenues grow in line with inflation, population, and real per-capita income growth, adjusted for the change in the composition of the population.
- The demographic (or ageing) profile is based on Statistics Canada's central scenario.
- Labour productivity grows at 1½ per cent per year, with annual inflation of 2 per cent.

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– There are no new policy initiatives beyond those announced in the 2003 budget. This means the federal government builds up surpluses over the medium term rather than continuing with its current policy of increasing spending (or cutting taxes) on a year-by-year basis in order to maintain a balanced budget (after subtracting the contingency reserve).

It is important to note that several potentially large but uncertain factors have not been taken into account. F*irst*, people are living longer and healthier lives which may postpone the point at which older people become heavy and expensive users of the health-care system. It may also lead to fewer prolonged illnesses that require lengthy hospital stays. S*econd*, labour shortages may push up real wages which in turn may encourage people to stay in the workforce for longer or to increase their participation rates throughout life. T*hird*, the revenue loss caused by having a smaller proportion of the population in work and paying taxes is likely to be partly offset by the net revenues from deferred-tax private pension plans (RRSPs and RPPs). At present, the government faces a net tax loss (or tax expenditure) of around 2½ percentage points of GDP through these schemes in which taxes on investment earnings are paid on withdrawal rather than at the time they are earned. This net tax loss (or expenditure) will decline in the future as withdrawals from those plans grow with an ageing population.

Under these assumptions, and with the assumed ageing profile of the population, provincial health-care spending is projected to rise to around 9½ per cent of GDP by 2040, an increase of around 3½ percentage points (Figure 40, Panel A). Federal elderly benefits (excluding the C/QPP) are projected to rise to 4.7 per cent of GDP, an increase of around 2½ percentage points.[91] A fall in the proportion of youngsters in the population lowers education costs by around 1 percentage point of GDP, while the overall impact of ageing on revenues is relatively small. Under this baseline scenario, combined federal and provincial net debt is more or less eliminated by 2040 (Figure 40, Panel B). This suggests that fiscal policy is probably not on an explosive or unsustainable path. One *caveat*, however, is that ageing pressures show up several years later than in most OECD countries, partly because the population is relatively young and partly because Canada's high life expectancy will delay the increase in health-care expenditures (which are heavily concentrated in the last few years of life). This implies that ageing-related spending will continue to mount significantly beyond the 2040 end-point of this analysis.

The key assumption in the baseline scenario is that real health-care costs on a per-capita per age-group basis grow no faster than real incomes (1.5 per cent per annum). This may be too optimistic. Real per-capita health spending grew at this rate over the past ten years, but that was a period of tight fiscal constraint. Over a longer period, King and Jackson (2000) estimate that real per-capita spending has increased by 1.8 per cent on average over 1975-2000 on an age-adjusted

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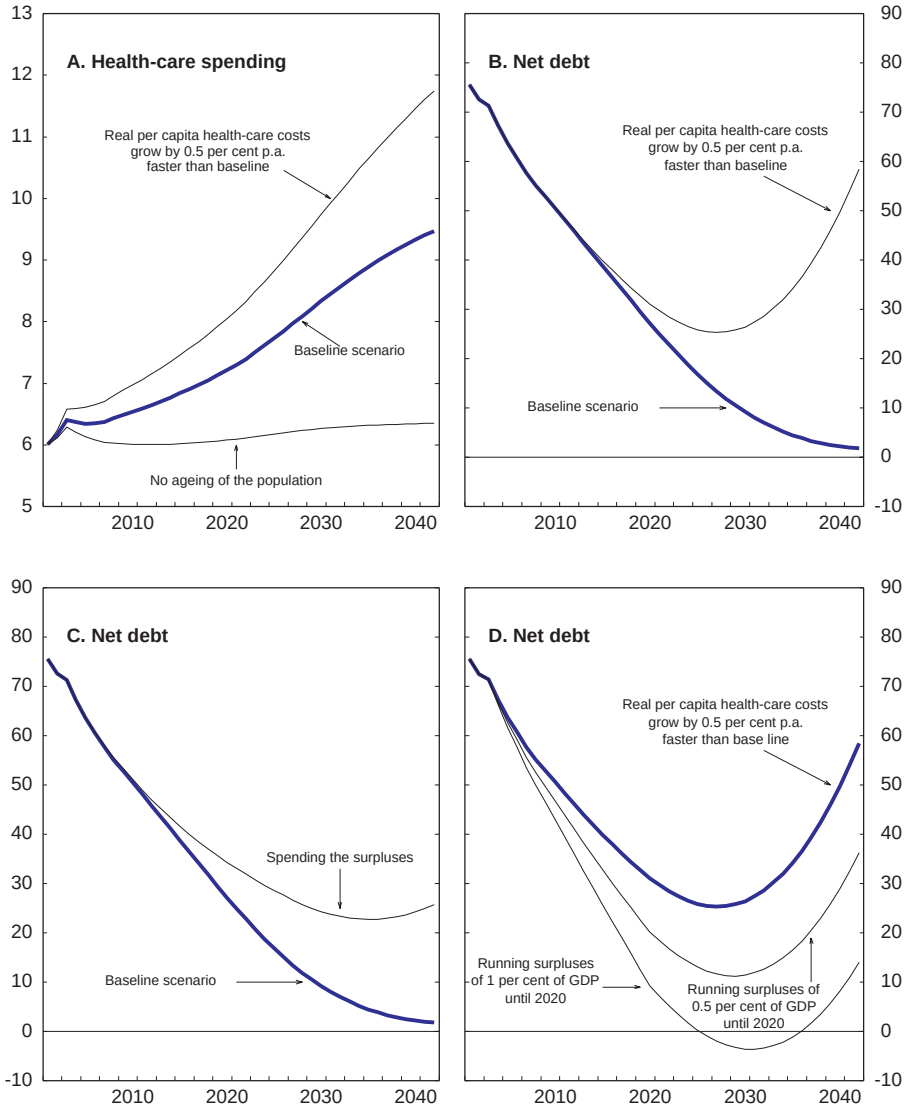
Figure 40. **Long-term fiscal scenarios**

Health-care spending and net public debt,[1] per cent of GDP



1. The figures show scenarios for total government (federal, provincial, territorial and local) excluding the Canada Quebec Pension Plan. Hence, initial net debt is higher than for the fully-consolidated general government.

Source: Finance Canada.

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basis. The pressures to enrich public health spending have already become apparent. The government has recently decided to widen the scope of the public health system by including more generous and comprehensive coverage for pharmaceuticals and home care (see below). These potentially expensive changes have not been incorporated in the baseline scenario.

An alternative scenario is to assume that health spending for each cohort grows at a rate that is ½ percentage point faster than productivity growth (*i.e.* by 2 per cent per annum). This is still well within the rates of increase that have been seen across the OECD over the past decade or so.[92] In this case, public health spending would be another 2 percentage points of GDP higher by 2040, rising to nearly 12 per cent of GDP, and would rise further to 15½ per cent by 2070. The resulting debt profile would begin to climb substantially after 2030 (Figure 40, Panel B). By 2040, net debt would be still slightly below current levels but rising rapidly and would be likely to peak at an uncomfortably high level.

Two alternative scenarios illustrate the importance of pre-emptively saving in advance of the ageing pressures. In the baseline scenario, sturdy primary surpluses over the next couple of decades give ample room to reduce debt as a proportion of GDP. But rather than reducing debt by maintaining surpluses, if federal and provincial governments instead eliminate them through one-off spending increases or tax cuts, then the resulting debt path will be noticeably higher, troughing at around 25 per cent of GDP at its most favourable point in 2028 (Figure 40, Panel C).

An additional question is to ask how big a surplus the government should run over the next 20 years if effective steps to control health-care costs are not put in place. Maintaining surpluses of ½ per cent of GDP until 2020 will make some difference, but a 1 per cent surplus allows net debt to be eliminated before health pressures begin to mount (Figure 40, Panel D).

The main policy conclusions to draw from this exercise include:

– The progress made over the past decade in eliminating the federal and provincial deficits, reducing debt levels and setting the bulk of the pension system on a sound financial basis has left the government well placed to cope with the fiscal pressures of an ageing population. It is certainly in a stronger position than many other OECD countries.

– This favourable outlook depends to a large extent on the ability of the government to control health-care spending. The baseline scenario, which is based on a relatively optimistic assumption about health costs, shows a substantial reduction in debt levels at both the federal and provincial levels. However, even a fairly small increase in the pace with which health spending rises – due to increased demand for services, the breadth of coverage or to a technology-driven rise in costs – can have a dramatic impact on the long-term fiscal position.

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– Unless effective steps are taken to ensure that health costs can be kept under control, it would be prudent for the government to increase the amount of pre-emptive saving that is done while the window of opportunity is open. While it is possible to construct even more benign scenarios, such as being able to maintain price rather than wage indexation of the pension safety net, there are big risks associated with the current policy of devoting planning surpluses to spending or tax cuts rather than to paying down debt more quickly, especially as expenditure increases may be difficult to reverse in the future. Running surpluses of ½-1 per cent of GDP would leave a comfortable margin to cope with unplanned long-term expenses. Adding more medium-term elements to the fiscal framework could help to achieve that goal.

The importance of controlling long-term costs in the health care sector is clear. Improving the performance, efficiency and sustainability of the health-care system is the focus of the next section.

Improving efficiency and controlling costs in the health-care system

Background and current problems

Health care is managed and delivered by the provinces, subject to the criteria and conditions of the Canada Health Act (CHA). The CHA provides for a reasonable degree of access, free of charge, to medically necessary hospital and physician services, which loosely speaking means most hospital-based treatment and visits to family doctors and specialists. It also forbids provinces to charge for these services, and its provisions have the effect of ruling out private insurance for anything covered under the CHA (see Box 6 for a more precise description). All provinces have additional plans that insure some of the areas not covered by the CHA, but they differ widely in scope. Each gives some level of home care, long-term care and outpatient drugs for specific population groups such as seniors and the poor, but coverage of other services such as dental care, rehabilitation and physiotherapy is less uniform.

With free access and a single payer, the CHA-based heart of the system is largely supply driven in the sense that, with demand exceeding supply, the volume of services provided is determined by supply constraints. It will be a challenge to control expenditure growth over the long term through any mechanism other than queuing. The problem may be less serious for the services that are not included under the CHA, as user fees and co-payments are common, at least for the majority of the population. Hence, there is currently more scope to control public spending in the areas of ambulatory care drugs and home care, which may be two of the fastest growing areas as the population ages and technologies improve. However, recently announced plans to bring both these areas

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Box 6. Key features of the health-care system

Health care is managed and delivered by the provinces, subject to the five principles of the Canada Health Act (CHA): that plans must be publicly administered, comprehensive, universally available on uniform terms and conditions, portable across provinces and with reasonable access without financial and other barriers. This last principle became interpreted as ruling out user charges and extra billing (a physician charging above the provincially set fee), and these are now specifically prohibited under the Act. The main features of the system include:

Community-based physician services

- Private insurance cover is effectively, although not explicitly, disallowed. More precisely, federal cash transfers to provinces can be reduced if provinces allow private payments for publicly insured services; as a result, most provinces have enacted regulations to prohibit the purchase of private health-care insurance for medically necessary services insured under provincial plans. Ninety per cent of physicians are paid on a fee-for-service basis and cannot charge above the publicly set fee. There are no out-of-pocket costs for CHA-mandated community-based physician services.

Hospital-based services

- Physician services: same as for community-based physician services. In addition, patients can choose their physician, and private insurance coverage is effectively ruled out (as for community-based physician services).
- Drugs and other health-care services: free of charge. Patient can pay for upgrades, such as a semi-private room.

Prescription drugs outside of hospitals

- Not covered by medicare. Provincial plans exist for defined populations (*e.g.* those in low incomes) and sometimes require co-payments. Private insurance coverage is permitted for people not covered by provincial plans (frequently via an employment plan). Private insurance to cover provincial co-payments is permitted but very rare.

Residential long-term care and home care

- Not covered by medicare. Private insurance permitted but rare.

Almost all hospitals are not-for-profit entities and are usually owned by community-based not-for-profit corporations or religious organisations. In all cases, the vast majority of funding comes from provincial governments, and the federal government provides funding through cash transfers to provinces. All provinces except Ontario channel most funding through 5-20 regional health authorities (RHAs). These are largely co-ordination bodies, trying to match supply and demand, but they do not raise revenues or make decisions about physician services.

Source: Hurley *et al.* (2002) and OECD.

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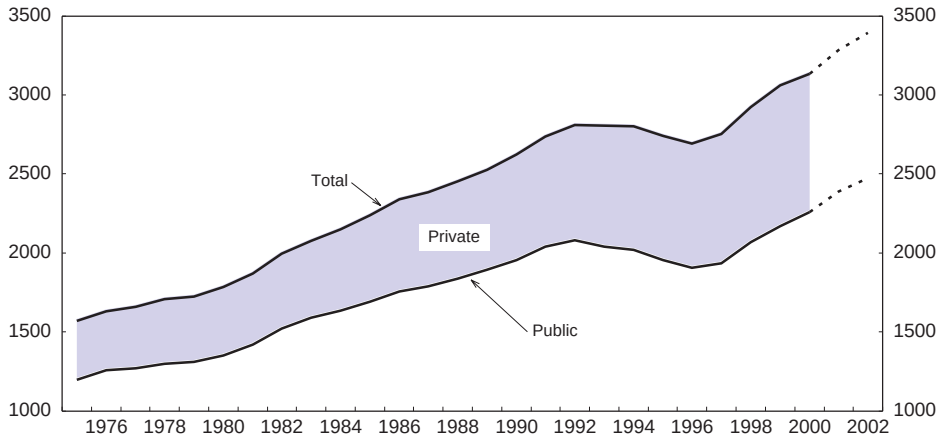
at least partly under the CHA – and therefore make them free – risks reducing the government's ability to control overall spending.

The cuts in public health budgets over the 1990s have now been largely reversed (Figure 41; Table 31). At current levels, it is difficult to gauge whether the system is under-funded or overly expensive. Expenditure per person is high but in line with per-capita incomes (high-income countries spend more of their income on health care). Offsetting this, the population is relatively young, which should reduce health costs; after adjusting for the age structure, spending per person is among the highest in the OECD.[93] One reason is that pay for medical personnel may be higher than average in order to compete with bountiful US salaries, although the magnitude of this latter effect is unclear.[94] The split between public and private funding (approximately 70/30)[95] is close to the G7 average, as is the share paid out of pocket (Table 32). Population health status, as measured by most health indicators, is generally good.[96]

While the system seems to perform well, there has been widespread public dissatisfaction and extensive debate about reform, although this has not been unique to Canada. This has led to several major reviews of the health care system, including the Commission led by Roy Romanow, and the Senate Committee led by Michael Kirby, both of which reported late last year (Annex V). The

Figure 41. **Health spending per person**
2002 C$[1]



1. Data for 2001 and 2002 are estimates.
Source: OECD Health Data, 2002.

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Table 31. **Health-care spending by category**

	2002 C$ per capita				Per cent of total			
	1980	1990	1995	2002	1980	1990	1995	2002
Hospitals	896	1 105	994	1 127	41.9	39.0	34.6	31.3
Other institutions	243	267	276	333	11.4	9.4	9.6	9.3
Physicians	316	429	412	481	14.7	15.2	14.4	13.4
Other professionals	217	301	332	417	10.1	10.6	11.6	11.6
Drugs	181	322	392	581	8.4	11.4	13.6	16.2
Capital	95	99	88	155	4.4	3.5	3.1	4.3
Public health and administration	91	119	147	214	4.3	4.2	5.1	6.0
Other	102	190	232	288	4.7	6.7	8.1	8.0
Total	**2 141**	**2 832**	**2 872**	**3 597**	**100**	**100**	**100**	**100**

Source: Canadian Institute for Health Information.

key concerns are related to access – problems finding a family doctor and long waiting lists to see specialists, undergo diagnostic tests and receive non-urgent treatment. More specifically:

– The shortage of doctors – in particular specialists (Table 33) – has several causes including sharp cuts in the number of places in medical

Table 32. **Health-care expenditure**

Per cent of trend GDP

	1980	1990	2000	Spending in 2000				
				Public	Private			Approximate public/private split[2]
					Total[1]	Health insurance	Out-of-pocket	
Canada	**7.0**	**9.0**	**9.1**	**6.4**	**2.7**	**1.0**	**1.5**	**70/30**
France	. .	8.7	9.5	7.2	2.3	1.2	1.0	75/25
Germany	8.8	8.7	10.6	8.0	2.6	1.3	1.1	75/25
Italy	. .	8.1	8.0	5.9	2.1	0.1	1.8	75/25
Japan	6.5	6.1	7.8	6.0	1.8	0.0	1.3	75/25
United Kingdom	5.5	6.2	7.3	5.9	1.4	. .	. .	80/20
United States	8.5	11.8	13.1	5.8	7.3	4.6	2.0	45/55
G7 average[3]	7.3	8.4	9.3	6.6	2.7	1.1	1.4	70/30
OECD average[3]	7.7	7.7	8.2	5.9	2.3	0.4	1.5	70/30

1. Total private expenditure includes health insurance, out-of-pocket and other spending (not shown).
2. Shares rounded to nearest 5 percentage points.
3. Unweighted average of countries for which data are available.

Source: OECD.

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schools (partly due to budget restrictions, but also because of poor forecasting and the absence of nation-wide manpower planning), a jump in emigration and a fall in immigration of medical personnel in the 1990s (see Chapter III), and a shift towards earlier retirement. Licensing restrictions can also hold back immigration; for example, there is a limit on the number of people who can take the doctor's certification exam in Ontario each year, and that number is set by the physicians' professional association. Shortages are worst in remote areas and have led to some drastic responses, such as Quebec's granting rural health boards the power to effectively draft doctors into staffing the emergency wards in their localities.

– Delays in diagnostic testing reflect under-investment in modern technologies such as MRI scanners, although there is plenty of older technology such as X-ray radiology machines (Table 33). It may also explain the slow uptake of some newer medical procedures, such as coronary angioplasty rather than bypass grafts as a treatment for ischaemic heart disease (Moïse, 2003). This under-investment in key technologies is surprising, since overall health-sector investment has been only slightly less than in other countries,[97] suggesting that funds have been poorly directed rather than insufficient. In particular, a large part of the capital budget may have gone to maintaining "bricks and mortar" as a result of over-building in the 1980s. In its 2003 Budget, the federal government provided an extra C$1.5 billion over three years to purchase diagnostic equipment. While this should ease the immediate problem, it does not address the deeper question of why every OECD country that invested less than Canada in the 1990s still has more to show for it in terms of access to new technology.

– While it is difficult to make reliable cross-country comparisons, Statistics Canada (2002) found that median waiting times for specialist visits, non-emergency surgery and diagnostic tests were around one month each. One patient in ten waited more than 17, 26 and 13 weeks, respectively, for these services. Waiting times have continued to increase despite the significant increase in funding in 2000.

Thus, the key problem appears to be one of waiting times and capacity. However, before increasing capacity policy makers need to ensure that existing systems are being used efficiently, so that increased spending leads to an expansion of supply rather than being solely absorbed in higher remuneration of health-care providers.

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Table 33. **Health resource indicators for various OECD countries**

2000 or latest available year

	Medical technology				Employment			Acute care beds	Investment
	Per million population				Per thousand population				Per cent of GDP
	Magnetic resonance imaging units	Computed tomography scanners	Radiation therapy machines	Lithotriptors	Practicing specialists	General practitioners	Practicing nurses	Total	Average, 1990-2000
Canada	**2.5**	**8.2**	**7.0**	**0.5**	**0.9**	**0.9**	**7.6**	**3.0**	**0.28**
Australia	4.5	20.8	4.9	. .	0.9	1.1	8.1	3.8	0.50
Austria	10.8	25.8	4.2	1.7	1.8	1.3	9.2	6.2	0.48
Czech Republic	1.7	9.6	9.8	2.8	2.3	0.7	8.4	6.6	. .
Denmark	6.6	11.4	5.4	. .	. .	0.6	7.3	3.3	0.22
Finland	11.0	13.5	14.3	0.4	1.4	1.7	. .	2.4	0.24
France	2.8	9.6	7.2	0.9	1.7	1.6	6.5	4.2	0.24
Germany	4.6	17.1	4.6	1.7	2.4	1.0	9.3	6.4	0.35
Iceland	10.7	21.3	14.2	3.6	. .	0.7	14.2	. .	0.22
Italy	6.7	19.6	3.7	2.9	. .	0.9	4.5	4.5	0.42
Japan	23.2	84.4	. .	. .	. .	. .	7.8	. .	0.30
Korea	5.4	28.2	5.3	4.4	. .	. .	1.4	5.2	0.28
Spain	4.9	12.2	3.8	1.8	. .	. .	3.7	. .	0.22
Sweden	7.9	14.2	. .	. .	2.2	0.6	8.4	2.4	0.33
Switzerland	13.0	18.5	11.2	3.9	2.1	0.4	. .	4.1	0.31
United Kingdom	3.9	3.6	4.8	. .	1.6	0.6	5.3	3.3	0.36
United States	8.1	13.6	4.1	2.7	1.4	0.8	8.3	3.0	0.22

Source: OECD Health Data 2002, 4th edition.

Lowering costs through supply-side reforms

Primary care reform

Progress in reforming primary care has been slow and fragmented, centring on pilot projects with short-term funding (Romanow, 2002). Most primary care is provided by individual physicians on a fee-for-service basis. Multidisciplinary clinics offering a range of primary care services on a "24/7" basis are rare, although Ontario is pushing for reforms in this area.[98] Barriers to innovation include regulations (for example, the role of nurse practitioners is restricted, except where there are too few doctors), union agreements (which require the use of particular types of professionals) and rigid funding rules (for example, "telephone triage" is not possible as doctors are paid only for face-to-face treatment).

Remuneration methods could also be reconsidered. A fee for service approach encourages a large volume of services but can lead to over-servicing. It also reduces control over the aggregate level of primary health expenditure. In contrast, a capitation approach improves incentives to spend more time with difficult cases and to provide comprehensive and preventive care. However, it reduces a patient's choice – at least temporarily – over his or her doctor and runs the risk that physicians may "cream skim" or, if given the option, register too many patients and under-serve them, although it is hard to know whether this occurs in practice. In any case, some form of capitation or salary-based payment – possibly mixed with fee for service – may be useful if provinces are to experiment with US-style managed care arrangements. These arrangements have shown that they can reduce costs, but with the attendant loss of patient choice. The slow progress towards alternative payment systems is surprising, since two-thirds of doctors would prefer to be paid in some way other than fee-for-service, although older doctors are much less likely to want to change (CMA, 2002). However, any shift away from fee-for-service should wait until the problem of supply shortages is solved since current arrangements are encouraging the maximum use of scarce resources.

More active purchasing and greater competition in the delivery of hospital services

Regional Health Authorities (RHAs) have the scope to play a much more direct role. In particular, some could become more active purchasers, rather than just passing on to hospitals the budget received from their provincial governments, based on historical spending patterns. They should be more involved in determining outputs, including both quality and price aspects. Efficiency could also be improved by increasing competition among providers, for example through greater use of internal markets and a cleaner split between purchaser and provider, with RHAs buying from the not-for-profit and for-profit sectors on a

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neutral basis. While some aspects of an internal market already exist, with RHAs buying some services on behalf of local residents, the move is far from complete. RHAs in all provinces except Ontario (where they are not used) provide hospital services, and some fund long-term care and home care through private not-for-profit or for-profit providers. However, the physician and drug budgets continue to be controlled directly by provincial governments. Health care would be better integrated if these were passed over to the RHAs.

Output-based funding mechanisms

Current cost-based funding mechanisms provide weak incentives to improve efficiency. Multi-year funding is virtually non-existent, hospitals typically submit budget requests before aggregate provincial funding is set (raising the incentives to "puff up" requests), and funding decisions often come too late (sometimes well into the fiscal year) for providers to be able to adopt a meaningful strategy for dealing with financial constraints (McKillop *et al.*, 2001; Deber, 2002). All provinces use cost-based funding, and some allocation approaches are extremely rigid, with little scope to re-allocate funds between institutions or between inputs when populations or health-care needs change. For example, every institution in Ontario receives the same percentage increase in its primary operating grant.[99] British Columbia and New Brunswick use highly bureaucratic institution-based line-by-line funding. Only Alberta and Saskatchewan use population-based funding methods (*i.e.* allocating funds to providers based on the population in their catchment area), although several other provinces are exploring this approach.

An alternative is to fund hospitals for the services they deliver rather than the inputs they use (*i.e.* money following the patient). The best-known approach is the diagnostic related treatment groups (DRGs) system that was introduced in the United States in the 1980s and has been used with varying degrees of success elsewhere. It has several favourable characteristics. It would: allocate budgets on the basis of output; give purchasers some control over treatment intensity (by altering prices); encourage hospitals to increase output where there is demand and capacity; create incentives to reduce costs per sickness episode; reduce the amount of micro-management by provincial health ministries; open the system to greater competition among providers; and place the risk of cost overruns on the provider, rather than the funder (at least in the short term). The key drawbacks are that it could reduce aggregate control of health-care spending, as suppliers have incentives to raise output or substitute among different treatments ("DRG creep") where there is spare capacity or where centrally set prices exceed marginal treatment costs. For this reason, some form of expenditure cap could be considered, drawing on experiences in other countries. Output based funding can also lead to over-servicing, especially

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if hospital-based specialists are also paid on a fee-for-service basis, so that changes to their remuneration schemes may be needed at the same time.

Improve the management information base

In the face of a tight budget constraint and waiting lists, inadequate information on costs per procedure and on patient preferences makes it difficult to allocate funding to priority areas through anything other than guesswork. Alberta and Ontario have taken some steps to try to fill the information void, but only a small number of institutions currently participate in the pilot projects. Cost control more generally may be an issue as well. For example, Esmail (2002) estimates that non-medical staff in British Columbia hospitals, such as painters, electricians and cooks, are paid up to 40 per cent more than their counterparts in non-medical enterprises. This may also reflect the sector's highly centralised and unionised employment system and insufficient contracting out.

Should more funding be drawn into the system by allowing private insurance for core services?

Canadians currently have broad insurance coverage that achieves one of the primary goals of policy: adequate access to essential health care. However, the system is unique in that people are not allowed to take out additional insurance in the areas covered by the CHA, although they can do so for the areas that are not covered. This means that individuals cannot obtain more health care than is provided under the current plan, even if they wish to do so, or are obliged to wait for it for much longer than desired.[100] In contrast, several other countries permit such duplicate private insurance plans, including the United Kingdom, Sweden (for private hospitals), Ireland, New Zealand and Australia (for hospital care). The key problem appears to be that individuals are prevented from insuring against the risk of less timely care than desired and that this may lead to a loss in consumer welfare.

The question of whether this restriction should be loosened, perhaps just for a limited range of services, is a complex one. The potential benefits include both financial and quality aspects: an increase in overall resources, and resultant easing in pressure on public health budgets, over a period in which the demand for health care will be pushed up by an ageing population and by technological change; improved information for suppliers as people are able to indicate their personal preferences more directly; potentially greater responsiveness to evolving customer preferences as a result of increased competition among insurers; and increased welfare for those who were previously unable to take out private insurance. The main potential drawback of introducing additional private insurance is the concern that it could create a two-tier system with privately insured patients getting faster service, and that this may harm those

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without such insurance. Whether this would be likely to occur is unclear both theoretically and empirically (see Cutler, 2002 and OECD, 2003b for reviews). Those left behind in the public queue could in fact be better off if the change frees up resources that are then ploughed back into more services for those who cannot afford to opt out. To ensure this occurs, private insurance would need to generate an increase in supply since current low levels of staffing are a major element in the waiting list problem. It would also need to be clear that people taking out extra insurance would receive no offset through lower taxes as this would just swap one fiscal pressure for another. Finally, any parallel Canadian private sector may need to be entirely separate from the existing system in order to avoid problems of physicians gaming the system. And since there is a risk that any resulting cost inflation in the private system could flow back into the public system, continued price regulation may be needed. In any case, calls for such private insurance would probably weaken if the problem of waiting lists in the CHA sector was resolved by other means.

Reviewing the coverage of the free publicly funded system and the call for greater cost sharing

As noted above, the distinctive feature of the medicare system in Canada is not only that it covers 100 per cent of the cost of doctor and hospital services, but that it also covers a limited number of services. This range of services has not kept pace with changes in medical practice, such as greater use of drugs and home care. This has also led to inconsistencies in coverage across provinces for some new techniques such as MRI scans. Both the Romanow and Kirby reports recommended expanding coverage to include catastrophic drug costs and post-acute and palliative home care. The 2003 *Accord on Health Care Renewal* between the federal, provincial and territorial governments agreed to incorporate some form of drug coverage by 2006 and first dollar coverage for some short-term acute home care services, with the details of the latter to be settled this year. In both cases, the level of coverage is to be decided by each provincial government, so differences across provinces may well remain large.[101]

With increasing pressure on governments to expand the basket of services covered under medicare, one option for covering the extra costs is to introduce some form of cost-sharing, as is done in many other countries. In principle, user fees, co-payments and deductibles can increase incentives to economise on usage of services. They can also reveal useful information about the relative value of specific services from the patient's point of view. The balance of evidence suggests they do restrain demand, especially when starting from a position of zero charges. Whether they significantly worsen health outcomes is less clear; most recent indicators appear insensitive to cost sharing, although measuring health outcomes is difficult (Box 7). While the private contribution to

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health spending (30 per cent) is already close to the OECD average, it consists of essentially no private payments for doctors' and hospital services combined with widespread insurance and out-of-pocket payments for everything else. It is far from obvious that this is the most efficient or equitable way of splitting the public and private contributions. The principle underlying the CHA is that it should cover all, and only, those services that are non-discretionary and for which cost sharing is inappropriate. In practice, however, there is a grey area in the middle which is likely to evolve over time because of changing social preferences as well as developing medical practices. Introducing broader cost sharing that spreads the cost more equally across the different types of care – accompanied by measures that ensured they do not limit access – would free up resources for other priorities such as expanding the range of services that is provided free of charge to the poor. The main risk is that across-the-board user charges may impact most on the chronically sick and those on low incomes. However, such problems can be reduced by setting stop-loss ceilings on individual

Box 7. The impact of user charges on health-care systems

The impact of user charges in health-care systems has been a subject of enormous debate, numerous studies and a handful of large-scale experiments. The evidence from the literature is reviewed in OECD (2003b). Regarding their impact on health-care usage, the most important information comes from the Health Insurance Experiment (HIE) conducted in the United States in the 1970s. Its results suggest that moving from a zero to a 25 per cent co-insurance rate for all care could lead to an average fall in spending of 20-25 per cent, but increases beyond that have little marginal effect. Unsurprisingly, the biggest impact is on ambulatory/outpatient care, with less impact on specialists' visits and hospital treatment. Measuring health outcomes remains difficult, but most indicators studied in the HIE were insensitive to cost-sharing. Manning *et al.* (1987) report that there were no benefits for the average patient from additional services received under a plan without co-payments. For poorer patients there was better control of blood pressure and vision problems, but they argue that these problems could potentially be resolved more cheaply through targeted programmes. Valdez *et al.* (1989) find, on the basis of the data from the HIE, that children with cost-sharing, fee-for-service plans had fewer medical contacts and received fewer preventive services than those assigned to an HMO. Nonetheless, children with the cost-sharing fee-for-service plans were perceived (by their mothers) to be in *better* health overall than those assigned to the HMOs. However, Solanki *et al.* (2000) find that cost-sharing reduces significantly the use of a range of preventive services. In addition, patients reduce "essential" and "unnecessary" care in equal proportions, which could lead to adverse health effects.

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expenditure or by integrating co-payments with the tax system so that those on low incomes pay less.[102] An alternative approach in this context is to offer more than one public plan, as is done in Denmark for example. In practice, there are many alternatives such as offering the choice of a free system and one that could have broader coverage but include co-payments and aspects of capitation funding and managed care.

A wider debate on alternatives such as these – drawing on the range of different approaches in other OECD countries, and not just regarding the US system as the only other option on the table – may help clarify the potential benefits and risks for both the Canadian health insurance system and its goal of equal access for equal need. The need will be all the more pressing as demand for health care rises. In the absence of systems to help regulate demand and improve resource reallocation, waiting lists and pressure on public spending will continue.

Improving fiscal management in other areas

The previous S*urvey* included a detailed review of public spending. It found that significant progress had been made in controlling the level of public spending and boosting value for money and concluded that the framework was in good shape overall. However, some problems were highlighted, even if they were described as "not enormous". They relate to the budget process, fiscal arrangements between federal and provincial governments, new ways of delivering social services, public management and social spending (see Box 8 of OECD, 2001c). Regarding *budget management*, the prudent budgeting process worked well when the deficit was being reigned in but has created its own problems now that the accounts have moved into surplus. There is a risk of spending fiscal windfalls in *ad hoc* ways at year-end, rather than carefully trading them off with other options as part of the budget process. Recommendations included pre-announcing a ranking of programmes that would be eligible for year-end spending, re-instating the Programme Review (value-for-money) process, and completing the move to a medium-term fiscal framework. Serious problems with some *arms-length agencies* needed to be fixed, including clarifying their objectives and monitoring their financial and policy performance. Recommendations were also made to improve *fiscal transparency*, such as clarifying the goals of the EI scheme and reviewing its rate-setting process. Finally, weaknesses in *public management* were discussed, such as archaic human-resources policies and insufficient resources for evaluating programmes.

The government has made progress on several of these recommendations. In 2003 it announced a new five-year programme review process, similar to the last one but with an increased focus on horizontal reallocation. Financial reporting has also been improved with the move to accrual accounting in the 2003 budget. The government has also solved the long-standing problem of main-

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taining an excessive EI premium rate, partly by gradually reducing the premium rate and partly by enriching benefit levels. The major extension in recent years has been to extend paid parental leave from six months to one year, and reducing the qualifying period to four months of work (600 hours) over the previous year. It is now one of the most generous schemes of its kind in the OECD.[103] On the premium side, the rate for employees will fall to 1.98 per cent in 2004, down from 2.20 per cent the previous year. Employers will pay 2.77 cents in the dollar. This new rate is probably actuarially fair, ignoring the hypothetical interest earnings from the scheme's notional C$40 billion surplus (HRDC Chief Actuary, 2002). The government has also reaffirmed its intention, first announced in 2000, to start reviewing the rate-setting process (rates are currently set by the Finance Minister, rather than by the EI Commission as they were until 2001). It intends to introduce a new more transparent regime by 2005.

Moving further towards a medium-term fiscal framework

It may be time to move further towards a medium-term fiscal framework now that the goal has switched from deficit reduction to maintaining budget balance. There is certainly a compelling political-economy argument in favour of using a short horizon when policy is focussed on reducing the deficit. For example, medium-term projections may show the deficit to be on target for elimination in, say, five years, in which case there may be a temptation to ease up on fiscal consolidation and delay balancing the budget by a year or two. But this argument no longer applies in an environment of rising expenditure. In this situation, showing just the short-term impact without the full medium-term consequences may give a misleading impression of how much fiscal room is available. A second problem with balancing the budget year by year is that policy risks becoming pro-cyclical. A cyclical upturn could lead to unsustainable increases in baseline fiscal commitments and may amplify aggregate demand at an inappropriate time, although if additional spending is channelled through the arms-length foundations its impact will most likely be spread out over future years. It is also the case that, in a cyclical downturn, the contingency and prudence reserves could prove insufficient as they can absorb a shock of only around 1½-2 per cent in the level of GDP. Under the current policy of avoiding a deficit, fiscal policy would then need to be tightened at the wrong time. The need for a more medium-term focus at the federal level is magnified by the pro-cyclical bias that most provinces have as a result of their own short-term balanced-budget restrictions. The federal government should therefore consider adding more medium-term elements to its fiscal framework by adopting a broader set of financial objectives over a longer planning horizon, including for example a medium-term debt-to-GDP objective. It is helpful to have both a stock and a flow target to ensure that one-off shocks, either positive or negative, do not put the fiscal position permanently off track.

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The controversial use of foundations

In 1999 the Auditor General highlighted problems with the governance and accountability of arms-length foundations that are increasingly being used by the federal government. These foundations, such as the Canada Millennium Scholarship Foundation (CMSF) and Genome Canada, receive advance funding through lump-sum grants that are then spent over subsequent years. In follow-up reports in 2001 and 2002, the Auditor General noted some improvement in this area, but inadequate progress overall. At least C$7.5 billion has been channelled through such agencies since 1990.

The main problem is that the foundations have been placed beyond the reach of effective ministerial oversight and parliamentary scrutiny. While delegating decisions to arms-length agencies can lift the quality of spending by shielding them from short-term political pressures, it must be matched by clear objectives and adequate reporting. Few agencies submit corporate plans to parliament or publish annual reports with a credible description of accomplishments. None go through broad audits of their compliance with their mandates, propriety, or value for money. Few arrangements allow a minister to effectively monitor an agency's spending and make adjustments if things go wrong or if the government's priorities change. The government recently announced its intention to deal with these issues for future foundations and clarified when it is appropriate to use them. It also plans to consult with existing foundations in an attempt to strengthen their governance agreements.

Aside from governance issues, the Auditor General has argued that these structures are being used to "hide surpluses," (OAG, 2002) thereby reducing the transparency of the public accounts (for example, most of the C$7 billion granted between 1999 and 2002 is still sitting in bank accounts but has been recorded as expenditure in the public accounts). The government argues that its accounting policies are consistent with guidelines in the accounting profession on the appropriate treatment of such vehicles, although judgement is required. It also argues that, because funds are spent by independent boards, they should not be considered part of government.

Conclusions

To sum up, Canada has done more than most countries to ensure that its old-age pension scheme is sustainable in the long term, but it still faces potentially large pressures from the health-care system. The recent review of health care by the Romanow Commission reaffirmed the principles of free and universal access and argued that such a system is financially sustainable in the long term. However, projections of the fiscal position over the next 40 years and beyond show that this optimistic assessment sits on a knife-edge: it assumes *i*) that the government allows healthy surpluses to build up over the next couple of decades,

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rather than "spending" them each year (up to the small contingency reserve); and *ii*) that health spending per person in each age group rises no faster than their incomes – an optimistic assumption. If either assumption does not materialise, Canada may face a debt blow-out when the baby boomers get old. Consequently, unless it finds ways to control health-care costs it would be prudent for the government to target something better than a balanced budget while it has the opportunity. Strengthening the medium-term focus of the fiscal framework is a necessary first step in this direction.

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Notes

1. The standard, national accounts measure of the saving rate shows an almost continuous downward trend since the early 1980s, but this trend disappears if a balance-sheet-based measure is used, which includes capital gains and losses. The latter measure would nevertheless show a significant decline in savings in the last three years as a result of the stock market decline.
2. If equity is measured at market value, the decline is even steeper until 1999, but there is some rise in the last three years due the fall in equity prices.
3. Part of this gap reflects measurement differences since passive job seekers (those who just scan a newspaper) are counted differently in the two countries. This widens the unemployment gap by around ¾ of a percentage point (Statistics Canada, 1998).
4. There is a large discrepancy between two measures of hours worked in the United States: 34.2 hours per week in 2001 according to the establishment survey, and 39.2 according to the household survey. The reasons for such a large discrepancy are not clear, but obviously have a large impact on the decomposition of the income gap (see Sharpe, 2003). All the US estimates in this Survey are based on the establishment data, which are thought to be more reliable, with some correction for multiple jobholding based on both establishment and household survey data.
5. In firm-level studies across several OECD countries, between 50 and 85 per cent of aggregate labour productivity growth was estimated to be due to productivity growth within individual firms, with a smaller contribution from entry and exit (low productivity firms dying, for example) and little being due to changes in market share (OECD, 2001a; Baldwin and Gu, 2002b). Firm-level growth accounted for a smaller – though still a majority – share of aggregate MFP growth, suggesting that differences in labour productivity growth at the level of the firm are partly due to differences in capital intensity developments.
6. There is a potential self-selection problem in making such a comparison, as only the most productive firms are likely to be able to export, so a non-exporter that lifts its productivity and then breaks into export markets will have its productivity growth attributed to the exporting group. Even so, three-quarters of exporters' labour productivity growth was generated by firms that exported continuously over this period, while only 10 per cent is due to previous non-exporters that began to export. The remaining 15 per cent is due to new firms that export. The MFP gap between exporters and domestically oriented firms is smaller than the labour productivity gap, indicating that exporters are more capital intensive on average, but even the MFP differential has widened over time.
7. If exporting leads to learning, then becoming an exporter should be of more benefit to domestic firms than to foreign-controlled firms, since foreign firms already have

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mechanisms for transferring know-how across borders. Baldwin and Gu (2002a) find evidence that this is in fact the case.

8. Baldwin *et al.* (2001) find that the implementation of NAFTA caused a significant increase in plant-level specialisation in manufacturing: *i.e.* greater product line specialisation and longer product runs.

9. This is one of several conclusions from the OECD's *Growth Study* and is supported by more recent work by Scarpetta and Tressel (2002) for a panel of 23 industries in 8 countries, and by Gust and Marquez (2002). Tang (2002) provides similar microeconomic evidence for Canada.

10. This is confirmed by Djankov *et al.* (2002), who found that barriers to business start-ups in Canada were among the lowest of the 85 countries in their sample.

11. These indicators are based on equity limits, screening and approval procedures, management or control restrictions, domestic content rules, etc. Canada's restrictions are highest in the banking, fixed telecommunications, air and road transport and electricity sectors.

12. Rao and Tang (2002) find that MFP levels of Canadian-controlled manufacturers were on average 16 per cent lower than those of foreign-controlled firms between 1989 and 1995, although there was some limited catch-up over time.

13. In 1997, large plants (*i.e.* those with more than 500 employees) accounted for 33 per cent of US manufacturing employment, compared with 23 per cent in Canada (Baldwin, Jarmin and Tang, 2002).

14. For example, see Green and Riddell (1997), Baker and Rea (1996) and Green and Sargent (1998).

15. The standard benefit replacement rate of 55 per cent was reduced by one percentage point for every 20 weeks of regular or fishing benefits collected in the previous five years, down to a floor of 50 per cent.

16. Even so, the impact was as expected. The 1998 EI *Monitoring and Assessment Report* concluded that "some elements of the reform are having their intended effect. The intensity rule has started to reduce benefits for repeat claimants." (page 26).

17. For countries that look back over the past year's work history, qualifying periods are around 26 weeks of work over the previous year in the United States, Japan and the Netherlands, 4 out of the previous 8 months in France and 39 weeks in Ireland. Other countries look back further (typically 2-3 years), which in some respects makes it easier to qualify, but the number of required weeks of work *per year* is never as low as 12 and seldom below 20 (amongst the richer OECD countries, the lowest are 13 weeks in Switzerland and 17 weeks in Germany). (For countries with look-back periods other than 12 months, the qualifying period has been converted to the number of weeks *per year*. For example, a German worker must have worked for 12 months in the past three years in order to qualify, which is equivalent to around 17 weeks per year). See OECD (2002a) for the precise qualifying requirements.

18. The correlation between regional unemployment rates in 1990 and 1997 was around 0.95. This is about the same as for similarly defined regions in Europe and well above levels in the United States (where the correlation was just 0.5) and Australia (0.6-0.7). The correlation for Canada is approximately the same for broadly defined regions (*i.e.* 10 provinces) and more narrowly defined ones (59 regions). See OECD (2000a) for details. A range of other evidence also points to greater labour mobility in the United States, which may go some way to explaining the unemployment gap between the two

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neighbours. For example, VAR models typically show that a region-specific shock has a larger and more persistent effect on the regional unemployment rate than is the case in the United States. Further evidence is surveyed in OECD (1999) and in Obstfeld and Peri (1998). Part of the problem in Canada can be attributed to older workers in low-skilled resource-based industries having difficulty reintegrating into the labour market after severe layoffs in the 1990s (*e.g.* forestry in British Columbia and fishing in the Atlantic provinces).

19. This section is one input into the Organisation's follow up on Sustainable Development as mandated by the Ministerial Council in May 2001. Other environmental-related topics follow at the end of the chapter.

20. The magnitude of the future public pension challenge depends on the policy with regard to first-tier safety net benefit (OAS/GIS). If this is indexed to prices, then the real level of benefits will gradually fall as people retire with more wealth, so lowering their entitlement to means-tested benefits. On this assumption, spending on this tier would rise from 2.4 per cent of GDP in 2002 to a peak of 3.2 per cent in 2030 and decline thereafter. Spending on the C/QPP would rise from 2.5 to 3.6 per cent of GDP, giving an overall rise of only 2 percentage points of GDP, with no further increase in the period to 2050. Such a rise would be well below the average for all OECD countries and in any case is being pre-funded. Thus, it is only the increased spending from the safety-net pension that will create a need for extra financing and then by only 0.8 percentage point of GDP. If these pensions were indexed to wages instead of prices, then spending would rise a further 2½ percentage points of GDP by 2050.

21. Contributions to RPPs and/or RRSPs are limited to a total of 18 per cent of earnings up to specified dollar limits. The pre-2003 budget RPP and RRSP dollar limits of C$14 500 and C$13 500 will be increased to C$18 000 by 2005 and 2006, respectively. Corresponding increases will be made to the maximum pension limit per year for defined RPPs, bringing it to C$2 000 by 2005. The limits will be indexed to average wage growth for subsequent years.

22. For example, Canadian students also performed well in the Third International Mathematics and Science Survey (TIMSS) in 1995-96. Scores were above average in mathematics and science at grade 8, and for mathematics at grade 4. In PISA, significant differences between provinces remained after controlling for differences in students' backgrounds and family circumstances (Corak and Lauzon, 2003).

23. Teachers in Alberta and Quebec undertake a probation period of 1-2 years before being certified. Ontario and British Columbia, in contrast, certify teachers automatically on completion of a Bachelor of Education (Epp and Epp, 2000).

24. In 1997-98, teachers in Quebec were paid around 30 per cent more the OECD average for lower secondary school and 20 per cent more for upper secondary (in PPP-adjusted terms, excluding Hungary, Mexico, the Czech Republic and Turkey, for those with 15 years of experience). Pay scales were also high compared with other workers, as the ratio of salaries to per-capita GDP was also high by world standards (Ministère de l'Education, 2000). Teachers' salaries in the rest of Canada tend to be higher than in Quebec (See BC Ministry of Education statistics at http://www.bced.gov.bc.ca/schools/interprovincial/).

25. Approximately 43 per cent of adult Canadians perform at proficiency levels 1 and 2 in document, prose and quantitative literacy. At the other end of the spectrum, an unusually large proportion – around a quarter – performs at the highest levels (*i.e.* levels 4 and 5). See Table 2.2 of IALS (2000).

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26. This can be seen from the unusually long tail in the distribution of scores in IALS; see Figure and Table 2.1 in IALS (2000). For example, the fifth percentile of test score distribution on both the prose and document literacy tests is lower in Canada than any other country except Poland and the United States. This problem is most serious amongst older Canadians.
27. Approximately 22 per cent of adult Canadians were in job-related education or training in 1997, with another 7 per cent in general training. This is well below rates in the United States, the United Kingdom and Germany, but better than in Italy (see OECD, 2002b; table C4.1). The average number of hours per participant in courses surveyed in the International Adult Literacy Survey was more than 300 in Canada, 50 per cent higher than the 18-country average.
28. In 2001, a 15 year-old Canadian could expect an additional 6.8 years of education, which is around the average for the G7 and slightly above the average for the OECD. However, at 0.8 per cent the graduation rate for advanced research programmes is below the OECD average (1.1 per cent), and well below the best performers (which can be as high as 2.5 per cent of people in the relevant age group; rates for the United States and the United Kingdom are about 60 per cent higher than Canada's). See OECD (2002e), Tables A12.1 and A2.1.
29. In 1997, young people from families earning C$25 000 or less were as likely to be attending university as those whose parents had C$25 000-C$50 000 in income, and not much less likely than those whose parents earned up to C$100 000 (Corak *et al.*, 2003).
30. The issues in this section are treated in considerably more depth in the OECD's *Review of Regulatory Reform* in Canada (OECD, 2002c).
31. Canada's restrictions amount to an effective 46.7 per cent cap on foreign ownership of a telecommunications carrier. Several other countries have restrictions on partially-state-owned incumbents, but few have restrictions on private companies. The other OECD countries with ownership limitations of privately owned telecommunications firms are Korea, Mexico and Turkey.
32. Gönenç and Nicoletti (2001) constructed synthetic indictors of the regulatory and market environment in the industry as an input to the OECD's regulatory database. Factors taken into account included, among others, ownership, route and fare regulations, authorisation of charter flights, market structure, number of international alliances and the number of third party (fifth and seventh freedom) carriers.
33. Early in 2001, Air Canada had a 90 per cent share of Canadian travel agency sales and a 75 per cent share of seat capacity in the domestic market. After merging with CAI, it became the sole carrier on the majority of the top 200 routes. In mid-2002, it was still estimated to have three-quarters of the market (Canada Transportation Act Review Panel, 2001 and Ward, 2002).
34. Cabotage is the right of a foreign airline to pick up local traffic in the course of international travel. The foreign ownership limit applies to voting shares, and the government has recently removed the 15 per cent restriction on ownership by a single party.
35. Trade between provinces was, not too long ago, at least ten times greater than between a province and a US state, after controlling for distance and differences in income (McCallum, 1995).
36. Tang and Rao (2001) found that R&D expenditures as a percentage of sales were around 50 per cent higher in Canadian-controlled manufacturing firms compared with foreign-controlled manufacturers. If foreign firms are assumed to be just as R&D intensive and if that differential is the same for non-manufacturers, the economy-wide R&D

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rate would be higher by around 0.2-0.3 percentage point of GDP – but would still be well short of the OECD average.

37. A government study, which was based on survey responses from firms that used the tax credits, found each dollar of credit led to an additional $1.38 of research expenditure (a cost-effectiveness ratio of 1.38). Other studies referenced in that paper put the ratio at 0.83-1.73 or 0.11-0.67 depending on the sector (Department of Finance and Revenue Canada, 1997). This dollar-for-dollar response is in line with international evidence (Hall and van Reenen, 1999).

38. Performance with respect to other air pollutants has been mixed. Emissions of mercury fell by two-thirds between 1990 and 1995 (Environment Canada, 2001), and by 1992 lead had disappeared from urban air following a 15-year effort to remove lead additives from gasoline (CESD, 2000). However, emissions of carbon monoxide were reduced by only 3.2 per cent over the past two decades, though average concentrations in Canadian cities declined sharply thanks to tighter emissions standards for cars. The ground-level ozone annual average has been continuously increasing over the past 20 years (Figure 20).

39. The Canadian commitments under the 1991 Agreement were to establish a permanent national cap for SO_2 emissions of 3.2 million tonnes by 2000 and to reduce emissions of nitrogen oxides by 100 000 tonnes below the forecast level of 970 000 tonnes in 2000.

40. The federal, provincial and territorial ministers of environment and of energy signed the Canada-Wide Acid Rain Strategy for Post-2000 in October 1998.

41. The Ozone Annex calls for annual caps by 2007 of 39 kilotonnes of nitrogen oxides from fossil-fuel power plants in central and southern Ontario and of 5 kilotonnes in southern Quebec. It also calls for the alignment of fuel and emissions standards for vehicles with those in the United States.

42. There are over 1 300 monitoring stations covering well over 200 rivers and lakes, but there has been no attempt to aggregate the data nationally. Some provinces have created aggregate water quality indices that average the differences in a number of pollutants from desirable levels, but these suffer from applying the same weight to each pollutant, regardless of the damage caused.

43. This is especially the case in the lower Fraser River, Southern Ontario and the St. Lawrence in Quebec.

44. In 2000, the implicit carbon tax was US$ 265 per tonne of carbon for gasoline and US$ 165 for diesel.

45. According to official figures, in 1996-2000 immigrant investors invested over C$2.5 billion in Canada and immigrant entrepreneurs created 41 000 jobs, about four for each entrepreneur (Government of Canada, 2001).

46. The planned range was maintained unchanged at 200 000-225 000 between 1998 and 2001, despite undershooting in 1998 and 1999. In the next two years actual inflows overshot the mark, and starting in 2002 the planned range is being raised gradually every year.

47. The overall cost of federal immigration programmes (including both immigrant selection and integration) was about C$1.2 billion in fiscal year 2001-02 (including the cost of services provided by other departments), 44 per cent of which was covered by revenue (mostly immigrant processing fees and entry fees). Thus, the net cost was about

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C$660 million, or about 0.5 per cent of total federal programme spending and 0.06 per cent of GDP (Citizenship and Immigration Canada, 2002).

48. These data refer to the population over 15 years old. However, these comparisons are not adjusted for age composition, unlike those presented in Figure 25.

49. Moreover, because of this interruption, studies that use potential experience (age minus years of schooling minus six) as a proxy for actual experience tend to overestimate both the initial earnings gap suffered by immigrants relative to comparable natives and the rate of earnings assimilation. Hum and Simpson (2002), using data on actual work experience from the Survey on Labour and Income Dynamics (SLID), find that it captures a significant part of the assimilation process.

50. In the Census data used in most empirical analyses of the effects of language skills, language proficiency is self-assessed. The respondent is asked only whether he/she is able to conduct a conversation in English and/or French and whether he/she normally speaks it at home. Information is also available regarding the respondent's mother tongue. Using 1991 Census data, in a study focussing on Canada's three main cities, Pendakur and Pendakur (1998) found that individuals whose mother tongue was neither English nor French, after controlling for education, place of birth and other demographic characteristics, had an earnings gap of between 10 and 17 per cent relative to those whose mother tongue was English.

51. These results may overestimate the degree of complementarity, since language learning behaviour responds to economic incentives. For example, Chiswick and Miller (1992) found that the immigrants whose other observable characteristics increase the likelihood of economic success are also those who invest more in improving their language proficiency. This is likely to be true also of unobservable characteristics.

52. Pendakur and Pendakur (2002) find an earnings gap of around 5 per cent in the 1970s and 1980s for Canadian-born visible minorities (excluding Aboriginals, for which it is much larger), rising to 15 per cent by 1996. They also show, however, that earnings differences across ethnic groups within the "white" population and across different visible minorities often exceed the gap between the white and the visible minority groups taken as a whole. On the other hand, Hum and Simpson (2000) find no significant differences except for Blacks (they do not consider Aboriginals). Wannell and Caron (1994) find that university graduates belonging to visible minorities have virtually the same earnings levels, but higher unemployment and lower participation rates than white Canadians.

53. Annual employment earnings are often used as a summary measure of economic success, but such comparisons may miss some important elements, since patterns of behaviour of immigrants and Canadian-born people with respect to the underlying determinants of earnings (participation and unemployment rates; proportion with full-time, full-year employment; hours worked; hourly wages) are likely to differ. It can be argued that wage differentials would provide a better measure of the economic opportunities available to immigrants than earnings, which could also reflect, for example, compensating variations in the number of hours worked. For this reason, some studies limit the analysis to full-time full-year workers.

54. To analyse this time dimension, either a sequence of Census data or a longitudinal database, which tracks a representative panel of immigrants over time, is usually required. The longitudinal Immigration Database (IMDB), which links immigrant entry records with their tax records in subsequent years, allows the economic performance of immigrants to be tracked, but does not permit comparisons with a Canadian-born

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counterpart population with controls for the various characteristics that can influence performance. Another longitudinal database, the Longitudinal Survey of Immigrants to Canada (LSIC), modelled on its Australian counterpart, is currently under development.

55. McDonald and Worswick (1998) find that immigrants' lower job tenure relative to natives partly explains their lower earnings. During recessions this effect may be amplified if recent immigrants experience even greater difficulties in finding non-temporary jobs, which could contribute to a slower catch-up.

56. The stipulation of occupation-specific Mutual Recognition Agreements and the so-called Red Seal Program for skilled trades should ensure that, at least for the professions and trades they cover, recognition obtained in one province becomes portable.

57. For example, CICIC publishes information regarding employment in Canada for 133 regulated professions and trades on its web site. CICIC was established in 1990 following Canada's ratification of the 1979 UNESCO Convention on the Recognition of Studies, Diplomas and Degrees.

58. For example, the Canadian Council of Professional Engineers conducts evaluations of foreign credentials, though these are not binding on the provincial licensing bodies. And the Medical Council of Canada offers an examination to evaluate the competency level of foreign-trained doctors (Mata, 1999).

59. For example, some Ontario community colleges have instituted a streamlined procedure for assessing prior learning of midwives and physiotherapists, in order to facilitate access by foreign-trained candidates.

60. The Ontario Ministry of Training, Colleges and Universities, with funding assistance from the federal government, has developed the Specific Terminology, Information and Counselling project (STIC), which is intended to provide foreign-trained professionals with training manuals and self-assessment tools.

61. There is some evidence that immigrants to the three large cities have higher unemployment rates and lower earnings, relative to the Canadian-born, than those who settle in smaller cities and in the rest of Canada (Citizenship and Immigration Canada, 2001), but this is due at least in part to selection bias. Toronto, Montreal and Vancouver have higher concentrations of visible minority immigrants, while those from Europe tend to spread more evenly. And within each ethnic group, immigrants with weaker labour market prospects are probably more attracted by the presence of large ethnic communities in cities. Partly as a result of this, while on average immigrants are better educated than the Canadian-born, this is not the case in large cities, especially Toronto.

62. This does not reflect a move toward suburban areas further away from the city centre, since Canadian census metropolitan areas (CMAs) are broadly defined and include not only the city as an administrative unit, but also most of the surrounding suburban region.

63. According to the US Current Population Survey, 542 000 Canadian-born individuals were living in the United States in 1997, equal to 1.8 per cent of the Canadian population.

64. Workers in a number of professional occupations need only show proof of their qualifications and a job offer in order to gain entry to the United States, with no need for the employer to prove that US workers will not be negatively affected. Moreover, processing times have been reduced, and permits can be renewed without limit. Thus, it is likely that workers who would otherwise have used other temporary migration channels

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(such as the US H1-B visas) or who intend to emigrate permanently have shifted to this new channel.

65. Canadian Census data seem to be consistent with this order of magnitude. The number of pre-1996 immigrants born in Hong Kong (China) had decreased to 198 000 by 2001, from 241 000 recorded by the 1996 census. Of course, the overall attrition also includes emigration to other countries and those who died in that period (probably a small number, considering that immigrants from Hong Kong arrived mainly after 1980 and few of them were old).

66. This would seem consistent with the pattern of self-selection found by Borjas and Bratsberg (1996) for return migrants from the United States to countries with higher returns to skills (*i.e.* with a more unequal income distribution than the United States, *e.g.* several Latin American countries).

67. Of course, there are also effects of emigration on the source country. Although they are potentially important, they are not discussed here.

68. Therefore, for example, if new immigrants earn more (or less) than existing residents over their lifetimes, the purely statistical effects on average income per capita should not be counted.

69. Counting also births to immigrants, the contribution would be even higher. Denton *et al.* (2000) estimate that over half of population growth between 1951 and 2001 can be attributed to immigrants in this broader sense. In principle, immigrants can contribute to population growth also by having more favourable vital statistics than the rest of the population, but in practice these effects are likely to be small. In Canada, the foreign-born tend to have similar fertility rates as the Canadian-born and slightly longer life expectancy.

70. In addition, a change in the supply of immigrant labour need not affect factor prices if it is absorbed by a shift in the product mix towards sectors more intensive in the type of labour now more abundant.

71. For the United States, economy-wide effects are usually found to be of the order of magnitude of a 1 per cent fall in non-immigrant wages for a 10 per cent increase in the proportion of the foreign-born (Friedberg and Hunt, 1995).

72. Using a production function approach, Laryea (1998) finds that while immigrant professionals tend to substitute for Canadian professionals, unskilled immigrants are complementary to Canadian-born skilled workers and professionals.

73. The higher propensity of immigrants to choose self-employment could be a reaction to the difficulties they encounter in obtaining skilled jobs matching their foreign qualifications. The fact that self-employed immigrants – whether or not they belong to a visible minority – earn substantially more than those working as employees would appear to be consistent with this interpretation. However, the difference in earnings is approximately the same as for the Canadian-born after controlling for age, education, language skills and sector of work (Li, 1997).

74. The requirement, under the immigrant investor programme, that the immigrant make an interest-free deposit with the government of C$400 000, recoverable after five years (see Annex IV), should be regarded as a tax.

75. Of course, immigrants influence the balance of payments also through their remittances, which are recorded as private current transfers. Such remittances reached C$2 billion in 2002, although that includes payments by all Canadian residents, not just immigrants.

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76. Green and Sparks (1999), using an estimated VAR model, find that the population increase permanently shifted the path of per capita income upward by 5.7 per cent.

77. According to these estimates, based on existing sectoral studies, the elasticity of output to inputs was 1.2 on average for the transport, communications and public utilities sectors, and 1.03 for manufacturing. Given the weights of these sectors in Canadian GDP (respectively, 11 and 19 per cent), the study found that for the Canadian economy as a whole the elasticity would be 1.03. In the case of manufacturing it could be argued that economies of scale, if they exist, could also be exploited *via* trade. However, the size of the domestic market remains important. As discussed in Chapter II, plant size is an important factor explaining the difference in productivity between Canadian and US manufacturing firms.

78. Looking only at public infrastructure (and assuming that all of it would be still needed even with a lower population), its cost of maintenance can be approximated looking at capital consumption allowances in national accounts. In 2001, these were equivalent to 1.9 per cent of GDP for the general government. A one per cent decline in population would therefore raise the annual per-capita cost by one per cent, or 0.019 per cent of GDP per capita (about C$6).

79. In its Innovation Strategy (Government of Canada, 2002) the government identifies attracting highly skilled immigrants as an essential element – together with continuing to raise the quality and number of new graduates and providing opportunities for life-long learning – to ensure that the labour force can meet the challenges of a knowledge-based economy.

80. See, for example, the studies by Gould (1994) for the United States and Girma and Yu (2002) for the United Kingdom.

81. The studies (starting with McCallum, 1995) showing that Canadian provinces trade much more with each other and much less with US states located at comparable geographical distance from them – and, therefore, presumably having comparable transport costs – indirectly confirm the importance of such transactions costs. However, transactions costs result not only from a lack of information, but also from regulatory impediments to trade and exchange-rate variability.

82. The problem is that the analysis does not separate the effects connected to the composition of each immigrant cohort from age and time-since-arrival effects, partly because each immigrant cohort is not compared with an age-adjusted non-immigrant benchmark. On the one hand, the impact of recent immigrants is negatively conditioned by the fact that they are younger than the average Canadian population and are still suffering from the "arrival effect". On the other hand, the present impact of the immigrant cohorts who arrived in the 1960s and 1970s and had better economic outcomes than more recent immigrants is not a good guide to the impact of the latter. Whether the overall result is biased upward or downward is hard to say, since it depends on the net effect of those two biases.

83. Assuming unchanged age-specific profiles of taxes and expenditures, the simulation did not take into account other factors that could affect these age profiles (*e.g.* on the one hand, rising health costs; on the other hand, less-than-full income indexation of pensions).

84. Assimilation and cohort effects similar to these were found in studies on the United States (Borjas and Trejo, 1991), where, however, immigrants' rates of participation in welfare programmes are always higher than natives' and are highest just after arrival.

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85. These OECD debt figures exclude the value of unfunded government pension liabilities in order to make them more internationally comparable. If these unfunded liabilities were to be included, as in the official Canadian figures, the corresponding debt reduction would be from 88 per cent of GDP in 1995 to 56 per cent in 2002.

86. The 2003 budget showed the five-year impacts of *some* policies, such as increased health-care spending, but did not have a fully revised medium-term projection that incorporated all budget measures.

87. These tax points represent changes made in 1977 in which the federal government dropped its income tax rates while all provinces increased theirs by an equivalent amount.

88. Provincial accounts of these trusts depend on individual accounting conventions at the provincial level.

89. Using Finance Canada's "rules of thumb" for cyclical sensitivity, a 1 per cent drop in the level of GDP would eliminate most of the contingency reserve – the fiscal balance would fall by around C$2.6 billion in the second year after the shock. The impact would be smaller if the monetary authorities respond by cutting interest rates by 50 basis points (which is the response encoded in the Taylor rule): the fiscal balance would then fall by C$2 billion. Thus, when the contingency and prudence reserve are combined, fiscal settings could absorb a GDP shock of perhaps 1½-2 per cent provided that monetary policymakers also respond.

90. An alternative definition of sustainable fiscal policy – the one used in fiscal gap analysis – is that the net present value of future primary surpluses at least covers the current level of net public debt. Several studies have concluded from fiscal gap calculations that baseline fiscal policy is sustainable, but with the gap falling to zero or into unsustainable territory under scenarios in which health-care spending continues to rise faster than wages: for example, see Matier, Wu and Jackson (2001), Kennedy and Matier (2003) and IMF (2003). Frederiksen (2002) shows an unsustainable fiscal gap under baseline policy, largely because his initial primary surplus is much lower than in other studies (it does not include resource royalties, such as oil and gas royalties in the western provinces).

91. Both these estimates are broadly in line with those of other studies, after taking account of differences in assumptions. See IMF (2003) and Robson (2001). The projected rise in pension costs in OECD (2001d) is much higher because it included the C/QPP but used a transitional 7.8 per cent contribution rate rather than the new rate of 9.9 per cent that will apply from 2003-04.

92. Real per-capita health care spending in the OECD has grown by an (unweighted) average of 3 per cent per year since 1980 (on a non-age-adjusted basis). The pace of growth picked up slightly in the 1990s, to 3.1 per cent (1990-2000).

93. IMF (2002) estimates per-capita health-care spending to be 20 per cent higher than would be expected given factors such as per-capita incomes, the age structure and life expectancy, although this estimate has a large standard error. Fraser Institute (2002) also concludes from several alternative age-adjustment methods that Canada is the highest spender among OECD countries with universal access systems.

94. Canadian physicians are paid less and work longer hours than their US counterparts. The average income of Canadian physicians was around C$105 000 in 2001 according to Statistics Canada, compared with US$195 000 (in 1998, according to the American Medical Association), which is 60 per cent higher in PPP-adjusted terms, though some other sources suggest a smaller gap. Studies from the early 1990s show that: *i*) fees for individual

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procedures in the United States can be several times higher than those in Canada; *ii*) practice expenses, including malpractice premiums, are approximately the same in both countries as a proportion of income; and *iii*) the gap in net income is much lower than the gap in fees for procedures, implying Canadian doctors see more patients in order to boost their incomes (Welch *et al.*, 1993; Fuchs and Hahn, 1990; Buske, 1997).

95. The 70 per cent public funding consists of 45 per cent for CHA services and 25 per cent for provincial coverage over and above the CHA-mandated services.

96. For example, life expectancies at birth and at age 65 are high, premature mortality is low, infant health is good and work absences due to illnesses are low.

97. Over the 1990s, Canada invested 0.28 per cent of GDP in medical facilities and equipment, close to the 0.31 per cent in the other G7 economies but a little less than the OECD average of 0.34 per cent.

98. Ontario is encouraging groups of five or more family doctors to form networks in order to improve access to primary care, especially after hours. However, the reform is limited, as none of the pilot networks include health professionals other than doctors.

99. The Ontario government is considering changing its funding approach, but in a way that may not improve incentives to manage costs. It has proposed a mixture of population-based and institution-based funding. Local population information would be used to estimate the expected number of cases of each procedure or service; this would then be multiplied up by expected *institution-specific* costs per procedure. Hence, there would be little incentive to reduce costs as that would reduce next year's funding by an equal amount. See McKillop *et al.* (2001).

100. This is not exactly true: Canadians can buy care individually in the United States or any other country if they are willing and able to pay for it out of pocket.

101. Home care accounts for around 4½ per cent of provincial and territorial health spending, but the range is from 2 per cent in Nunavut to 10 per cent in British Columbia.

102. Of course, this would reduce their revenue-raising capacity and their effectiveness in curbing demand and increase administration costs. Rough estimates by Evans (2002) show that either 20 per cent coinsurance or a C$300 deductible could raise C$9 billion per annum, provided health-care spending was distributed evenly across the population However, in any year a high proportion of spending is accounted for by a small number of individuals, so many will not reach the C$300 deductible level, reducing the total amount raised. Similarly, exempting those in the lowest income decile and the permanently institutionalised population – who have minimal income but very high per-person health costs – would reduce coinsurance revenue by around a third, to C$6 billion. This is still a significant sum (6 per cent of total health-care spending) and assumes no change in behaviour by households, but is based on a fairly meagre exemption.

103. In December 2000, parental leave benefits were extended from 10 weeks to 35 weeks, effectively increasing the total maternity and parental leave time from six months to one year. As a consequence, the number of parents receiving EI parental benefits each month almost quadrupled to 123 000 by January 2002, while programme expenditure more than doubled to C$2.7 billion per year. See OECD (2000b) for details of parental and related benefits in other countries.

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Provincial and territorial social assistance benefits

	Constant 2001 Canadian dollars									Benefit level as percentage of ⅔ average income[1]		Percentage change in benefit
	1986	1990	1995	1996	1997	1998	1999	2000	2001	1991	2001	1995-2001
Newfoundland												
Single employable	5 056	4 851	4 836	2 752	1 200	1 206	1 204	1 679	3 070	31	15	–36.5
Person with a disability	–	9 727	9 291	9 141	9 068	9 022	8 938	8 807	8 656	61	43	–6.8
Single parent, one child	12 676	12 400	12 591	12 388	12 303	12 271	12 122	11 904	11 704	78	58	–7.0
Couple, two children	14 663	14 339	13 624	13 405	13 439	13 387	13 153	12 813	12 596	91	62	–7.5
Prince Edward Island												
Single employable	9 391	9 049	6 300	5 770	5 757	5 704	5 603	5 744	5 640	65	33	–10.5
Person with a disability	–	10 473	9 836	9 065	8 889	8 807	8 651	8 711	8 532	76	50	–13.3
Single parent, one child	12 945	12 631	11 811	11 266	10 800	10 375	9 778	9 844	9 564	91	56	–19.0
Couple, two children	18 969	18 665	17 735	16 360	16 409	15 715	14 715	15 005	14 521	135	86	–18.1
Nova Scotia												
Single employable	6 902	7 347	6 601	6 514	4 796	4 751	4 610	4 488	4 611	46	23	–30.1
Person with a disability	–	10 072	9 579	9 425	9 279	9 193	9 031	8 791	8 067	62	41	–15.8
Single parent, one child	11 952	12 230	11 806	11 616	11 436	11 140	10 700	10 304	9 284	76	47	–21.4
Couple, two children	14 387	14 976	13 944	14 962	15 153	14 696	13 486	13 195	13 474	93	68	–3.4
New Brunswick												
Single employable	3 402	3 627	3 461	3 445	3 431	3 399	3 339	3 250	3 168	23	16	–8.5
Person with a disability	–	9 497	7 164	7 131	7 216	7 185	7 058	6 870	6 696	59	35	–6.5
Single parent, one child	10 217	9 827	10 594	10 530	10 657	10 648	10 460	10 180	9 922	61	51	–6.3
Couple, two children	11 053	10 617	11 860	11 782	12 091	12 159	11 944	11 622	11 328	66	59	–4.5
Quebec												
Single employable	3 581	6 531	6 708	6 600	6 401	6 309	6 349	6 238	6 209	36	30	–7.4
Person with a disability	–	8 583	9 042	9 095	9 084	9 168	9 182	9 074	9 065	47	44	0.3
Single parent, one child	12 049	11 915	12 888	12 681	12 012	11 613	11 110	10 558	10 351	66	50	–19.7
Couple, two children	15 573	14 290	15 120	14 876	14 116	13 617	12 840	12 298	12 041	79	58	–20.4
Ontario												
Single employable	7 652	8 987	8 829	7 242	7 173	7 107	6 981	6 795	6 623	47	29	–25.0
Person with a disability	–	12 677	12 819	12 613	12 418	12 303	12 085	11 764	11 466	67	50	–10.6
Single parent, one child	13 706	16 553	15 994	13 134	12 955	12 508	11 879	11 381	10 861	87	47	–32.1
Couple, two children	17 060	21 719	20 595	16 971	16 741	16 036	15 041	14 277	13 452	115	58	–34.7

Provincial and territorial social assistance benefits (*cont.*)

	Constant 2001 Canadian dollars									Benefit level as percentage of ⅔ average income[1]		Percentage change in benefit
	1986	1990	1995	1996	1997	1998	1999	2000	2001	1991	2001	1995-2001
Manitoba												
Single employable	7 593	7 984	7 220	6 677	5 796	5 743	5 641	5 491	5 352	48	27	–25.9
Person with a disability	–	8 458	8 940	8 797	8 661	8 581	8 502	8 328	8 117	50	41	–9.2
Single parent, one child	11 370	11 362	10 773	10 600	10 436	10 015	9 424	9 204	9 363	68	48	–13.1
Couple, two children	17 773	20 083	18 753	16 800	15 543	14 670	13 688	13 181	12 847	120	65	–31.5
Saskatchewan												
Single employable	6 356	6 220	6 440	6 336	5 709	5 674	5 839	5 797	5 772	36	29	–10.4
Person with a disability	–	10 042	9 257	9 372	8 379	8 380	8 601	8 473	8 424	58	42	–9.0
Single parent, one child	13 042	12 627	11 606	11 419	11 243	9 966	9 995	9 700	9 401	73	47	–19.0
Couple, two children	18 295	17 511	16 550	16 283	15 097	14 526	14 495	13 813	13 332	101	67	–19.4
Alberta												
Single employable	9 045	5 999	5 286	5 201	5 151	5 176	5 084	4 949	4 824	34	22	–8.7
Person with a disability	–	7 419	7 365	7 247	7 165	7 172	7 228	7 572	7 380	42	33	0.2
Single parent, one child	13 43	11 248	10 277	10 110	10 037	9 854	9 569	9 201	8 741	64	40	–14.9
Couple, two children	19 690	16 573	16 347	16 084	15 911	15 522	14 739	14 233	13 425	94	61	–17.9
British Columbia												
Single employable	6 460	7 249	7 420	6 744	6 640	6 579	6 462	6 342	6 251	38	28	–15.8
Person with a disability	–	10 122	10 439	10 271	10 112	10 019	9 841	9 659	9 522	53	43	–8.8
Single parent, one child	11 635	12 945	13 376	13 160	12 903	12 459	11 837	11 446	11 103	68	50	–17.0
Couple, two children	15 891	16 110	17 058	16 784	16 416	15 723	14 748	14 109	13 534	85	61	–20.7
Yukon												
Single employable	7 676	9 381	8 827	8 685	11 853	11 744	11 536	11 230	12 045	44	55	36.5
Person with a disability	–	10 430	10 504	10 335	13 478	13 353	13 117	12 769	13 545	49	62	29.0
Single parent, one child	13 786	15 418	14 659	14 423	17 832	17 342	16 622	15 989	16 449	72	75	12.2
Couple, two children	21 122	22 849	22 097	21 742	25 683	24 904	23 741	22 732	22 786	107	104	3.1
Northwest Territories												
Single employable	–	–	12 638	12 352	7 798	8 005	8 958	8 720	8 731	–	34	–30.9
Person with a disability	–	–	14 315	14 332	10 332	10 301	11 213	10 915	11 091	–	43	–22.5
Single parent, one child	–	–	21 425	20 981	18 363	18 164	19 099	18 401	17 824	–	69	–16.8
Couple, two children	–	–	25 396	24 856	24 618	24 640	25 056	24 012	23 021	–	89	–9.4

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Provincial and territorial social assistance benefits (*cont.*)

	Constant 2001 Canadian dollars									Benefit level as percentage of ⅔ average income[1]		Percentage change in benefit
	1986	1990	1995	1996	1997	1998	1999	2000	2001	1991	2001	1995-2001
Nunavut												
Single employable	–	–	–	–	–	–	10 738	10 453	10 188	–	41	–
Person with a disability	–	–	–	–	–	–	12 952	12 607	12 288	–	50	–
Single parent, one child	–	–	–	–	–	–	27 099	26 189	25 290	–	102	–
Couple, two children	–	–	–	–	–	–	32 048	30 818	29 567	–	120	–

1. Average income calculated as average hourly earnings of employees paid by the hour for each province, multiplied by 2 000 hours per year. "Employable" beneficiaries covers everyone except the disabled and others with young children.

Source: National Council of Welfare and Statistics Canada.

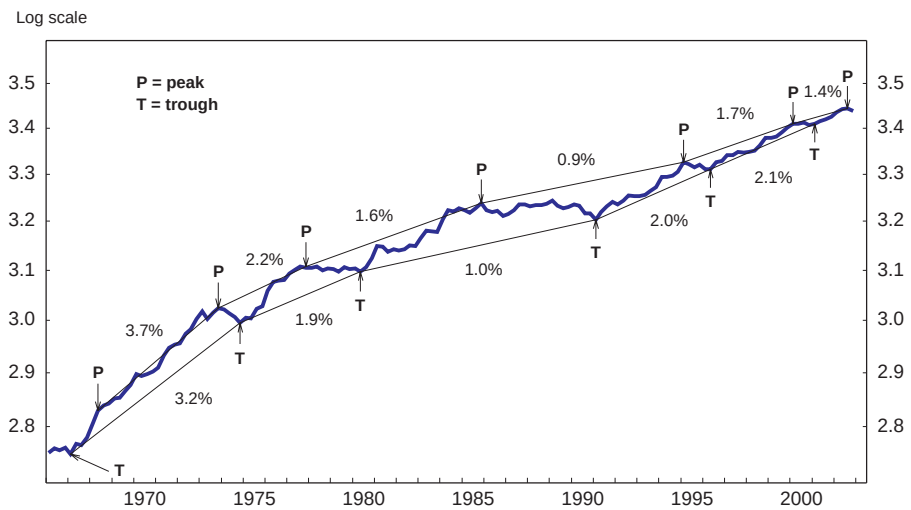
Annex I

Has trend productivity growth picked up?

Figure A1 shows Canadian labour productivity, measured as the logarithm of real output per hour worked in the business sector. In the decade to 2002, business-sector output grew at an average rate of 4.4 per cent per annum, employment rose by 2.3 per cent a year, and annual hours worked per employee increased by around 0.1 per cent per annum. The net result is labour productivity growth averaging a healthy 2 per cent, almost matching the United States' 2.1 per cent (for the non-farm business sector).

At least some of this good performance reflects a cyclical rebound because productivity growth is pro-cyclical, and the economy in 1992 was in the depths of recession while output in 2002 was probably around or above trend. There are many ways to split a time series into its cyclical and trend components, ranging from simply eye-balling the data to more sophisticated econometric approaches. This Annex discusses several approaches in order to

Figure A1. **Labour productivity in the business sector**



Source: Statistics Canada; OECD.

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see whether productivity growth picked up in the 1990s, and – more speculatively – whether it accelerated further in the second half of the decade.[1]

Perhaps the simplest way to correct for the business cycle is to measure growth rates from peak to peak or trough to trough. This is done in Figure A1, which clearly shows the slowdown in productivity growth after the first oil shock (from around 3½ to 2 per cent), a further deceleration from the early 1980s and a partial recovery in the 1990s. This approach would suggest a trend productivity growth rate around 1¾ to 2 per cent since the second half of the 1990s.

Statistical filters

An alternative approach is to use a statistical filter such as the omnipresent Hodrick-Prescott (HP) filter. While the HP filter is simple and well understood, it has at least two practical disadvantages. *First*, the results depend almost entirely on the arbitrary smoothing parameter, λ. *Second*, it has a tendency to "chase the data" at each end of the sample, so if the last few observations are part of a cyclical upswing then the estimated trend will be twisted upwards.

The most common approach to choosing the smoothing parameter is to use the industry-standard value of 1 600, which works well for US GNP but not for much else. In principle, the aim is to smooth the data just enough until all traces of the business cycle have been removed. One way of doing this would be to use additional information, such as the capacity utilisation rate, to help identify the cycle and then choose λ so the resulting trend growth rate is not correlated with that independent measure of the cycle. Table A1 shows the results of such an exercise in which the change in trend growth rate, estimated with an HP filter, was regressed on three lags of the change in the industrial capacity utilisation rate. Trend productivity growth calculated with the standard value of $\lambda = 1\ 600$ is still highly correlated with capacity utilisation, and therefore retains a significant amount of cyclical variation. The smoothing parameter has to be pushed towards 8 000 before the cycle is fully removed.[2] A range of trend productivity growth rates using different values of λ are shown in Figure A2. They all show productivity recovering to a little under 2 per cent per annum, starting around 1990 and flattening off after 1995. Visually, however, they all still appear to be affected by the cyclical slump in the first half of the 1990s.

The second problem with the HP filter – the trend being biased at the end of the sample – is usually dealt with by extending the data, usually with forecast values, and filtering over the longer sample period. This has the obvious problem that optimistic forecasts will lead to an optimistic estimate of the *current* trend rate of growth. The impact of the end-point problem is illustrated in Figure A3. The simple HP filter is run until the end of 2002; the

Table A1. **Choosing the smoothing parameter**

Smoothing parameter (λ)	Test of whether lags of capacity utilisation are significant (p-value)
100	0.0000*
1 600	0.0002*
3 200	0.007*
4 800	0.03*
6 400	0.08*
8 000	0.19
10 000	0.25

Note: A * indicates the lags are significant at the 10 per cent level.
Source: OECD.

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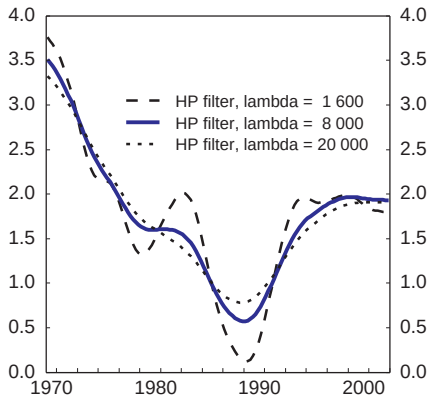
Figure A2. **Alternative HP filters**

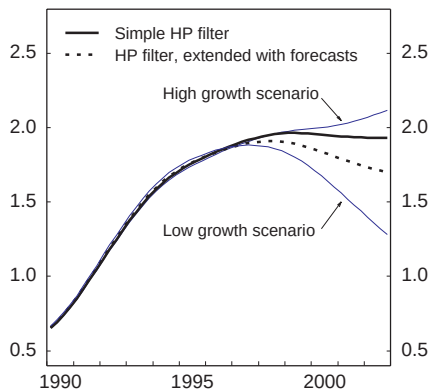
Trend labour productivity growth, annual percentage change



Source: OECD.

Figure A3. **End-point adjustments**

Trend labour productivity growth, annual percentage change



Source: OECD.

extended filter assumes productivity growth of 1¾ per cent from 2002 to 2006, while the high and low growth alternatives assume that productivity growth is 1 per cent higher or lower respectively out to 2006 (this range of 1 per cent is relatively wide, so can be thought of as likely upper and lower bounds). The medium-term forecast clearly has a large impact on the estimate of the current trend growth rate: a 1 per cent increase in assumed productivity growth from 2003-07 raises the estimate of current trend productivity growth by around 0.5 percentage point.

Aside from these practical problems, the HP filter has the conceptual problem that it tries to identify the cycle by assuming that none exists. More specifically, if p is log productivity, and p^* the estimated trend, then the HP filter can be written as the problem of finding p^* subject to the constraints:

$$\Delta p_t^* = \Delta p_{t-1}^* + \eta_t,$$
$$p - p^* = \varepsilon_t$$

where ε_t is a random error with a variance θ, and η is a random error with variance θ/λ, where λ is the smoothing parameter. The first equation makes the plausible assumption that trend growth is a random walk, but the second assumes the "cyclical" component is random, or white noise.[3] All univariate filters have to make some identifying assumptions in order to split trend from cycle, but the HP assumptions make little economic sense in the context of a normal business cycle. A more plausible alternative is to assume that the cyclical component is autoregressive; *i.e.* to replace the second equation with

$$p - p^* = \alpha(p - p^*)_{t-1} + \beta(p - p^*)_{t-2} + \varepsilon_t$$

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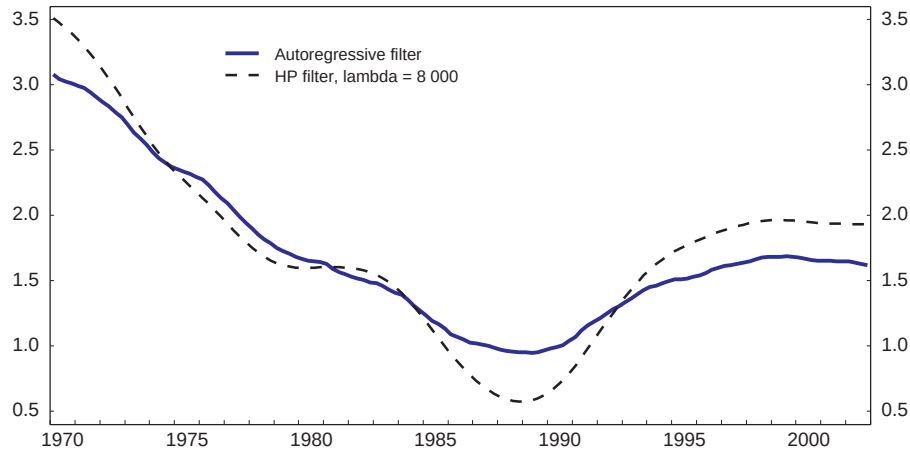
Figure A4. **HP *versus* autoregressive filter**

Trend labour productivity growth, annual percentage change



Source: OECD.

and to freely estimate the variances rather than impose the smoothness constraint. This variant can be estimated using the Kalman Filter. The result is shown in Figure A4. It shows the slide in productivity growth halting around 1990, rising to around 1¾ per cent per annum and then levelling off towards the end of the decade.

Econometric approaches

The main drawback of all single-variable filters is that they ignore other information that may help separate trend from cycle. An alternative approach is to use an econometric model to incorporate this sort of extra information. For example, the behaviour of inflation should give some clues regarding the state of the business cycle: if inflation is rising then the output gap is probably positive, other things equal. This sort of approach can be implemented by estimating a Phillips curve for inflation and using the Kalman filter to find potential output (or, equivalently, the output gap). In its simplest form, this involves estimating a model such as:

$$\pi_t = \pi_t^e + \beta(y - y^*)_t + \gamma(\pi_t^m - \pi_{t-1}) + \delta . lags \, \Delta \pi_t^m ,$$

$$y_t^* = \mu + y_{t-1}^* + \varepsilon_t .$$

where π_t is the rate of core consumer price inflation, π^e is inflation expectations, π^m is import price inflation, y is (log) real output, and y^* is the (unknown) trend output. The first equation is a standard Phillips curve in which inflation depends on import prices and the output gap, (y-y^*). The second equation assumes that potential output is a random walk with drift

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(although other assumptions are possible). The Kalman filter is then used to estimate the parameters of the equation and the unobserved variable, y^*. A similar approach was used by Richardson *et al.* (2001) to estimate NAIRUs for 21 OECD countries, except they used the unemployment gap, (u-u^*), rather than the output gap, (y-y^*), as their cyclical indicator.

Although this model can be used to estimate potential output, more information is needed to identify potential labour productivity. Output is equal to labour input (hours worked) multiplied by labour productivity, so the output gap can be split into a labour input gap and a productivity gap:

$$y - y^* = (n - n^*) + (p - p^*)$$

Both gaps can be put into the Phillips curve (and the simplifying assumption made that the coefficient, β, is the same for each gap). However, more information is needed to separately identify trend labour input, n^*, and trend productivity, p^*, because the output gap can be split in an unlimited number of ways (for example, an enormously negative employment gap (n-n^*) plus an enormously positive productivity gap (p-p^*), or *vice versa*, both of which will be observationally equivalent in the Phillips curve). To pin the two gaps down, the industrial capacity utilisation rate (*capu*) is used as extra identifying information. The assumption is made that the employment gap and the productivity gap are both pro-cyclical, *i.e.* positively correlated with capacity utilisation. With these assumptions, the full model becomes:

$$\pi_t = \pi_t^e + \beta(n - n^*)_t + \beta(p - p^*)_t + \gamma(\pi_t^m - \pi_{t-1}) + \delta.lags\,\Delta\pi_t^m,$$

$$n_t^* = \mu_1 + n_{t-1}^* + \varepsilon_{1t}$$

$$p_t^* = \mu_2 + p_{t-1}^* + \varepsilon_{2t}$$

$$n - n^* = \varphi_1.capu + \eta_1$$

$$p - p^* = \varphi_2.capu + \eta_2$$

The detailed specification of the variables is the same as in Richardson *et al.* (2001), except for the use of business sector employment and productivity in place of the unemployment gap. The general estimation approach is also discussed in that paper,[4] and the results are summarised in Figure A5. The left-hand panel shows a similar story to the other estimates: trend productivity growth picks up in the first half of the 1990s, then stabilises around 1¾ per cent per annum. There does not appear to be a further acceleration in the second half of the decade. The right-hand panel adds on trend hours growth (n^*) to get an estimate of overall business sector potential output growth. This estimate is broadly consistent with (but smoother than[5]) the OECD's production-function-based estimate of potential output for the business sector. It shows that the decline in potential output growth throughout the 1970s and 1980s can be largely attributed to a decline in trend productivity growth, with only a small contribution coming from trend hours or employment. Since 1992, however, the pickup in trend labour input growth has played a bigger role than in the earlier period, reflecting the fall in structural unemployment and a recovery in the trend participation rate.

Production-function-based estimates

The OECD uses a production function approach to estimate potential (trend) labour productivity and potential output. The constant-returns-to-scale Cobb-Douglas function can be written in log form as:

$$y = \theta + \alpha\, n + (1 - \alpha)k$$

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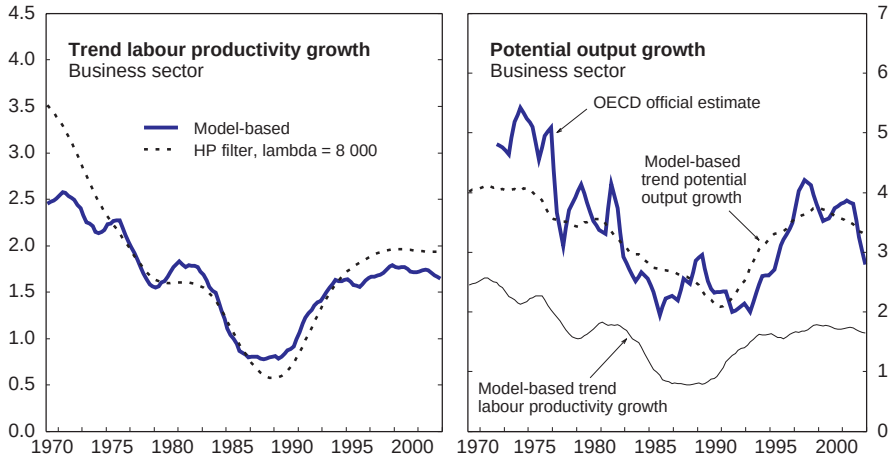
Figure A5. **Model-based estimates**
Annual percentage change



Source: OECD.

where α is the average wage share of GDP, k is the log of the capital stock, and θ is the log of multifactor productivity (MFP). Potential output, y^*, is calculated using this formula by replacing the actual inputs with their trend values: θ^*, n^*, and k^*. Trend MFP is estimated by running an HP filter through actual MFP, while trend capital, k^*, is assumed to be equal to actual capital, k, since the cyclical component of the capital stock is usually small. The labour input is more complicated. It can be written as:

$$N = H.E.\rho.P - E^{govt}$$

where H is hours worked per person, E is the economy-wide employment rate, ρ is the participation rate, P is the population of working age, and E^{govt} is government employment. Trend hours and participation, H^* and ρ^*, are estimated by running an HP filter through actual H and ρ; the working-age population and E^{govt} are not detrended since they do not have significant cyclical components; and potential employment is calculated as $(1 - u^*)$, where u^* is the estimated NAIRU.

The results of this exercise are shown in Figure A6.[6] Ignoring the short term volatility,[7] the production function estimates show a slightly different pattern to most of the others. The productivity decline stops a little earlier – in the mid-1980s, when the reform process began in earnest – then is low but stable for the following decade before picking up from the mid-1990s, a few years later than the other estimates. The production function approach appears to be less distorted by the cyclical downturn from 1990-94, possibly because that downturn had a bigger cyclical impact on labour productivity than on multifactor productivity.[8]

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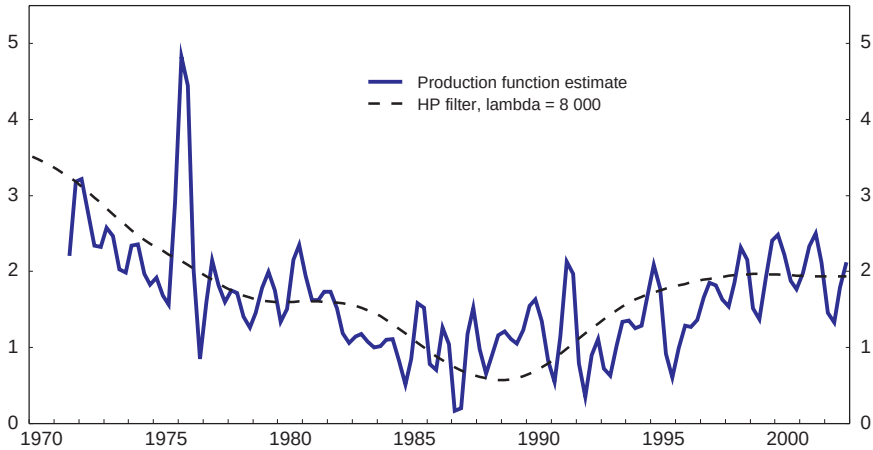
Figure A6. **Production function estimate**

Trend labour productivity growth, annual percentage change



Source: OECD.

Multifactor (residual) productivity

Labour productivity can pick up for many reasons, including an increase in capital per worker (capital deepening) and an improvement in the composition of the workforce (a greater proportion of high-productivity workers), as well as individuals and firms increasing their technical efficiency. Multifactor productivity can be calculated as the residual after the first two factors are taken into account. Measures of MFP, however, are subject to wide margins of error as both the capital stock and labour quality are difficult to measure. Hence, estimates of trend MFP growth are much more speculative than trend labour productivity growth.

The estimates shown in Figure A7 are based on Statistics Canada's MFP estimates,[9] which is measured in a reasonably comparable way to the Bureau of Labor Statistics' estimate for the United States. The business-sector capital stock is measured on a net basis using geometric depreciation and excluding owner-occupied dwellings. Labour quality is measured in a conceptually similar way to the chain-linked capital stock, with labour inputs weighted by relative wage rates. Both countries have seen a shift to higher-wage occupations – which is assumed to imply higher-quality workers – and therefore the labour input grows faster than if no quality adjustment was made. That in turn lowers estimated MFP (residual productivity) growth.

Two estimates of the trend are shown: an HP filter through the annual data, with $\lambda = 500$; and the alternative univariate filter in which the cyclical component is assumed to be autoregressive rather than white noise. There is a clear productivity pickup, more so for the alternative filter, but trend MFP growth rises by less than trend labour productivity growth, so at

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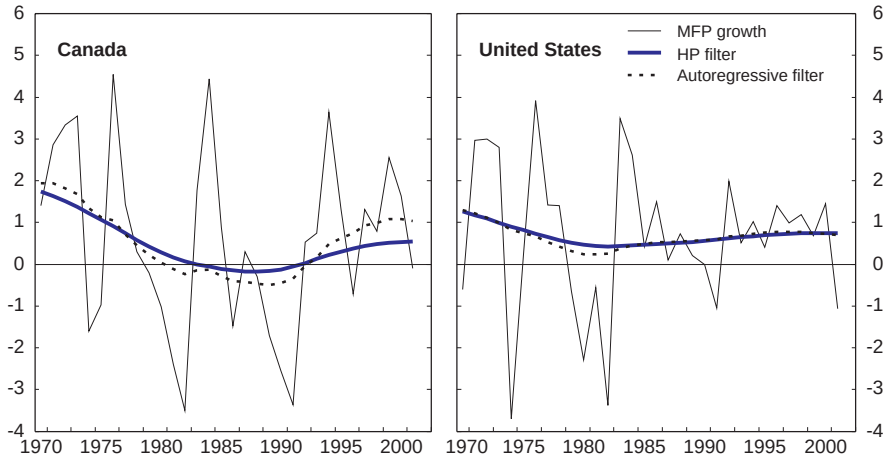
Figure A7. **MFP growth**
Annual percentage change



Source: Statistics Canada; US Bureau of Labor Statistics; OECD.

least some of the improvement is due to a shift to a better-quality workforce and to investment in physical capital. By the end of the period, Canadian and United States trend MFP growth rates were similar – at a little under 1 per cent per year.

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Notes

1. These issues are also investigated in Robidoux and Wong (2003).
2. The Secretariat also uses a value of 8 000 when filtering MFP as part of its process of estimating potential output.
3. In principle, the optimal value of λ can be estimated at the same time as the trend, p*, by using a Kalman filter. In this case, doing so results in a λ of just 1 (*i.e.* virtually no smoothing), which is not surprising given the HP filter's underlying assumption that there is no business cycle at all.
4. The main practical problem is choosing the variances in the state equation, since it is usually impossible to econometrically estimate them in small samples. For this exercise, they were chosen so that trend productivity and employment growth ended up approximately as smooth as the OECD's official trend estimates, although several sensitivity tests were performed.
5. The short term variability in the production-function-based estimate comes from changes in growth of the working-age population. The production function estimates are constructed on a bottom-up basis, being built up from smoothed (detrended) values of the inputs. However, the population is not smoothed on the grounds that variations in working age population are not cyclical in nature.
6. These estimates differ slightly from the Secretariat's published productivity estimates because, while the methodological approach is identical, this Annex uses Statistics Canada's definition of the business sector, and hence uses slightly different series for output, employment, and hours worked.
7. Trend labour productivity has some short-term volatility because, as noted above, the working-age population is not smoothed. Hence, if the working-age population jumps by, for example, 1 per cent, then the available business-sector workforce also increases by 1 per cent. However, with an unchanged capital stock, potential business-sector output will rise by around 0.67 per cent (since the wage share is 0.67). Potential labour productivity therefore falls by 0.33 per cent, at least until the new workers are provided with capital.
8. A potential drawback of the production function approach is that it requires estimates of the capital stock, which can be highly uncertain. In practice, however, it is unlikely to significantly affect estimates of potential labour productivity, since errors in measuring the capital stock will tend to be soaked up by (offset by) MFP. In any case, both gross and net capital stocks were used as a sensitivity check and produced very similar results, despite large differences in their respective growth rates over the 1990s.
9. For Canada, two different MFP series are spliced together in 1981. There are methodological differences between the two halves of the sample, the most important of which is that no adjustment for changes in labour quality is made before 1981. Note that these MFP estimates are not the same as the implicit MFP in the previous sections (which are calculated as a simple residual from the production function).

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Annex II

An experiment in "making work pay": the self sufficiency project

The Self Sufficiency Project (SSP) was an experimental study of the impact of wage subsidies on single parents in receipt of social assistance. Around 6 000 parents from two provinces (British Columbia and New Brunswick) were randomly recruited to the study between 1992 and 1995, half of whom were offered the subsidy while the other half were set aside in a control group. It was designed to test whether financial incentives would encourage parents to move off, and stay off, welfare. The subsidy was paid to single parents who, within twelve months of signing up for the project, moved into full-time work. The financial incentive was large, roughly doubling earnings for someone working full time at the minimum wage. However, it was a limited-time offer. It could be collected for a maximum of three years, provided the recipient kept working and stayed off social assistance.

The SSP consisted of three linked studies. The main study was as described above, but targeted at long-term benefit recipients so that it focussed on one of the more difficult labour-market groups. To be eligible, a parent had to have been receiving income assistance for more than a year. The "SSP Plus" experiment was the same, except that this small sample of 299 New Brunswick recipients was also offered extra employment support services. These included help with preparing a résumé, job coaching, job leads and the preparation of an employment plan. The third group consisted of those new to welfare. They were enrolled in SSP when they first began receiving social assistance, becoming eligible for the wage subsidy if they stayed on welfare for a year and then found a full-time job. This was done to test whether in-work benefits had a "revolving door" effect whereby someone might stay on welfare longer in order to get the wage subsidy.

The SSP *increased employment*, at least initially (Figure A8). One-third of the main group left welfare and took up full time work. On average, they received the supplement for 22 months over their three years of eligibility and received C$18 000 in supplement payments. At the peak of its impact, the SSP doubled the proportion of sample members working full time (relative to the control group). The results were consistent across a range of family types, such as the age of the youngest child. The key question is what happened after the money was turned off. Although there was *no long-run impact on employment compared with the control group*, it is not obvious how to interpret this. The impact fell back to zero because employment rates of the control group steadily rose, not because parents who took up the subsidy lost their job or quit after the subsidy ended. It is impossible to know whether the differential impact would have been different if employment rates had not been recovering strongly throughout the economy. A second factor is that the policy environment was not constant: benefit provisions for the control group were toughened in the mid-1990s in both provinces. However, even though there was no permanent impact on the level of employment, the faster integration of programme group members into full-time paid work represents a significant welfare gain. A further result is that, unsurprisingly, the supplement *raised income and reduced poverty*. In

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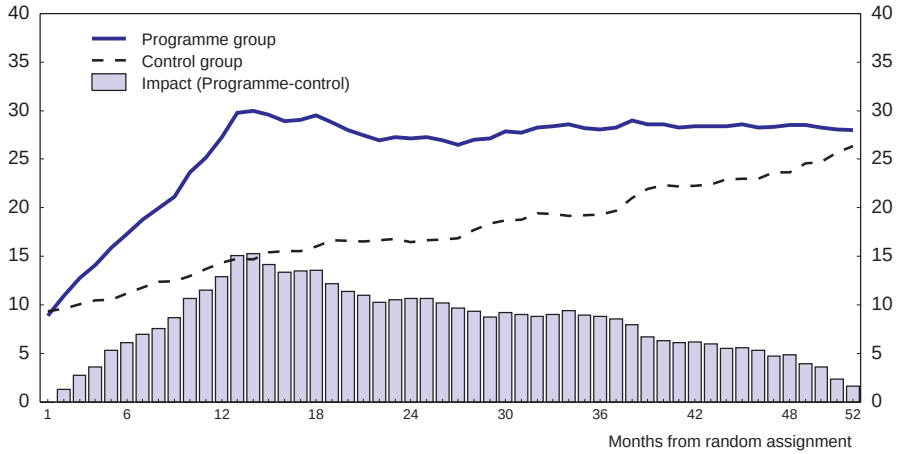
Figure A8. **In-work benefits had a large but temporary impact**

Percentage employed full-time, by months from random assignment



Source: Greenwood (2002).

addition, the *performance of elementary school-age children improved*. Amongst the SSP-plus sample, who were offered extra employment assistance, many more found jobs (half in the first year, compared with a third in the regular sample), but they were more likely to lose them, perhaps because the extra initial success rate meant that the experiment was digging deeper into the welfare pool. But there was a more noticeable long-term impact than for the main study, suggesting that the *employment support measures had a lasting impact* on behaviour, although the small sample size weakens this conclusion. The impact was even greater for the *new entrants to welfare*, but there was no evidence of them lengthening their welfare spells in order to get the wage subsidy. There was, however, a small *net fiscal cost*: increased tax revenue, and welfare savings were not enough to offset the higher transfers and the administration costs.

A related project was the *Earnings Supplement Project* (ESP) which was aimed at the unemployed and repeat users of EI. It offered a top-up to wages for those who went back to full-time work quickly (26 weeks for displaced workers; only 12 weeks for repeat EI users) but only if they did not go back to their previous employer. Impacts were negligible for repeat users, mainly because 90 per cent of those assigned to the programme expected to be able to return to their usual job. The incentive was not enough to encourage people to break out of their pattern of seasonal work and drawing EI in the off-season. Impacts were positive but very small and short-lived for the non-regular users.

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Annex III

The innovation strategy

In February 2002 the government released its *Innovation Strategy*, which lays out an ambitious agenda for closing the innovation gap with the United States. The first part, encouraging innovation, has four goals: *i*) creating and using knowledge commercially; *ii*) maintaining the supply of knowledge workers; *iii*) creating the right regulatory environment; and *iv*) strengthening communities so that innovation is spread across the country. The second part, improving skills, focuses on: *i*) lifelong learning; *ii*) tertiary education; *iii*) sharpening work-related skills; and *iv*) helping immigrants into the workforce. Below that are at least 30 specific targets. These include those that are clear and measurable (albeit very ambitious), such as "by 2010, rank among the top five countries in the world in terms of research and development (R&D) performance." They also include more nebulous and hard-to-measure goals, such as "by 2010, significantly improve the innovation performance of communities across Canada," and "support the development and dissemination of knowledge and information on adult learning".

The *Strategy* itself does not lay out the policy measures that would be required in order to hit these targets nor does it resolve tensions between the goals. For instance, it argues that innovation thrives in industrial clusters, which tend to be in or near the big cities, and these need "the right kind of support." Yet it also calls for more innovation in smaller communities. If this simply says that more innovation, everywhere, is a good thing then the strategy is vacuous. The difficult question about whether research should localised or diffuse is not addressed. To be fair, however, few governments have clear views on trade-offs like this, and there will be a great deal of learning-by-doing as different approaches are tried.

A national summit was held in November 2002, attended by representatives from business, academia, non-government organisations and provincial and federal governments. Its task was to prioritise the goals and give specific recommendations on how to achieve them. Its priority recommendations were grouped into five areas, and are listed below.

Improving research, development and commercialisation

1. Improve the relationship between the receptor community and universities, colleges and researchers. Strengthen receptor capacity (*e.g.* by establishing clusters and providing incentives for the private sector to commercialise new products).
2. Continue to expand university-based research by strengthening research infrastructure (for example, establish a permanent programme for the reimbursement of indirect costs; expand research funding to the granting councils, to the Canada Foundation for Innovation and to the Canada Research Chairs; and ensure participation across universities of all sizes and disciplines).

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3. Expand successful programmes that support commercialisation by broadening and deepening the mandates of programmes (for example, the Industrial Research Assistance Program and Technology Partnerships Canada).

Enhancing the innovation environment

4. Reduce or eliminate capital taxes at the federal and provincial levels.
5. Improve the functioning of the Scientific Research and Experimental Development (SR&ED) tax credit (*e.g.* make it fully refundable, broaden the definition of R&D to include commercialisation, and simplify the application process).
6. Develop new tax-based instruments to stimulate seed and early-stage investments (for example, tax credits for angel investors).
7. Accelerate regulatory reform (to 2005).

Strengthening the learning culture

8. Establish a pan-Canadian literacy and essential skills development system, supported by federal, provincial and territorial governments. Establish programmes to improve literacy and basic skills based on individual and community needs and interests.
9. Integrate innovation-related skills in curricula (including interdisciplinary, cross-curricular, risk-taking, problem-solving approaches to learning).
10. Adjust the system of student financial assistance to meet the changing needs of students, the post-secondary education sector and the knowledge-based economy (for example, assistance levels, debt and repayment issues, under-represented groups, e-learning).
11. Expand capacity in the post-secondary system by increasing infrastructure (physical, human, financial) using cost-effective design principles.

Building an inclusive and skilled workforce

12. Increase participation levels of under-employed groups (including women, youth, people with disabilities, visible minorities and Aboriginal people).
    - Encourage the use of Prior Learning Assessment and Recognition (PLAR) through occupation-based projects to include non-formal and informal learning and skills.
    - Improve access to training, apprenticeship and post-secondary internship programmes to target Aboriginal people, persons with disabilities, new Canadians and other under-employed groups.
13. Provide incentives and programmes (for individuals and organisations) to increase in-house training and apprenticeship training carried out by industry.
    - Use sector councils to channel workplace training programmes.
    - Create a new industry-led training corporation or a number of sectoral training bodies to champion and oversee training for trades and technical skills that are in high demand; make use of college and university programmes as well.
    - Expand apprenticeship programmes, and create more relevant industry training programmes through partnerships between industry, government and academic institutions.

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14. Improve the process for recognising foreign credentials (for example, allow the process to begin overseas; coordinate credential evaluation processes; set up a single source of information on licensing requirements; establish norms for work experience; develop resources for employers; and use programmes such as the Canadian Council for Human Resources in the Environment Industry).

Strengthening communities

15. Collaborate with local (rural, urban and Aboriginal) social, economic and community development stakeholders across municipal, business, voluntary and NGO sectors to prepare long-term community innovation plans and strategies.
16. Support the growth of clusters by fostering the development of local "kernel" research institutions and schools, facilitating cross-community partnerships, providing information on best practices for building clusters and creating entrepreneurial networks; and capitalising on the benefits and synergies of geography to encourage spin-offs, innovation and the dissemination of ideas.
17. Extend broadband access across the country into rural and remote areas (according to community-specific access needs) to bring cultural, social (e-health), economic (e-business) and learning (e-learning) communities of interest together to encourage the dissemination, implementation and customisation of community solutions (e-everything).
18. Enhance the learning capacity of children, youth and adults from rural and Aboriginal communities by providing support programmes and educational opportunities tailored to the needs of the local community: develop content to match technology; establish local centres of excellence; set up strong schools; establish community access centres; increase broadband to support distance education; engage young people in actual innovation; and increase access to capital (for example, tax-free bonds).

Source: The *Innovation Strategy* is available at *www.innovationstrategy.gc.ca*.

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Annex IV

Main categories of immigrants, refugees and temporary residents

This Annex describes the main categories of permanent immigrants, refugees and temporary residents envisaged under the existing Canadian legislation. The latest legislation, together with the regulations for its application, came into force in June 2002. The main broad categories of immigrants envisaged in it are the same as defined by the 1962 Immigration Act, although the conditions for admission have evolved over time, and some specific programmes are more recent (for example, the three business immigration programmes were introduced in 1978). A summary of the evolution of the points system for the selection of skilled workers since it was first introduced in 1967 is presented in Table A2.

In Canada, immigration is a shared jurisdiction between the federal and the provincial/territorial governments. As discussed in Chapter II, the degree to which responsibilities in this field are decentralised differs across provinces/territories according to the content of their individual immigration agreements with the federal government. The province of Quebec has the broadest degree of autonomy in this field, including powers of selection of its economic immigrants, for which its uses its own points system.

Permanent immigrants and refugees

Family class immigrants

They include close relatives (spouses or common-law partners, dependent children, parents and grandparents) sponsored by a permanent resident or citizen of Canada who is at least 18 years of age. The sponsored immigrant can be accompanied by his/her dependent children. The sponsor must commit to provide for the maintenance of the sponsored immigrant for 10 years (3 years for a spouse; 10 years or until age 22, whichever is longer, for dependent children).

Economic immigrants

Skilled workers: immigrants selected for their suitability for the Canadian labour force based on an assessment of their skills. Assessment is based on a system that assigns points for age, education, work experience, knowledge of Canadian languages and "adaptability"; additional points are given if the applicant already has a permanent job offer (see Table A3). Spouses and dependent children are admitted along with the principal applicant, without being individually skills-tested, but the principal applicant can obtain extra points for the spouse's education level, regarded as an indicator of adaptability. This category also includes assisted relatives (formerly also called "nominated relatives"), whose age and/or relation to a permanent resident or citizen of Canada does not qualify them for family class

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Table A2. **The evolution of the Canadian points system**[1]

Assessment factors	Points					
	1967	1978	1986	1993	1993[2]	2002
Education	20	12	12	16	14	25
Specific vocational preparation[3]	10	15	15	18	16	. .
Work experience	. .	8	8	8	7	21
Occupational demand[4]	15	15	10	10	9	. .
Arranged employment	10	10	10	10	9	10
Levels[5]	. .	. .	10	10	9	. .
Knowledge of official languages	10	10	15	15	13	24
Age	10	10	10	10	9	10
Adaptability/personal suitability	15	10	10	10	9	10
Relative in Canada[6]	5	5	. .	5	5	. .
Intended destination in Canada	5	5	. .	. .	. .	. .
Total	100	100	100	112	100	100
Points required for admission	50	50	70	70	. .	75

1. The points indicated are the maximum possible for each factor. Criteria for assigning points within each factor were changed over time. Those for 2002 are shown in Table A3.
2. In this column, the points awarded under the 1993 regime are artificially re-scaled to make them comparable with other years (*i.e.* to add up to 100). The actual maximum number of points under the 1993 regime are listed to the left of this column.
3. Indicates the skill level of the applicant's intended occupation.
4. Points awarded on the basis of the applicant's intended occupation, for the occupations on the General Occupational List (which included occupations considered to be in strong demand). Applicants whose intended occupation was not on the list could qualify only if they had arranged employment.
5. The number of points awarded to all applicants for the "Levels" factor was set by the Minister at a value from zero to 10 points. From time to time, the points allotted to this factor were adjusted upwards or downwards to increase or decrease the number of applicants who would otherwise meet the selection requirements. This factor was introduced for the first time in 1986 and its value was set at 5 points. In 1993 its value was increased to 8 points.
6. Prior to the 1986 regime, up to 5 points were awarded to independent applicants with relatives in Canada. Those who were able to get an undertaking of assistance from a relative in Canada ("Assisted Relatives") were instead awarded from 15 to 30 points (depending upon the relationship of the applicant to the sponsor and the latter's citizenship status), but were not assessed against the factors reflecting ability to become self-sufficient in the short term (*i.e.* Arranged Employment and Knowledge of Official Languages). From 1986 to 1993 all this was replaced by a bonus of 10 points to all applicants sponsored by a relative in Canada who submitted an undertaking of assistance. From 1993 to 2002 the mandatory requirement for an undertaking of assistance was eliminated and 5 points were given to all applicants with relatives in Canada. From 2002, 5 points are awarded under the "Adaptability" factor when either the applicant or the accompanying spouse or common-law partner has relatives in Canada.

Source: Green and Green (1999); Citizenship and Immigration Canada.

immigration, but who get extra points for that (also under the "adaptability" heading) if they apply as skilled workers.

B*usiness immigrants*: they include entrepreneurs, investors and self-employed persons, with their spouses and children. E*ntrepreneurs* are admitted on the condition that they own and manage a qualifying business in Canada for at least one year in the three years following arrival, employing at least one Canadian citizen or permanent resident other than the entrepreneur and his/her dependants. I*nvestors* must have business experience, prove they have a net worth of at least C$800 000 and are required to make a deposit of C$400 000 with the Receiver General of Canada, to be repaid (without interest) after five years. Finally, *self-employed* persons include farmers and some categories of professionals (*e.g.* artists, athletes) and must have the experience and ability to create their own employment and contribute

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Table A3. **Assessment grid for skilled workers under the 2002 regulations**

Assessment factors and criteria for assigning points	Points
Education	**25 maximum**
Ph.D. or Masters and 17 years of full-time study	25
Two Bachelor's degrees or 3-year tertiary diploma/trade certificate/apprenticeship and 15 years of full-time study	22
2-year university degree/tertiary diploma/trade certificate/apprenticeship and 14 years of full-time study	20
1-year university degree/tertiary diploma/trade certificate/apprenticeship and 13 years of full-time study	15
1-year tertiary diploma/trade certificate/apprenticeship and 12 years of full-time study	12
Completed secondary school	5
Knowledge of official languages	**24 maximum**
1st official language	16 maximum
Points per ability[1] according to proficiency level: high 4; moderate 2; basic 1-2; no proficiency 0.	
2nd official language	8 maximum
Points per ability[1] according to proficiency level: high or moderate 2; basic 1-2; no proficiency 0.	
Work experience	**21 maximum**
1 year	15
2 years	17
3 years	19
4 year	21
Age	**10 maximum**
21-49 years at time of application (less 2 points for each year over 49 or below 21)	10
Arranged employment in Canada	**10 maximum**
Permanent employment offer confirmed by HRDC[2]	10
Temporary worker applying from within Canada[3]	10
Adaptability	**10 maximum**
Spouse's or common-law partner's education	3 to 5
Minimum one year full-time authorised work in Canada	5
Minimum two year full-time authorised post-secondary study in Canada	5
Having received points under the Arranged employment in Canada factor	5
Family relationship in Canada	5
Total	**100**
Points required for admission	**75**

1. The four language abilities are: speaking, listening, reading and writing.
2. The Department of Human Resource Development Canada (HRDC) has to confirm that the job offer is genuine, that the wages and working conditions would be sufficient to attract and retain Canadians and that the employment is not seasonal or part-time.
3. To obtain points, a temporary worker applying from within Canada must have obtained HRDC validation or be exempt from the requirement to obtain it (see text of this Annex under "Temporary Workers").

Source: Citizenship and Immigration Canada.

significantly to Canada in certain defined areas. Applicants as business immigrants are also assessed for their skills based on a modified, much less selective variant of the points system used for skilled workers (Table A4); they are given points for relevant business experience as owners or managers.

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Table A4. **Assessment grid for business immigrants under the 2002 regulations**

Assessment factors	Maximum points
Education	25
Knowledge of official languages	24
Relevant work or business experience	35
Age	10
Adaptability	6
Total	**100**
Points required for admission	**35**

Source: Citizenship and Immigration Canada.

Provincial and territorial nominees: Immigrants who meet provincially/territorially specific labour market needs based on agreements between the latter and the federal government. All provinces (except Ontario) and the territory of Yukon have agreements envisaging such nominee programmes. The nominees must meet federal admissibility requirements related to health and security.

Refugees

They include refugees selected from abroad (sponsored by the government, private organisations or groups of citizens) and asylum seekers who have claimed refugee status after landing in Canada and whose claim has been accepted by the Immigration and Refugee Board. The main objective of the refugee protection policy is humanitarian. Although in principle refugees must demonstrate an ability to resettle in Canada, this requirement can be waived in cases where there is an urgent need for protection. The criteria for the determination of refugee status are established in the Immigration Act. In addition to Convention refugees, defined on the basis of the 1951 Geneva Convention, Canada's Immigration Act considers some additional categories admitted for humanitarian reasons. Refugees selected from abroad can belong to any of these categories. Refugee claimants landed in Canada need to be recognised as Convention refugees. To be admitted, all refugees also need to pass medical, security and criminal checks.

Other

They include: live-in caregivers who have moved from temporary to permanent immigrant status (see below); retirees; refugees whose claim has been accepted but have not yet obtained permanent resident status (the Post-Determination Refugee Claimants in Canada); and some refugee claimants admitted on special humanitarian grounds (the Deferred Removal Order Class).

Temporary residents

Temporary workers

Foreign nationals normally need a work permit to work temporarily in Canada. Permits can be issued for up to three years and are renewable. To obtain a work permit an applicant

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needs a job offer validated by Human Resources Development Canada (HRDC). HRDC has to confirm that, although the wage and working conditions offered are standard, the job cannot be filled by a Canadian worker, and that employing a foreign worker will not negatively affect the Canadian economy and labour force. Some categories of workers (*e.g.* business visitors, performing artists, medical students authorised by the Canadian regulatory body) do not need a work permit. Others (professionals, traders and investors entering under NAFTA and other international agreements; some types of entrepreneurs and intra-company transferees; and spouses of foreign students) need the work permit but do not need HRDC confirmation. Simplified authorisation procedures exist for information technology workers, for seasonal agricultural workers from Mexico and the Caribbean, and in Ontario for certain occupations in the manufacturing and construction sectors. A special temporary work permit programme exists for live-in caregivers.

Under the 2002 legislation temporary workers can apply for permanent residence status from within Canada and receive their landed status without leaving the country (previously, this possibility existed only for live-in caregivers). They can obtain extra points under the arranged employment and adaptability factors if they had obtained HRDC validation or were exempt from the requirement to obtain it.

Foreign students

Foreign nationals who intend to study at a post-secondary institution in Canada for more than six months need a study permit. To obtain it they need to have a letter of acceptance from the Canadian educational institution and to prove that they are able to pay the tuition fees and to financially support themselves. Foreign students are allowed to work on-campus while studying, and may be allowed to work off-campus to support themselves in special cases. Their spouses can obtain a work permit. After graduation, students can work in Canada for up to one year (two in New Brunswick) in a field related to their studies.

Refugee claimants

They are persons who arrive in Canada and seek refugee status according to the 1951 Geneva Convention. They can stay in Canada as temporary residents and are allowed to work to support themselves while their claim is being assessed.

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Annex V

Three reports on health-care reform: Romanow, Kirby and Mazankowski

Romanow – Building on Values: The future of health care in Canada

The final report of the *Commission on the Future of Health Care in Canada* was presented to the government in November 2002. It broadly supported the current universal, publicly funded single-payer system for core health services and doctors visits. It also recommended extending the Canada Health Act (CHA) to provide free access to several services that are not currently covered (notably, home care and drugs) and saw no role for private for-profit delivery or for user charges.

It estimated that its recommendations would cost the federal government C$8.5 billion in one-off funding spread over the next three years, plus an increase of at least C$6.5 billion in annual spending (equivalent to a 9 per cent increase in public health expenditure). The key recommendations included:

- *Coverage and delivery*. Reaffirm the five principles of the CHA and expand its coverage to include diagnostic services and priority home care services (post-acute home care, palliative and mental health home care). Consider banning workers' compensation agencies from contracting with private providers to deliver fast-track diagnostic services (such as MRI scans) for potential claimants as it is "contrary to the spirit of medicare". But spend more on diagnostic machinery, initially *via* a dedicated fund, so that waiting lists for such services are dealt with by the public rather than private sector.
- *Funding*. Provide *federal* funding *via* a dedicated ("tagged") transfer to provinces rather than through the CHST block transfer. Increase federal funding and set it five years in advance, with an escalator provision to cover cost increases.
- *User charges*. Services covered under the CHA should remain free of charge.
- *The role of the private sector*. Continue with the regulatory framework that effectively rules out private insurance for services covered under the CHA and consider whether private for-profit providers should be allowed to continue operating.
- *Drugs*. Establish a national drug formulary for evaluating drugs; consider integrating coverage for prescription drugs within medicare but, as this will be expensive, in the meantime provide federal funding for "catastrophic" drug coverage (*i.e.* where a person faces a drug bill for many thousands of dollars).
- *Improve health information* by, for example, establishing personal electronic health records and co-ordinating health technology assessment across the country. A Health Council would act as a nation-wide information and advisory body.
- Change the approach to *primary health care*. The report does not recommend a single best approach, except to say that federal transfers to provinces for primary care should be conditional on reform and progress in various areas.

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The report also had recommendations regarding access to health care in rural communities, Aboriginal health issues, managing waiting lists and global issues (such as not "poaching" physicians from developing countries, and pre-empting any NAFTA or WTO-related healthcare issues).

Kirby – The Health of Canadians: The Federal Role

This Senate report on the federal role in health care, tabled a month before Romanow's, had many similar recommendations. In particular, it re-affirmed the principles of the CHA and recommended additional federal health funding of around C$5 billion per year. Like Romanow, it recommended a national drug formulary, coverage for catastrophic drug bills, expanding the CHA to cover post-acute and palliative home care, federal funding for high-tech diagnostic equipment, encouraging "24/7" health care groups as a way to reform primary care and providing extra "earmarked" federal funding conditional on reforms by provincial governments (rather than being delivered through the CHST block grant). Areas where the Kirby report had a different slant included:

- Creating a progressive earmarked health tax in which the tax rate depends on income.
- Instituting a national maximum waiting time, after which a person can get immediate treatment in another area (including the United States) paid for by the provincial government.
- Devolving more power to regional health authorities so they become the main purchaser of health services on behalf of residents in their region. These powers would include setting physician fees, controlling drug spending and choosing providers.
- Funding hospitals on an output (fee-for-service) basis, rather than through the current input-based system.
- Making the federal government pay half of community hospitals' capital costs in fast-growing areas.
- Providing federal funding for more places in medical schools.

Mazankowski – A Framework for Reform

The Mazankowski report on Alberta's provincial health care system, tabled in 2001, was one of many provincial reviews over the past decade. It differs from the others, and from the Romanow and Kirby reports, by taking a much more market-based approach. It highlighted access problems (especially waiting lists), serious shortages of health professionals and "unsustainable" expenditure growth. It rejected rationing by queuing as a solution, opting instead for broadening of the funding options and a structural reorganisation to increase choice and competition. Recommendations included:

- Increase customer choice.
- Consider new revenue sources, including medical savings accounts, increased health-care premiums, user fees (in some areas such as long-term care), co-payments, deductibles, taxable benefits and supplementary insurance.
- Review (with a view to reducing) what is covered free of charge under the provincial plan.
- Encourage more competition in healthcare delivery by, for example, giving regional health authorities more power and encouraging them to buy services from a wide range of providers, including those from other regions.
- Move away from fee-for-service remuneration for physicians.

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- Implement a maximum 90-day waiting time.
- Split funding from provision.
- Establish multi-year contracts between the provincial government and regional health authorities that set out performance targets to be achieved and budgets to be provided.
- Create a health debit card to track individual use of health-care services.

Source: Romanow (2002), Kirby (2002) and Mazankowski (2001).

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Annex VI

Calendar of main economic events

2001

May

The Bank of Canada lowered its target for the overnight rate by 25 basis points, to 4.5 per cent.

July

An agreement was reached among 180 countries, including Canada, on the implementation the 1997 Kyoto Protocol.

The Bank of Canada lowered its target for the overnight rate by 25 basis points, to 4.25 per cent.

August

The Bank of Canada lowered its target for the overnight rate by 25 basis points, to 4.0 per cent.

September

The Bank of Canada, in parallel with other central banks following the terrorist attacks of 11 September 2001, lowered its target for the overnight rate by 50 basis points, to 3.5 per cent.

October

The Bank of Canada lowered its target for the overnight rate by 75 basis points, to 2.75 per cent.

November

The Bank of Canada lowered its target for the overnight rate by 50 basis points, to 2.25 per cent.

December

In its Budget 2001 the federal government announced additional spending of C$7.7 billion on various security measures (border and air security, policing and intelligence). It forecast

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that balanced budgets would continue to be achieved, despite the economic slowdown, and confirmed that previously announced tax cuts would continue to be implemented.

2002

January

The Bank of Canada lowered its target for the overnight rate by 25 basis points, to 2.0 per cent.

February

The government launched Canada's Innovation Strategy, which sets out long-term goals for creating a strong framework for innovation and to improving work-related skill levels.

April

The Bank of Canada raised its target for the overnight rate by 25 basis points, to 2.25 per cent.

May

Ontario's wholesale and the retail electricity markets were liberalised.

June

The new *Immigration and Refugee Protection Act* came into effect with the related regulations. The new legislation focuses immigrant selection on the types of skills required in a knowledge-based economy, streamlines the procedures for assessing refugee claims, and strengthens border security.

The Bank of Canada raised its target for the overnight rate by 25 basis points, to 2.5 per cent.

July

The Bank of Canada raised its target for the overnight rate by 25 basis points, to 2.75 per cent.

September

New financial services sector legislation took effect.

October

The federal government published its Climate Change Plan, outlining policy options for meeting its commitments to reducing greenhouse gas emissions. The focus was on a range of voluntary and regulatory measures with a limited permit-trading scheme for large emitters.

The federal government presented its Economic and Fiscal Update 2002, showing budget surpluses (after subtracting contingency and prudence reserves) rising gradually from C$1 billion in 2002-03 to C$14.6 billion in 2007-08.

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November

The Final Report of the Romanow Commission on Health Reform was released. It broadly supported the current universal, publicly-funded, single-payer system for core health services and doctors visits. It recommended extending the Canada Health Act to provide free access to home care and catastrophic drug coverage and recommended a substantial increase in federal funding for health care.

A national summit fleshed out the Innovation Strategy, setting out 18 priority policy recommendations.

The Ontario government ended the liberalisation of the retail electricity market, setting a fixed price of 4.3 cents a kilowatt hour for residential and small commercial customers and rebating electricity costs in excess of the new fixed price since the market was liberalised in May.

December

A WTO arbitration panel report ended the Canada-Brazil dispute on the terms of Canada's financing of aircraft exports. The arbitration panel determined that Brazil should be authorised to impose $385 million in countermeasures against Canada, less than Brazil's original request.

Parliament ratified the Kyoto Protocol, under which Canada is committed to reducing greenhouse gas emissions to 6 per cent below 1990 levels during the first commitment period (2008-12).

2003

February

Canada's First Ministers agreed on the outlines of a new health plan, under which the federal government would provide an extra C$35 billion to the provinces over five years. They also made commitments to setting a timetable for expanding coverage to include home care and catastrophic drug bills, and to accelerate primary care reform.

The federal government released its Budget 2003, increasing spending in 2003-04 and 2004-05 by around 0.5 per cent of GDP. Health-care funding accounted for around half of the increase. It also announced that the Canada Health and Social Transfer (CHST) would be split into separate health and social transfers from 1 April 2004. Various tax initiatives were undertaken, including phasing out the corporate capital tax by 2008.

Canada and the United States agreed to the creation of a WTO arbitration panel for the final resolution of the dispute arising from the imposition by the United States of anti-dumping duties on Canadian softwood lumber.

March

The first cases of Severe Acute Respiratory Syndrome (SARS) were reported in Canada.

The Bank of Canada raised its target for the overnight rate by 25 basis points, to 3.0 per cent.

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April

Air Canada filed for protection from creditors.

The House of Commons industry committee recommended abolishing foreign ownership restrictions on telecommunications and cable TV companies.

The Bank of Canada raised its target for the overnight rate by 25 basis points, to 3.25 per cent.

The government closed most of the cod fisheries after a collapse in fish stocks.

May

Canada agreed to comply with a WTO finding that its dairy export practices constituted an export subsidy.

A case of Mad Cow Disease (BSE) was discovered in Alberta. Various trading partners immediately closed their markets for beef and by-products from Canada.

June

The House of Commons heritage committee recommended maintaining existing foreign ownership restrictions on broadcasting and telecommunications companies. It also recommended greater Canadian content in programming and increased financial support for the broadcasting industry.

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